



**TSMC 0.18 μ m Process 1.8-Volt
SAGE-X™ Standard Cell Library
Databook**

September 2003

Release 4.1

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Printed in the United States of America.

DB-SX-TSM003-4.1/18

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Release History

This section contains the release history for the TSMC 0.18 μ m Process 1.8-Volt SAGE-X™ Standard Cell Library Databook.

Part Number	Release Number	Date of Release	Updates
DB-SX-TSM003-1.0/18	1.0	March 2000	• Initial Release
DB-SX-TSM003-2.2/18	2.2	September 2000	• Document updated
DB-SX-TSM003-3.0/18	3.0	July 2001	• A*CSH* cells redesigned
DB-SX-TSM003-3.1/18	3.1	October 2001	• Update adder cells
DB-SX-TSM003-4.0/18	4.0	February 2002	• Library upgrade to EDA package 3
DB-SX-TSM003-4.1/18	4.1	September 2003	• Ktemp derating value updated.

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- International 408-548-3298
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Artisan's SAGE-X™ standard cell library builds upon our SAGE architecture, producing the optimum combination of high-density with high-performance. The cell line-up is derived from extensive customer design, synthesis, and place-and-route benchmark analysis. Library optimization is achieved by carefully matching the library functions and drive strengths to leading synthesis and place-and-route tools, producing superior RTL-to-GDSII results.

How This Book Is Organized

This introduction is organized into three sections:

- **Global Parameters** provides an overview of parameters specific to your SAGE-X library.
- **Special Cells** details the types of special cells you will find included in the library.
- **Reading the Standard Cell Datasheet** describes what you will find in each datasheet.

Datasheets for each cell in this library are provided after the introduction. The datasheets are included in alphabetical order within the following categories:

- Base Cells
- Advanced Arithmetic Cells
- Register File Cells
- Synthesis Optimized Arithmetic Cells

Global Parameters

This section specifies global parameters for the TSMC 0.18μm Process 1.8-Volt SAGE-X™ Standard Cell Library Databook. It covers physical specifications, electrical specifications, derating factors, propagation delay calculation, timing constraints, power calculation, and power-rail strapping.

Physical Specifications

Table 1 shows the physical design specifications of this SAGE-X Standard Cell Library.

Table 1. Physical Specifications

Drawn Gate Length (μm)	0.18
Layers of Metal	4, 5, or 6
Layout Grid (μm)	0.005
Vertical Pin Grid (μm)	0.56
Horizontal Pin Grid (μm)	0.66
Cell Power and Ground Rail Width (μm)	0.8

In this library, all pins are located on the vertical and horizontal pin grids. Most place-and-route tools work more efficiently with all pins on grids, and some tools even require it.

This library also supports designs with 4, 5, or 6 layers of metal. The designer may need to change the design rules in the technology file, because the top-level metal has a greater minimum width and greater minimum spacing requirement. See "Design Rule 0.18 μm LOGIC Salicide 1.8V/3.3V Process" design rule manual. The designer must define these correctly for the place-and-route tool.

Table 2 describes the electrical specifications for this library.

Table 2. Electrical Specifications

Parameter	Minimum	Maximum
DC Supply Voltage (Vdd)	1.62 V	1.98 V
Junction Temperature	-40°C	125°C

Table 3 shows the derating factors for this SAGE-X Standard Cell Library.

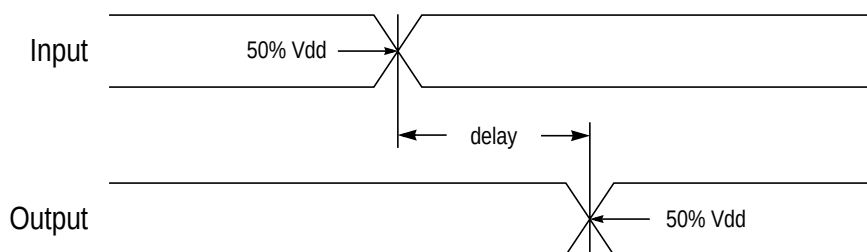
Table 3. Derating Factors

K_{Process} (slow)	1.27
K_{Process} (typical)	1.000 (by definition)
K_{Process} (fast)	0.793
K_{Volt} (1.8V to 1.62V)	-0.709/V
K_{Volt} (1.8V to 1.98V)	-0.492/V
K_{Temp} (25°C to -40°C)	0.00122/°C
K_{Temp} (25°C to 0°C)	0.00124/°C
K_{Temp} (25°C to 125°C)	0.00131/°C

Propagation Delay

The propagation delay through a cell is the sum of the intrinsic delay, the load-dependent delay, and the input-slew dependent delay. Delays are defined as the time interval between the input stimulus crossing 50% of V_{dd} and the output crossing 50% of V_{dd} . Figure 1 illustrates propagation delay.

Figure 1. Propagation Delay



Factors that affect propagation delays include: temperature, supply voltage, process variations, fanout loading, interconnect loading, input-transition time, input-signal polarity, and timing constraints (see below). The timing models provided with this library include the effects of input-transition time on propagation delays. Also, all timing models use a table lookup method to calculate accurate timing. To simplify calculations, the standard cell datasheets provide all timing numbers for an input slew of 0.03ns and a linearized load factor, K_{load} , which is not as accurate as the timing models. All cells have been characterized with a fully populated metal2 (0.66μm horizontal pitch) and metal3 (0.56μm vertical pitch) routing grid across the entire cell layout.

The SAGE-X Standard Cell Library may contain negative propagation delays. Although most third-party verification tools can handle negative propagation delays, some tools will turn negative delays into a zero value.

Derating Factors

Derating factors are coefficients that the typical process characterization data is multiplied by to arrive at timing data that reflects appropriate operating conditions. The derating factor table on page 12 provides derating factors for variations in process case, temperature, and voltage.

Derating factors are derived by averaging the performance of many different cells in the library. A particular combination of cells may perform better or worse than indicated by these derating factors.

Delay Calculation

Using the delay data in the datasheets ($t_{\text{intrinsic}}$, K_{load} , and C_{load}) and the delay derating factors, the estimated total propagation delay is

$$t_{\text{TPD}} = (K_{\text{Process}}) \cdot [1 + (K_{\text{Volt}} \cdot \Delta V_{\text{dd}})] \cdot [1 + (K_{\text{Temp}} \cdot \Delta T)] \cdot t_{\text{typical}}$$

$$t_{\text{typical}} = t_{\text{intrinsic}} + (K_{\text{load}} \cdot C_{\text{load}})$$

where:

t_{TPD} = total propagation delay (ns);

t_{typical} = delay at typical corner—1.8V, 25°C, typical process (ns);

$t_{\text{intrinsic}}$ = delay through the cell when there is no output load (ns);

K_{load} = load delay multiplier (ns/pF);

C_{load} = total output load capacitance (pF);

K_{Process} = process derating factor, where process is slow, typical, or fast;

K_{Volt} = voltage derating factor (/V);

ΔV_{dd} = $V_{\text{dd}} - 1.8\text{V}$;

K_{Temp} = temperature derating factor (/°C);

ΔT = junction temperature — 25°C.

Timing Constraints

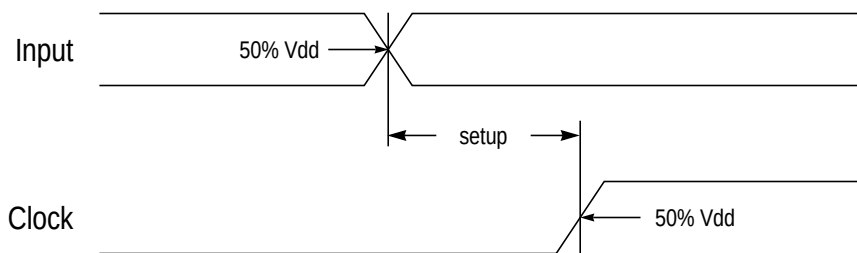
Timing constraints define minimum time intervals during which specific signals must be held steady in order to ensure the correct functioning of any given cell. Timing constraints include: setup time, hold time, recovery time, and minimum pulse width.

The sequential-cell timing models provided with this library include the effects of input-transition time and data-signal and clock-signal polarity on timing constraints. To simplify calculations, the datasheets specify timing constraint values for 0.03ns data slew and 0.03ns clock slew. Other factors that affect timing constraints include temperature, supply voltage, and process case variations. All cells have been characterized with a fully populated metal2 (0.66 μ m horizontal pitch) and metal3 (0.56 μ m vertical pitch) routing grid across the entire cell layout.

Timing constraints can affect propagation delays. The intrinsic delays given in the datasheets are measured with relaxed timing constraints (longer than necessary setup times, hold times, recovery times, and pulse widths). The use of shorter timing constraint intervals may increase delay. Each cell is considered functional as long as the actual delay does not exceed the delay given in the datasheets by more than 10%.

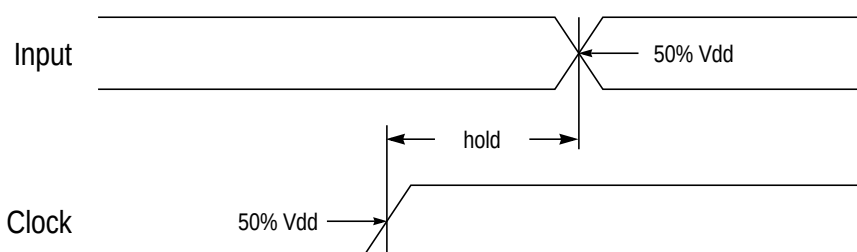
Setup Time

The setup time for a sequential cell is the minimum length of time the data-input signal must remain stable before the active edge of the clock (or other specified signal) to ensure correct functioning of the cell. The cell is considered functional as long as the delay for the output reaching its expected value does not exceed the reference delay (measured with a large setup time) by more than 10%. Setup-constraint values are measured as the interval between the data signal crossing 50% of Vdd and the clock signal crossing 50% of Vdd. For the measurement of setup time, the data input signal is kept stable after the active clock edge for an infinite hold time. Figure 2 illustrates setup time for a positive-edge-triggered sequential cell.

Figure 2. Setup Time

Hold Time

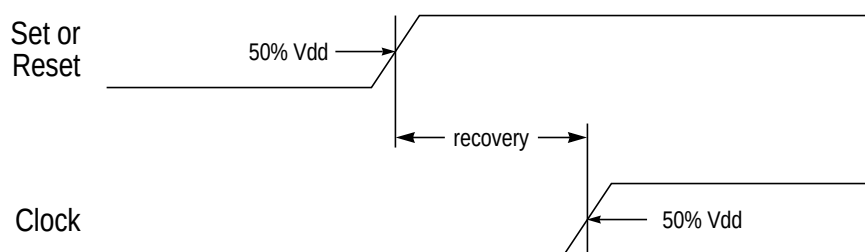
The hold time for a sequential cell is the minimum length of time the data-input signal must remain stable after the active edge of the clock (or other specified signal) to ensure correct functioning of the cell. The cell is considered functional as long as the delay for the output reaching its expected value does not exceed the reference delay (measured with a large hold time) by more than 10%. Hold-constraint values are measured as the interval between the clock signal crossing 50% of Vdd and the data signal crossing 50% of Vdd. For the measurement of hold time, the data input signal is held stable before the active clock edge for an infinite setup time. Figure 3 illustrates hold time for a positive-edge-triggered sequential cell.

Figure 3. Hold Time

Recovery Time

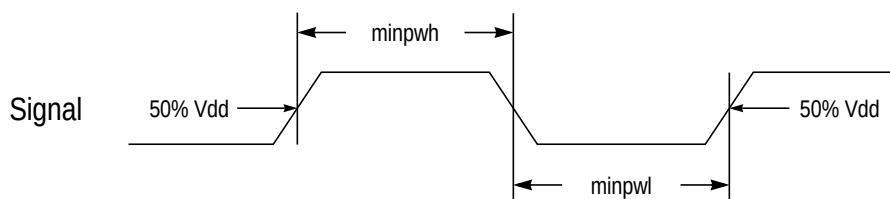
Recovery time for sequential cells is the minimum length of time that the active-low set or reset signal must remain high before the active edge of the clock to ensure correct functioning of the cell. The cell is considered functional as long as the delay for the output reaching its expected value does not exceed the reference delay (measured with a large recovery time) by more than 10%. Recovery constraint values are measured as the interval between the set or reset signal crossing 50% of V_{dd} and the clock signal crossing 50% of V_{dd} . For the measurement of recovery time, the set or reset signal is held stable after the active clock edge for an infinite hold time. Figure 4 illustrates recovery time.

Figure 4. Recovery Time



Minimum Pulse Width

Minimum pulse width is the minimum length of time between the leading and trailing edges of a pulse waveform. Minimum pulse width high (minpwh) is measured as the interval between the rising edge of the signal crossing 50% of V_{dd} and the falling edge of the signal crossing 50% of V_{dd} . Minimum pulse width low (minpwl) is measured as the interval between the falling edge of the signal crossing 50% of V_{dd} and the rising edge of the signal crossing 50% of V_{dd} . Figure 5 illustrates minimum pulse width.

Figure 5. Minimum Pulse Width

Minimum pulse width is defined to be 0.50ns for all set/reset pins (SN, RN) and 0.18ns for all clock pins (G, GN, CK, CKN). These are the largest minimum pulse widths measured from all the cells in the library. An input pulse of shorter duration will produce unpredictable results.

Power Dissipation

The SAGE-X Standard Cell Library is designed to dissipate only AC power, except for the small reverse-bias leakage currents which are normally present in all CMOS circuits.

The power dissipation internal to a cell when a given input switches is primarily dependent upon the cell design itself. The power dissipation of a complete design, or part of a design, using cells from the SAGE-X Library is primarily a function of the switching frequency of the design's internal nets. These nets include the inputs and outputs of each cell and the capacitive load associated with the outputs of each cell.

The SAGE-X Library datasheets contain both an AC power table which documents the internal energy consumption of each cell and a pin capacitance table which gives input-pin capacitance data used to compute output loading. This information, coupled with design-specific information, can be used to estimate the total power dissipation of a cell within a design.

The AC power tables specify the amount of energy consumed within a cell ($\mu\text{W}/\text{MHz}$) when the corresponding pin changes state at 25°C, 1.8V, and typical process. The energy data in the tables were measured for an input slew of 0.03ns and no loading at the outputs.

For combinatorial cells, energy values are provided for only input pins. The energy value for each input pin is the average of energies associated with the input transitions which result in an output transition.

For sequential cells, the energy associated with each input pin is the average energy of those input transitions which *do not* result in an output transition. The energy associated with the output pin of a sequential cell is the average energy of all cases where an output transition is the result of a clock-input transition, minus the energy associated with the clock input pin. In the event that a sequential cell has multiple outputs, all output energy data will be associated with only one output pin.

Power Calculation

Power dissipation is dependent upon the power-supply voltage, frequency of operation, internal capacitance, and output load. The power dissipated by each cell is:

$$P_{avg} = \sum_{n=1}^x (E_{in} \cdot f_{in}) + \sum_{n=1}^y \left(C_{on} \cdot V_{dd}^2 \cdot \frac{1}{2} f_{on} \right) + E_{os} \cdot f_{o1}$$

where:

- P_{avg} = average power (μ W);
- x = number of input pins;
- E_{in} = energy associated with the n th input pin (μ W/MHz);
- f_{in} = frequency at which the n th input pin changes state during the normal operation of the design (MHz);
- y = number of output pins;
- C_{on} = external capacitive loading on the n th output pin, including the capacitance of each input pin connected to the output driver, plus the route wire capacitance, actual or estimated (pF);
- V_{dd} = operating voltage = 1.8V;
- f_{on} = frequency at which the n th output pin changes state during the normal operation of the design (MHz);
- E_{os} = energy associated with the output pin for sequential cells only (μ W/MHz).

The switching frequency of inputs and outputs of a particular cell in a design can be obtained from a gate-level logic simulator, for example Verilog, by applying typical input stimuli and measuring the activity on each node of interest. The total average power for the design can be computed by adding the average power for each cell.

For example, for a DFFXL cell with clock switching at 133MHz, input and output pins switching at 20MHz, an external capacitive loading on the output pin of 0.02pF, and using the power table in the DFF datasheet shown on page 27, the power dissipated by the DFFXL is given by the equation:

$$P_{avg} = \sum_{n=1}^x (E_{in} \cdot f_{in}) + \sum_{n=1}^y \left(C_{on} \cdot Vdd^2 \cdot \frac{1}{2} f_{on} \right) + E_{os} \cdot f_{o1}$$

Substituting,

$$x = 2;$$

$$E_{i1} = 0.0342 \mu\text{W/MHz};$$

$$E_{i2} = 0.0359 \mu\text{W/MHz};$$

$$f_{i1} = 20 \text{ MHz};$$

$$f_{i2} = 133 \text{ MHz};$$

$$y = 2;$$

$$C_{o1} = 0.02 \text{ pF};$$

$$C_{o2} = 0.02 \text{ pF};$$

$$Vdd = 1.8\text{V};$$

$$f_{o1} = 20 \text{ MHz};$$

$$f_{o2} = 20 \text{ MHz};$$

$$E_{os} = 0.0358 \mu\text{W/MHz}$$

we have:

$$P_{avg} = \sum_{n=1}^2 (E_{in} \cdot f_{in}) + \sum_{n=1}^2 \left(C_{on} \cdot Vdd^2 \cdot \frac{1}{2} f_{on} \right) + E_{os} \cdot f_{o1}$$

$$\begin{aligned} P_{avg} &= (E_{i1} \cdot f_{i1}) + (E_{i2} \cdot f_{i2}) \\ &\quad + \left(C_{o1} \cdot 1.8^2 \cdot \frac{1}{2} f_{o1} \right) + \left(C_{o2} \cdot 1.8^2 \cdot \frac{1}{2} f_{o2} \right) \\ &\quad + (E_{os} \cdot f_{o1}) \end{aligned}$$

$$\begin{aligned}
 P_{\text{avg}} &= (0.0342 \bullet 20) + (0.0359 \bullet 133) \\
 &\quad + \left(0.02 \bullet 3.24 \bullet \frac{1}{2}(20)\right) + \left(0.02 \bullet 3.24 \bullet \frac{1}{2}(20)\right) \\
 &\quad + (0.0358 \bullet 20) \\
 P_{\text{avg}} &= 7.54 \text{ } \mu\text{W}
 \end{aligned}$$

Power-Rail Strapping

The designer must determine the required amount of vertical power-rail strapping to satisfy all requirements imposed by the design methodology for a given design. Power-rail strapping should be sized small enough to optimize standard cell height and maximize router efficiency, yet it must be large enough to provide sufficient power to the cells.

The following guidelines provide a rough estimate with many simplifying assumptions. For a given module design, the designer can estimate the amount of vertical power-rail strapping that is required to fulfill electromigration requirements.

Given:

$$\begin{aligned}
 I_{\text{avg}} &= \text{total average current for the module, calculated from previous section (mA);} \\
 w_{\text{m1}} &= \text{VSS/VDD metal1 wire width (}\mu\text{m), see Physical Specifications;} \\
 r &= \text{number of rows in module;} \\
 d_{\text{m1}} &= \text{maximum metal1 current density allowed for the process (mA/}\mu\text{m);} \\
 d_{\text{m2}} &= \text{maximum metal2 current density allowed for the process (mA/}\mu\text{m);} \\
 I_{\text{m1}} &= \text{maximum current that can be supported by all horizontal metal1 wires (mA);} \\
 I_{\text{strap}} &= \text{total current that must be supported by the vertical metal2 strapping (mA);} \\
 w_{\text{m2}} &= \text{metal2 wire width required for vertical strapping (}\mu\text{m);} \\
 c &= \text{minimum number of metal2 straps;}
 \end{aligned}$$

we have:

$$I_{m1} = w_{m1} \bullet r \bullet 2 \bullet d_{m1},$$

where multiplying by 2 assumes metal1 wires are supplied from both ends;

$$I_{strap} = \frac{(I_{avg} - I_{m1})}{2},$$

where dividing by 2 assumes the metal2 vertical strap wires are supplied from both ends;

$$w_{m2} = \frac{I_{strap}}{d_{m2}},$$

It is recommended that the metal2 wire width, w_{m2} , be divided into c equal portions which are spaced equidistant across the module, where

$$c = \frac{I_{avg}}{I_{m1}}, \text{ rounded up to the next integer.}$$

The same consideration must be given to the number of vias used to connect the metal1 and metal2 straps.

Adding Routing Channels

In the SAGE-X Standard Cell Library, each cell is designed with a uniform cell height of 5.0 μ m (i.e., 9 tracks tall with 0.56 μ m per track). The cell layouts allow neighboring rows of cells to share common power or ground rails when cells abut each other at the top and bottom edges of the cell bounding box. The sea-of-cells layout with no channels between rows will usually yield the minimum area. In case of extremely congested areas, the designer may want to separate some rows of cells to increase the number of routing channels within a particular layout region. Because geometries must overlap cell boundaries, a particular spacing between the rows may result in DRC violations for layer spacing. It is recommended that you do not use spacings that cause DRC violations. If these spacings must be used, the DRC violations must be fixed manually by filling the void between the rows with the appropriate layer(s). Table 4 indicates which

DRC violations to expect and how to correct them for a separation between rows of cells.

Table 4. Correcting DRC Violations

Row Separation in Number of Grids	Expected DRC Violations	Action to Correct DRC Violations
0 (Rows Abut)	None	None
1	None	None
2	NWELL space < 0.6 μm	Draw NWELL layer between rows to merge NWELL regions above and below row separation.
3	None	None
4	None	None
5 or more	None	None

Special Cells

This section discusses special cells in the SAGE-X Standard Cell Library.

Antenna-Fix Cell

The library contains an antenna-fix cell which must be inserted manually. However, most place and route tools will indicate which nets require the antenna-fix cell. The 0.18 TSMC antenna effect prevention guideline, "Design Rule 0.18mm LOGIC Salicide 1.8V/3.3V Process," specifies a maximum wire length. During place and route, the router may connect wires to the input gates of cells that are longer than the maximum length allowable by the guideline. The antenna cell can be used in this case to add an optional diode on the net close to the input gates which do not meet the guideline. Pin A on the antenna cell connects to a diode, reverse biased to ground. A diode can be added to either P or N.

Fill Cells

The library contains several FILL cells: FILL1, FILL2, FILL4, FILL8, FILL16, FILL32, FILL64. The number appended to "FILL" in the cell name denotes the width of the cell in tracks.

During place and route, the FILL cells are used to connect power and ground rails across an area containing no cells. The FILL cells are also used to ensure gaps do not occur between well or implant layers which could cause design rule violations. Using wider cells where appropriate reduces the size of the layout database.

Low-Power (XL) Cells

The library contains a wide variety of cells, denoted by an "XL" suffix in the cell name, that are designed specifically for low-power applications. Input capacitance for the XL cells is much lower than that for corresponding X1 (1x drive strength) cells. Because XL cells have been designed for the sole purpose of reducing power consumption, output rise and fall times for these cells may not be equal, and due to the low-drive capability of the XL cells, these cells are not intended for use in critical timing paths, or to drive heavily loaded nets.

TIEHI/LO Cells

The library contains a TIEHI cell and a TIELO cell. The outputs of the TIEHI and TIELO cells are driven through diffusion to provide isolation from the power and ground rails for better ESD protection. The standard cell abstract methodology assumes that the TIEHI and TIELO cells are used to tie off any inputs to power and ground. If these cells are not used and the router is allowed to drop vias on the power rail, DRC errors or shorts may result.

Delay Cells

The library contains delay cells that have the same width. These delay cells allow the designer to adjust a given delay path with a simple cell substitution after place and route.

Reading the Standard Cell Datasheet

Please refer to the datasheet for DFF on pages 26 and 27 for the arrangement of each of the following datasheet sections.

1. Cell Name

The cell name field contains the cell name. The datasheets are presented alphabetically by cell name.

2. Cell Description

The cell description gives the function of the cell. When applicable, the equation(s) for the output pins are provided.

3. Function Table

The function table gives all possible combinations of input and output signals for the cell. Table 5 defines the symbols used in datasheet function tables.

Table 5. Function Table Key

Symbol	Description
0	Logic Low
1	Logic High
	High to Low Transition
	Low to High Transition
x	Don't Care
IL	Illegal/Undefined
Z	High Impedance

4. Logic Symbol

The logic symbol is a graphical representation of the cell, similar to the view in the schematic editor when the cell is instantiated. The symbol shows the name and location of the input and output pins.

5. Cell Size

This cell size table gives the height and width (μm) for each drive strength of the cell.

6. Functional Schematic

The functional schematic provides a functional representation of the cell.

7. Drive Strength

The drive strength of each cell is indicated by an “X” followed by the unit strength.

8. AC Power

The AC power table shows the amount of energy consumed ($\mu\text{W}/\text{MHz}$) within the cell when the corresponding pin changes state. The energy data for each drive strength of the cell are calculated at 25°C, 1.8V, typical process, input slew of 0.03ns, and no external load at the output pins.

9. Delay

The delay table shows the intrinsic delay (ns) which is the delay through the cell when there is no load on the output, and the load multiplier for load dependent delay, K_{load} (ns/pF). The delays and load multiplier for each drive strength of the cell are calculated at 25°C, 1.8V, typical process, and input slew of 0.03ns.

10. Timing Constraints

The timing constraints table shows the timing conditions (ns) required at 25°C, 1.8V, and typical process to maintain proper functionality. Setup constraint values are measured for 0.03ns data slew and 0.03ns clock slew. Hold constraint values are measured for 0.03ns data slew and 0.03ns clock slew. Minimum pulse width is defined to be 0.50ns for all set/reset pins and 0.18ns for all clock pins. These are the largest minimum pulse widths measured from all the cells in the library.

11. Pin Capacitance

The pin capacitance table shows the typical loading at the input pins of the cell (pF) for each drive strength of the cell.



DFF

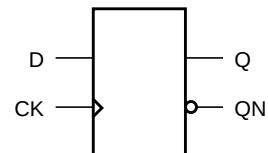
Cell Description

The DFF cell is a positive-edge triggered, static D-type flip-flop.

Functions

D	CK	Q[n+1]	QN[n+1]
0		0	1
1		1	0
x		Q[n]	QN[n]

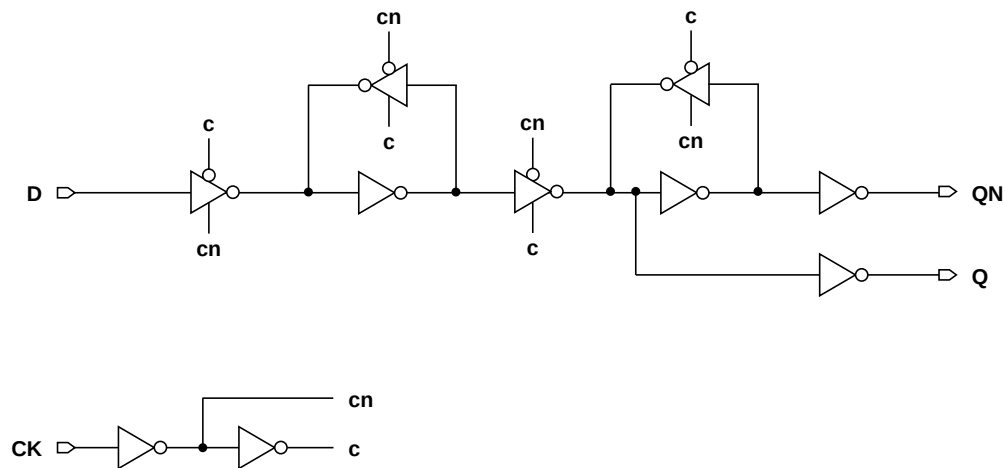
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFXL	5.04	11.22
DFFX1	5.04	11.22
DFFX2	5.04	13.86
DFFX4	5.04	16.50

Functional Schematic





DFF

AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0342	0.0302	0.0412	0.0651
CK	0.0359	0.0329	0.0431	0.0687
Q	0.0358	0.0417	0.0666	0.1127

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0033	0.0020	0.0025	0.0039
CK	0.0021	0.0027	0.0039	0.0062

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2436	0.2104	0.1891	0.1712	6.1380	4.4127	2.2692	1.1474
CK $\rightarrow$ Q $\downarrow$	0.1979	0.1740	0.1601	0.1470	3.5312	2.4480	1.2227	0.6489
CK $\rightarrow$ QN $\uparrow$	0.2676	0.2386	0.2149	0.1932	6.2662	4.5329	2.2658	1.1460
CK $\rightarrow$ QN $\downarrow$	0.3346	0.2971	0.2669	0.2423	3.4466	2.4126	1.2035	0.6415

Timing Constraints at 25°C, 1.8V, Typical Process

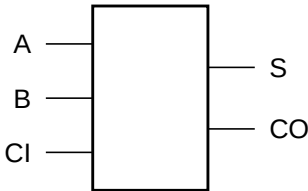
Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0352	0.0742	0.0742	0.0586
	setup $\downarrow$ $\rightarrow$ CK	0.1016	0.1680	0.1680	0.1523
	hold $\uparrow$ $\rightarrow$ CK	-0.0234	-0.0469	-0.0469	-0.0391
	hold $\downarrow$ $\rightarrow$ CK	0.0156	-0.0469	-0.0469	-0.0391
CK	minpwh	0.1238	0.1140	0.0995	0.0898
	minpwl	0.1820	0.1674	0.1577	0.1286

Cell Description

The ADDF cell provides the arithmetic sum (S) and carry out (CO) of two operands (A, B) with carry in (CI). The two outputs (S, CO) are represented by the logic equations:

$S = (A \oplus B \oplus CI)$
 $CO = (A \oplus B) \bullet CI + (A \bullet B)$

Logic Symbol



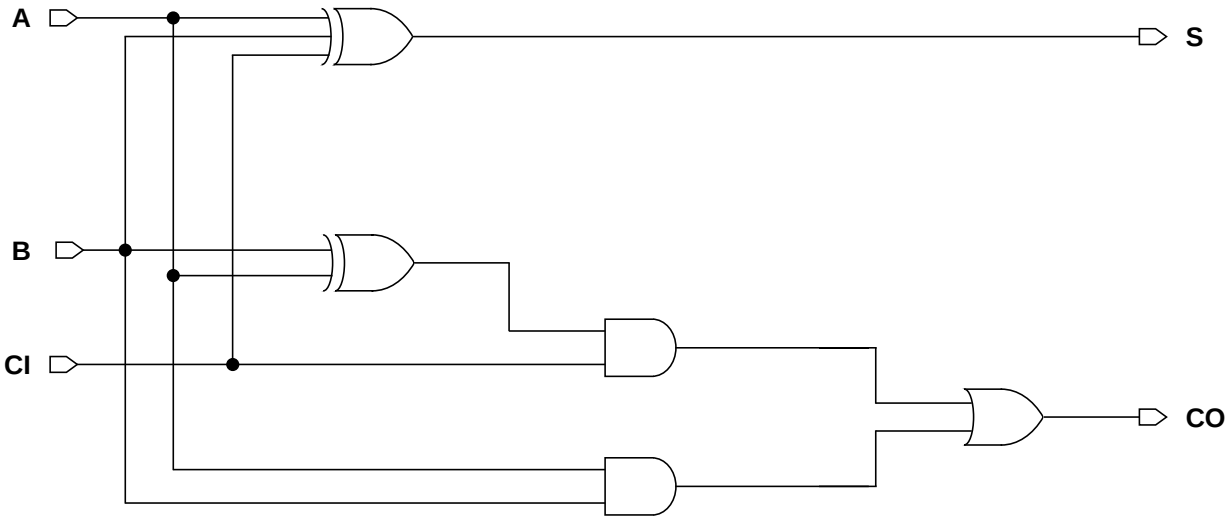
Functions

CI	A	B	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
ADDFXL	5.04	13.86
ADDFX1	5.04	13.86
ADDFX2	5.04	13.86
ADDFX4	5.04	15.18

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.1126	0.1126	0.1326	0.2001
B	0.1450	0.1450	0.1829	0.2582
CI	0.0608	0.0608	0.0835	0.1498

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0072	0.0072	0.0072	0.0072
B	0.0071	0.0071	0.0071	0.0071
CI	0.0065	0.0065	0.0065	0.0065

Delays (25°C, 1.8V, Typical Process)

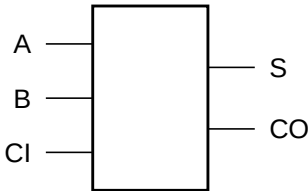
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ S $\uparrow$	0.2179	0.2193	0.2374	0.2873	4.5789	4.5458	2.2458	1.1130
A $\rightarrow$ S $\downarrow$	0.2849	0.2975	0.3354	0.4038	2.8268	2.5634	1.4023	0.7216
B $\rightarrow$ S $\uparrow$	0.2549	0.2566	0.2714	0.3131	4.5882	4.5491	2.2478	1.1155
B $\rightarrow$ S $\downarrow$	0.3284	0.3410	0.3792	0.4476	2.8267	2.5633	1.4023	0.7215
CI $\rightarrow$ S $\uparrow$	0.1742	0.1756	0.2020	0.2653	4.5793	4.5461	2.2468	1.1144
CI $\rightarrow$ S $\downarrow$	0.1428	0.1554	0.1913	0.2568	2.8274	2.5665	1.4122	0.7319
A $\rightarrow$ CO $\uparrow$	0.2726	0.2725	0.3001	0.3630	4.5311	4.5263	2.2360	1.1065
A $\rightarrow$ CO $\downarrow$	0.2591	0.2665	0.3023	0.3604	2.6311	2.4760	1.3619	0.7003
B $\rightarrow$ CO $\uparrow$	0.3156	0.3154	0.3431	0.4059	4.5302	4.5259	2.2358	1.1064
B $\rightarrow$ CO $\downarrow$	0.2822	0.2862	0.3156	0.3642	2.5175	2.4285	1.3290	0.6726
CI $\rightarrow$ CO $\uparrow$	0.1391	0.1400	0.1662	0.2297	4.5634	4.5394	2.2437	1.1116
CI $\rightarrow$ CO $\downarrow$	0.1750	0.1838	0.2205	0.2812	2.6856	2.4978	1.3721	0.7054

Cell Description

The ADDFH cell is a high-speed cell providing the arithmetic sum (S) and carry out (CO) of two operands (A, B) with carry in (CI). The two outputs (S, CO) are represented by the logic equations:

$S = (A \oplus B \oplus CI)$
 $CO = (A \oplus B) \bullet CI + (A \bullet B)$

Logic Symbol



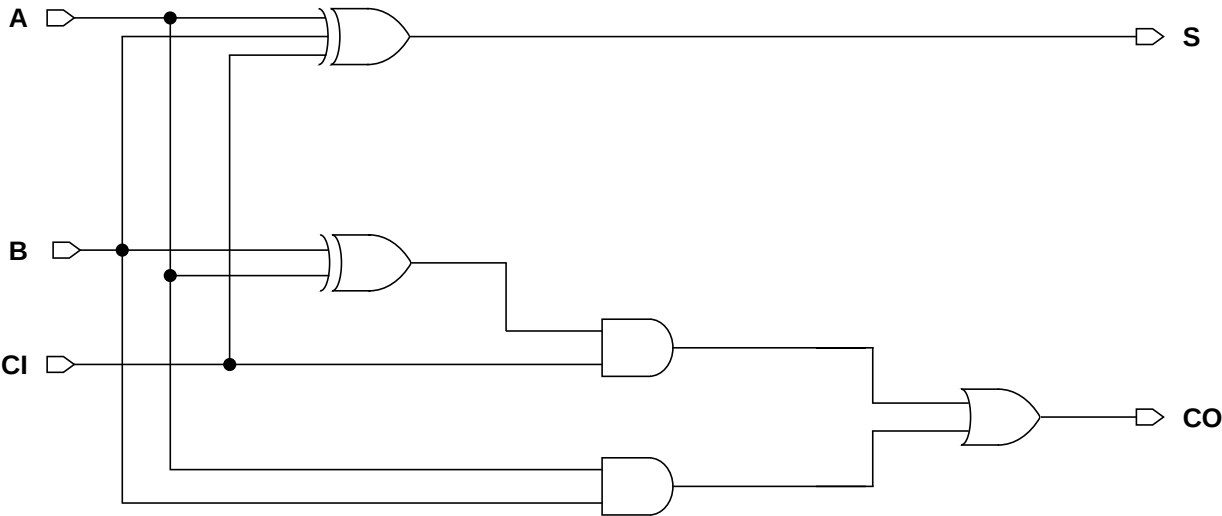
Functions

CI	A	B	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
ADDFHXL	5.04	14.52
ADDFHX1	5.04	15.18
ADDFHX2	5.04	22.44
ADDFHX4	5.04	23.10

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A	0.1118	0.1367	0.2578	0.2957
B	0.0980	0.1236	0.2213	0.2618
CI	0.0611	0.0710	0.1209	0.1619

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0042	0.0063	0.0118	0.0118
B	0.0087	0.0145	0.0261	0.0262
CI	0.0023	0.0044	0.0081	0.0081

Delays (25°C, 1.8V, Typical Process)

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ S $\uparrow$	0.3717	0.2431	0.2348	0.2516	6.2639	4.5315	2.2640	1.1057
A $\rightarrow$ S $\downarrow$	0.3539	0.2693	0.2582	0.2860	3.6127	2.4498	1.2979	0.6592
B $\rightarrow$ S $\uparrow$	0.2591	0.1746	0.1590	0.1795	6.2956	4.5350	2.2654	1.1069
B $\rightarrow$ S $\downarrow$	0.3334	0.2109	0.1948	0.2254	3.6094	2.4491	1.2979	0.6591
CI $\rightarrow$ S $\uparrow$	0.3202	0.1928	0.1694	0.1943	6.2797	4.5353	2.2651	1.1068
CI $\rightarrow$ S $\downarrow$	0.2835	0.1909	0.1685	0.2016	3.6375	2.4560	1.3005	0.6606
A $\rightarrow$ CO $\uparrow$	0.3774	0.2437	0.2350	0.2510	6.2729	4.5307	2.2642	1.1059
A $\rightarrow$ CO $\downarrow$	0.3751	0.2661	0.2580	0.2836	3.7122	2.4468	1.2979	0.6587
B $\rightarrow$ CO $\uparrow$	0.2232	0.1546	0.1443	0.1611	6.3032	4.5322	2.2648	1.1061
B $\rightarrow$ CO $\downarrow$	0.3505	0.1988	0.1855	0.2101	3.7333	2.4267	1.2927	0.6539
CI $\rightarrow$ CO $\uparrow$	0.1465	0.1064	0.0929	0.1050	6.2972	4.5351	2.2651	1.1066
CI $\rightarrow$ CO $\downarrow$	0.2314	0.1549	0.1408	0.1643	3.8361	2.4729	1.3074	0.6628

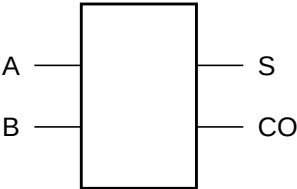
Cell Description

The ADDH cell provides the arithmetic sum (S) and carry out (CO) of two operands (A, B). The two outputs (S, CO) are represented by the logic equations:

$S = (\bar{A} \bullet B) + (A \bullet \bar{B})$

$CO = A \bullet B$

Logic Symbol



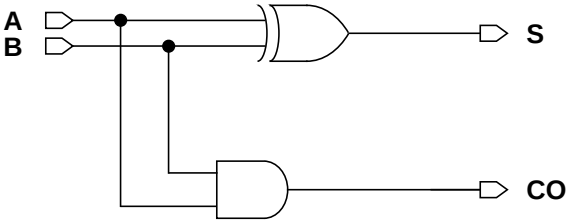
Functions

A	B	S	CO
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
ADDHXL	5.04	7.26
ADDHX1	5.04	7.92
ADDHX2	5.04	11.88
ADDHX4	5.04	18.48

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A	0.0611	0.1066	0.1930	0.3687
B	0.0451	0.0597	0.1037	0.1922

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0048	0.0108	0.0217	0.0423
B	0.0062	0.0086	0.0129	0.0249

Delays (25°C, 1.8V, Typical Process)

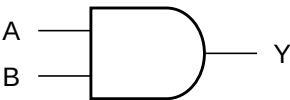
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ S $\uparrow$	0.1351	0.0883	0.0771	0.0734	7.0655	2.8422	1.3802	0.6874
A $\rightarrow$ S $\downarrow$	0.1420	0.0943	0.0854	0.0810	3.9661	1.7254	0.8447	0.4181
B $\rightarrow$ S $\uparrow$	0.0682	0.0589	0.0555	0.0504	6.9671	2.8255	1.3693	0.6818
B $\rightarrow$ S $\downarrow$	0.0853	0.0789	0.0737	0.0674	3.7285	1.6436	0.8072	0.4014
A $\rightarrow$ CO $\uparrow$	0.0778	0.0920	0.0770	0.0771	6.2674	4.5354	2.2094	1.1053
A $\rightarrow$ CO $\downarrow$	0.1029	0.1307	0.1124	0.1101	3.4089	2.5815	1.2864	0.6436
B $\rightarrow$ CO $\uparrow$	0.0750	0.0927	0.0735	0.0734	6.2678	4.5355	2.2084	1.1054
B $\rightarrow$ CO $\downarrow$	0.0931	0.1255	0.1035	0.0994	3.4058	2.5786	1.2856	0.6433

Cell Description

The AND2 cell provides the logical AND of two inputs (A, B). The output (Y) is represented by the logic equation:

$Y = (A \bullet B)$

Logic Symbol



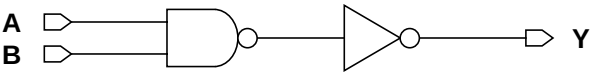
Functions

A	B	Y
0	x	0
x	0	0
1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
AND2XL	5.04	2.64
AND2X1	5.04	2.64
AND2X2	5.04	2.64
AND2X4	5.04	3.30

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0177	0.0190	0.0286	0.0506
B	0.0200	0.0216	0.0339	0.0613

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0021	0.0020	0.0034	0.0060
B	0.0022	0.0021	0.0034	0.0065

Delays at 25°C, 1.8V, Typical Process

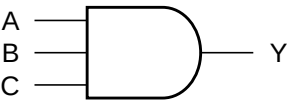
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0841	0.0895	0.0720	0.0689	6.2748	4.5339	2.3525	1.0984
A $\rightarrow$ Y $\downarrow$	0.1006	0.1223	0.0970	0.1069	3.4200	2.4200	1.2829	0.6719
B $\rightarrow$ Y $\uparrow$	0.0923	0.0969	0.0763	0.0742	6.2745	4.5375	2.3524	1.0984
B $\rightarrow$ Y $\downarrow$	0.1126	0.1364	0.1091	0.1231	3.4246	2.4229	1.2842	0.6726

Cell Description

The AND3 cell provides the logical AND of three inputs (A, B, C). The output (Y) is represented by the logic equation:

$Y = (A \bullet B \bullet C)$

Logic Symbol



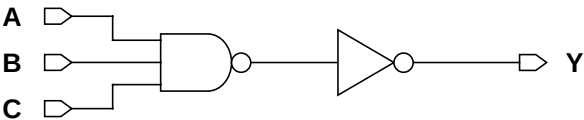
Functions

A	B	C	Y
0	x	x	0
x	0	x	0
x	x	0	0
1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
AND3XL	5.04	3.30
AND3X1	5.04	3.30
AND3X2	5.04	3.30
AND3X4	5.04	3.96

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0187	0.0205	0.0316	0.0528
B	0.0218	0.0235	0.0374	0.0616
C	0.0245	0.0255	0.0420	0.0700

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0027	0.0025	0.0037	0.0061
B	0.0026	0.0024	0.0037	0.0061
C	0.0026	0.0024	0.0041	0.0065

Delays at 25°C, 1.8V, Typical Process

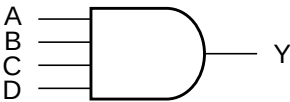
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.1028	0.1150	0.0909	0.0943	6.2840	4.5448	2.2136	1.1213
A $\rightarrow$ Y $\downarrow$	0.1231	0.1426	0.1164	0.1030	3.1825	2.5960	1.2905	0.6344
B $\rightarrow$ Y $\uparrow$	0.1103	0.1229	0.1006	0.1016	6.2843	4.5447	2.2135	1.1214
B $\rightarrow$ Y $\downarrow$	0.1381	0.1573	0.1308	0.1150	3.1899	2.5997	1.2918	0.6350
C $\rightarrow$ Y $\uparrow$	0.1145	0.1275	0.1061	0.1064	6.2845	4.5448	2.2135	1.1214
C $\rightarrow$ Y $\downarrow$	0.1515	0.1711	0.1454	0.1265	3.2001	2.6039	1.2940	0.6359

Cell Description

The AND4 cell provides the logical AND of four inputs (A, B, C, D). The output (Y) is represented by the logic equation:

$Y = (A \bullet B \bullet C \bullet D)$

Logic Symbol



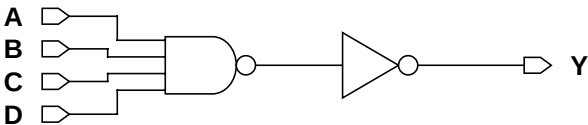
Functions

A	B	C	D	Y
0	x	x	x	0
x	0	x	x	0
x	x	0	x	0
x	x	x	0	0
1	1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
AND4XL	5.04	3.96
AND4X1	5.04	3.96
AND4X2	5.04	3.96
AND4X4	5.04	7.26

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0179	0.0197	0.0333	0.0573
B	0.0208	0.0228	0.0393	0.0691
C	0.0239	0.0259	0.0446	0.0803
D	0.0271	0.0283	0.0491	0.0900

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0026	0.0024	0.0039	0.0069
B	0.0025	0.0022	0.0037	0.0072
C	0.0027	0.0024	0.0038	0.0079
D	0.0027	0.0025	0.0040	0.0086

Delays at 25°C, 1.8V, Typical Process

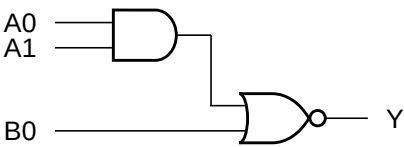
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.1030	0.1254	0.1069	0.0948	6.2963	4.5541	2.3019	1.1304
A $\rightarrow$ Y $\downarrow$	0.1184	0.1375	0.1180	0.1181	3.4528	2.4340	1.2903	0.6464
B $\rightarrow$ Y $\uparrow$	0.1157	0.1385	0.1169	0.1059	6.2964	4.5542	2.3019	1.1296
B $\rightarrow$ Y $\downarrow$	0.1345	0.1529	0.1334	0.1358	3.4608	2.4375	1.2918	0.6472
C $\rightarrow$ Y $\uparrow$	0.1246	0.1471	0.1242	0.1136	6.2964	4.5542	2.3019	1.1296
C $\rightarrow$ Y $\downarrow$	0.1515	0.1684	0.1478	0.1525	3.4711	2.4425	1.2940	0.6485
D $\rightarrow$ Y $\uparrow$	0.1294	0.1526	0.1293	0.1194	6.2948	4.5536	2.3019	1.1296
D $\rightarrow$ Y $\downarrow$	0.1653	0.1825	0.1614	0.1687	3.4852	2.4485	1.2969	0.6503

Cell Description

The AOI21 cell provides the logical inverted OR of one AND group and an additional input. The output (Y) is represented by the logic equation:

$Y = \overline{(A0 \bullet A1) + B0}$

Logic Symbol



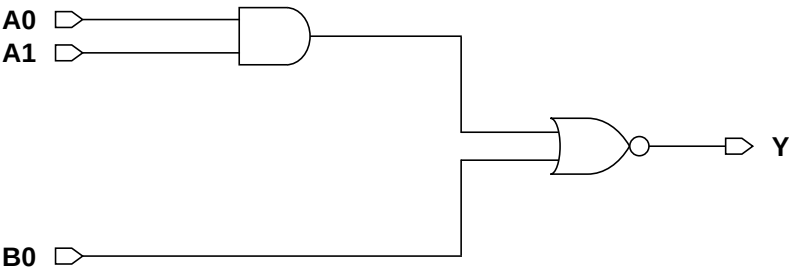
Functions

A0	A1	B0	Y
0	x	0	1
x	0	0	1
x	x	1	0
1	1	x	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI21XL	5.04	2.64
AOI21X1	5.04	2.64
AOI21X2	5.04	4.62
AOI21X4	5.04	6.60

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0	0.0152	0.0214	0.0431	0.0811
A1	0.0191	0.0270	0.0520	0.0992
B0	0.0150	0.0208	0.0413	0.0758

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0033	0.0049	0.0098	0.0179
A1	0.0033	0.0046	0.0091	0.0179
B0	0.0033	0.0046	0.0084	0.0157

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0703	0.0653	0.0638	0.0582	9.4748	6.7344	3.3680	1.7049
A0 $\rightarrow$ Y $\downarrow$	0.0331	0.0329	0.0319	0.0304	3.8677	2.7993	1.4000	0.7243
A1 $\rightarrow$ Y $\uparrow$	0.0862	0.0805	0.0784	0.0739	9.4597	6.7276	3.3640	1.7034
A1 $\rightarrow$ Y $\downarrow$	0.0386	0.0379	0.0366	0.0356	3.8690	2.8009	1.4008	0.7247
B0 $\rightarrow$ Y $\uparrow$	0.0577	0.0587	0.0551	0.0538	9.4730	6.7342	3.3665	1.7046
B0 $\rightarrow$ Y $\downarrow$	0.0216	0.0209	0.0207	0.0194	3.3436	2.3722	1.2683	0.6342

Cell Description

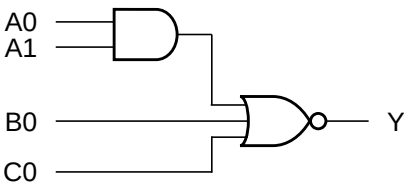
The AOI211 cell provides the logical inverted OR of one AND group and two additional inputs. The output (Y) is represented by the logic equation:

$Y = \overline{(A0 \bullet A1) + B0 + C0}$

Functions

A0	A1	B0	C0	Y
0	x	0	0	1
x	0	0	0	1
x	x	x	1	0
x	x	1	x	0
1	1	x	x	0

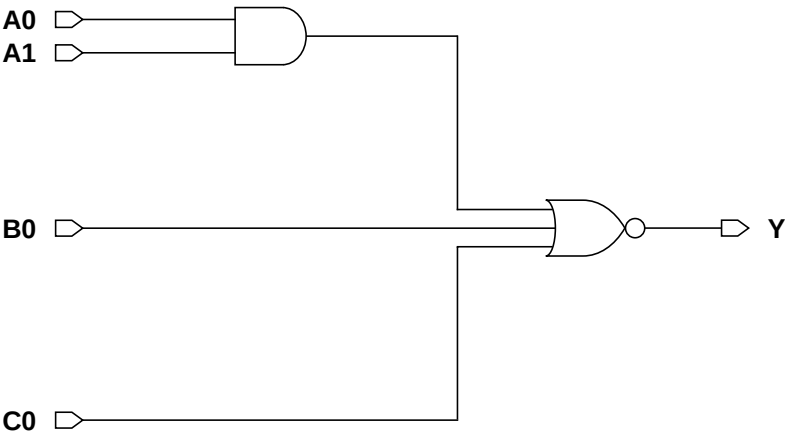
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI211XL	5.04	3.30
AOI211X1	5.04	3.30
AOI211X2	5.04	5.94
AOI211X4	5.04	6.60

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0	0.0229	0.0330	0.0676	0.0775
A1	0.0269	0.0384	0.0780	0.0817
B0	0.0180	0.0256	0.0514	0.0736
C0	0.0221	0.0313	0.0641	0.0774

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0036	0.0051	0.0102	0.0033
A1	0.0036	0.0049	0.0097	0.0033
B0	0.0037	0.0051	0.0091	0.0035
C0	0.0033	0.0046	0.0097	0.0032

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y↑	0.1142	0.1124	0.1108	0.2895	12.2928	8.5519	4.2999	1.1043
A0 → Y↓	0.0409	0.0443	0.0433	0.1738	3.9164	2.8270	1.4133	0.6419
A1 → Y↑	0.1347	0.1321	0.1298	0.3113	12.2743	8.5434	4.2953	1.1043
A1 → Y↓	0.0467	0.0497	0.0480	0.1792	3.9211	2.8311	1.4161	0.6418
B0 → Y↑	0.0821	0.0820	0.0773	0.2609	12.2937	8.5527	4.2986	1.1042
B0 → Y↓	0.0251	0.0249	0.0243	0.1484	3.3469	2.3709	1.2687	0.6415
C0 → Y↑	0.1108	0.1098	0.1098	0.2894	12.2802	8.5465	4.2975	1.1042
C0 → Y↓	0.0320	0.0322	0.0342	0.1551	3.3719	2.3793	1.2738	0.6416

Cell Description

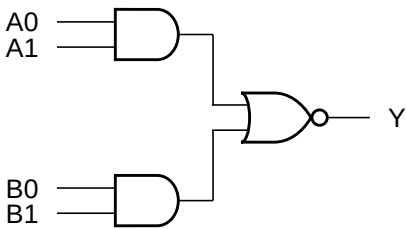
The AOI22 cell provides the logical inverted OR of two AND groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + (B0 \bullet B1)}$$

Functions

A0	A1	B0	B1	Y
0	x	0	x	1
0	x	x	0	1
x	0	0	x	1
x	0	x	0	1
x	x	1	1	0
1	1	x	x	0

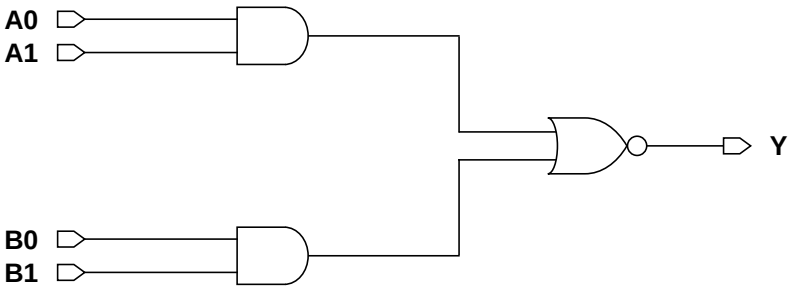
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI22XL	5.04	3.30
AOI22X1	5.04	3.30
AOI22X2	5.04	5.94
AOI22X4	5.04	9.24

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0	0.0171	0.0232	0.0447	0.0882
A1	0.0207	0.0282	0.0550	0.1084
B0	0.0222	0.0301	0.0587	0.1168
B1	0.0258	0.0353	0.0690	0.1364

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0036	0.0048	0.0092	0.0180
A1	0.0036	0.0049	0.0100	0.0185
B0	0.0034	0.0045	0.0090	0.0178
B1	0.0032	0.0045	0.0093	0.0179

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y↑	0.0643	0.0648	0.0644	0.0647	9.4778	6.7361	3.3680	1.6880
A0 → Y↓	0.0260	0.0253	0.0249	0.0252	3.8203	2.7842	1.3935	0.7216
A1 → Y↑	0.0806	0.0811	0.0811	0.0778	9.4680	6.7314	3.3657	1.6392
A1 → Y↓	0.0313	0.0310	0.0306	0.0306	3.8360	2.7906	1.3951	0.7225
B0 → Y↑	0.1071	0.0971	0.0931	0.0893	9.4733	6.7336	3.3669	1.6399
B0 → Y↓	0.0484	0.0455	0.0437	0.0442	3.8671	2.7996	1.3985	0.7238
B1 → Y↑	0.1220	0.1127	0.1087	0.1042	9.4611	6.7282	3.3642	1.6388
B1 → Y↓	0.0535	0.0512	0.0493	0.0498	3.8745	2.8036	1.4005	0.7248

Cell Description

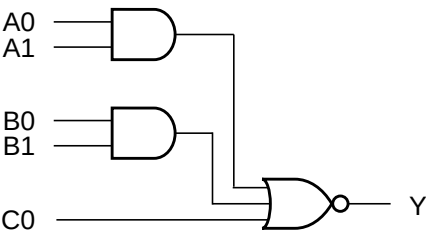
The AOI221 cell provides the logical inverted OR of two AND groups and a third input. The output (Y) is represented by the logic equation:

$Y = \overline{(A0 \bullet A1) + (B0 \bullet B1) + C0}$

Functions

A0	A1	B0	B1	C0	Y
0	x	0	x	0	1
0	x	x	0	0	1
x	0	0	x	0	1
x	0	x	0	0	1
x	x	x	x	1	0
x	x	1	1	x	0
1	1	x	x	x	0

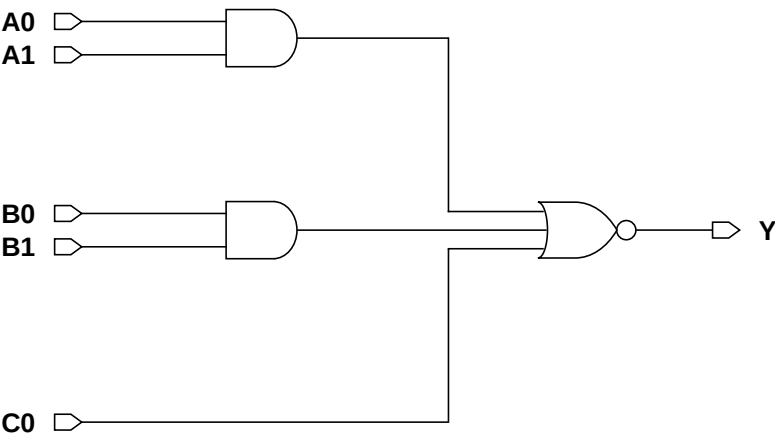
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI221XL	5.04	4.62
AOI221X1	5.04	4.62
AOI221X2	5.04	7.92
AOI221X4	5.04	7.26

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0242	0.0329	0.0628	0.0750
A1	0.0279	0.0380	0.0734	0.0808
B0	0.0305	0.0413	0.0795	0.0807
B1	0.0341	0.0462	0.0901	0.0844
C0	0.0230	0.0311	0.0595	0.0774

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0037	0.0051	0.0098	0.0034
A1	0.0038	0.0052	0.0101	0.0033
B0	0.0037	0.0050	0.0098	0.0033
B1	0.0037	0.0049	0.0099	0.0033
C0	0.0036	0.0049	0.0094	0.0034

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1400	0.1279	0.1196	0.3113	12.0620	8.6004	4.3006	1.1462
A0 $\rightarrow$ Y $\downarrow$	0.0452	0.0429	0.0398	0.1753	3.8748	2.8020	1.4015	0.6414
A1 $\rightarrow$ Y $\uparrow$	0.1602	0.1477	0.1391	0.3322	12.0466	8.5930	4.2964	1.1462
A1 $\rightarrow$ Y $\downarrow$	0.0505	0.0479	0.0444	0.1808	3.8809	2.8053	1.4036	0.6414
B0 $\rightarrow$ Y $\uparrow$	0.1670	0.1513	0.1437	0.3376	12.0614	8.5995	4.3004	1.1462
B0 $\rightarrow$ Y $\downarrow$	0.0513	0.0507	0.0475	0.1853	3.9833	2.8525	1.4272	0.6417
B1 $\rightarrow$ Y $\uparrow$	0.1863	0.1704	0.1636	0.3587	12.0450	8.5920	4.2964	1.1462
B1 $\rightarrow$ Y $\downarrow$	0.0564	0.0559	0.0529	0.1904	3.9892	2.8552	1.4293	0.6417
C0 $\rightarrow$ Y $\uparrow$	0.0946	0.0981	0.0897	0.2697	12.0573	8.5991	4.2995	1.1462
C0 $\rightarrow$ Y $\downarrow$	0.0268	0.0257	0.0239	0.1498	3.3548	2.3736	1.2711	0.6412

Cell Description

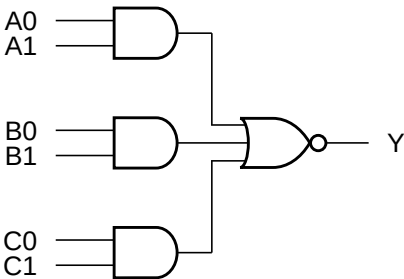
The AOI222 cell provides the logical inverted OR of three AND groups. The output (Y) is represented by the logic equation:

$Y = \overline{(A0 \bullet A1) + (B0 \bullet B1) + (C0 \bullet C1)}$

Functions

A0	A1	B0	B1	C0	C1	Y
0	x	0	x	0	x	1
0	x	0	x	x	0	1
0	x	x	0	0	x	1
0	x	x	0	x	0	1
x	0	0	x	0	x	1
x	0	0	x	x	0	1
x	0	x	0	0	x	1
x	0	x	0	x	0	1
x	x	x	x	1	1	0
x	x	1	1	x	x	0
1	1	x	x	x	x	0

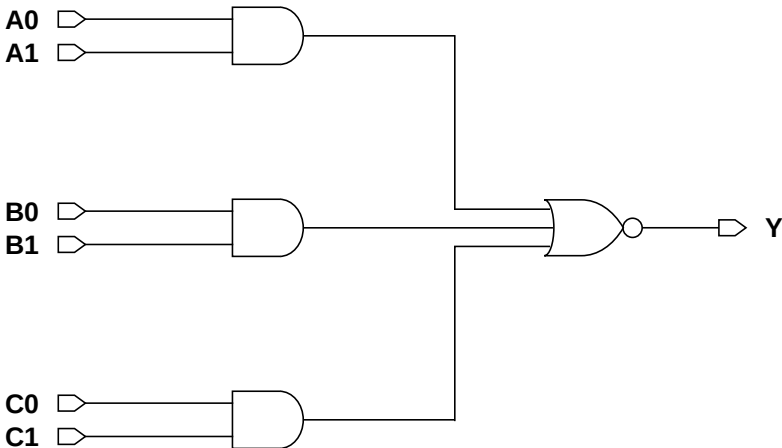
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI222XL	5.04	5.28
AOI222X1	5.04	5.28
AOI222X2	5.04	9.24
AOI222X4	5.04	7.92

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0	0.0258	0.0340	0.0649	0.0761
A1	0.0295	0.0394	0.0750	0.0819
B0	0.0311	0.0423	0.0805	0.0814
B1	0.0349	0.0474	0.0908	0.0868
C0	0.0370	0.0506	0.0974	0.0867
C1	0.0407	0.0558	0.1080	0.0908

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0038	0.0053	0.0103	0.0036
A1	0.0039	0.0053	0.0107	0.0037
B0	0.0036	0.0051	0.0098	0.0033
B1	0.0036	0.0049	0.0101	0.0032
C0	0.0036	0.0050	0.0097	0.0033
C1	0.0036	0.0050	0.0099	0.0033

Delays at 25°C, 1.8V, Typical Process

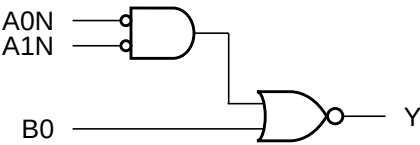
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1062	0.1040	0.1024	0.2678	12.5412	8.6016	4.3012	1.1248
A0 $\rightarrow$ Y $\downarrow$	0.0331	0.0312	0.0301	0.1673	3.8303	2.7903	1.3955	0.6418
A1 $\rightarrow$ Y $\uparrow$	0.1284	0.1259	0.1226	0.2913	12.5313	8.5964	4.2982	1.1248
A1 $\rightarrow$ Y $\downarrow$	0.0388	0.0378	0.0360	0.1721	3.8450	2.7949	1.3979	0.6418
B0 $\rightarrow$ Y $\uparrow$	0.1969	0.1729	0.1629	0.3611	12.5390	8.5995	4.2996	1.1249
B0 $\rightarrow$ Y $\downarrow$	0.0593	0.0546	0.0515	0.1935	3.8676	2.8033	1.4007	0.6420
B1 $\rightarrow$ Y $\uparrow$	0.2185	0.1919	0.1833	0.3820	12.5246	8.5928	4.2966	1.1248
B1 $\rightarrow$ Y $\downarrow$	0.0653	0.0608	0.0583	0.1992	3.8759	2.8035	1.4009	0.6420
C0 $\rightarrow$ Y $\uparrow$	0.2241	0.1969	0.1879	0.3871	12.5393	8.5995	4.2998	1.1248
C0 $\rightarrow$ Y $\downarrow$	0.0715	0.0698	0.0670	0.2078	3.9580	2.8432	1.4209	0.6420
C1 $\rightarrow$ Y $\uparrow$	0.2441	0.2166	0.2073	0.4078	12.5245	8.5934	4.2964	1.1248
C1 $\rightarrow$ Y $\downarrow$	0.0778	0.0760	0.0733	0.2139	3.9529	2.8413	1.4199	0.6420

Cell Description

The AOI2BB1 cell provides the logical inverted OR of one AND group of two inverted inputs (A0N, A1N) and an additional non-inverted input (B0). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{A0N} \bullet \overline{A1N})} + B0$$

Logic Symbol



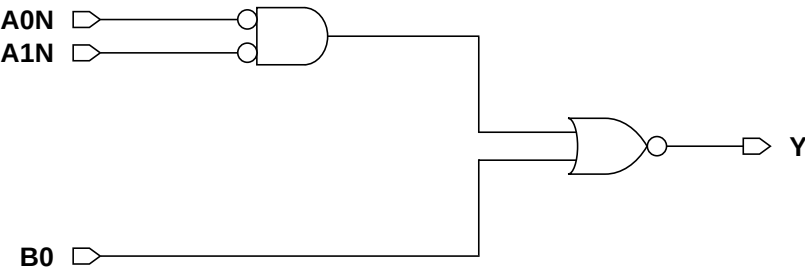
Functions

A0N	A1N	B0	Y
1	x	0	1
x	1	0	1
x	x	1	0
0	0	x	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI2BB1XL	5.04	3.30
AOI2BB1X1	5.04	3.30
AOI2BB1X2	5.04	4.62
AOI2BB1X4	5.04	7.26

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0N	0.0190	0.0224	0.0353	0.0704
A1N	0.0218	0.0251	0.0403	0.0824
B0	0.0145	0.0184	0.0374	0.0718

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0N	0.0026	0.0025	0.0043	0.0081
A1N	0.0024	0.0025	0.0045	0.0083
B0	0.0033	0.0042	0.0088	0.0160

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0N $\rightarrow$ Y $\uparrow$	0.0877	0.0871	0.0761	0.0762	9.4858	6.8461	3.3671	1.6402
A0N $\rightarrow$ Y $\downarrow$	0.1442	0.1617	0.1427	0.1386	3.2325	2.5324	1.3008	0.6492
A1N $\rightarrow$ Y $\uparrow$	0.0914	0.0948	0.0842	0.0842	9.4881	6.8474	3.3677	1.6406
A1N $\rightarrow$ Y $\downarrow$	0.1570	0.1770	0.1575	0.1522	3.2324	2.5323	1.3008	0.6492
B0 $\rightarrow$ Y $\uparrow$	0.0563	0.0533	0.0499	0.0471	9.4631	6.8378	3.3652	1.6388
B0 $\rightarrow$ Y $\downarrow$	0.0234	0.0248	0.0248	0.0240	3.0828	2.4581	1.2720	0.6359

Cell Description

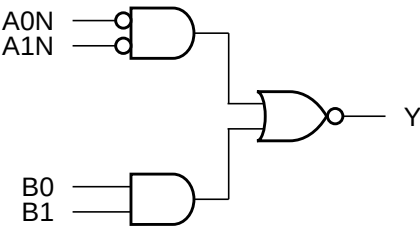
The AOI2BB2 cell provides the logical inverted OR of one AND group of two inverted inputs (A0N, A1N) and one AND group of two non-inverted inputs (B0, B1). The output (Y) is represented by the logic equation:

$$Y = \overline{(A0N \bullet A1N)} + (B0 \bullet B1)$$

Functions

A0N	A1N	B0	B1	Y
1	x	0	x	1
1	x	x	0	1
x	1	0	x	1
x	1	x	0	1
x	x	1	1	0
0	0	x	x	0

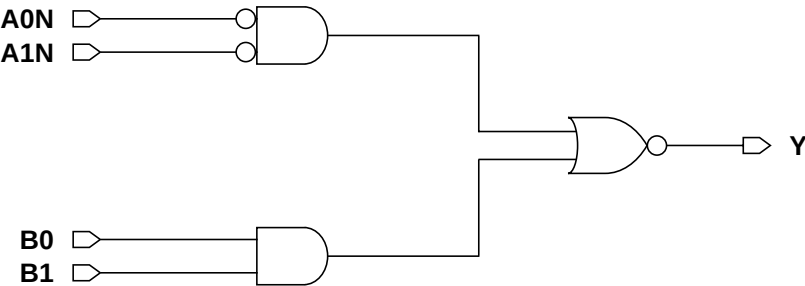
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI2BB2XL	5.04	4.62
AOI2BB2X1	5.04	4.62
AOI2BB2X2	5.04	5.94
AOI2BB2X4	5.04	9.90

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0N	0.0198	0.0217	0.0351	0.0659
A1N	0.0219	0.0244	0.0404	0.0790
B0	0.0161	0.0216	0.0433	0.0820
B1	0.0201	0.0271	0.0536	0.1045

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0N	0.0027	0.0025	0.0044	0.0079
A1N	0.0023	0.0022	0.0043	0.0084
B0	0.0035	0.0047	0.0096	0.0177
B1	0.0034	0.0047	0.0100	0.0183

Delays at 25°C, 1.8V, Typical Process

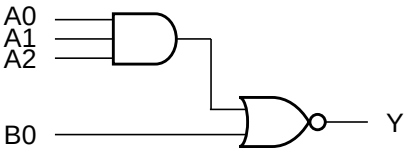
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0N $\rightarrow$ Y $\uparrow$	0.0820	0.0798	0.0730	0.0717	7.0716	5.0199	2.5087	1.2227
A0N $\rightarrow$ Y $\downarrow$	0.1538	0.1591	0.1449	0.1319	3.5140	2.4476	1.3387	0.6470
A1N $\rightarrow$ Y $\uparrow$	0.0849	0.0832	0.0805	0.0807	7.0743	5.0212	2.5096	1.2231
A1N $\rightarrow$ Y $\downarrow$	0.1639	0.1700	0.1582	0.1463	3.5139	2.4476	1.3387	0.6470
B0 $\rightarrow$ Y $\uparrow$	0.0712	0.0671	0.0619	0.0595	9.4766	6.7343	3.3677	1.6401
B0 $\rightarrow$ Y $\downarrow$	0.0334	0.0329	0.0306	0.0314	3.8582	2.8005	1.3986	0.7243
B1 $\rightarrow$ Y $\uparrow$	0.0868	0.0830	0.0774	0.0749	9.4596	6.7277	3.3641	1.6387
B1 $\rightarrow$ Y $\downarrow$	0.0387	0.0385	0.0358	0.0368	3.8592	2.8014	1.3994	0.7247

Cell Description

The AOI31 cell provides the logical inverted OR of one AND group and an additional input. The output (Y) is represented by the logic equation:

$Y = \overline{(A0 \bullet A1 \bullet A2) + B0}$

Logic Symbol



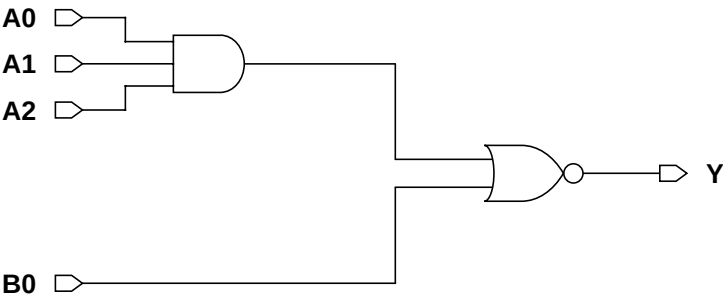
Functions

A0	A1	A2	B0	Y
0	x	x	0	1
x	0	x	0	1
x	x	0	0	1
x	x	x	1	0
1	1	1	x	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI31XL	5.04	3.30
AOI31X1	5.04	3.30
AOI31X2	5.04	5.94
AOI31X4	5.04	5.94

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0	0.0166	0.0227	0.0485	0.0698
A1	0.0211	0.0293	0.0602	0.0761
A2	0.0251	0.0350	0.0720	0.0799
B0	0.0203	0.0282	0.0567	0.0756

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0037	0.0049	0.0112	0.0035
A1	0.0036	0.0050	0.0102	0.0033
A2	0.0035	0.0048	0.0099	0.0033
B0	0.0033	0.0045	0.0085	0.0033

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y↑	0.0739	0.0704	0.0736	0.2192	9.4820	6.7373	3.3696	1.1041
A0 → Y↓	0.0378	0.0373	0.0388	0.1729	4.4283	3.1747	1.5894	0.6417
A1 → Y↑	0.0932	0.0908	0.0918	0.2399	9.4668	6.7313	3.3660	1.1041
A1 → Y↓	0.0463	0.0461	0.0459	0.1792	4.4301	3.1763	1.5909	0.6418
A2 → Y↑	0.1099	0.1078	0.1097	0.2589	9.4724	6.7347	3.3679	1.1041
A2 → Y↓	0.0498	0.0497	0.0501	0.1827	4.4300	3.1762	1.5909	0.6417
B0 → Y↑	0.0823	0.0828	0.0788	0.2339	9.4859	6.7415	3.3706	1.1041
B0 → Y↓	0.0216	0.0206	0.0210	0.1476	3.3537	2.3795	1.2718	0.6414

Cell Description

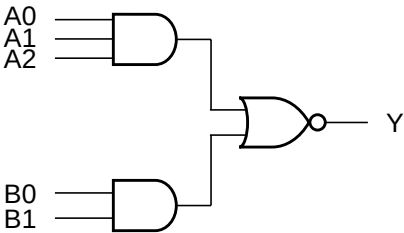
The AOI32 cell provides the logical inverted OR of two AND groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1 \bullet A2) + (B0 \bullet B1)}$$

Functions

A0	A1	A2	B0	B1	Y
0	x	x	0	x	1
0	x	x	x	0	1
x	0	x	0	x	1
x	0	x	x	0	1
x	x	0	0	x	1
x	x	0	x	0	1
x	x	x	1	1	0
1	1	1	x	x	0

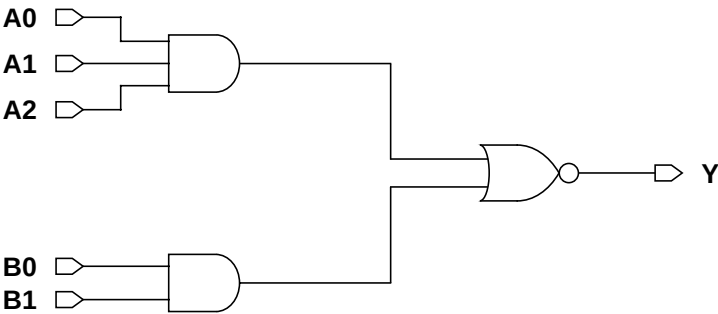
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI32XL	5.04	4.62
AOI32X1	5.04	4.62
AOI32X2	5.04	7.26
AOI32X4	5.04	6.60

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0238	0.0317	0.0629	0.0757
A1	0.0283	0.0380	0.0752	0.0812
A2	0.0322	0.0437	0.0873	0.0850
B0	0.0226	0.0307	0.0602	0.0771
B1	0.0263	0.0357	0.0702	0.0798

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0037	0.0050	0.0097	0.0034
A1	0.0037	0.0049	0.0102	0.0034
A2	0.0035	0.0048	0.0105	0.0032
B0	0.0036	0.0047	0.0094	0.0035
B1	0.0038	0.0048	0.0098	0.0035

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1089	0.1049	0.1024	0.2523	9.4767	6.7360	3.3683	1.1042
A0 $\rightarrow$ Y $\downarrow$	0.0542	0.0536	0.0522	0.1867	4.4255	3.1749	1.5873	0.6275
A1 $\rightarrow$ Y $\uparrow$	0.1284	0.1247	0.1221	0.2735	9.4655	6.7309	3.3657	1.1042
A1 $\rightarrow$ Y $\downarrow$	0.0631	0.0624	0.0609	0.1962	4.4329	3.1787	1.5893	0.6275
A2 $\rightarrow$ Y $\uparrow$	0.1451	0.1420	0.1402	0.2921	9.4703	6.7339	3.3671	1.1041
A2 $\rightarrow$ Y $\downarrow$	0.0665	0.0661	0.0653	0.1997	4.4330	3.1787	1.5893	0.6275
B0 $\rightarrow$ Y $\uparrow$	0.0907	0.0898	0.0879	0.2374	9.4892	6.7424	3.3716	1.1041
B0 $\rightarrow$ Y $\downarrow$	0.0269	0.0255	0.0247	0.1615	3.8347	2.7941	1.3970	0.6274
B1 $\rightarrow$ Y $\uparrow$	0.1077	0.1060	0.1041	0.2548	9.4793	6.7375	3.3690	1.1041
B1 $\rightarrow$ Y $\downarrow$	0.0325	0.0315	0.0306	0.1658	3.8503	2.7987	1.3994	0.6274

Cell Description

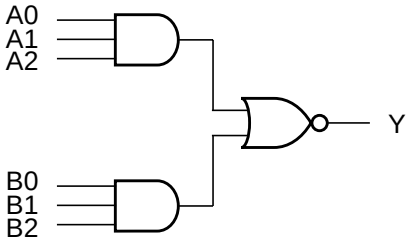
The AOI33 cell provides the logical inverted OR of two AND groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1 \bullet A2) + (B0 \bullet B1 \bullet B2)}$$

Functions

A0	A1	A2	B0	B1	B2	Y
0	x	x	0	x	x	1
0	x	x	x	0	x	1
0	x	x	x	x	0	1
x	0	x	0	x	x	1
x	0	x	x	0	x	1
x	0	x	x	x	0	1
x	x	0	0	x	x	1
x	x	0	x	0	x	1
x	x	0	x	x	0	1
x	x	x	1	1	1	0
1	1	1	x	x	x	0

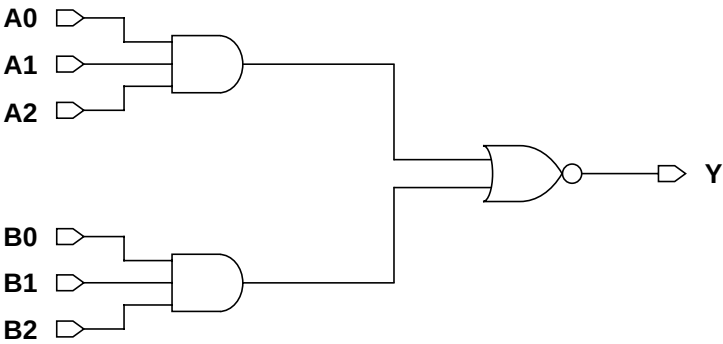
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI33XL	5.04	5.28
AOI33X1	5.04	5.28
AOI33X2	5.04	8.58
AOI33X4	5.04	7.26

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0	0.0240	0.0336	0.0660	0.0768
A1	0.0282	0.0395	0.0775	0.0815
A2	0.0322	0.0451	0.0897	0.0865
B0	0.0302	0.0427	0.0849	0.0836
B1	0.0346	0.0489	0.0967	0.0883
B2	0.0385	0.0546	0.1091	0.0920

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0037	0.0051	0.0100	0.0035
A1	0.0037	0.0052	0.0105	0.0035
A2	0.0038	0.0052	0.0114	0.0037
B0	0.0036	0.0050	0.0096	0.0033
B1	0.0036	0.0050	0.0101	0.0033
B2	0.0035	0.0049	0.0107	0.0032

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1040	0.1047	0.0984	0.2525	9.4877	6.7423	3.3714	1.1042
A0 $\rightarrow$ Y $\downarrow$	0.0344	0.0342	0.0322	0.1808	4.4074	3.1685	1.6142	0.6420
A1 $\rightarrow$ Y $\uparrow$	0.1238	0.1251	0.1177	0.2733	9.4785	6.7380	3.3689	1.1041
A1 $\rightarrow$ Y $\downarrow$	0.0437	0.0437	0.0411	0.1882	4.4154	3.1724	1.6162	0.6420
A2 $\rightarrow$ Y $\uparrow$	0.1425	0.1433	0.1382	0.2950	9.4822	6.7398	3.3704	1.1041
A2 $\rightarrow$ Y $\downarrow$	0.0481	0.0478	0.0464	0.1932	4.4159	3.1725	1.6166	0.6420
B0 $\rightarrow$ Y $\uparrow$	0.1450	0.1457	0.1409	0.2991	9.4845	6.7407	3.3706	1.1042
B0 $\rightarrow$ Y $\downarrow$	0.0680	0.0693	0.0682	0.2088	4.4241	3.1763	1.6189	0.6419
B1 $\rightarrow$ Y $\uparrow$	0.1647	0.1656	0.1601	0.3199	9.4745	6.7359	3.3682	1.1041
B1 $\rightarrow$ Y $\downarrow$	0.0778	0.0791	0.0778	0.2185	4.4244	3.1764	1.6189	0.6418
B2 $\rightarrow$ Y $\uparrow$	0.1816	0.1831	0.1788	0.3383	9.4776	6.7379	3.3692	1.1041
B2 $\rightarrow$ Y $\downarrow$	0.0813	0.0828	0.0829	0.2218	4.4243	3.1763	1.6189	0.6418

Cell Description

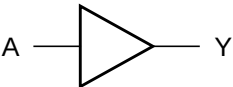
The BUF cell provides the logical buffer of a single input (A). The output (Y) is represented by the logic equation:

$Y = A$

Functions

A	Y
0	0
1	1

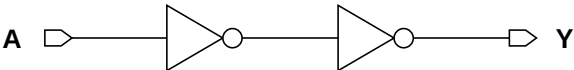
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
BUFXL	5.04	2.64
BUFX1	5.04	2.64
BUFX2	5.04	2.64
BUFX3	5.04	2.64
BUFX4	5.04	3.30
BUFX8	5.04	5.94
BUFX12	5.04	6.60
BUFX16	5.04	8.58
BUFX20	5.04	10.56

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0147	0.0169	0.0283	0.0382	0.0508	0.0946	0.1414	0.1901	0.2437

Pin Capacitance

Pin	Capacitance (pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0023	0.0023	0.0034	0.0046	0.0060	0.0111	0.0164	0.0216	0.0270

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	0.0575	0.0618	0.0610	0.0590	0.0592	0.0544	0.0551	0.0561	0.0551
A $\rightarrow$ Y $\downarrow$	0.0826	0.0926	0.0911	0.0889	0.0876	0.0813	0.0815	0.0816	0.0791

Description	K_{load} (ns/pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	6.2560	4.5278	2.2620	1.5104	1.1315	0.5518	0.3679	0.2759	0.2207
A $\rightarrow$ Y $\downarrow$	3.3931	2.3978	1.2812	0.8359	0.6413	0.3202	0.2135	0.1589	0.1280

Cell Description

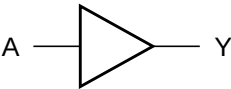
The CLKBUF cell provides the logical buffer of a single input (A), with balanced delays for clock signals. The output (Y) is represented by the logic equation:

$Y = A$

Functions

A	Y
0	0
1	1

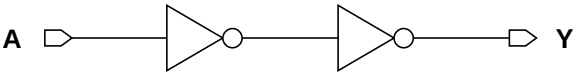
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
CLKBUFXL	5.04	2.64
CLKBUFX1	5.04	2.64
CLKBUFX2	5.04	2.64
CLKBUFX3	5.04	2.64
CLKBUFX4	5.04	3.30
CLKBUFX8	5.04	4.62
CLKBUFX12	5.04	10.56
CLKBUFX16	5.04	12.54
CLKBUFX20	5.04	15.84

Functional Schematic



AC Power

Pin	Power (μ W/MHz)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0154	0.0208	0.0258	0.0314	0.0398	0.0754	0.1887	0.2507	0.3213

Pin Capacitance

Pin	Capacitance (pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0021	0.0035	0.0031	0.0034	0.0042	0.0079	0.0191	0.0229	0.0315

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	0.0544	0.0632	0.0715	0.0805	0.0782	0.0727	0.0730	0.0761	0.0713
A $\rightarrow$ Y $\downarrow$	0.1221	0.0661	0.0845	0.0922	0.0915	0.0858	0.0840	0.0875	0.0817

Description	K_{load} (ns/pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	4.0976	4.1719	2.2621	1.5634	1.1321	0.5521	0.2214	0.1703	0.1383
A $\rightarrow$ Y $\downarrow$	4.6737	4.6217	2.2482	1.3957	1.0677	0.5241	0.2132	0.1638	0.1279

Cell Description

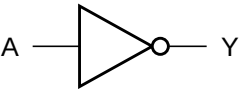
The CLKINV cell provides the logical inversion of a single input (A), with balanced delays for clock signals. The output (Y) is represented by the logic equation:

$Y = \overline{A}$

Functions

A	Y
0	1
1	0

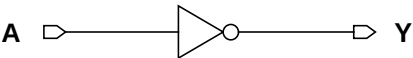
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
CLKINVXL	5.04	1.98
CLKINVX1	5.04	1.98
CLKINVX2	5.04	1.98
CLKINVX3	5.04	1.98
CLKINVX4	5.04	2.64
CLKINVX8	5.04	3.96
CLKINVX12	5.04	12.54
CLKINVX16	5.04	16.50
CLKINVX20	5.04	19.14

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0092	0.0106	0.0189	0.0257	0.0344	0.0707	0.2369	0.3154	0.3896

Pin Capacitance

Pin	Capacitance (pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0027	0.0032	0.0059	0.0086	0.0111	0.0230	0.0068	0.0088	0.0107

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	0.0268	0.0247	0.0194	0.0184	0.0179	0.0182	0.1486	0.1486	0.1479
A $\rightarrow$ Y $\downarrow$	0.0233	0.0213	0.0183	0.0163	0.0162	0.0168	0.1462	0.1486	0.1490

Description	K_{load} (ns/pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	5.7442	4.5230	2.2616	1.5139	1.1226	0.5510	0.2215	0.1661	0.1331
A $\rightarrow$ Y $\downarrow$	4.9202	3.8768	2.2393	1.3871	1.0613	0.5335	0.2193	0.1645	0.1316

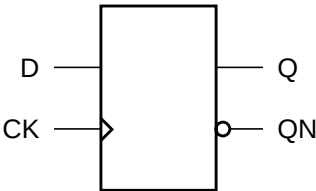
Cell Description

The DFF cell is a positive-edge triggered, static D-type flip-flop.

Function Table

D	CK	Q[n+1]	QN[n+1]
0		0	1
1		1	0
x		Q[n]	QN[n]

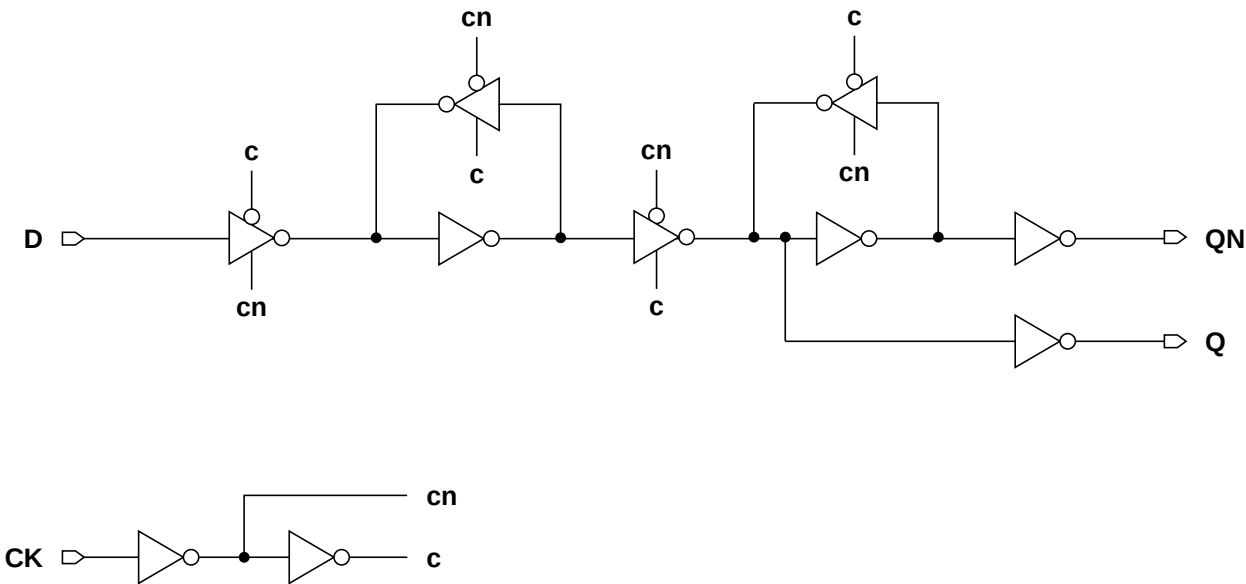
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFXL	5.04	11.22
DFFX1	5.04	11.22
DFFX2	5.04	13.86
DFFX4	5.04	16.50

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0342	0.0302	0.0412	0.0651
CK	0.0359	0.0329	0.0431	0.0687
Q	0.0358	0.0417	0.0666	0.1127

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0033	0.0020	0.0025	0.0039
CK	0.0021	0.0027	0.0039	0.0062

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2436	0.2104	0.1891	0.1712	6.1380	4.4127	2.2692	1.1474
CK $\rightarrow$ Q $\downarrow$	0.1979	0.1740	0.1601	0.1470	3.5312	2.4480	1.2227	0.6489
CK $\rightarrow$ QN $\uparrow$	0.2676	0.2386	0.2149	0.1932	6.2662	4.5329	2.2658	1.1460
CK $\rightarrow$ QN $\downarrow$	0.3346	0.2971	0.2669	0.2423	3.4466	2.4126	1.2035	0.6415

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0352	0.0742	0.0742	0.0586
	setup $\downarrow$ $\rightarrow$ CK	0.1016	0.1680	0.1680	0.1523
	hold $\uparrow$ $\rightarrow$ CK	-0.0234	-0.0469	-0.0469	-0.0391
	hold $\downarrow$ $\rightarrow$ CK	0.0156	-0.0469	-0.0469	-0.0391
CK	minpwh	0.1238	0.1140	0.0995	0.0898
	minpwl	0.1820	0.1674	0.1577	0.1286

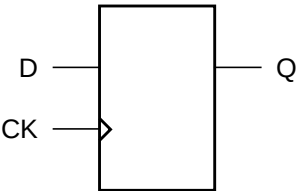
Cell Description

The DFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop. The cell has a single output (Q) and fast clock-to-out path.

Functions

D	CK	Q[n+1]
0		0
1		1
x		Q[n]

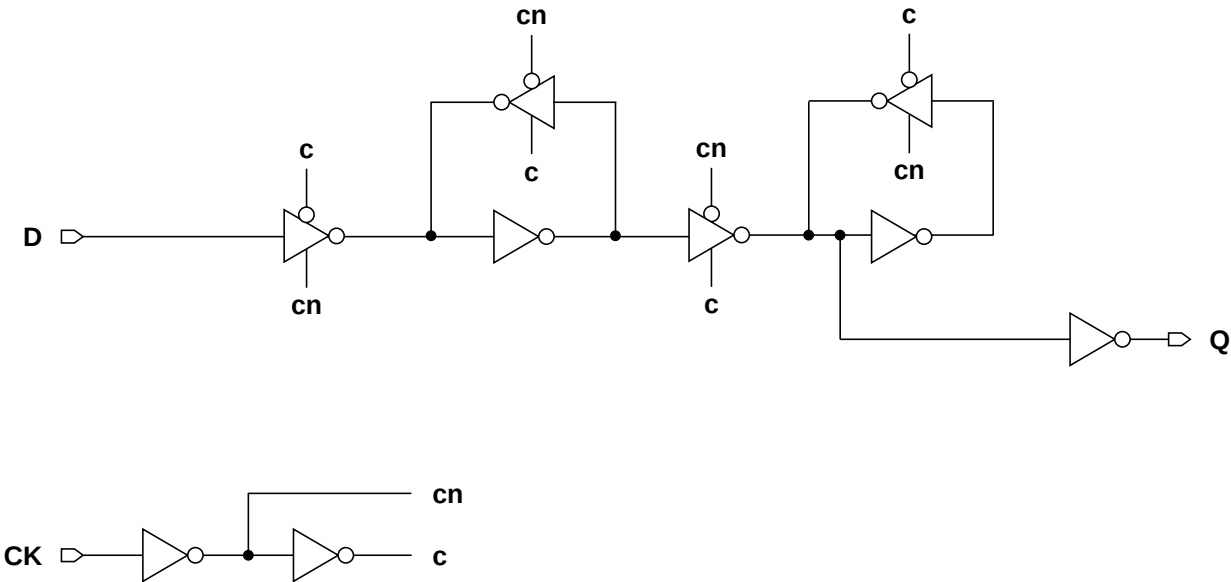
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFHQXL	5.04	10.56
DFFHQX1	5.04	10.56
DFFHQX2	5.04	13.20
DFFHQX4	5.04	14.52

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0382	0.0396	0.0548	0.0769
CK	0.0356	0.0340	0.0400	0.0518
Q	0.0248	0.0268	0.0367	0.0530

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0035	0.0022	0.0023	0.0031
CK	0.0023	0.0031	0.0038	0.0054

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2233	0.1824	0.1745	0.1571	6.2753	4.4089	2.2672	1.1252
CK $\rightarrow$ Q $\downarrow$	0.2270	0.1706	0.1627	0.1454	3.6601	2.5636	1.2812	0.6482

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0508	0.1016	0.0977	0.0859
	setup $\downarrow$ $\rightarrow$ CK	0.1523	0.2109	0.2148	0.1953
	hold $\uparrow$ $\rightarrow$ CK	-0.0156	-0.0508	-0.0508	-0.0430
	hold $\downarrow$ $\rightarrow$ CK	0.0117	-0.0469	-0.0469	-0.0430
CK	minpwh	0.1238	0.1043	0.0995	0.0898
	minpwl	0.1626	0.1674	0.1577	0.1286

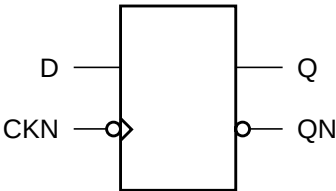
Cell Description

The DFFN cell is a negative-edge triggered, static D-type flip-flop.

Function

D	CKN	Q[n+1]	QN[n+1]
0		0	1
1		1	0
x		Q[n]	QN[n]

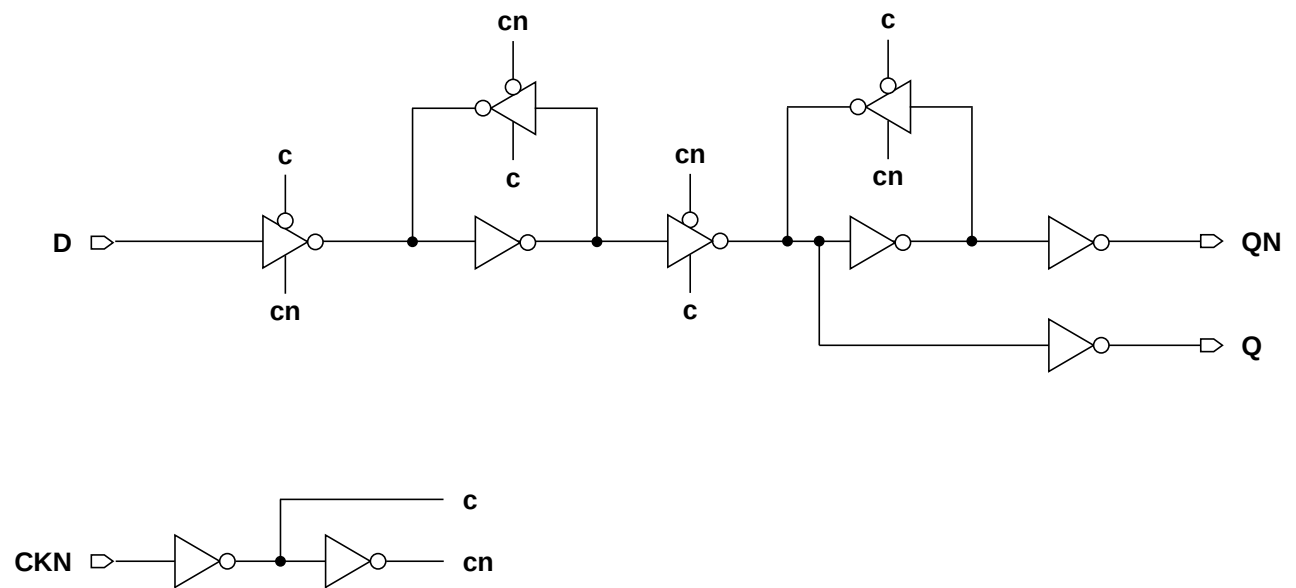
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFNXL	5.04	11.22
DFFNX1	5.04	11.22
DFFNX2	5.04	13.86
DFFNX4	5.04	15.84

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0338	0.0306	0.0413	0.0611
CKN	0.0362	0.0389	0.0537	0.0833
Q	0.0369	0.0442	0.0731	0.1195

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0030	0.0018	0.0022	0.0029
CKN	0.0023	0.0030	0.0038	0.0063

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CKN $\rightarrow$ Q $\uparrow$	0.1916	0.1533	0.1390	0.1202	6.1488	4.4153	2.2705	1.1854
CKN $\rightarrow$ Q $\downarrow$	0.3383	0.2882	0.2588	0.2217	3.5456	2.5240	1.2608	0.6492
CKN $\rightarrow$ QN $\uparrow$	0.3996	0.3518	0.3129	0.2756	6.2689	4.5328	2.2657	1.1839
CKN $\rightarrow$ QN $\downarrow$	0.2767	0.2408	0.2171	0.1939	3.4415	2.4131	1.2036	0.6423

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CKN	0.0430	0.0859	0.0859	0.0859
	setup $\downarrow$ $\rightarrow$ CKN	0.0664	0.1406	0.1328	0.1328
	hold $\uparrow$ $\rightarrow$ CKN	0.0938	0.0391	0.0352	0.0234
	hold $\downarrow$ $\rightarrow$ CKN	-0.0508	-0.1094	-0.0977	-0.0977
CKN	minpwl	0.1577	0.1286	0.1189	0.0995
	minpwh	0.1335	0.1577	0.1383	0.1238

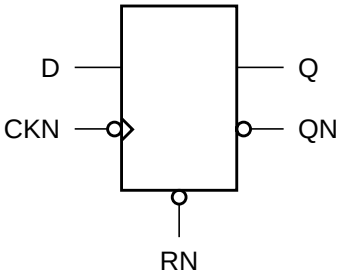
Cell Description

The DFFNR cell is a negative-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN).

Function

RN	D	CKN	Q[n+1]	QN[n+1]
0	x	x	0	1
1	0		0	1
1	1		1	0
1	x		Q[n]	QN[n]

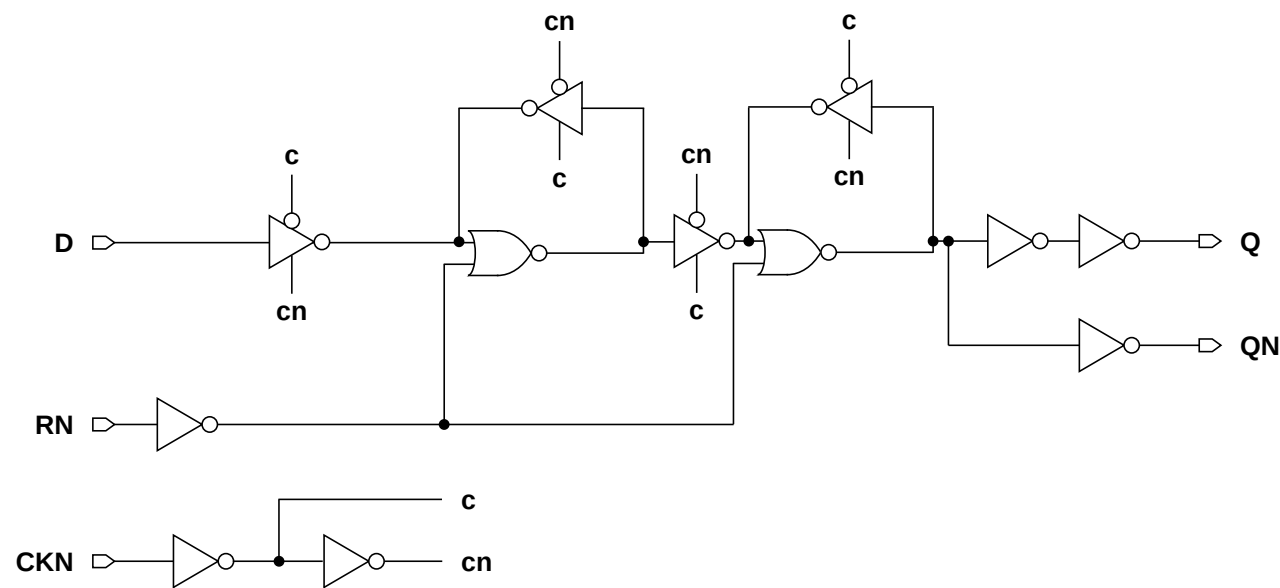
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFNRXL	5.04	15.84
DFFNRX1	5.04	15.18
DFFNRX2	5.04	17.16
DFFNRX4	5.04	20.46

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0359	0.0287	0.0334	0.0437
CKN	0.0377	0.0358	0.0437	0.0563
RN	0.0157	0.0174	0.0215	0.0326
Q	0.0445	0.0534	0.0830	0.1420

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0034	0.0021	0.0021	0.0029
CKN	0.0025	0.0029	0.0030	0.0039
RN	0.0022	0.0025	0.0033	0.0057

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CKN $\rightarrow$ Q $\uparrow$	0.3902	0.3444	0.3313	0.3040	6.2624	4.5307	2.2077	1.1318
CKN $\rightarrow$ Q $\downarrow$	0.4709	0.4475	0.4480	0.3948	3.4021	2.4023	1.3197	0.6467
RN $\rightarrow$ Q $\downarrow$	0.2638	0.2582	0.2475	0.2250	3.4024	2.4022	1.3197	0.6467
CKN $\rightarrow$ QN $\uparrow$	0.4144	0.3793	0.3691	0.3215	6.2586	4.5294	2.2089	1.1324
CKN $\rightarrow$ QN $\downarrow$	0.3555	0.3013	0.2828	0.2503	3.6306	2.4662	1.3448	0.6582
RN $\rightarrow$ QN $\uparrow$	0.2074	0.1908	0.1692	0.1523	6.2686	4.5327	2.2105	1.1334

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CKN	0.0391	0.0820	0.1172	0.0938
	setup $\downarrow$ $\rightarrow$ CKN	0.0742	0.1445	0.1602	0.1250
	hold $\uparrow$ $\rightarrow$ CKN	0.1055	0.0391	0.0352	0.0352
	hold $\downarrow$ $\rightarrow$ CKN	-0.0469	-0.1016	-0.1094	-0.0859
CKN	minpwl	0.1820	0.1529	0.1383	0.1189
	minpwh	0.1432	0.1577	0.1820	0.1480
RN	minpwl	0.2062	0.2014	0.2305	0.3469
	recovery	0.0430	0.0781	0.1016	0.0820
	removal	0.1211	0.0625	0.0664	0.0703

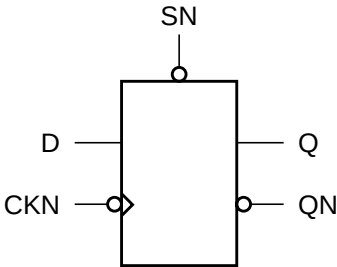
Cell Description

The DFFNS cell is a negative-edge triggered, static D-type flip-flop with asynchronous active-low set (SN).

Function

SN	D	CKN	Q[n+1]	QN[n+1]
0	x	x	1	0
1	0		0	1
1	1		1	0
1	x		Q[n]	QN[n]

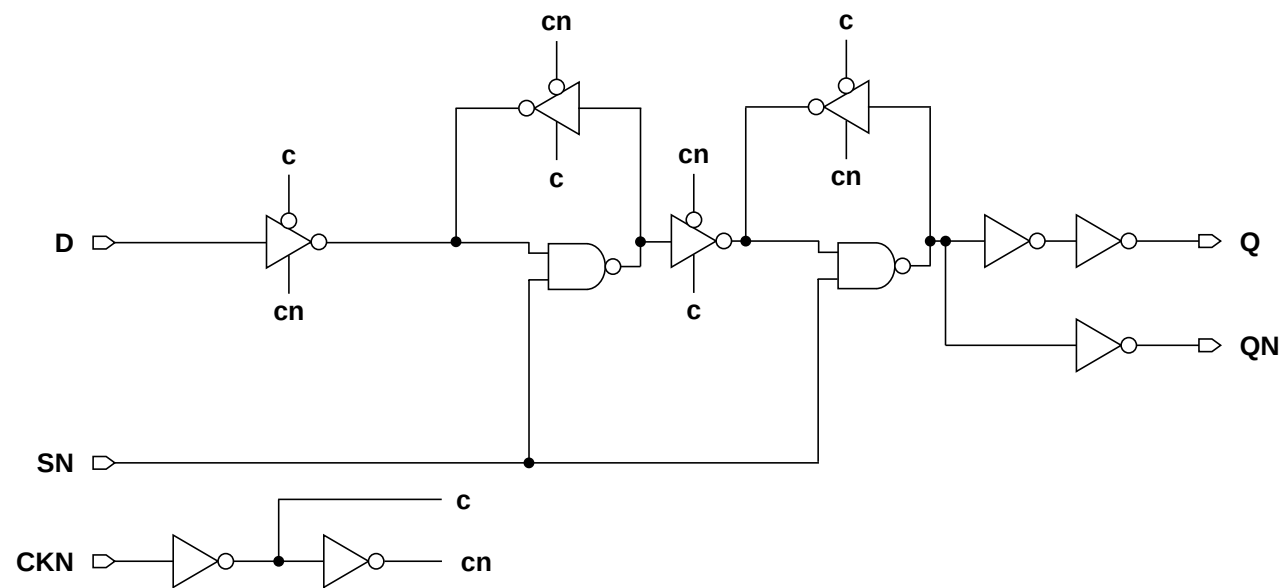
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFNSXL	5.04	13.86
DFFNSX1	5.04	13.86
DFFNSX2	5.04	13.86
DFFNSX4	5.04	19.14

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0349	0.0270	0.0305	0.0397
CKN	0.0354	0.0342	0.0377	0.0499
SN	0.0066	0.0074	0.0103	0.0173
Q	0.0420	0.0494	0.0788	0.1359

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0030	0.0018	0.0018	0.0021
CKN	0.0021	0.0027	0.0029	0.0035
SN	0.0048	0.0053	0.0071	0.0122

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CKN $\rightarrow$ Q $\uparrow$	0.3155	0.2871	0.2823	0.2708	6.2648	4.5307	2.2911	1.1246
CKN $\rightarrow$ Q $\downarrow$	0.4409	0.4169	0.4142	0.3957	3.3986	2.3990	1.2826	0.6417
SN $\rightarrow$ Q $\uparrow$	0.1741	0.1596	0.1624	0.1562	6.2642	4.5308	2.2911	1.1246
CKN $\rightarrow$ QN $\uparrow$	0.3906	0.3546	0.3401	0.3225	6.2742	4.5338	2.2952	1.1267
CKN $\rightarrow$ QN $\downarrow$	0.2825	0.2452	0.2326	0.2201	3.4714	2.4104	1.2878	0.6440
SN $\rightarrow$ QN $\downarrow$	0.1411	0.1179	0.1131	0.1057	3.4571	2.4107	1.2902	0.6464

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CKN	0.0195	0.0703	0.0703	0.0703
	setup $\downarrow$ $\rightarrow$ CKN	0.0742	0.1406	0.1484	0.1328
	hold $\uparrow$ $\rightarrow$ CKN	0.0977	0.0352	0.0391	0.0391
	hold $\downarrow$ $\rightarrow$ CKN	-0.0586	-0.1094	-0.1172	-0.1094
CKN	minpwl	0.1577	0.1383	0.1286	0.1238
	minpwh	0.1286	0.1480	0.1529	0.1383
SN	minpwl	0.1383	0.1238	0.1286	0.1577
	recovery	-0.0977	-0.0469	-0.0469	-0.0469
	removal	0.1055	0.0547	0.0547	0.0586

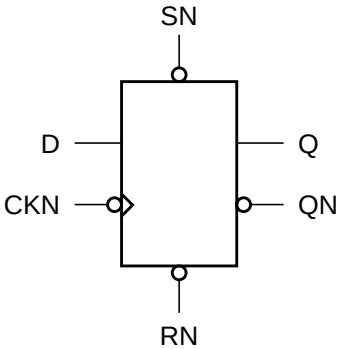
Cell Description

The DFFNSR cell is a negative-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN) and set (SN), and set dominating reset.

Functions

RN	SN	D	CKN	Q[n+1]	QN[n+1]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0
1	1	0		0	1
1	1	1		1	0
1	1	x		Q[n]	QN[n]

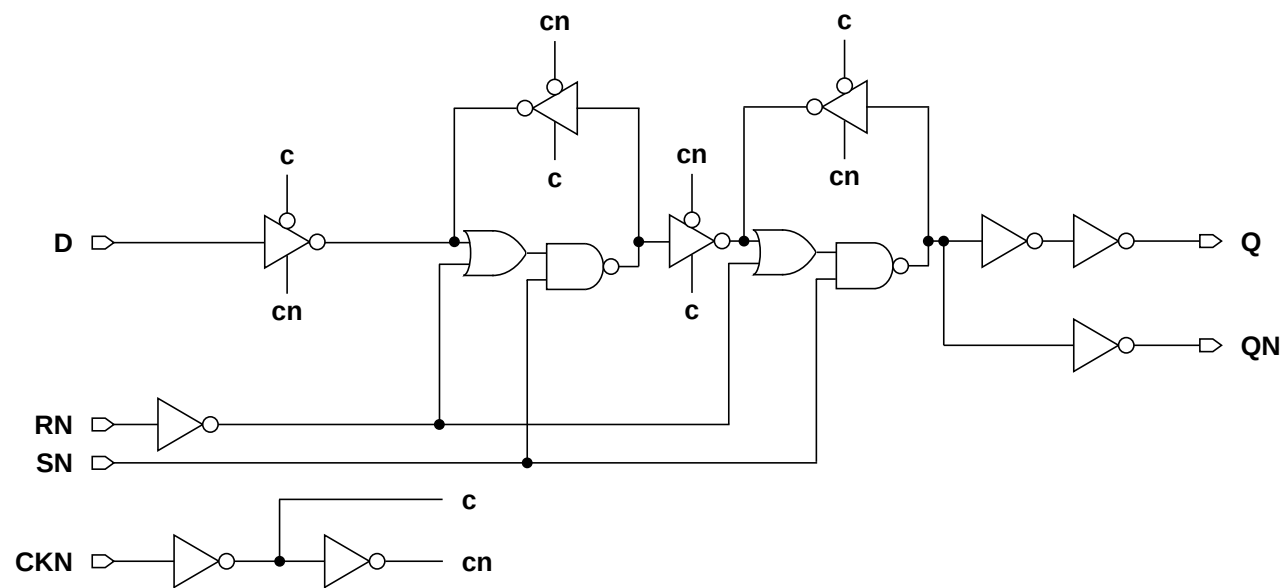
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFNSRXL	5.04	17.16
DFFNSRX1	5.04	16.50
DFFNSRX2	5.04	17.16
DFFNSRX4	5.04	23.10

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0414	0.0334	0.0373	0.0534
CKN	0.0377	0.0373	0.0419	0.0600
SN	0.0084	0.0086	0.0117	0.0182
RN	0.0176	0.0193	0.0244	0.0410
Q	0.0473	0.0537	0.0860	0.1491

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0033	0.0022	0.0021	0.0033
CKN	0.0022	0.0027	0.0029	0.0041
SN	0.0062	0.0066	0.0089	0.0153
RN	0.0025	0.0027	0.0038	0.0059

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CKN $\rightarrow$ Q $\uparrow$	0.3945	0.3479	0.3301	0.3152	6.2603	4.5292	2.2913	1.1317
CKN $\rightarrow$ Q $\downarrow$	0.4778	0.4518	0.4308	0.4007	3.1374	2.4048	1.2834	0.6418
SN $\rightarrow$ Q $\uparrow$	0.2039	0.1881	0.1809	0.1721	6.2603	4.5291	2.2912	1.1317
SN $\rightarrow$ Q $\downarrow$	0.2034	0.1858	0.1789	0.1712	3.1384	2.4051	1.2833	0.6417
RN $\rightarrow$ Q $\downarrow$	0.2962	0.2752	0.2551	0.2442	3.1383	2.4052	1.2833	0.6418
CKN $\rightarrow$ QN $\uparrow$	0.4121	0.3749	0.3556	0.3276	6.2867	4.5372	2.2954	1.1344
CKN $\rightarrow$ QN $\downarrow$	0.3509	0.2982	0.2768	0.2658	3.7062	2.4779	1.3105	0.6578
SN $\rightarrow$ QN $\uparrow$	0.1382	0.1098	0.1044	0.0985	6.3111	4.5480	2.3011	1.1380
SN $\rightarrow$ QN $\downarrow$	0.1584	0.1365	0.1271	0.1226	3.5037	2.4300	1.2953	0.6519
RN $\rightarrow$ QN $\uparrow$	0.2307	0.1990	0.1804	0.1712	6.3086	4.5475	2.3009	1.1378

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CKN	0.0703	0.1602	0.1367	0.1094
	setup $\downarrow$ $\rightarrow$ CKN	0.1094	0.1836	0.1953	0.1445
	hold $\uparrow$ $\rightarrow$ CKN	0.0781	0.0117	0.0156	0.0312
	hold $\downarrow$ $\rightarrow$ CKN	-0.0781	-0.1250	-0.1445	-0.1016
CKN	minpwl	0.1674	0.1480	0.1335	0.1189
	minpwh	0.1383	0.1723	0.1723	0.1335
SN	minpwl	0.1577	0.1432	0.1432	0.1868
	recovery	-0.0820	-0.0391	-0.0430	-0.0469
	removal	0.0898	0.0469	0.0508	0.0586
RN	minpwl	0.2111	0.1965	0.2305	0.3663
	recovery	0.0703	0.1328	0.0977	0.1016
	removal	0.0898	0.0195	0.0508	0.0469

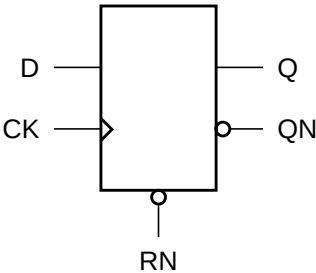
Cell Description

The DFFR cell is a positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN).

Functions

RN	D	CK	Q[n+1]	QN[n+1]
0	x	x	0	1
1	0		0	1
1	1		1	0
1	x		Q[n]	QN[n]

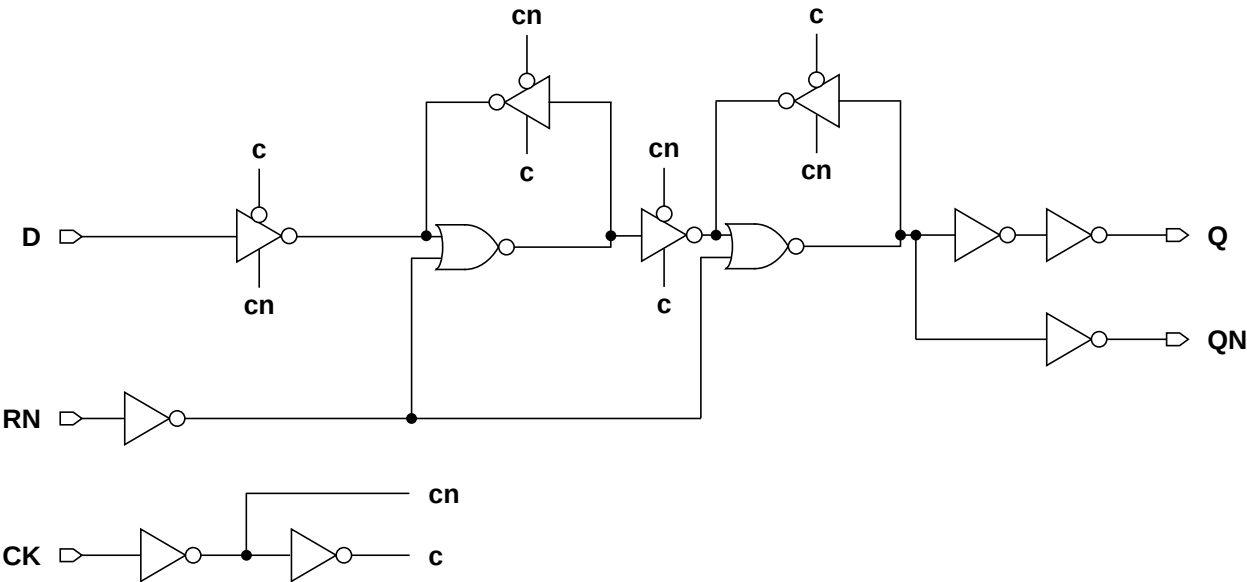
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFRXL	5.04	15.18
DFFRX1	5.04	15.18
DFFRX2	5.04	17.16
DFFRX4	5.04	19.80

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0355	0.0286	0.0327	0.0435
CK	0.0366	0.0343	0.0378	0.0440
RN	0.0149	0.0171	0.0212	0.0322
Q	0.0469	0.0541	0.0845	0.1469

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0033	0.0022	0.0020	0.0029
CK	0.0023	0.0029	0.0029	0.0041
RN	0.0020	0.0025	0.0033	0.0057

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.4475	0.4098	0.4005	0.3552	6.2679	4.5307	2.2078	1.1319
CK $\rightarrow$ Q $\downarrow$	0.3163	0.3266	0.3114	0.2832	3.4016	2.4024	1.2811	0.6467
RN $\rightarrow$ Q $\downarrow$	0.2608	0.2577	0.2475	0.2249	3.4025	2.4025	1.2810	0.6467
CK $\rightarrow$ QN $\uparrow$	0.2620	0.2585	0.2331	0.2100	6.2620	4.5294	2.2087	1.1323
CK $\rightarrow$ QN $\downarrow$	0.4148	0.3668	0.3511	0.3017	3.6343	2.4668	1.3062	0.6582
RN $\rightarrow$ QN $\uparrow$	0.2069	0.1902	0.1697	0.1522	6.2712	4.5329	2.2104	1.1334

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0391	0.0703	0.0859	0.0625
	setup $\downarrow$ $\rightarrow$ CK	0.0000	0.0898	0.1094	0.0781
	hold $\uparrow$ $\rightarrow$ CK	-0.0352	-0.0625	-0.0742	-0.0547
	hold $\downarrow$ $\rightarrow$ CK	0.0234	-0.0508	-0.0547	-0.0430
CK	minpwh	0.0316	0.0316	0.0316	0.0316
	minpwl	0.1965	0.1820	0.2111	0.1626
RN	minpwl	0.0316	0.0316	0.0316	0.0316
	recovery	0.0430	0.0625	3.9922	3.8594
	removal	-0.0273	-0.0430	-0.0469	-0.0234

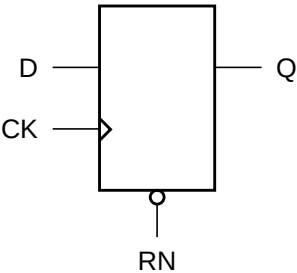
Cell Description

The DFFRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN). The cell has a single output (Q) and fast clock-to-out path.

Functions

RN	D	CK	Q[n+1]
0	x	x	0
1	0		0
1	1		1
1	x		Q[n]

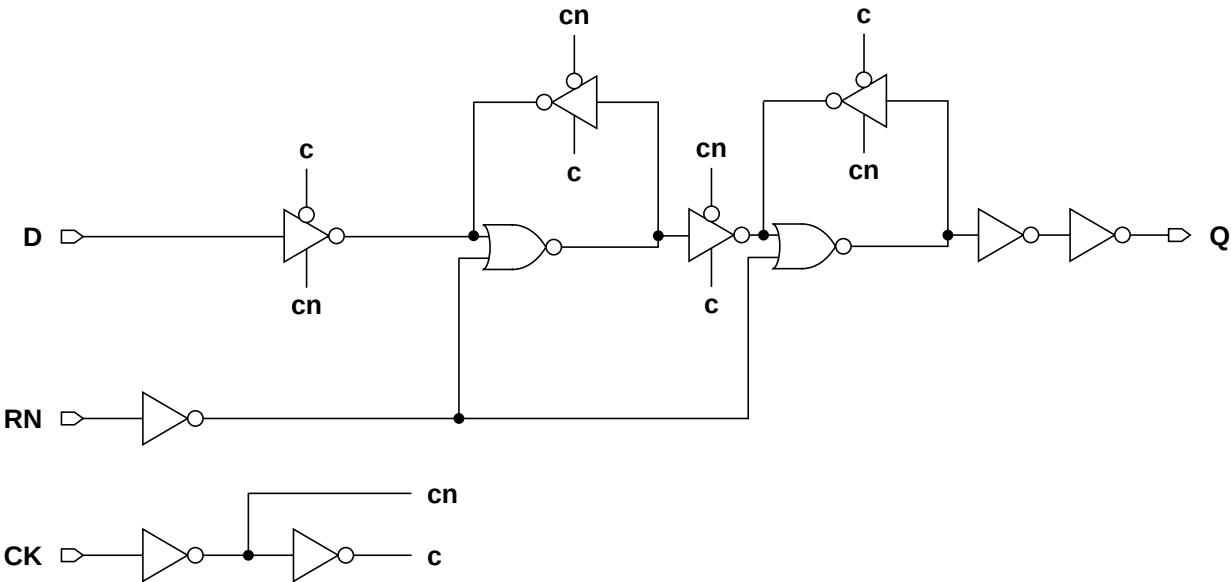
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFRHQXL	5.04	13.86
DFFRHQX1	5.04	13.86
DFFRHQX2	5.04	17.16
DFFRHQX4	5.04	21.12

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0360	0.0423	0.0660	0.1015
CK	0.0364	0.0364	0.0467	0.0654
RN	0.0197	0.0248	0.0341	0.0543
Q	0.0305	0.0337	0.0481	0.0760

Pin Capacitance

Pin	XL	X1	X2	X4
	XL	X1	X2	X4
D	XL	X1	X2	X4
CK	XL	X1	X2	X4
RN	XL	X1	X2	X4

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2523	0.2029	0.1795	0.1684	9.4911	6.7358	3.3676	1.8473
CK $\rightarrow$ Q $\downarrow$	0.2670	0.1789	0.1606	0.1487	3.7684	2.4441	1.2977	0.7102
RN $\rightarrow$ Q $\downarrow$	0.1876	0.1393	0.1255	0.1056	3.1023	2.0320	1.1784	0.6903

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0781	0.1328	0.1328	0.1172
	setup $\downarrow$ $\rightarrow$ CK	0.1641	0.2266	0.2188	0.1914
	hold $\uparrow$ $\rightarrow$ CK	-0.0195	-0.0586	-0.0508	-0.0430
	hold $\downarrow$ $\rightarrow$ CK	0.0195	-0.0391	-0.0273	-0.0234
CK	minpwh	0.1286	0.1140	0.0995	0.0898
	minpwl	0.1723	0.1723	0.1529	0.1286
RN	minpwl	0.2839	0.2936	0.4003	0.6089
	recovery	0.1055	0.1211	0.1289	0.1133
	removal	-0.0312	-0.0469	-0.0352	-0.0156

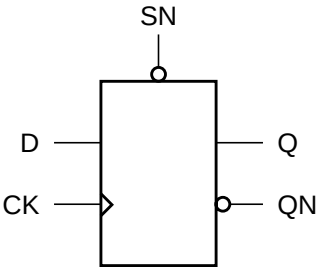
Cell Description

The DFFS cell is a positive-edge triggered, static D-type flip-flop with asynchronous active-low set (SN).

Functions

SN	D	CK	Q[n+1]	QN[n+1]
0	x	x	1	0
1	0		0	1
1	1		1	0
1	x		Q[n]	QN[n]

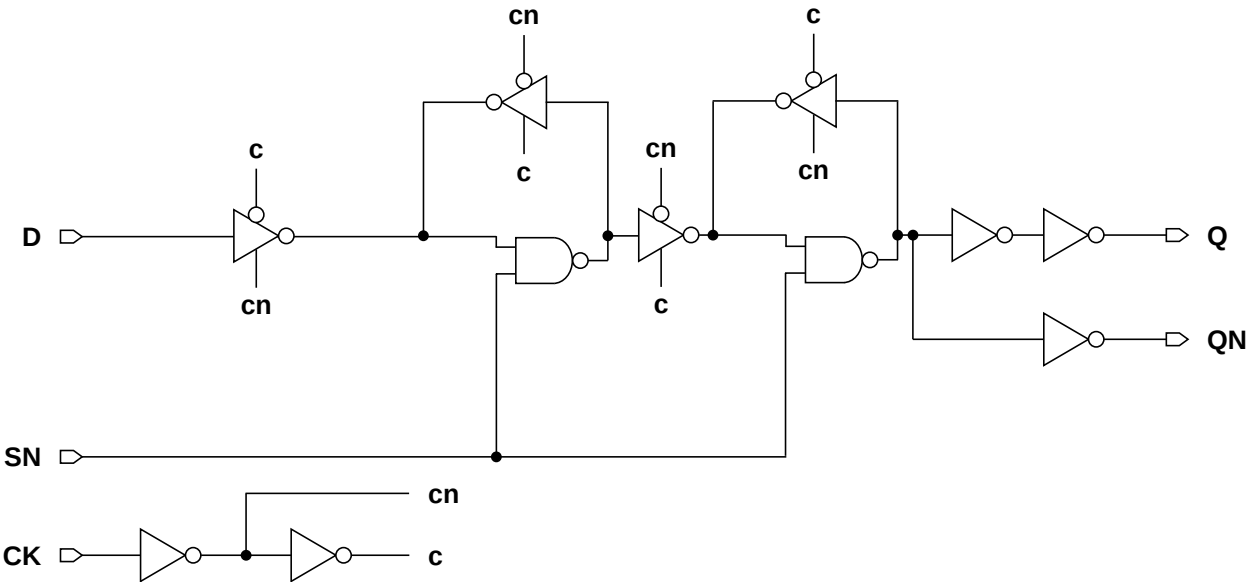
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFSXL	5.04	12.54
DFFSX1	5.04	12.54
DFFSX2	5.04	13.86
DFFSX4	5.04	18.48

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0340	0.0262	0.0301	0.0377
CK	0.0340	0.0306	0.0332	0.0409
SN	0.0068	0.0075	0.0114	0.0178
Q	0.0430	0.0468	0.0804	0.1364

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0033	0.0020	0.0020	0.0023
CK	0.0022	0.0029	0.0030	0.0036
SN	0.0051	0.0055	0.0077	0.0127

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.3662	0.3367	0.3453	0.3252	6.2652	4.5308	2.2071	1.1315
CK $\rightarrow$ Q $\downarrow$	0.2964	0.2983	0.3109	0.2861	3.4014	2.4005	1.2829	0.6413
SN $\rightarrow$ Q $\uparrow$	0.1751	0.1562	0.1641	0.1563	6.2649	4.5309	2.2071	1.1315
CK $\rightarrow$ QN $\uparrow$	0.2448	0.2371	0.2353	0.2142	6.2729	4.7324	2.2115	1.1059
CK $\rightarrow$ QN $\downarrow$	0.3323	0.2966	0.2962	0.2755	3.4724	2.5712	1.2883	0.6438
SN $\rightarrow$ QN $\downarrow$	0.1414	0.1162	0.1155	0.1070	3.4617	2.5717	1.2911	0.6464

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0352	0.0703	0.0664	0.0625
	setup $\downarrow$ $\rightarrow$ CK	0.1016	0.1602	0.1680	0.1562
	hold $\uparrow$ $\rightarrow$ CK	-0.0273	-0.0508	-0.0508	-0.0430
	hold $\downarrow$ $\rightarrow$ CK	0.0039	-0.0625	-0.0625	-0.0508
CK	minpwh	0.1189	0.1189	0.1140	0.1043
	minpwl	0.1674	0.1577	0.1674	0.1577
SN	minpwl	0.1383	0.1189	0.1335	0.1577
	recovery	-0.0195	-0.0039	0.0000	0.0039
	removal	0.1484	0.1055	0.1094	0.1055

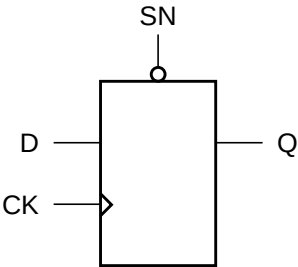
Cell Description

The DFFSHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with asynchronous active-low set (SN). The cell has a single output (Q) and fast clock-to-out path.

Functions

SN	D	CK	Q[n+1]
0	x	x	1
1	0		0
1	1		1
1	x		Q[n]

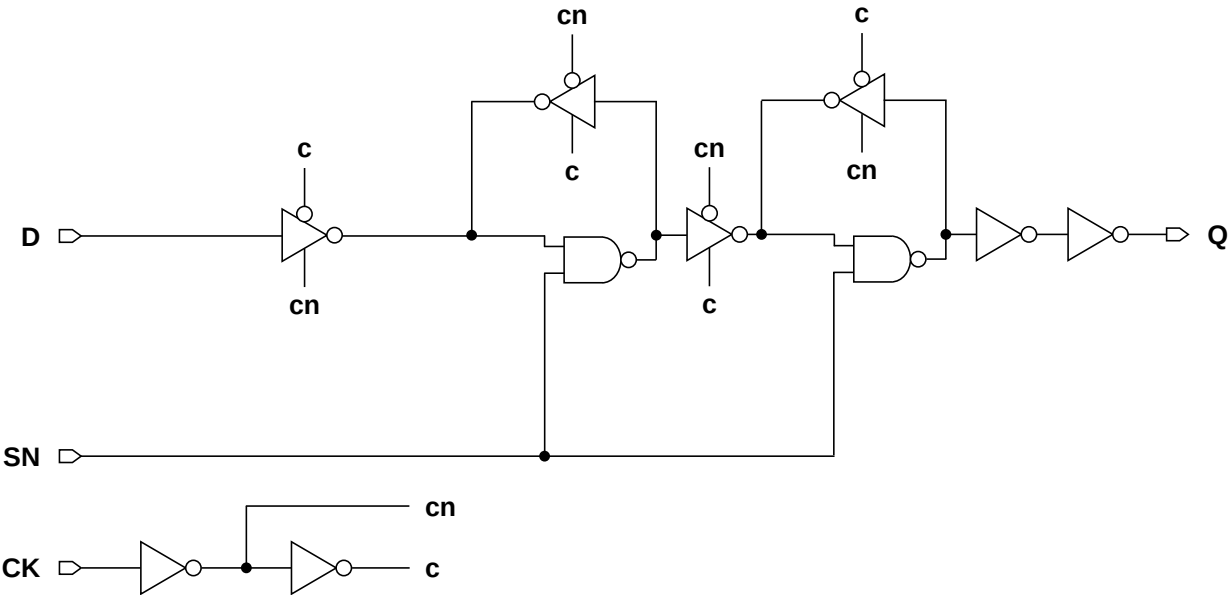
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFSHQXL	5.04	13.20
DFFSHQX1	5.04	13.20
DFFSHQX2	5.04	15.84
DFFSHQX4	5.04	18.48

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0375	0.0415	0.0582	0.0907
CK	0.0354	0.0361	0.0428	0.0601
SN	0.0106	0.0114	0.0194	0.0311
Q	0.0298	0.0309	0.0399	0.0670

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0034	0.0019	0.0026	0.0039
CK	0.0023	0.0030	0.0039	0.0058
SN	0.0084	0.0090	0.0143	0.0224

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2475	0.2054	0.1837	0.1664	6.3012	4.5378	2.2658	1.1328
CK $\rightarrow$ Q $\downarrow$	0.2570	0.1803	0.1578	0.1473	4.1870	2.8468	1.4175	0.7307
SN $\rightarrow$ Q $\uparrow$	0.0780	0.0895	0.0911	0.0951	3.5399	2.4699	1.2841	0.6890

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0508	0.1016	0.0898	0.0859
	setup $\downarrow$ $\rightarrow$ CK	0.1797	0.2500	0.2148	0.2031
	hold $\uparrow$ $\rightarrow$ CK	-0.0195	-0.0547	-0.0430	-0.0391
	hold $\downarrow$ $\rightarrow$ CK	0.0078	-0.0703	-0.0430	-0.0547
CK	minpwh	0.1286	0.1189	0.1043	0.0898
	minpwl	0.1674	0.1723	0.1480	0.1286
SN	minpwl	0.1238	0.1432	0.1723	0.2450
	recovery	0.0508	0.0664	0.0703	0.0664
	removal	0.1562	0.1094	0.1172	0.1172

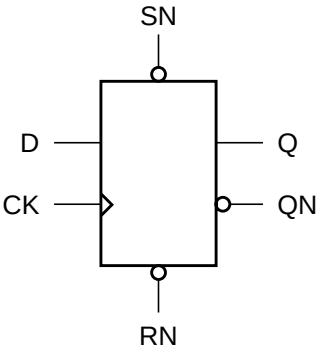
Cell Description

The DFFSR cell is a positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN) and set (SN), and set dominating reset.

Functions

RN	SN	D	CK	Q[n+1]	QN[n+1]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0
1	1	0		0	1
1	1	1		1	0
1	1	x		Q[n]	QN[n]

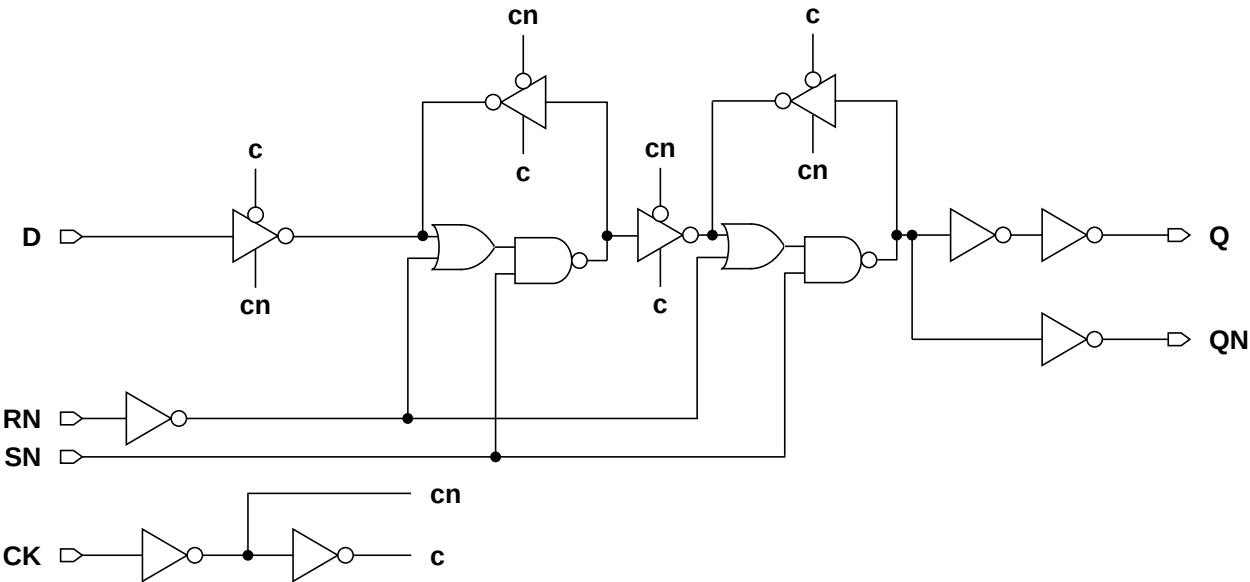
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFSRXL	5.04	17.16
DFFSRX1	5.04	17.16
DFFSRX2	5.04	17.16
DFFSRX4	5.04	23.76

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0406	0.0332	0.0381	0.0522
CK	0.0388	0.0367	0.0378	0.0493
SN	0.0092	0.0095	0.0128	0.0194
RN	0.0169	0.0185	0.0236	0.0409
Q	0.0483	0.0558	0.0876	0.1511

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0032	0.0020	0.0020	0.0032
CK	0.0023	0.0029	0.0031	0.0042
SN	0.0063	0.0067	0.0090	0.0154
RN	0.0023	0.0027	0.0038	0.0061

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.4400	0.3985	0.3810	0.3651	6.2614	4.5296	2.2912	1.1318
CK $\rightarrow$ Q $\downarrow$	0.3390	0.3362	0.3135	0.3022	3.4161	2.4071	1.2834	0.6419
SN $\rightarrow$ Q $\uparrow$	0.2041	0.1900	0.1803	0.1735	6.2607	4.5294	2.2911	1.1318
SN $\rightarrow$ Q $\downarrow$	0.2051	0.1889	0.1782	0.1728	3.4167	2.4073	1.2835	0.6419
RN $\rightarrow$ Q $\downarrow$	0.3063	0.2798	0.2545	0.2463	3.4170	2.4072	1.2833	0.6419
CK $\rightarrow$ QN $\uparrow$	0.2711	0.2575	0.2385	0.2280	6.2876	4.5371	2.2953	1.1344
CK $\rightarrow$ QN $\downarrow$	0.3961	0.3480	0.3278	0.3148	3.7149	2.4820	1.3103	0.6579
SN $\rightarrow$ QN $\uparrow$	0.1377	0.1111	0.1038	0.0989	6.3124	4.5478	2.3011	1.1380
SN $\rightarrow$ QN $\downarrow$	0.1583	0.1377	0.1266	0.1231	3.5084	2.4312	1.2952	0.6520
RN $\rightarrow$ QN $\uparrow$	0.2386	0.2017	0.1800	0.1721	6.3096	4.5472	2.3009	1.1379

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup↑ → CK	0.0781	0.1602	0.1289	0.0938
	setup↓ → CK	0.1289	0.2031	0.2148	0.1719
	hold↑ → CK	-0.0508	-0.1055	-0.0898	-0.0586
	hold↓ → CK	0.0039	-0.0430	-0.0625	-0.0391
CK	minpwh	0.1189	0.1140	0.1092	0.0995
	minpwl	0.1965	0.2208	0.2062	0.1626
SN	minpwl	0.1577	0.1432	0.1432	0.1868
	recovery	-0.0078	0.0195	0.0195	0.0117
	removal	0.1367	0.0977	0.1055	0.1094
RN	minpwl	0.2256	0.1965	0.2305	0.3663
	recovery	0.0703	0.1328	0.0938	0.0820
	removal	-0.0430	-0.0938	-0.0625	-0.0391

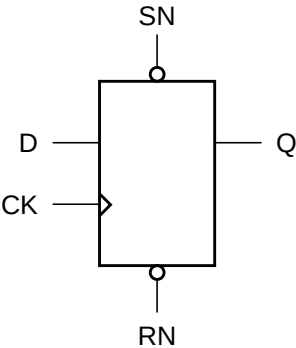
Cell Description

The DFFSRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN) and set (SN), and set dominating reset. The cell has a single output (Q) and fast clock-to-out path.

Functions

RN	SN	D	CK	Q[n+1]
0	1	x	x	0
1	0	x	x	1
0	0	x	x	1
1	1	0		0
1	1	1		1
1	1	x		Q[n]

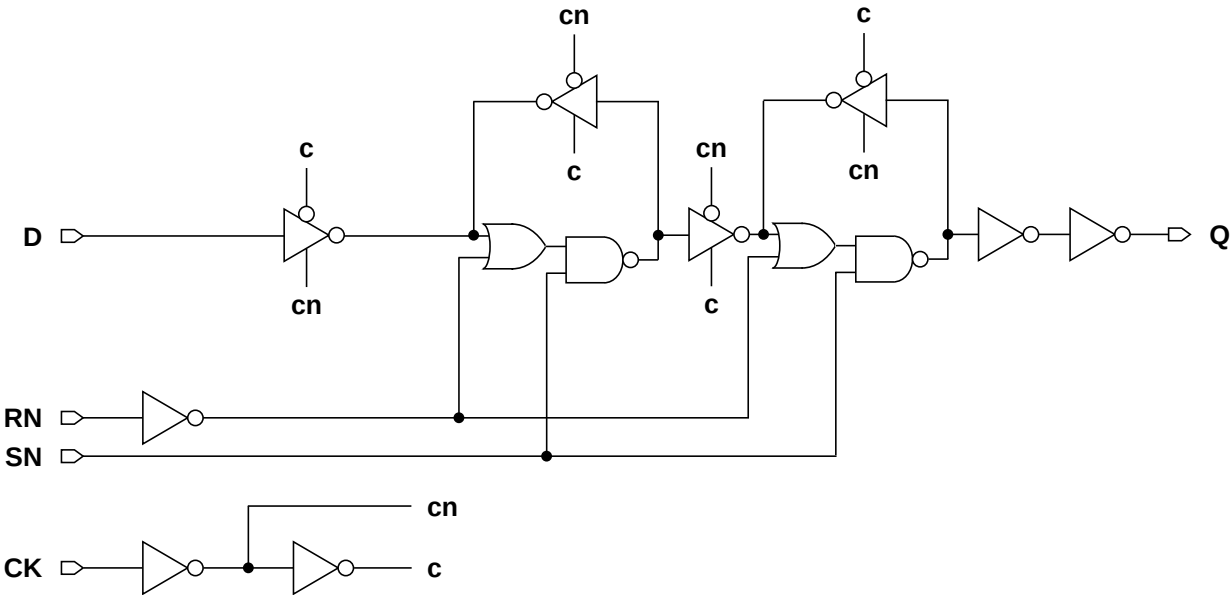
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFSRHQXL	5.04	15.84
DFFSRHQX1	5.04	15.84
DFFSRHQX2	5.04	22.44
DFFSRHQX4	5.04	30.36

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0392	0.0443	0.0689	0.1158
CK	0.0376	0.0375	0.0492	0.0738
SN	0.0130	0.0142	0.0198	0.0330
RN	0.0239	0.0289	0.0429	0.0694
Q	0.0347	0.0329	0.0531	0.0931

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0035	0.0023	0.0033	0.0051
CK	0.0023	0.0033	0.0046	0.0069
SN	0.0110	0.0120	0.0173	0.0286
RN	0.0027	0.0041	0.0060	0.0104

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2628	0.2132	0.1973	0.1875	9.4904	6.7390	3.3696	1.6848
CK $\rightarrow$ Q $\downarrow$	0.2578	0.1748	0.1638	0.1551	4.1375	2.8312	1.4285	0.7338
SN $\rightarrow$ Q $\uparrow$	0.0844	0.0928	0.0978	0.0996	4.0579	2.8729	1.4782	0.7939
SN $\rightarrow$ Q $\downarrow$	0.0499	0.0546	0.0496	0.0441	3.7424	2.5920	1.3861	0.7321
RN $\rightarrow$ Q $\downarrow$	0.1919	0.1405	0.1225	0.1114	3.7862	2.5894	1.3847	0.7322

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0898	0.1406	0.1406	0.1328
	setup $\downarrow$ $\rightarrow$ CK	0.1758	0.2227	0.2188	0.2109
	hold $\uparrow$ $\rightarrow$ CK	-0.0234	-0.0586	-0.0508	-0.0391
	hold $\downarrow$ $\rightarrow$ CK	0.0195	-0.0195	-0.0156	-0.0156
CK	minpwh	0.1335	0.1189	0.0995	0.0946
	minpwl	0.1820	0.1771	0.1529	0.1335
SN	minpwl	0.1383	0.1674	0.2014	0.2887
	recovery	0.0625	0.0742	0.0820	0.0820
	removal	0.1484	0.1016	0.1094	0.1172
RN	minpwl	0.2936	0.2208	0.3324	0.5701
	recovery	0.1094	0.1367	0.1406	0.1289
	removal	-0.0273	-0.0586	-0.0430	-0.0156

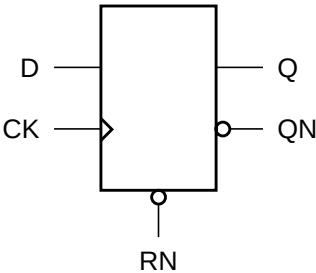
Cell Description

The DFFTR cell is a positive-edge triggered, static D-type flip-flop with synchronous active-low reset (RN).

Functions

RN	D	CK	Q[n+1]	QN[n+1]
0	x		0	1
x	x		Q[n]	QN[n]
1	0		0	1
1	1		1	0

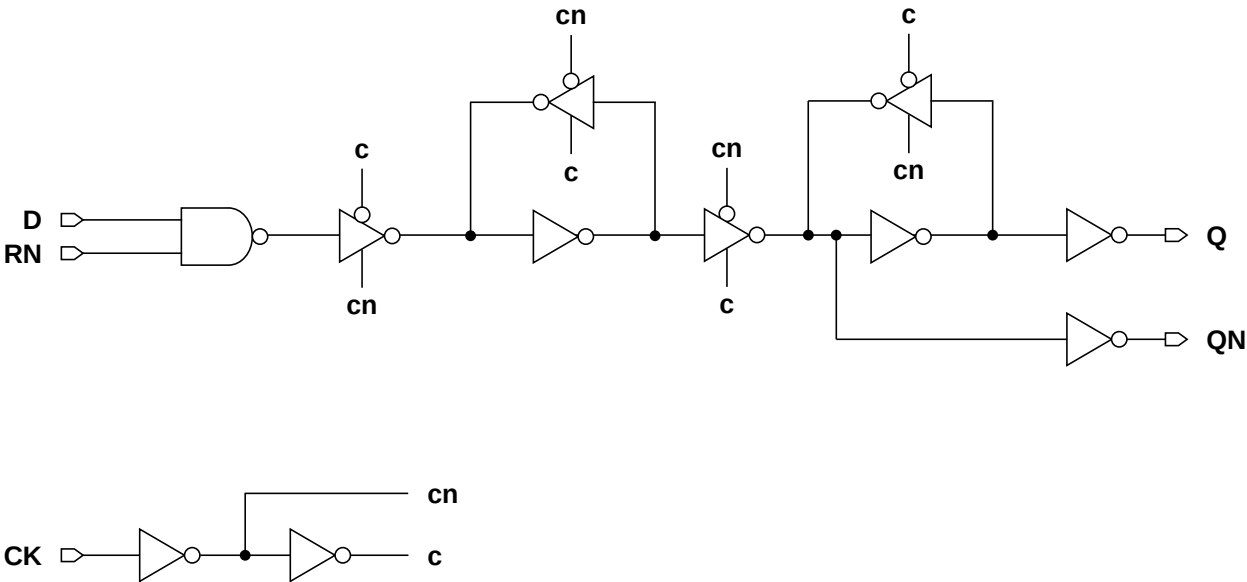
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFTRXL	5.04	11.22
DFFTRX1	5.04	11.22
DFFTRX2	5.04	13.86
DFFTRX4	5.04	16.50

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0321	0.0310	0.0392	0.0621
CK	0.0364	0.0361	0.0432	0.0640
RN	0.0346	0.0328	0.0412	0.0648
Q	0.0316	0.0380	0.0673	0.1160

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0029	0.0023	0.0023	0.0033
CK	0.0023	0.0032	0.0042	0.0065
RN	0.0025	0.0019	0.0019	0.0029

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2339	0.2146	0.2063	0.1735	6.2651	4.5322	2.2656	1.1331
CK $\rightarrow$ Q $\downarrow$	0.1860	0.1661	0.1588	0.1478	3.5117	2.4360	1.3019	0.6491
CK $\rightarrow$ QN $\uparrow$	0.2381	0.2219	0.2136	0.1993	6.2711	4.5337	2.2631	1.1317
CK $\rightarrow$ QN $\downarrow$	0.3084	0.2909	0.2890	0.2454	3.4342	2.4108	1.2856	0.6416

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0703	0.1055	0.1133	0.0938
	setup $\downarrow$ $\rightarrow$ CK	0.1562	0.2617	0.2461	0.2031
	hold $\uparrow$ $\rightarrow$ CK	-0.0586	-0.0820	-0.0859	-0.0664
	hold $\downarrow$ $\rightarrow$ CK	-0.0469	-0.1406	-0.1250	-0.0977
CK	minpwh	0.1189	0.1140	0.1092	0.0898
	minpwl	0.2111	0.2014	0.1965	0.1626
RN	setup $\uparrow$ $\rightarrow$ CK	0.0781	0.1133	0.1172	0.0977
	setup $\downarrow$ $\rightarrow$ CK	0.1758	0.2812	0.2656	0.2188
	hold $\uparrow$ $\rightarrow$ CK	-0.0586	-0.0820	-0.0898	-0.0703
	hold $\downarrow$ $\rightarrow$ CK	-0.0508	-0.1562	-0.1367	-0.1055

Cell Description

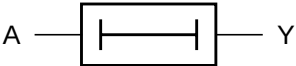
The DLY1 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$Y = A$

Functions

A	Y
0	0
1	1

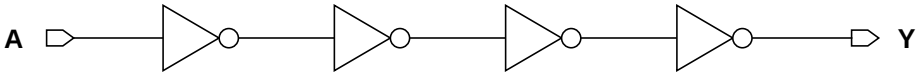
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DLY1X1	5.04	3.96

Functional Schematic



AC Power

Pin	Power (μ W/MHz)
	X1
A	0.0320

Pin Capacitance

Pin	Capacitance (pF)
	X1
A	0.0019

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)	K _{load} (ns/pF)
	X1	X1
A $\rightarrow$ Y $\uparrow$	0.1555	4.5302
A $\rightarrow$ Y $\downarrow$	0.1824	2.4028

Cell Description

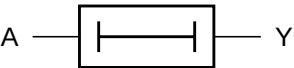
The DLY2 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$Y = A$

Functions

A	Y
0	0
1	1

Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DLY2X1	5.04	3.96

Functional Schematic



AC Power

Pin	Power (μ W/MHz)
	X1
A	0.0368

Pin Capacitance

Pin	Capacitance (pF)
	X1
A	0.0020

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)	K _{load} (ns/pF)
	X1	X1
A $\rightarrow$ Y $\uparrow$	0.3018	4.5393
A $\rightarrow$ Y $\downarrow$	0.3316	2.4878

Cell Description

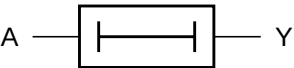
The DLY3 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$Y = A$

Functions

A	Y
0	0
1	1

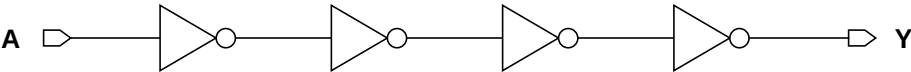
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DLY3X1	5.04	4.62

Functional Schematic



AC Power

Pin	Power (μ W/MHz)
	X1
A	0.0434

Pin Capacitance

Pin	Capacitance (pF)
	X1
A	0.0020

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)	K _{load} (ns/pF)
	X1	X1
A $\rightarrow$ Y $\uparrow$	0.4952	4.5522
A $\rightarrow$ Y $\downarrow$	0.4939	2.6125

Cell Description

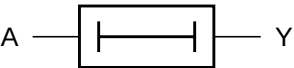
The DLY4 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$Y = A$

Functions

A	Y
0	0
1	1

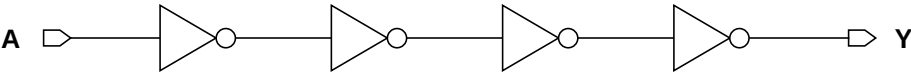
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DLY4X1	5.04	4.62

Functional Schematic



AC Power

Pin	Power (μ W/MHz)
	X1
A	0.0513

Pin Capacitance

Pin	Capacitance (pF)
	X1
A	0.0020

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)	K _{load} (ns/pF)
	X1	X1
A $\rightarrow$ Y $\uparrow$	0.7190	4.5772
A $\rightarrow$ Y $\downarrow$	0.6667	2.7621

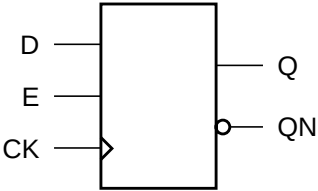
Cell Description

The EDFF cell is a positive-edge triggered, static D-type flip-flop with synchronous active-high enable (E).

Functions

E	D	CK	Q[n+1]	QN[n+1]
0	x	x	Q[n]	QN[n]
1	0		0	1
1	1		1	0
1	x		Q[n]	QN[n]

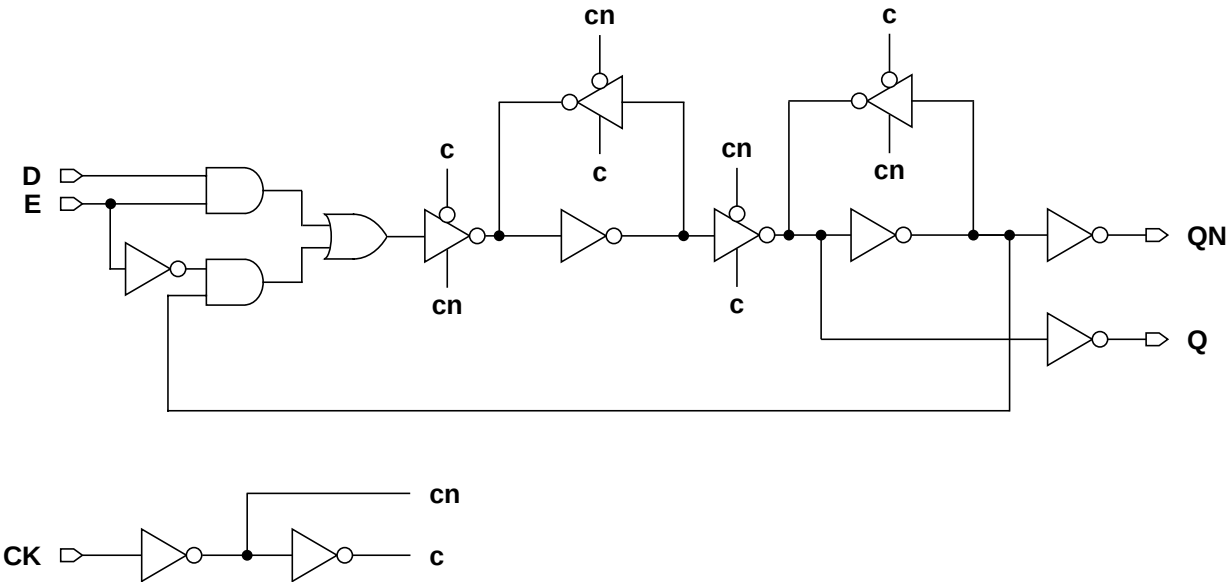
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
EDFFXL	5.04	15.18
EDFFX1	5.04	15.18
EDFFX2	5.04	17.82
EDFFX4	5.04	20.46

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0371	0.0365	0.0494	0.0749
CK	0.0433	0.0408	0.0479	0.0694
E	0.0533	0.0490	0.0625	0.0917
Q	0.0389	0.0449	0.0743	0.1272

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0029	0.0019	0.0023	0.0037
CK	0.0023	0.0030	0.0042	0.0069
E	0.0054	0.0045	0.0049	0.0060

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2495	0.2233	0.1928	0.1768	6.2597	5.0667	2.2241	1.1336
CK $\rightarrow$ Q $\downarrow$	0.1975	0.1694	0.1590	0.1487	3.5119	2.6868	1.3295	0.6502
CK $\rightarrow$ QN $\uparrow$	0.2823	0.2360	0.2197	0.2069	6.2758	4.5345	2.2222	1.1324
CK $\rightarrow$ QN $\downarrow$	0.3670	0.3138	0.2848	0.2576	3.5353	2.4260	1.3199	0.6436

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0938	0.1211	0.1406	0.1211
	setup $\downarrow$ $\rightarrow$ CK	0.2539	0.4648	0.3828	0.3242
	hold $\uparrow$ $\rightarrow$ CK	-0.0742	-0.0898	-0.1055	-0.0898
	hold $\downarrow$ $\rightarrow$ CK	-0.1211	-0.3086	-0.2422	-0.1992
CK	minpwh	0.1286	0.1189	0.0995	0.0946
	minpwl	0.2208	0.2062	0.2062	0.1771
E	setup $\uparrow$ $\rightarrow$ CK	0.2930	0.5039	0.4219	0.3711
	setup $\downarrow$ $\rightarrow$ CK	0.1602	0.3672	0.2930	0.2344
	hold $\uparrow$ $\rightarrow$ CK	-0.0898	-0.1094	-0.1250	-0.1016
	hold $\downarrow$ $\rightarrow$ CK	-0.1172	-0.1523	-0.1680	-0.1641

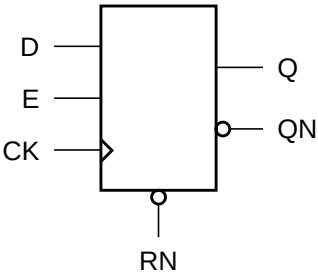
Cell Description

The EDFFTR cell is a positive-edge triggered, static D-type flip-flop with synchronous active-high enable (E) and synchronous active-low reset (RN).

Functions

RN	E	D	CK	Q[n+1]	QN[n+1]
0	x	x		0	1
x	x	x		Q[n]	QN[n]
1	0	x		Q[n]	QN[n]
1	1	0		0	1
1	1	1		1	0

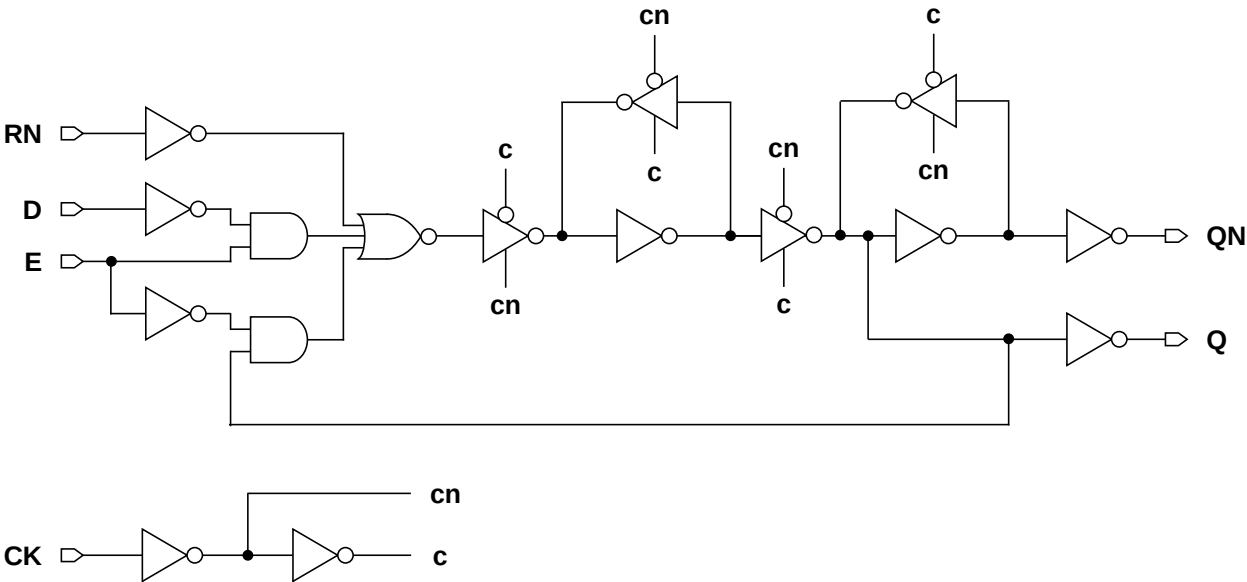
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
EDFFTRXL	5.04	16.50
EDFFTRX1	5.04	16.50
EDFFTRX2	5.04	17.82
EDFFTRX4	5.04	21.12

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0408	0.0387	0.0524	0.0820
CK	0.0476	0.0452	0.0541	0.0751
E	0.0580	0.0519	0.0674	0.0997
RN	0.0470	0.0435	0.0569	0.0886
Q	0.0430	0.0487	0.0739	0.1227

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0030	0.0019	0.0023	0.0038
CK	0.0024	0.0033	0.0044	0.0066
E	0.0060	0.0048	0.0053	0.0066
RN	0.0027	0.0020	0.0023	0.0037

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2597	0.2285	0.2096	0.1852	6.2758	4.5350	2.4018	1.1334
CK $\rightarrow$ Q $\downarrow$	0.2087	0.1772	0.1594	0.1470	3.2798	2.4474	1.3005	0.6500
CK $\rightarrow$ QN $\uparrow$	0.3085	0.2530	0.2201	0.2056	6.2807	4.5358	2.4000	1.1323
CK $\rightarrow$ QN $\downarrow$	0.3782	0.3269	0.2954	0.2658	3.5301	2.4291	1.2896	0.6437

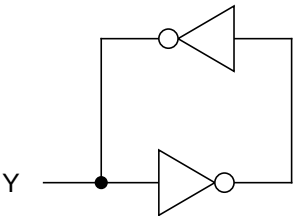
Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.1289	0.1680	0.1797	0.1445
	setup $\downarrow$ $\rightarrow$ CK	0.2812	0.5117	0.4062	0.3359
	hold $\uparrow$ $\rightarrow$ CK	-0.1016	-0.1328	-0.1406	-0.1094
	hold $\downarrow$ $\rightarrow$ CK	-0.1445	-0.3477	-0.2695	-0.2109
CK	minpwh	0.1286	0.1189	0.1092	0.0995
	minpwl	0.2693	0.2548	0.2450	0.1965
E	setup $\uparrow$ $\rightarrow$ CK	0.3242	0.5430	0.4453	0.3828
	setup $\downarrow$ $\rightarrow$ CK	0.1953	0.4258	0.3320	0.2578
	hold $\uparrow$ $\rightarrow$ CK	-0.1250	-0.1602	-0.1641	-0.1289
	hold $\downarrow$ $\rightarrow$ CK	-0.1445	-0.1953	-0.2148	-0.1992
RN	setup $\uparrow$ $\rightarrow$ CK	0.1445	0.1875	0.1953	0.1602
	setup $\downarrow$ $\rightarrow$ CK	0.2422	0.4141	0.3438	0.2891
	hold $\uparrow$ $\rightarrow$ CK	-0.1211	-0.1562	-0.1562	-0.1250
	hold $\downarrow$ $\rightarrow$ CK	-0.0820	-0.2461	-0.1992	-0.1523

Cell Description

The HOLD cell holds data at a known value. This cell is often used for holding data on a tri-state bus.

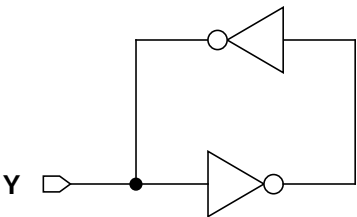
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
HOLDX1	5.04	2.64

Functional Schematic



AC Power

Pin	Power (μ W/MHz)
	X1
Y	0.0199

Pin Capacitance

Pin	Capacitance (pF)
	X1
Y	0.1045

Cell Description

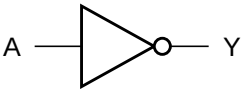
The INV cell provides the logical inversion of a single input (A). The output (Y) is represented by the logic equation:

$Y = \overline{A}$

Functions

A	Y
0	1
1	0

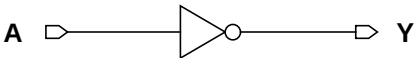
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
INVXL	5.04	1.32
INVX1	5.04	1.32
INVX2	5.04	1.98
INVX3	5.04	2.64
INVX4	5.04	2.64
INVX8	5.04	3.96
INVX12	5.04	8.58
INVX16	5.04	11.22
INVX20	5.04	12.54

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0087	0.0117	0.0218	0.0329	0.0394	0.0773	0.1706	0.2260	0.2820

Pin Capacitance

Pin	Capacitance (pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0027	0.0036	0.0071	0.0104	0.0136	0.0271	0.0068	0.0090	0.0110

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	0.0261	0.0253	0.0228	0.0243	0.0206	0.0198	0.1303	0.1276	0.1265
A $\rightarrow$ Y $\downarrow$	0.0154	0.0146	0.0140	0.0146	0.0125	0.0125	0.1235	0.1232	0.1183

Description	K_{load} (ns/pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	6.2539	4.5257	2.2629	1.5216	1.1447	0.5513	0.3680	0.2760	0.2209
A $\rightarrow$ Y $\downarrow$	3.3414	2.3675	1.2661	0.8247	0.6333	0.3211	0.2194	0.1647	0.1316

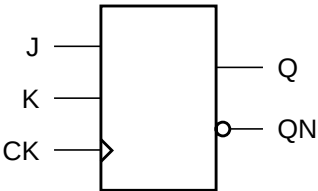
Cell Description

The JKFF cell is a positive-edge triggered JK-type flip-flop.

Function

J	K	CK	Q[n+1]	QN[n+1]
x	x		Q[n]	QN[n]
0	0		Q[n]	QN[n]
0	1		0	1
1	0		1	0
1	1		QN[n]	Q[n]

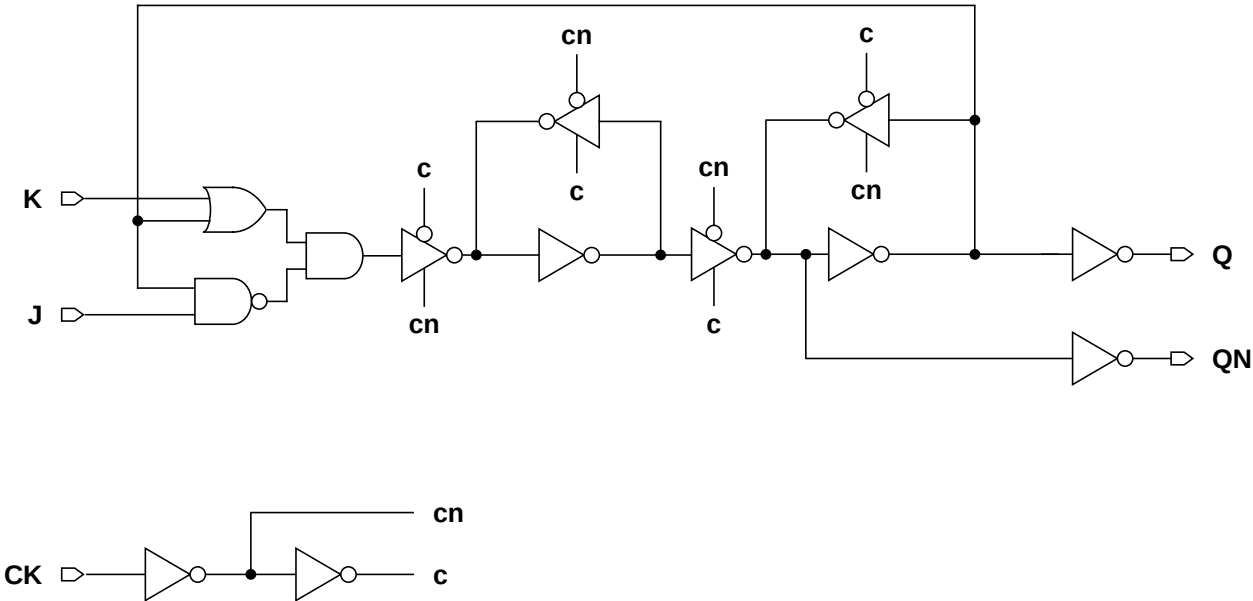
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
JKFFXL	5.04	13.86
JKFFX1	5.04	13.86
JKFFX2	5.04	16.50
JKFFX4	5.04	19.80

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
J	0.0316	0.0307	0.0420	0.0689
K	0.0295	0.0274	0.0385	0.0596
CK	0.0386	0.0369	0.0466	0.0665
Q	0.0475	0.0496	0.0832	0.1361

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
J	0.0016	0.0014	0.0015	0.0016
K	0.0029	0.0019	0.0023	0.0035
CK	0.0021	0.0029	0.0038	0.0070

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2874	0.2325	0.2275	0.1953	6.2840	4.5364	2.2089	1.1046
CK $\rightarrow$ Q $\downarrow$	0.3731	0.3152	0.2961	0.2586	3.5723	2.4341	1.2882	0.6436
CK $\rightarrow$ QN $\uparrow$	0.2387	0.2140	0.2083	0.1777	6.2652	4.9498	2.2106	1.1058
CK $\rightarrow$ QN $\downarrow$	0.1852	0.1606	0.1662	0.1426	3.5142	2.8861	1.3026	0.6494

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
J	setup $\uparrow$ $\rightarrow$ CK	0.1680	0.2734	0.2500	0.2422
	setup $\downarrow$ $\rightarrow$ CK	0.1328	0.1602	0.1719	0.1836
	hold $\uparrow$ $\rightarrow$ CK	-0.0977	-0.2070	-0.1758	-0.1641
	hold $\downarrow$ $\rightarrow$ CK	-0.1250	-0.1484	-0.1602	-0.1758
K	setup $\uparrow$ $\rightarrow$ CK	0.0859	0.1094	0.1250	0.1094
	setup $\downarrow$ $\rightarrow$ CK	0.1484	0.3672	0.2695	0.2188
	hold $\uparrow$ $\rightarrow$ CK	-0.0820	-0.1016	-0.1172	-0.0977
	hold $\downarrow$ $\rightarrow$ CK	-0.1250	-0.3242	-0.2344	-0.1875
CK	minpwh	0.1383	0.1092	0.1189	0.0995
	minpwl	0.2062	0.2645	0.2305	0.1820

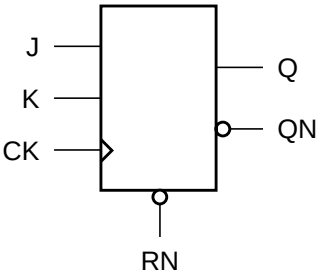
Cell Description

The JKFFR cell is a positive-edge triggered JK-type flip-flop with asynchronous active-low reset (RN).

Function

RN	J	K	CK	Q[n+1]	QN[n+1]
1	x	x		Q[n]	QN[n]
0	x	x	x	0	1
1	0	0		Q[n]	QN[n]
1	0	1		0	1
1	1	0		1	0
1	1	1		QN[n]	Q[n]

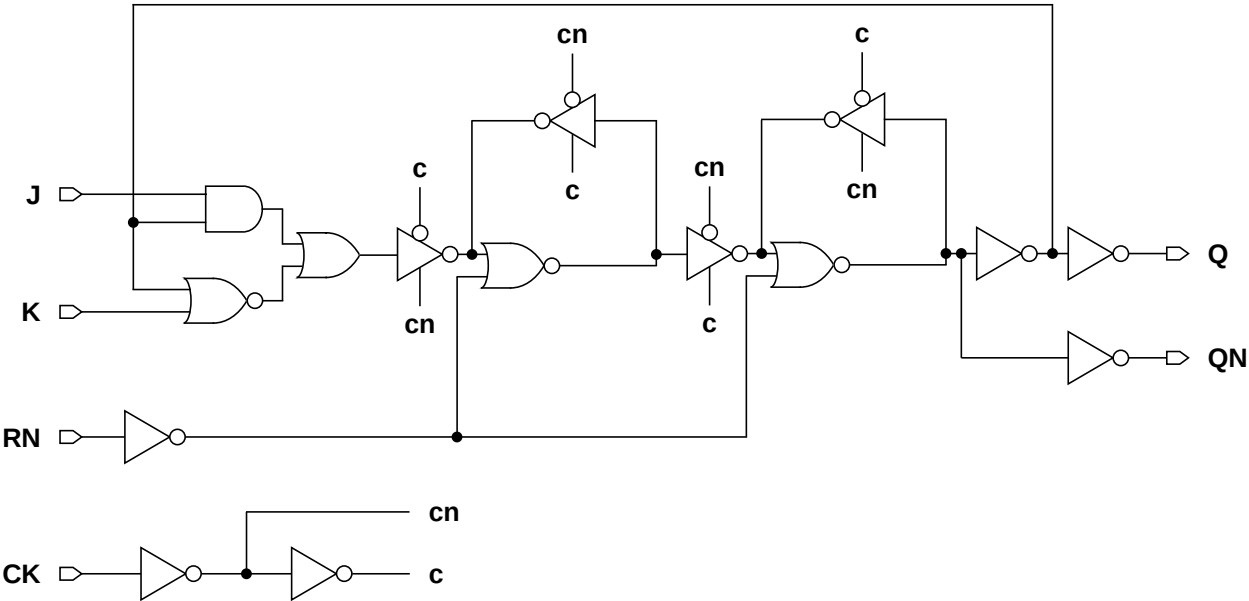
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
JKFFRXL	5.04	17.16
JKFFRX1	5.04	18.48
JKFFRX2	5.04	18.48
JKFFRX4	5.04	23.10

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
J	0.0311	0.0267	0.0288	0.0416
K	0.0378	0.0336	0.0364	0.0498
CK	0.0404	0.0387	0.0406	0.0484
RN	0.0168	0.0191	0.0227	0.0365
Q	0.0712	0.0746	0.1085	0.1722

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
J	0.0029	0.0019	0.0018	0.0024
K	0.0026	0.0025	0.0025	0.0026
CK	0.0021	0.0027	0.0030	0.0041
RN	0.0023	0.0028	0.0035	0.0059

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.5096	0.4578	0.4227	0.3677	6.2793	4.5376	2.2103	1.1052
CK $\rightarrow$ Q $\downarrow$	0.3940	0.3827	0.3453	0.3009	3.5751	2.4594	1.2957	0.6350
RN $\rightarrow$ Q $\downarrow$	0.3553	0.3199	0.2833	0.2577	3.5690	2.4565	1.2943	0.6343
CK $\rightarrow$ QN $\uparrow$	0.2477	0.2486	0.2363	0.2125	6.2595	4.5302	2.2095	1.1052
CK $\rightarrow$ QN $\downarrow$	0.3928	0.3551	0.3484	0.2994	3.6324	2.5774	1.3119	0.6438
RN $\rightarrow$ QN $\uparrow$	0.2051	0.1833	0.1729	0.1682	6.2637	4.5325	2.2116	1.1063

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
J	setup $\uparrow$ $\rightarrow$ CK	0.1133	0.1328	0.1484	0.1602
	setup $\downarrow$ $\rightarrow$ CK	0.1250	0.3047	0.3164	0.2383
	hold $\uparrow$ $\rightarrow$ CK	-0.1055	-0.1211	-0.1367	-0.1406
	hold $\downarrow$ $\rightarrow$ CK	-0.0938	-0.2539	-0.2578	-0.1797
K	setup $\uparrow$ $\rightarrow$ CK	0.1719	0.2812	0.3086	0.2617
	setup $\downarrow$ $\rightarrow$ CK	0.1719	0.1641	0.1797	0.1953
	hold $\uparrow$ $\rightarrow$ CK	-0.0938	-0.2227	-0.2344	-0.1797
	hold $\downarrow$ $\rightarrow$ CK	-0.1602	-0.1523	-0.1602	-0.1836
CK	minpwh	0.1286	0.1238	0.1238	0.1043
	minpwl	0.2450	0.2256	0.2353	0.2208
RN	minpwl	0.1965	0.1917	0.2402	0.3469
	recovery	0.0547	0.0703	0.0898	0.0820
	removal	-0.0391	-0.0547	-0.0625	-0.0508

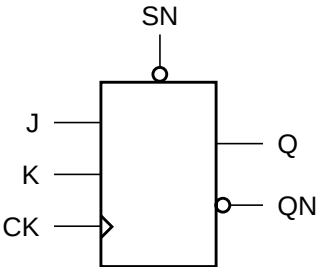
Cell Description

The JKFFS cell is a positive-edge triggered JK-type flip-flop with asynchronous active-low set (SN).

Function

SN	J	K	CK	Q[n+1]	QN[n+1]
1	x	x		Q[n]	QN[n]
0	x	x	x	1	0
1	0	0		Q[n]	QN[n]
1	0	1		0	1
1	1	0		1	0
1	1	1		QN[n]	Q[n]

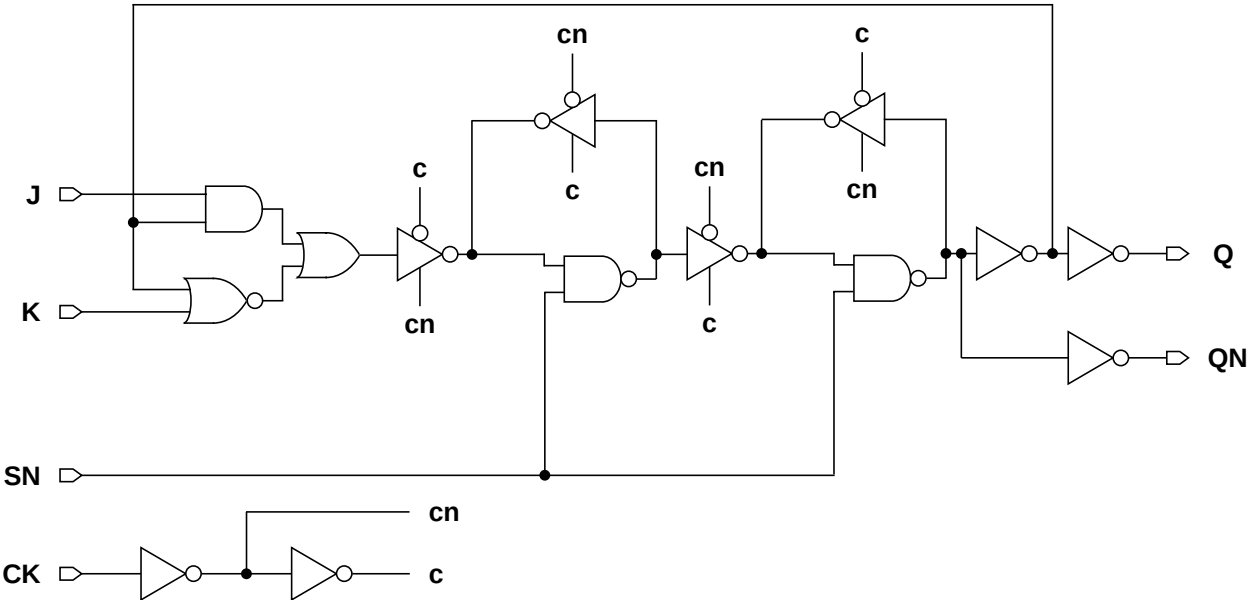
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
JKFFSXL	5.04	16.50
JKFFSX1	5.04	17.16
JKFFSX2	5.04	17.82
JKFFSX4	5.04	20.46

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
J	0.0296	0.0246	0.0273	0.0313
K	0.0425	0.0358	0.0397	0.0466
CK	0.0408	0.0381	0.0390	0.0444
SN	0.0052	0.0065	0.0089	0.0157
Q	0.0674	0.0770	0.1036	0.1589

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
J	0.0027	0.0017	0.0017	0.0020
K	0.0025	0.0025	0.0025	0.0024
CK	0.0023	0.0027	0.0030	0.0037
SN	0.0053	0.0063	0.0082	0.0128

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.4593	0.3983	0.3793	0.3558	6.2817	4.5351	2.3089	1.1046
CK $\rightarrow$ Q $\downarrow$	0.4160	0.3639	0.3391	0.2998	3.5803	2.4325	1.2960	0.6437
SN $\rightarrow$ Q $\uparrow$	0.2516	0.1976	0.1877	0.1673	6.2773	4.5345	2.3087	1.1046
CK $\rightarrow$ QN $\uparrow$	0.2616	0.2548	0.2295	0.2160	6.2909	4.7400	2.3112	1.1064
CK $\rightarrow$ QN $\downarrow$	0.3490	0.3186	0.3039	0.2946	3.5046	2.5011	1.2898	0.6448
SN $\rightarrow$ QN $\downarrow$	0.1444	0.1191	0.1136	0.1070	3.4852	2.5010	1.2921	0.6473

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
J	setup $\uparrow$ $\rightarrow$ CK	0.1055	0.1367	0.1406	0.1484
	setup $\downarrow$ $\rightarrow$ CK	0.1523	0.3672	0.3867	0.2656
	hold $\uparrow$ $\rightarrow$ CK	-0.1016	-0.1289	-0.1289	-0.1367
	hold $\downarrow$ $\rightarrow$ CK	-0.1328	-0.3320	-0.3438	-0.2305
K	setup $\uparrow$ $\rightarrow$ CK	0.2031	0.3477	0.3711	0.2812
	setup $\downarrow$ $\rightarrow$ CK	0.1680	0.1641	0.1680	0.1797
	hold $\uparrow$ $\rightarrow$ CK	-0.1094	-0.2773	-0.2891	-0.2031
	hold $\downarrow$ $\rightarrow$ CK	-0.1562	-0.1523	-0.1562	-0.1680
CK	minpwh	0.1286	0.1238	0.1238	0.1140
	minpwl	0.2208	0.2208	0.2111	0.2111
SN	minpwl	0.1626	0.1383	0.1383	0.1577
	recovery	-0.0352	-0.0078	0.0000	-0.0078
	removal	0.1484	0.1094	0.1055	0.1133

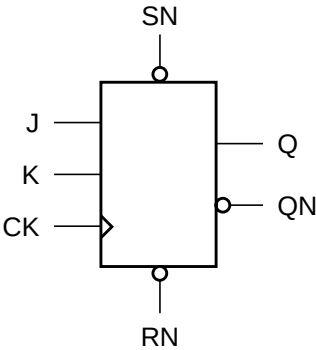
Cell Description

The JKFFSR cell is a positive-edge triggered JK-type flip-flop with asynchronous active-low reset (RN) and set (SN), and set dominating reset.

Function

RN	SN	J	K	CK	Q[n+1]	QN[n+1]
1	1	x	x		Q[n]	QN[n]
1	0	x	x	x	1	0
0	1	x	x	x	0	1
0	0	x	x	x	1	0
1	1	0	0		Q[n]	QN[n]
1	1	0	1		0	1
1	1	1	0		1	0
1	1	1	1		QN[n]	Q[n]

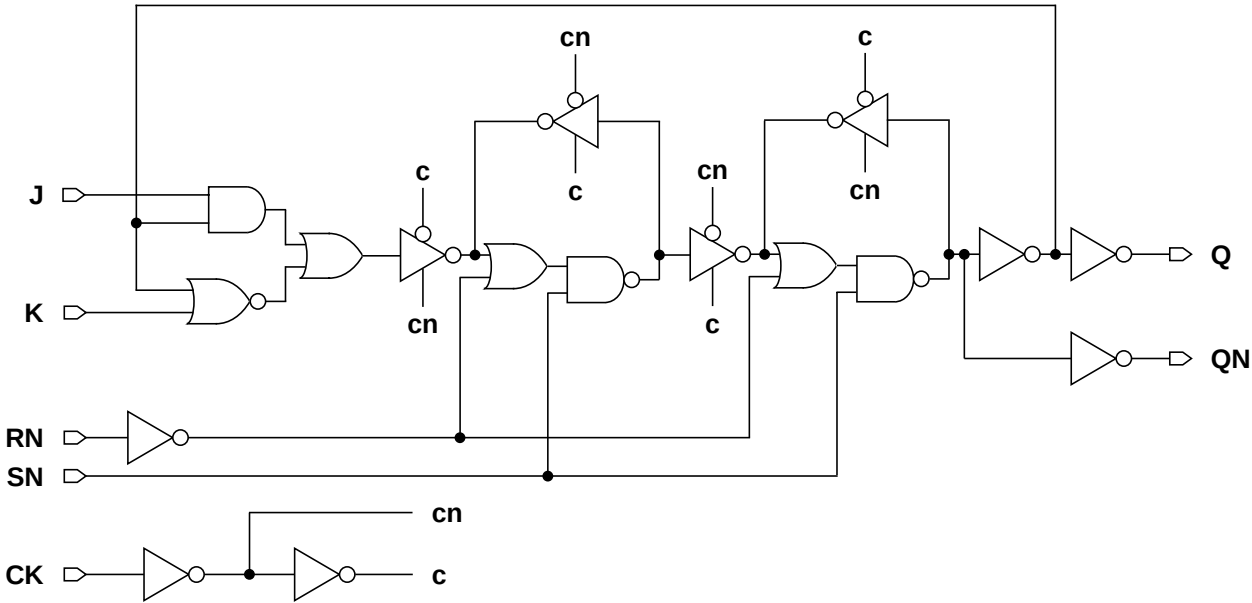
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
JKFFSRXL	5.04	19.80
JKFFSRX1	5.04	19.80
JKFFSRX2	5.04	20.46
JKFFSRX4	5.04	21.78

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
J	0.0340	0.0289	0.0322	0.0321
K	0.0408	0.0348	0.0409	0.0407
CK	0.0409	0.0388	0.0406	0.0411
SN	0.0051	0.0058	0.0092	0.0092
RN	0.0153	0.0171	0.0225	0.0225
Q	0.0745	0.0802	0.1147	0.1635

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
J	0.0028	0.0018	0.0020	0.0020
K	0.0027	0.0027	0.0027	0.0027
CK	0.0020	0.0025	0.0030	0.0030
SN	0.0045	0.0051	0.0073	0.0073
RN	0.0025	0.0030	0.0041	0.0041

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.5705	0.4735	0.4512	0.5020	6.2868	4.5362	2.2648	1.1120
CK $\rightarrow$ Q $\downarrow$	0.4529	0.3806	0.3668	0.4021	3.6003	2.4331	1.2917	0.6291
SN $\rightarrow$ Q $\uparrow$	0.2670	0.2178	0.1956	0.2301	6.2838	4.5352	2.2642	1.1117
SN $\rightarrow$ Q $\downarrow$	0.2999	0.2228	0.2225	0.2560	3.5824	2.4264	1.2895	0.6284
RN $\rightarrow$ Q $\downarrow$	0.3838	0.3002	0.2922	0.3259	3.5827	2.4265	1.2895	0.6284
CK $\rightarrow$ QN $\uparrow$	0.3009	0.2781	0.2676	0.2910	6.2886	5.0730	2.2695	1.1169
CK $\rightarrow$ QN $\downarrow$	0.4525	0.3978	0.3832	0.4247	3.7220	2.7759	1.3180	0.6537
SN $\rightarrow$ QN $\uparrow$	0.1465	0.1194	0.1228	0.1447	6.3119	5.0829	2.2744	1.1191
SN $\rightarrow$ QN $\downarrow$	0.1651	0.1458	0.1299	0.1573	3.5069	2.7315	1.2973	0.6359
RN $\rightarrow$ QN $\uparrow$	0.2304	0.1967	0.1923	0.2145	6.3104	5.0824	2.2743	1.1190

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
J	setup↑ → CK	0.1289	0.1875	0.1680	0.1680
	setup↓ → CK	0.1211	0.3203	0.2578	0.2539
	hold↑ → CK	-0.1172	-0.1680	-0.1523	-0.1523
	hold↓ → CK	-0.0742	-0.2461	-0.1992	-0.1953
K	setup↑ → CK	0.1875	0.3242	0.2812	0.2734
	setup↓ → CK	0.1836	0.2148	0.1953	0.1953
	hold↑ → CK	-0.0898	-0.2344	-0.1953	-0.1875
	hold↓ → CK	-0.1758	-0.1875	-0.1758	-0.1758
CK	minpwh	0.1480	0.1432	0.1383	0.1432
	minpwl	0.2693	0.2742	0.2450	0.2450
SN	minpwl	0.1771	0.1577	0.1432	0.1771
	recovery	-0.0234	0.0039	0.0000	-0.0039
	removal	0.1562	0.1094	0.1172	0.1172
RN	minpwl	0.2208	0.1965	0.2305	0.2499
	recovery	0.0664	0.1328	0.0859	0.0820
	removal	-0.0508	-0.1055	-0.0625	-0.0664

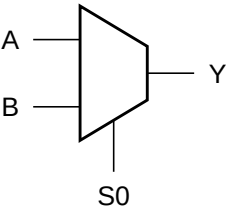
Process Technology:
TSMC CL018G

Cell Description

The MX2 cell is a 2-to-1 multiplexer. The state of the select input (S0) determines which data input (A, B) is presented to the output (Y).The output (Y) is represented by the logic equation:

$Y = (\overline{S0} \bullet A) + (S0 \bullet B)$

Logic Symbol



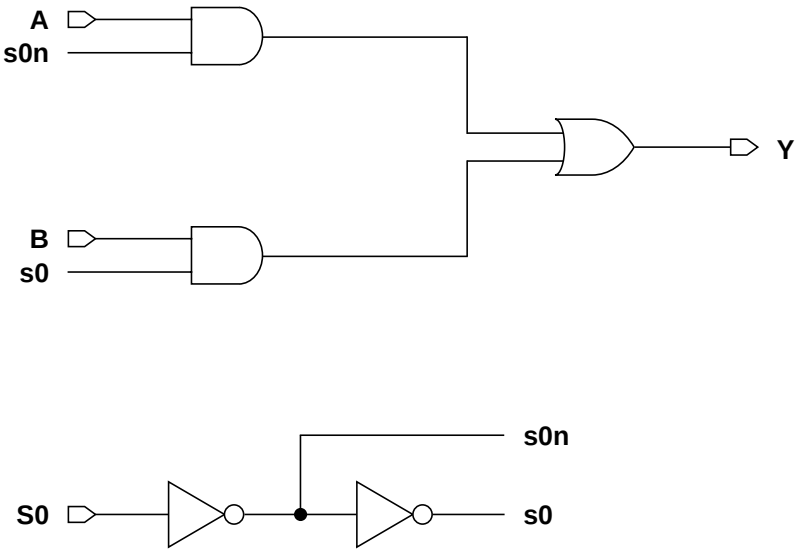
Functions

S0	A	B	Y
0	0	x	0
0	1	x	1
1	x	0	0
1	x	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
MX2XL	5.04	5.28
MX2X1	5.04	5.28
MX2X2	5.04	5.94
MX2X4	5.04	6.60

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
S0	0.0368	0.0378	0.0625	0.0890
A	0.0229	0.0276	0.0475	0.0735
B	0.0256	0.0303	0.0536	0.0795

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
S0	0.0063	0.0059	0.0089	0.0103
A	0.0024	0.0035	0.0059	0.0073
B	0.0024	0.0034	0.0053	0.0062

Delay Tables at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
S0 $\rightarrow$ Y $\uparrow$	0.1386	0.1318	0.1339	0.1421
S0 $\rightarrow$ Y $\downarrow$	0.1438	0.1250	0.1252	0.1458
A $\rightarrow$ Y $\uparrow$	0.1121	0.0938	0.0884	0.1010
A $\rightarrow$ Y $\downarrow$	0.1697	0.1314	0.1282	0.1402
B $\rightarrow$ Y $\uparrow$	0.1113	0.0913	0.0892	0.0975
B $\rightarrow$ Y $\downarrow$	0.1729	0.1347	0.1331	0.1557

Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
S0 $\rightarrow$ Y $\uparrow$	6.2719	4.5326	2.2505	1.1335
S0 $\rightarrow$ Y $\downarrow$	3.6246	2.4427	1.3034	0.6783
A $\rightarrow$ Y $\uparrow$	6.2737	4.5328	2.2506	1.1340
A $\rightarrow$ Y $\downarrow$	3.6188	2.4494	1.3039	0.6744
B $\rightarrow$ Y $\uparrow$	6.2741	4.5332	2.2509	1.1338
B $\rightarrow$ Y $\downarrow$	3.6257	2.4505	1.3036	0.6784

Cell Description

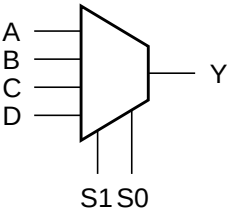
The MX4 cell is a 4-to-1 multiplexer. The state of the select inputs (S1, S0) determines which data input (A, B, C, D) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = (\overline{S0} \bullet \overline{S1} \bullet A) + (S0 \bullet \overline{S1} \bullet B) + (\overline{S0} \bullet S1 \bullet C) + (S0 \bullet S1 \bullet D)$$

Functions

S1	S0	A	B	C	D	Y
0	0	0	x	x	x	0
0	0	1	x	x	x	1
0	1	x	0	x	x	0
0	1	x	1	x	x	1
1	0	x	x	0	x	0
1	0	x	x	1	x	1
1	1	x	x	x	0	0
1	1	x	x	x	1	1

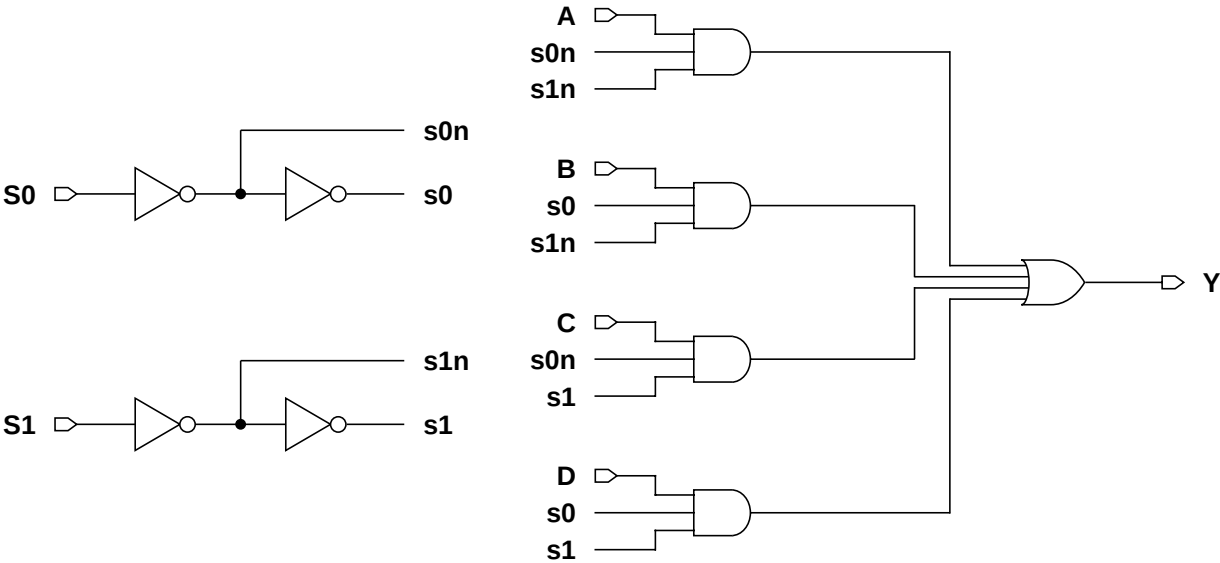
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
MX4XL	5.04	13.20
MX4X1	5.04	13.20
MX4X2	5.04	15.18
MX4X4	5.04	15.84

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
S0	0.0610	0.0763	0.1213	0.1422
S1	0.0351	0.0435	0.0678	0.0823
A	0.0339	0.0454	0.0740	0.1038
B	0.0372	0.0498	0.0831	0.1128
C	0.0381	0.0506	0.0847	0.1144
D	0.0425	0.0553	0.0934	0.1231

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
S0	0.0108	0.0140	0.0228	0.0228
S1	0.0062	0.0070	0.0110	0.0110
A	0.0021	0.0044	0.0068	0.0068
B	0.0021	0.0043	0.0067	0.0067
C	0.0020	0.0043	0.0069	0.0069
D	0.0022	0.0044	0.0068	0.0068

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
S0 $\rightarrow$ Y $\uparrow$	0.2499	0.1856	0.1766	0.1910	6.3153	4.5421	2.2558	1.1097
S0 $\rightarrow$ Y $\downarrow$	0.2795	0.2093	0.2130	0.2436	4.0854	2.5636	1.3553	0.7094
S1 $\rightarrow$ Y $\uparrow$	0.1493	0.1460	0.1344	0.1480	6.3127	4.5407	2.2546	1.1094
S1 $\rightarrow$ Y $\downarrow$	0.1389	0.1367	0.1338	0.1646	4.0007	2.5478	1.3475	0.7049
A $\rightarrow$ Y $\uparrow$	0.1890	0.1403	0.1374	0.1534	6.3159	4.5426	2.2550	1.1094
A $\rightarrow$ Y $\downarrow$	0.2777	0.1941	0.1925	0.2224	4.0668	2.5601	1.3508	0.7071
B $\rightarrow$ Y $\uparrow$	0.1883	0.1389	0.1381	0.1543	6.3154	4.5424	2.2553	1.1097
B $\rightarrow$ Y $\downarrow$	0.2817	0.1978	0.1977	0.2276	4.0732	2.5616	1.3505	0.7069
C $\rightarrow$ Y $\uparrow$	0.1949	0.1422	0.1414	0.1565	6.3205	4.5431	2.2561	1.1099
C $\rightarrow$ Y $\downarrow$	0.2890	0.2025	0.2046	0.2344	4.1063	2.5671	1.3562	0.7096
D $\rightarrow$ Y $\uparrow$	0.1942	0.1401	0.1408	0.1559	6.3193	4.5429	2.2561	1.1100
D $\rightarrow$ Y $\downarrow$	0.2939	0.2062	0.2084	0.2382	4.1108	2.5678	1.3557	0.7093

Cell Description

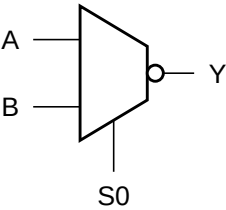
The MXI2 cell is a 2-to-1 multiplexer with inverted output. The state of the select input (S0) determines which data input (A, B) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = (\overline{S0} \bullet A) + (S0 \bullet B)$$

Functions

S0	A	B	Y
0	0	x	1
0	1	x	0
1	x	0	1
1	x	1	0

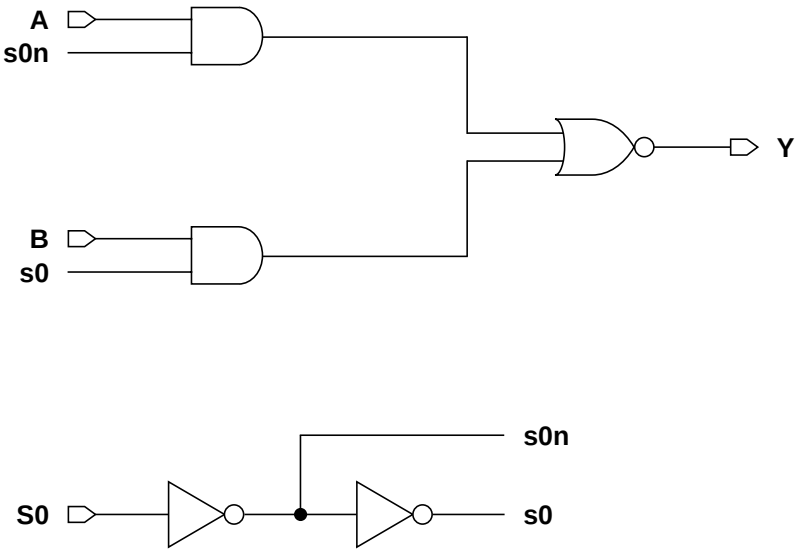
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
MXI2XL	5.04	4.62
MXI2X1	5.04	4.62
MXI2X2	5.04	5.28
MXI2X4	5.04	9.24

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
S0	0.0272	0.0316	0.0536	0.1004
A	0.0173	0.0222	0.0368	0.0790
B	0.0187	0.0262	0.0441	0.0955

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
S0	0.0064	0.0070	0.0100	0.0210
A	0.0027	0.0039	0.0070	0.0137
B	0.0021	0.0041	0.0063	0.0138

Delay Tables at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
S0 $\rightarrow$ Y $\uparrow$	0.0622	0.0604	0.0614	0.0576
S0 $\rightarrow$ Y $\downarrow$	0.0609	0.0771	0.0673	0.0703
A $\rightarrow$ Y $\uparrow$	0.0684	0.0569	0.0557	0.0548
A $\rightarrow$ Y $\downarrow$	0.0473	0.0415	0.0448	0.0442
B $\rightarrow$ Y $\uparrow$	0.0936	0.0602	0.0584	0.0607
B $\rightarrow$ Y $\downarrow$	0.0540	0.0381	0.0444	0.0407

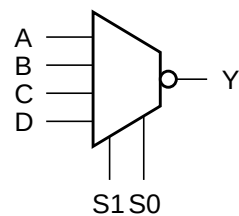
Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
S0 $\rightarrow$ Y $\uparrow$	9.7822	4.8874	2.8189	1.3693
S0 $\rightarrow$ Y $\downarrow$	4.7414	2.7786	1.8772	0.8381
A $\rightarrow$ Y $\uparrow$	7.0444	4.9229	2.8272	1.3758
A $\rightarrow$ Y $\downarrow$	3.9398	2.8465	1.7447	0.8551
B $\rightarrow$ Y $\uparrow$	9.9930	4.9225	2.8748	1.3735
B $\rightarrow$ Y $\downarrow$	5.0681	2.8447	1.9623	0.8528

Cell Description

The MXI4 cell is a 4-to-1 multiplexer with inverted output. The state of the select inputs (S1, S0) determines which data input (A, B, C, D) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = (\overline{S0} \bullet \overline{S1} \bullet A) + (S0 \bullet \overline{S1} \bullet B) + (\overline{S0} \bullet S1 \bullet C) + (S0 \bullet S1 \bullet D)$$

Logic Symbol



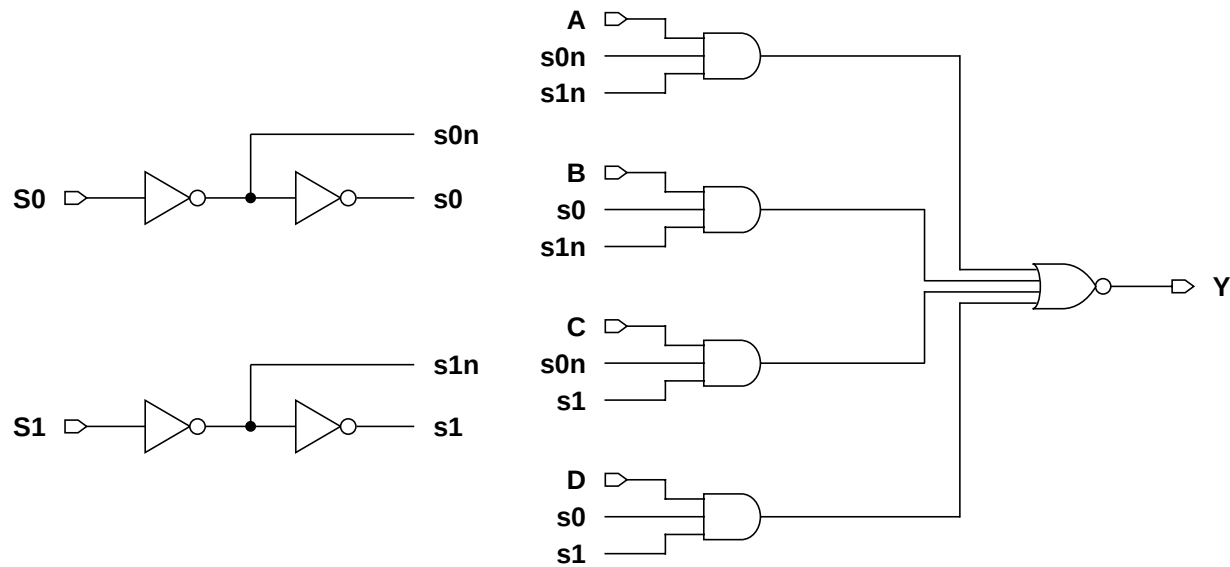
Functions

S1	S0	A	B	C	D	Y
0	0	0	x	x	x	1
0	0	1	x	x	x	0
0	1	x	0	x	x	1
0	1	x	1	x	x	0
1	0	x	x	0	x	1
1	0	x	x	1	x	0
1	1	x	x	x	0	1
1	1	x	x	x	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
MXI4XL	5.04	15.18
MXI4X1	5.04	15.18
MXI4X2	5.04	15.84
MXI4X4	5.04	16.50

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
S0	0.0679	0.0646	0.1109	0.1475
S1	0.0381	0.0390	0.0612	0.0849
A	0.0454	0.0484	0.0864	0.1187
B	0.0492	0.0510	0.0918	0.1229
C	0.0402	0.0426	0.0728	0.0997
D	0.0426	0.0449	0.0777	0.1060

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
S0	0.0106	0.0089	0.0144	0.0172
S1	0.0044	0.0043	0.0053	0.0062
A	0.0022	0.0027	0.0046	0.0056
B	0.0021	0.0026	0.0046	0.0058
C	0.0024	0.0028	0.0049	0.0058
D	0.0021	0.0026	0.0046	0.0055

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
S0 $\rightarrow$ Y $\uparrow$	0.2882	0.2428	0.2197	0.2271	6.2797	4.5358	2.3236	1.1270
S0 $\rightarrow$ Y $\downarrow$	0.3361	0.2804	0.2555	0.2644	3.6596	2.4624	1.3064	0.6581
S1 $\rightarrow$ Y $\uparrow$	0.1508	0.1420	0.1361	0.1387	6.2762	4.5338	2.3229	1.1266
S1 $\rightarrow$ Y $\downarrow$	0.1586	0.1309	0.1231	0.1347	3.6491	2.4595	1.3013	0.6561
A $\rightarrow$ Y $\uparrow$	0.2998	0.2429	0.2238	0.2310	6.2799	4.5358	2.3237	1.1270
A $\rightarrow$ Y $\downarrow$	0.2784	0.2254	0.2104	0.2245	3.6591	2.4623	1.3064	0.6581
B $\rightarrow$ Y $\uparrow$	0.3014	0.2442	0.2287	0.2354	6.2799	4.5358	2.3236	1.1270
B $\rightarrow$ Y $\downarrow$	0.2767	0.2223	0.2096	0.2215	3.6591	2.4623	1.3063	0.6581
C $\rightarrow$ Y $\uparrow$	0.2798	0.2265	0.2017	0.2017	6.2773	4.5352	2.3231	1.1266
C $\rightarrow$ Y $\downarrow$	0.2672	0.2172	0.1959	0.2100	3.6506	2.4606	1.3017	0.6573
D $\rightarrow$ Y $\uparrow$	0.2804	0.2271	0.2027	0.2067	6.2773	4.5352	2.3232	1.1265
D $\rightarrow$ Y $\downarrow$	0.2649	0.2141	0.1965	0.2076	3.6506	2.4604	1.3017	0.6573

Cell Description

The NAND2 cell provides the logical NAND of two inputs (A, B). The output (Y) is represented by the logic equation:

$$Y = \overline{(A \bullet B)}$$

Logic Symbol



Functions

A	B	Y
0	x	1
x	0	1
1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND2XL	5.04	1.98
NAND2X1	5.04	1.98
NAND2X2	5.04	3.30
NAND2X4	5.04	4.62

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0106	0.0139	0.0258	0.0501
B	0.0135	0.0181	0.0351	0.0683

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0033	0.0042	0.0082	0.0163
B	0.0029	0.0039	0.0086	0.0158

Delay at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0324	0.0313	0.0286	0.0295
A $\rightarrow$ Y $\downarrow$	0.0198	0.0195	0.0179	0.0181
B $\rightarrow$ Y $\uparrow$	0.0409	0.0400	0.0389	0.0395
B $\rightarrow$ Y $\downarrow$	0.0241	0.0242	0.0239	0.0235

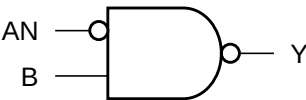
Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	6.2614	4.5288	2.1944	1.1638
A $\rightarrow$ Y $\downarrow$	3.8144	2.8429	1.3901	0.7200
B $\rightarrow$ Y $\uparrow$	6.2527	4.5253	2.1935	1.1631
B $\rightarrow$ Y $\downarrow$	3.8226	2.8470	1.3923	0.7210

Cell Description

The NAND2B cell provides the logical NAND of one inverted input (AN) and one non-inverted input (B). The output (Y) is represented by the logic equation:

$$Y = \overline{(AN \bullet B)}$$

Logic Symbol



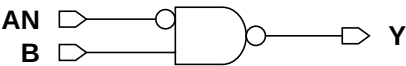
Functions

AN	B	Y
1	x	1
x	0	1
0	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND2BXL	5.04	2.64
NAND2BX1	5.04	2.64
NAND2BX2	5.04	3.96
NAND2BX4	5.04	5.28

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0172	0.0201	0.0366	0.0697
B	0.0113	0.0145	0.0282	0.0546

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0022	0.0022	0.0036	0.0065
B	0.0031	0.0041	0.0086	0.0160

Delays at 25°C, 1.8V, Typical Process

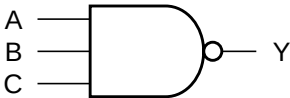
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0677	0.0714	0.0691	0.0707	6.2671	4.5324	2.2190	1.1649
AN $\rightarrow$ Y $\downarrow$	0.0902	0.1002	0.0956	0.0905	3.8602	2.8062	1.4027	0.7257
B $\rightarrow$ Y $\uparrow$	0.0409	0.0394	0.0382	0.0385	6.2588	4.5291	2.2532	1.1641
B $\rightarrow$ Y $\downarrow$	0.0264	0.0264	0.0262	0.0265	3.8340	2.7914	1.3959	0.7228

Cell Description

The NAND3 cell provides the logical NAND of three inputs (A, B, C). The output (Y) is represented by the logic equation:

$$Y = \overline{(A \bullet B \bullet C)}$$

Logic Symbol



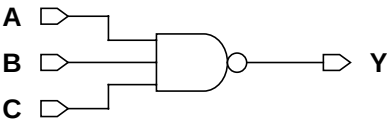
Functions

A	B	C	Y
0	x	x	1
x	0	x	1
x	x	0	1
1	1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND3XL	5.04	2.64
NAND3X1	5.04	2.64
NAND3X2	5.04	4.62
NAND3X4	5.04	6.60

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0122	0.0170	0.0299	0.0510
B	0.0161	0.0226	0.0409	0.0672
C	0.0200	0.0284	0.0512	0.0866

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0036	0.0050	0.0087	0.0151
B	0.0033	0.0048	0.0091	0.0144
C	0.0033	0.0046	0.0098	0.0147

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0453	0.0381	0.0351	0.0377
A $\rightarrow$ Y $\downarrow$	0.0303	0.0270	0.0228	0.0241
B $\rightarrow$ Y $\uparrow$	0.0565	0.0491	0.0481	0.0497
B $\rightarrow$ Y $\downarrow$	0.0379	0.0352	0.0320	0.0323
C $\rightarrow$ Y $\uparrow$	0.0669	0.0586	0.0595	0.0613
C $\rightarrow$ Y $\downarrow$	0.0414	0.0388	0.0365	0.0366

Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	6.2638	4.0570	2.1806	1.4261
A $\rightarrow$ Y $\downarrow$	4.3758	3.1524	1.5760	1.0130
B $\rightarrow$ Y $\uparrow$	6.2557	4.0544	2.1797	1.4254
B $\rightarrow$ Y $\downarrow$	4.3830	3.1561	1.5780	1.0139
C $\rightarrow$ Y $\uparrow$	6.2631	4.0587	2.1822	1.4266
C $\rightarrow$ Y $\downarrow$	4.3827	3.1561	1.5782	1.0140

Cell Description

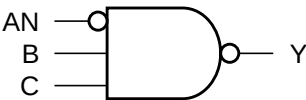
The NAND3B cell provides the logical NAND of one inverted input (AN) and two non-inverted inputs (B, C). The output (Y) is represented by the logic equation:

$Y = \overline{AN} \bullet B \bullet C$

Functions

AN	B	C	Y
1	x	x	1
x	0	x	1
x	x	0	1
0	1	1	0

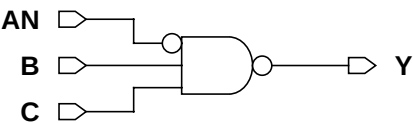
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND3BXL	5.04	3.30
NAND3BX1	5.04	3.30
NAND3BX2	5.04	5.28
NAND3BX4	5.04	7.26

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0244	0.0294	0.0434	0.0726
B	0.0137	0.0184	0.0334	0.0529
C	0.0171	0.0239	0.0449	0.0708

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0024	0.0025	0.0039	0.0066
B	0.0034	0.0047	0.0092	0.0143
C	0.0033	0.0047	0.0100	0.0150

Delays at 25°C, 1.8V, Typical Process

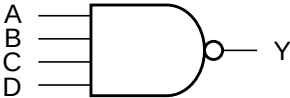
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0869	0.0898	0.0752	0.0748	6.2698	4.0995	2.1819	1.3884
AN $\rightarrow$ Y $\downarrow$	0.1116	0.1215	0.0970	0.0903	4.4100	3.1715	1.5841	1.0226
B $\rightarrow$ Y $\uparrow$	0.0556	0.0507	0.0470	0.0472	6.2629	4.0972	2.1817	1.3880
B $\rightarrow$ Y $\downarrow$	0.0392	0.0401	0.0347	0.0349	4.3921	3.1621	1.5807	1.0213
C $\rightarrow$ Y $\uparrow$	0.0669	0.0616	0.0591	0.0590	6.2667	4.0995	2.1831	1.3940
C $\rightarrow$ Y $\downarrow$	0.0439	0.0449	0.0397	0.0393	4.3930	3.1625	1.5809	1.0213

Cell Description

The NAND4 cell provides a logical NAND of four inputs (A, B, C, D). The output (Y) is represented by the logic equation:

$$Y = \overline{(A \bullet B \bullet C \bullet D)}$$

Logic Symbol



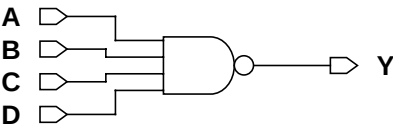
Functions

A	B	C	D	Y
0	x	x	x	1
x	0	x	x	1
x	x	0	x	1
x	x	x	0	1
1	1	1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND4XL	5.04	3.30
NAND4X1	5.04	3.30
NAND4X2	5.04	5.94
NAND4X4	5.04	11.22

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A	0.0137	0.0181	0.0341	0.0693
B	0.0185	0.0241	0.0454	0.0917
C	0.0226	0.0310	0.0591	0.1191
D	0.0264	0.0366	0.0714	0.1447

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0034	0.0050	0.0092	0.0194
B	0.0037	0.0049	0.0094	0.0196
C	0.0036	0.0049	0.0098	0.0200
D	0.0033	0.0047	0.0104	0.0215

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0449	0.0423	0.0410	0.0414
A $\rightarrow$ Y $\downarrow$	0.0302	0.0297	0.0280	0.0284
B $\rightarrow$ Y $\uparrow$	0.0608	0.0559	0.0549	0.0549
B $\rightarrow$ Y $\downarrow$	0.0437	0.0417	0.0399	0.0400
C $\rightarrow$ Y $\uparrow$	0.0727	0.0678	0.0674	0.0674
C $\rightarrow$ Y $\downarrow$	0.0503	0.0486	0.0472	0.0472
D $\rightarrow$ Y $\uparrow$	0.0815	0.0776	0.0794	0.0802
D $\rightarrow$ Y $\downarrow$	0.0527	0.0523	0.0525	0.0532

Delays at 25°C, 1.8V, Typical Process

Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	6.2621	4.2594	2.1807	1.0904
A $\rightarrow$ Y $\downarrow$	5.0643	3.6636	1.8317	0.9159
B $\rightarrow$ Y $\uparrow$	6.2599	4.2572	2.1796	1.0898
B $\rightarrow$ Y $\downarrow$	5.0740	3.6676	1.8337	0.9169
C $\rightarrow$ Y $\uparrow$	6.2676	4.2619	2.1821	1.0911
C $\rightarrow$ Y $\downarrow$	5.0738	3.6677	1.8337	0.9169
D $\rightarrow$ Y $\uparrow$	6.2858	4.2722	2.1877	1.0939
D $\rightarrow$ Y $\downarrow$	5.0726	3.6675	1.8341	0.9171

Cell Description

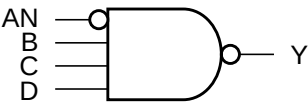
The NAND4B cell provides a logical NAND of one inverted input (AN) and three non-inverted inputs (B, C, D). The output (Y) is represented by the logic equation:

$Y = \overline{(AN \bullet B \bullet C \bullet D)}$

Functions

AN	B	C	D	Y
1	x	x	x	1
x	0	x	x	1
x	x	0	x	1
x	x	x	0	1
0	1	1	1	0

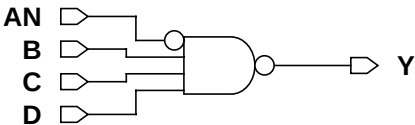
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND4BXL	5.04	3.96
NAND4BX1	5.04	4.62
NAND4BX2	5.04	6.60
NAND4BX4	5.04	11.88

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
AN	0.0230	0.0281	0.0501	0.0982
B	0.0144	0.0214	0.0372	0.0751
C	0.0183	0.0269	0.0492	0.0977
D	0.0223	0.0326	0.0619	0.1235

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0024	0.0022	0.0040	0.0071
B	0.0036	0.0049	0.0094	0.0196
C	0.0034	0.0048	0.0098	0.0200
D	0.0033	0.0046	0.0106	0.0217

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0873	0.0915	0.0820	0.0845	6.2726	4.2620	2.1820	1.0912
AN $\rightarrow$ Y $\downarrow$	0.1105	0.1189	0.1034	0.1006	5.0901	3.6754	1.8373	0.9185
B $\rightarrow$ Y $\uparrow$	0.0592	0.0569	0.0537	0.0534	6.2676	4.2619	2.1818	1.0910
B $\rightarrow$ Y $\downarrow$	0.0457	0.0467	0.0425	0.0421	5.0819	3.6724	1.8355	0.9178
C $\rightarrow$ Y $\uparrow$	0.0716	0.0695	0.0668	0.0666	6.2708	4.2639	2.1830	1.0915
C $\rightarrow$ Y $\downarrow$	0.0526	0.0544	0.0503	0.0502	5.0815	3.6724	1.8356	0.9178
D $\rightarrow$ Y $\uparrow$	0.0820	0.0800	0.0793	0.0798	6.2871	4.2722	2.1877	1.0940
D $\rightarrow$ Y $\downarrow$	0.0561	0.0582	0.0556	0.0562	5.0814	3.6721	1.8357	0.9179

Cell Description

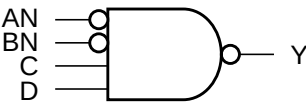
The NAND4BB cell provides a logical NAND of two inverted inputs (AN, BN) and two non-inverted inputs (C, D). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} \bullet \overline{BN} \bullet C \bullet D)}$$

Functions

AN	BN	C	D	Y
1	x	x	x	1
x	1	x	x	1
x	x	0	x	1
x	x	x	0	1
0	0	1	1	0

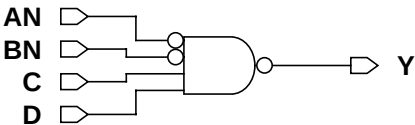
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND4BBXL	5.04	5.28
NAND4BBX1	5.04	5.28
NAND4BBX2	5.04	7.26
NAND4BBX4	5.04	12.54

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0234	0.0286	0.0514	0.1003
BN	0.0268	0.0333	0.0621	0.1187
C	0.0170	0.0228	0.0420	0.0840
D	0.0209	0.0283	0.0534	0.1067

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0023	0.0022	0.0040	0.0071
BN	0.0023	0.0023	0.0038	0.0072
C	0.0035	0.0049	0.0099	0.0200
D	0.0034	0.0048	0.0106	0.0214

Delays at 25°C, 1.8V, Typical Process

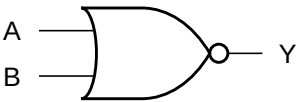
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0886	0.0917	0.0830	0.0853	6.2658	4.2623	2.1821	1.0913
AN $\rightarrow$ Y $\downarrow$	0.1101	0.1187	0.1039	0.1007	5.0827	3.6750	1.8370	0.9183
BN $\rightarrow$ Y $\uparrow$	0.1020	0.1056	0.0983	0.0966	6.2683	4.2638	2.1826	1.0913
BN $\rightarrow$ Y $\downarrow$	0.1203	0.1308	0.1169	0.1119	5.1000	3.6853	1.8409	0.9201
C $\rightarrow$ Y $\uparrow$	0.0736	0.0684	0.0665	0.0661	6.2754	4.2679	2.1851	1.0926
C $\rightarrow$ Y $\downarrow$	0.0557	0.0559	0.0525	0.0522	5.0847	3.6757	1.8372	0.9185
D $\rightarrow$ Y $\uparrow$	0.0845	0.0793	0.0793	0.0794	6.2898	4.2750	2.1890	1.0946
D $\rightarrow$ Y $\downarrow$	0.0596	0.0600	0.0583	0.0585	5.0843	3.6755	1.8374	0.9186

Cell Description

The NOR2 cell provides a logical NOR of two inputs (A, B). The output (Y) is represented by the logic equation:

$$Y = \overline{(A + B)}$$

Logic Symbol



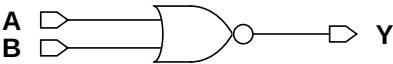
Functions

A	B	Y
0	0	1
x	1	0
1	x	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR2XL	5.04	1.98
NOR2X1	5.04	1.98
NOR2X2	5.04	3.30
NOR2X4	5.04	4.62

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A	0.0110	0.0143	0.0275	0.0545
B	0.0139	0.0182	0.0365	0.0728

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0033	0.0045	0.0083	0.0169
B	0.0029	0.0040	0.0085	0.0160

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0426	0.0403	0.0347	0.0351
A $\rightarrow$ Y $\downarrow$	0.0213	0.0196	0.0183	0.0187
B $\rightarrow$ Y $\uparrow$	0.0536	0.0519	0.0485	0.0473
B $\rightarrow$ Y $\downarrow$	0.0259	0.0244	0.0247	0.0245

Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	9.4704	6.7329	3.3656	1.6550
A $\rightarrow$ Y $\downarrow$	3.5015	2.3672	1.2659	0.6330
B $\rightarrow$ Y $\uparrow$	9.4589	6.7278	3.3647	1.6542
B $\rightarrow$ Y $\downarrow$	3.5234	2.3779	1.2717	0.6357

Cell Description

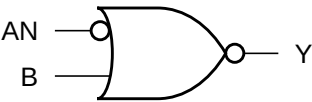
The NOR2B cell provides a logical NOR of one inverted input (AN) and one non-inverted input (B). The output (Y) is represented by the logic equation:

$Y = \overline{AN + B}$

Functions

AN	B	Y
1	0	1
x	1	0
0	x	0

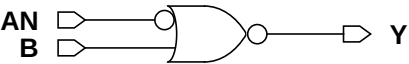
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR2BXL	5.04	2.64
NOR2BX1	5.04	2.64
NOR2BX2	5.04	3.96
NOR2BX4	5.04	5.28

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0171	0.0206	0.0326	0.0668
B	0.0137	0.0188	0.0365	0.0728

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0023	0.0023	0.0038	0.0067
B	0.0032	0.0043	0.0087	0.0162

Delays at 25°C, 1.8V, Typical Process

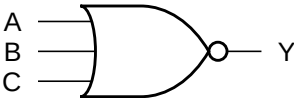
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0768	0.0803	0.0706	0.0725	9.4747	6.7347	3.3665	1.6479
AN $\rightarrow$ Y $\downarrow$	0.0968	0.1095	0.0958	0.0959	3.4031	2.4035	1.2815	0.6406
B $\rightarrow$ Y $\uparrow$	0.0547	0.0538	0.0496	0.0491	9.4633	6.7296	3.3653	1.6468
B $\rightarrow$ Y $\downarrow$	0.0247	0.0242	0.0242	0.0242	3.3643	2.3790	1.2722	0.6361

Cell Description

The NOR3 cell provides a logical NOR of three inputs (A, B, C). The output (Y) is represented by the logic equation:

$$Y = \overline{(A + B + C)}$$

Logic Symbol



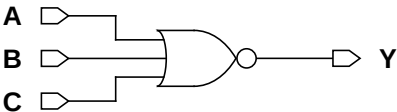
Functions

A	B	C	Y
0	0	0	1
x	x	1	0
x	1	x	0
1	x	x	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR3XL	5.04	2.64
NOR3X1	5.04	2.64
NOR3X2	5.04	4.62
NOR3X4	5.04	6.60

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0142	0.0188	0.0364	0.0573
B	0.0177	0.0239	0.0468	0.0735
C	0.0213	0.0296	0.0584	0.0923

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0036	0.0049	0.0092	0.0163
B	0.0034	0.0046	0.0097	0.0154
C	0.0033	0.0044	0.0098	0.0153

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0591	0.0550	0.0488	0.0484
A $\rightarrow$ Y $\downarrow$	0.0241	0.0230	0.0217	0.0210
B $\rightarrow$ Y $\uparrow$	0.0890	0.0844	0.0794	0.0772
B $\rightarrow$ Y $\downarrow$	0.0313	0.0305	0.0309	0.0292
C $\rightarrow$ Y $\uparrow$	0.1004	0.0961	0.0930	0.0900
C $\rightarrow$ Y $\downarrow$	0.0341	0.0340	0.0364	0.0342

Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	12.2929	8.5987	4.2983	2.6086
A $\rightarrow$ Y $\downarrow$	3.3371	2.3652	1.2664	0.7358
B $\rightarrow$ Y $\uparrow$	12.2809	8.5936	4.2966	2.6072
B $\rightarrow$ Y $\downarrow$	3.3580	2.3756	1.2704	0.7377
C $\rightarrow$ Y $\uparrow$	12.2779	8.5923	4.2972	2.6075
C $\rightarrow$ Y $\downarrow$	3.4066	2.3995	1.2809	0.7427

Cell Description

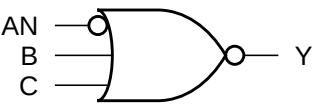
The NOR3B cell provides a logical NOR of one inverted input (AN) and two non-inverted inputs (B, C). The output (Y) is represented by the logic equation:

$Y = \overline{AN + B + C}$

Functions

AN	B	C	Y
1	0	0	1
x	x	1	0
x	1	x	0
0	x	x	0

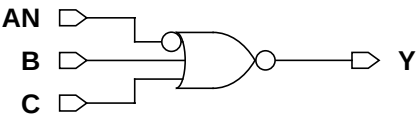
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR3BXL	5.04	3.96
NOR3BX1	5.04	3.96
NOR3BX2	5.04	5.28
NOR3BX4	5.04	7.26

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
AN	0.0190	0.0231	0.0374	0.0683
B	0.0181	0.0249	0.0454	0.0774
C	0.0215	0.0299	0.0576	0.0963

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0021	0.0023	0.0040	0.0064
B	0.0034	0.0046	0.0091	0.0159
C	0.0033	0.0045	0.0100	0.0163

Delays at 25°C, 1.8V, Typical Process

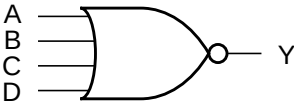
Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0967	0.0952	0.0832	0.0854	12.2974	8.6003	4.2991	2.4621
AN $\rightarrow$ Y $\downarrow$	0.1063	0.1101	0.1018	0.1049	3.4004	2.3978	1.2798	0.7428
B $\rightarrow$ Y $\uparrow$	0.0920	0.0873	0.0797	0.0777	12.2827	8.5945	4.2967	2.4604
B $\rightarrow$ Y $\downarrow$	0.0315	0.0305	0.0301	0.0300	3.3600	2.3769	1.2711	0.7381
C $\rightarrow$ Y $\uparrow$	0.1028	0.0983	0.0938	0.0909	12.2791	8.5931	4.2979	2.4607
C $\rightarrow$ Y $\downarrow$	0.0344	0.0342	0.0361	0.0360	3.4056	2.3992	1.2811	0.7429

Cell Description

The NOR4 cell provides a logical NOR of four inputs (A, B, C, D). The output (Y) is represented by the logic equation:

$$Y = \overline{(A + B + C + D)}$$

Logic Symbol



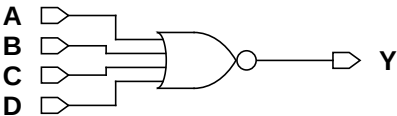
Functions

A	B	C	D	Y
0	0	0	0	1
x	x	x	1	0
x	x	1	x	0
x	1	x	x	0
1	x	x	x	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR4XL	5.04	3.96
NOR4X1	5.04	3.96
NOR4X2	5.04	5.94
NOR4X4	5.04	11.22

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0166	0.0218	0.0412	0.0757
B	0.0209	0.0280	0.0539	0.0989
C	0.0254	0.0350	0.0681	0.1205
D	0.0296	0.0403	0.0807	0.1435

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0041	0.0054	0.0100	0.0190
B	0.0039	0.0052	0.0103	0.0200
C	0.0038	0.0051	0.0109	0.0197
D	0.0036	0.0049	0.0113	0.0200

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0641	0.0559	0.0474	0.0524
A $\rightarrow$ Y $\downarrow$	0.0268	0.0251	0.0239	0.0244
B $\rightarrow$ Y $\uparrow$	0.1120	0.1041	0.0959	0.1032
B $\rightarrow$ Y $\downarrow$	0.0368	0.0362	0.0368	0.0373
C $\rightarrow$ Y $\uparrow$	0.1416	0.1326	0.1244	0.1296
C $\rightarrow$ Y $\downarrow$	0.0421	0.0417	0.0424	0.0420
D $\rightarrow$ Y $\uparrow$	0.1522	0.1433	0.1385	0.1444
D $\rightarrow$ Y $\downarrow$	0.0415	0.0421	0.0441	0.0438

Delays at 25°C, 1.8V, Typical Process

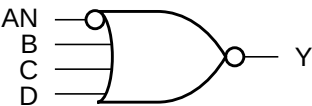
Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	13.9924	9.6051	4.8006	2.7817
A $\rightarrow$ Y $\downarrow$	3.3443	2.3689	1.2667	0.6982
B $\rightarrow$ Y $\uparrow$	13.9804	9.6020	4.8013	2.7821
B $\rightarrow$ Y $\downarrow$	3.3601	2.3748	1.2688	0.6992
C $\rightarrow$ Y $\uparrow$	13.9813	9.6023	4.8011	2.7815
C $\rightarrow$ Y $\downarrow$	3.4118	2.4023	1.2842	0.7067
D $\rightarrow$ Y $\uparrow$	13.9782	9.6012	4.8024	2.7821
D $\rightarrow$ Y $\downarrow$	3.4758	2.4345	1.3018	0.7154

Cell Description

The NOR4B cell provides a logical NOR of one inverted input (AN) and three non-inverted inputs (B, C, D). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{A}N + B + C + D)}$$

Logic Symbol



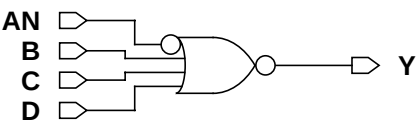
Functions

AN	B	C	D	Y
1	0	0	0	1
x	x	x	1	0
x	x	1	x	0
x	1	x	x	0
0	x	x	x	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR4BXL	5.04	3.96
NOR4BX1	5.04	3.96
NOR4BX2	5.04	6.60
NOR4BX4	5.04	11.88

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
AN	0.0217	0.0263	0.0446	0.0793
B	0.0202	0.0264	0.0542	0.0980
C	0.0247	0.0329	0.0686	0.1202
D	0.0290	0.0383	0.0801	0.1446

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0024	0.0027	0.0046	0.0071
B	0.0039	0.0051	0.0102	0.0197
C	0.0038	0.0050	0.0109	0.0198
D	0.0038	0.0049	0.0110	0.0203

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.1010	0.0983	0.0841	0.0904	13.9928	10.3136	4.8009	2.7821
AN $\rightarrow$ Y $\downarrow$	0.1196	0.1130	0.1081	0.1102	3.4092	2.3972	1.2997	0.7048
B $\rightarrow$ Y $\uparrow$	0.1102	0.1064	0.0950	0.1035	13.9843	10.3104	4.8009	2.7820
B $\rightarrow$ Y $\downarrow$	0.0359	0.0336	0.0354	0.0360	3.3625	2.3773	1.2911	0.7001
C $\rightarrow$ Y $\uparrow$	0.1386	0.1342	0.1261	0.1308	13.9808	10.3090	4.8018	2.7815
C $\rightarrow$ Y $\downarrow$	0.0410	0.0390	0.0433	0.0419	3.4119	2.4004	1.3043	0.7067
D $\rightarrow$ Y $\uparrow$	0.1497	0.1452	0.1383	0.1456	13.9795	10.3084	4.8019	2.7822
D $\rightarrow$ Y $\downarrow$	0.0407	0.0394	0.0448	0.0438	3.4745	2.4303	1.3214	0.7153

Cell Description

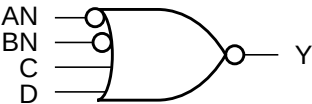
The NOR4BB cell provides a logical NOR of two inverted inputs (AN, BN) and two non-inverted inputs (C, D). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} + \overline{BN} + C + D)}$$

Functions

AN	BN	C	D	Y
1	1	0	0	1
x	x	x	1	0
x	x	1	x	0
x	0	x	x	0
0	x	x	x	0

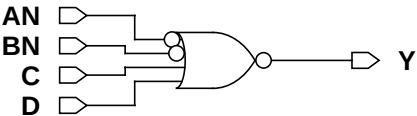
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR4BBXL	5.04	5.28
NOR4BBX1	5.04	5.28
NOR4BBX2	5.04	7.26
NOR4BBX4	5.04	13.20

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0213	0.0268	0.0451	0.0792
BN	0.0247	0.0314	0.0570	0.1000
C	0.0272	0.0369	0.0709	0.1221
D	0.0318	0.0426	0.0836	0.1445

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0023	0.0027	0.0046	0.0071
BN	0.0025	0.0028	0.0048	0.0070
C	0.0039	0.0052	0.0108	0.0198
D	0.0038	0.0051	0.0113	0.0201

Delays at 25°C, 1.8V, Typical Process

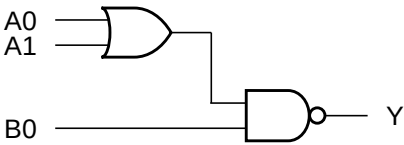
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.1074	0.0997	0.0885	0.0914	13.9900	9.6043	4.8021	2.7823
AN $\rightarrow$ Y $\downarrow$	0.1190	0.1153	0.1087	0.1108	3.3989	2.3944	1.2794	0.7047
BN $\rightarrow$ Y $\uparrow$	0.1641	0.1529	0.1375	0.1462	13.9807	9.6019	4.8010	2.7834
BN $\rightarrow$ Y $\downarrow$	0.1332	0.1293	0.1230	0.1284	3.4013	2.3958	1.2793	0.7053
C $\rightarrow$ Y $\uparrow$	0.1517	0.1401	0.1305	0.1335	13.9812	9.6020	4.8010	2.7818
C $\rightarrow$ Y $\downarrow$	0.0438	0.0427	0.0430	0.0418	3.4119	2.4021	1.2846	0.7070
D $\rightarrow$ Y $\uparrow$	0.1630	0.1515	0.1440	0.1477	13.9789	9.6016	4.8019	2.7822
D $\rightarrow$ Y $\downarrow$	0.0437	0.0437	0.0450	0.0440	3.4742	2.4333	1.3009	0.7151

Cell Description

The OAI21 cell provides the logical inverted AND of one OR group and an additional input. The output (Y) is represented by the logic equation:

$Y = \overline{(A0 + A1)} \bullet B0$

Logic Symbol



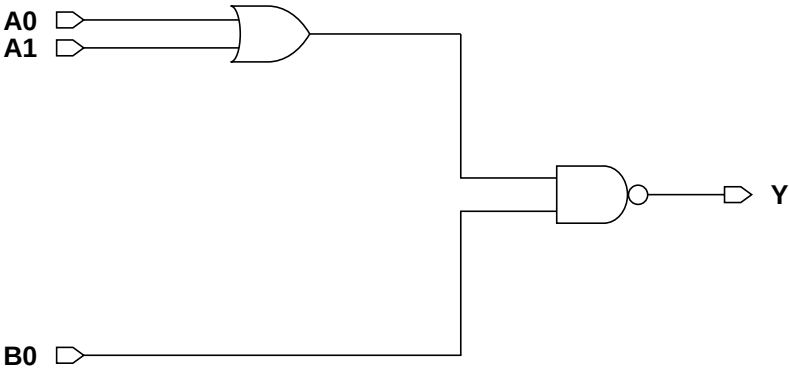
Functions

A0	A1	B0	Y
0	0	x	1
x	x	0	1
x	1	1	0
1	x	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI21XL	5.04	2.64
OAI21X1	5.04	3.30
OAI21X2	5.04	5.28
OAI21X4	5.04	7.26

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0185	0.0235	0.0448	0.0834
A1	0.0212	0.0277	0.0526	0.0980
B0	0.0153	0.0188	0.0349	0.0688

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0035	0.0046	0.0097	0.0186
A1	0.0034	0.0045	0.0089	0.0179
B0	0.0032	0.0042	0.0080	0.0156

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0761	0.0710	0.0721	0.0624	9.4687	6.7324	3.3666	1.6398
A0 $\rightarrow$ Y $\downarrow$	0.0343	0.0311	0.0322	0.0291	4.1217	2.7854	1.4083	0.7173
A1 $\rightarrow$ Y $\uparrow$	0.0891	0.0848	0.0845	0.0747	9.4637	6.7301	3.3653	1.6392
A1 $\rightarrow$ Y $\downarrow$	0.0424	0.0392	0.0401	0.0371	4.1345	2.7923	1.4114	0.7188
B0 $\rightarrow$ Y $\uparrow$	0.0377	0.0336	0.0333	0.0306	6.2747	4.5384	2.2689	1.1474
B0 $\rightarrow$ Y $\downarrow$	0.0334	0.0292	0.0293	0.0278	4.1293	2.7891	1.4098	0.7180

Cell Description

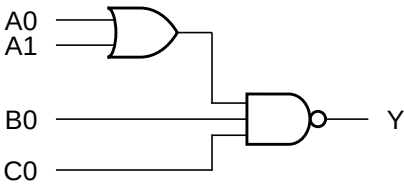
The OAI211 cell provides the logical inverted OR of one OR group and two additional inputs. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet B0 \bullet C0}$$

Functions

A0	A1	B0	C0	Y
0	0	x	x	1
x	x	0	x	1
x	x	x	0	1
x	1	1	1	0
1	x	1	1	0

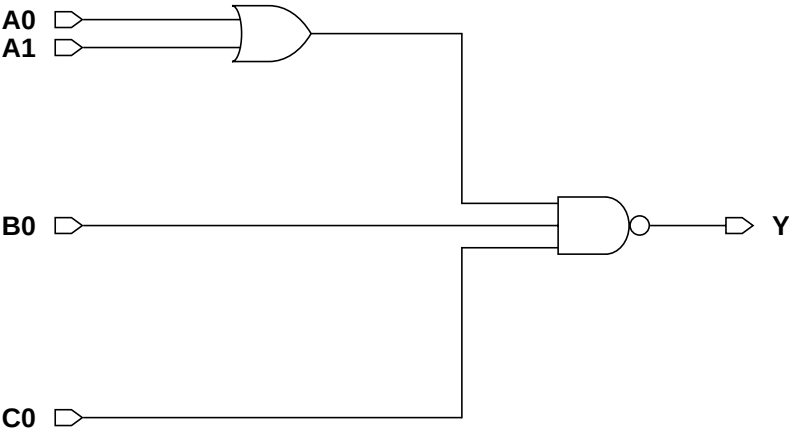
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI211XL	5.04	3.96
OAI211X1	5.04	3.96
OAI211X2	5.04	6.60
OAI211X4	5.04	5.94

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0	0.0245	0.0330	0.0630	0.0793
A1	0.0279	0.0374	0.0710	0.0847
B0	0.0156	0.0209	0.0384	0.0688
C0	0.0199	0.0269	0.0500	0.0715

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0036	0.0050	0.0104	0.0034
A1	0.0035	0.0048	0.0095	0.0033
B0	0.0034	0.0045	0.0085	0.0034
C0	0.0033	0.0044	0.0088	0.0029

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y↑	0.1105	0.1086	0.1053	0.2671	9.4862	6.7411	3.3711	1.1041
A0 → Y↓	0.0467	0.0463	0.0447	0.1824	4.3857	3.1786	1.5833	0.6418
A1 → Y↑	0.1243	0.1222	0.1175	0.2802	9.4827	6.7393	3.3702	1.1042
A1 → Y↓	0.0559	0.0559	0.0540	0.1933	4.3991	3.1851	1.5910	0.6418
B0 → Y↑	0.0437	0.0430	0.0413	0.1735	6.2786	4.5393	2.3564	1.1041
B0 → Y↓	0.0382	0.0384	0.0359	0.1768	4.3935	3.1823	1.5895	0.6418
C0 → Y↑	0.0568	0.0564	0.0553	0.1854	6.2725	4.5369	2.3551	1.1041
C0 → Y↓	0.0474	0.0475	0.0452	0.1846	4.4008	3.1859	1.5914	0.6418

Cell Description

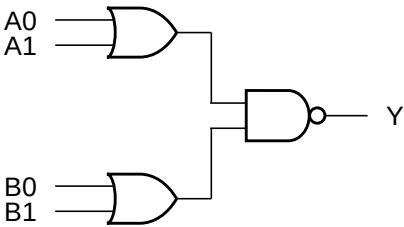
The OAI22 cell provides the logical inverted AND of two OR groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet (B0 + B1)}$$

Functions

A0	A1	B0	B1	Y
0	0	x	x	1
x	x	0	0	1
x	1	x	1	0
x	1	1	x	0
1	x	x	1	0
1	x	1	x	0

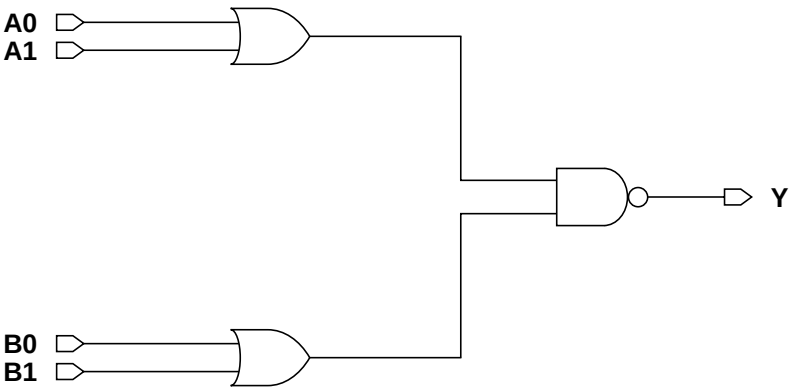
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI22XL	5.04	3.96
OAI22X1	5.04	3.96
OAI22X2	5.04	5.94
OAI22X4	5.04	9.24

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0	0.0166	0.0223	0.0441	0.0882
A1	0.0195	0.0269	0.0517	0.1060
B0	0.0249	0.0328	0.0666	0.1225
B1	0.0282	0.0367	0.0742	0.1381

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0035	0.0048	0.0099	0.0187
A1	0.0036	0.0048	0.0089	0.0180
B0	0.0036	0.0048	0.0099	0.0187
B1	0.0033	0.0044	0.0089	0.0180

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y↑	0.0543	0.0518	0.0559	0.0524	9.5107	6.7547	3.3776	1.6450
A0 → Y↓	0.0328	0.0329	0.0358	0.0361	3.8339	2.7899	1.4265	0.7350
A1 → Y↑	0.0692	0.0659	0.0676	0.0644	9.5108	6.7535	3.3758	1.6443
A1 → Y↓	0.0414	0.0418	0.0437	0.0446	3.8470	2.7956	1.4292	0.7363
B0 → Y↑	0.1000	0.0950	0.1002	0.0878	9.4722	6.7336	3.3668	1.6397
B0 → Y↓	0.0463	0.0455	0.0490	0.0467	3.8402	2.7930	1.4279	0.7357
B1 → Y↑	0.1119	0.1071	0.1120	0.1000	9.4665	6.7307	3.3655	1.6391
B1 → Y↓	0.0539	0.0535	0.0571	0.0553	3.8486	2.7972	1.4300	0.7368

Cell Description

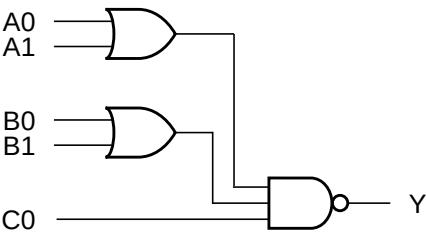
The OAI221 cell provides the logical inverted AND of two OR groups and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet (B0 + B1) \bullet C0}$$

Functions

A0	A1	B0	B1	C0	Y
0	0	x	x	x	1
x	x	0	0	x	1
x	x	x	x	0	1
x	1	x	1	1	0
x	1	1	x	1	0
1	x	x	1	1	0
1	x	1	x	1	0

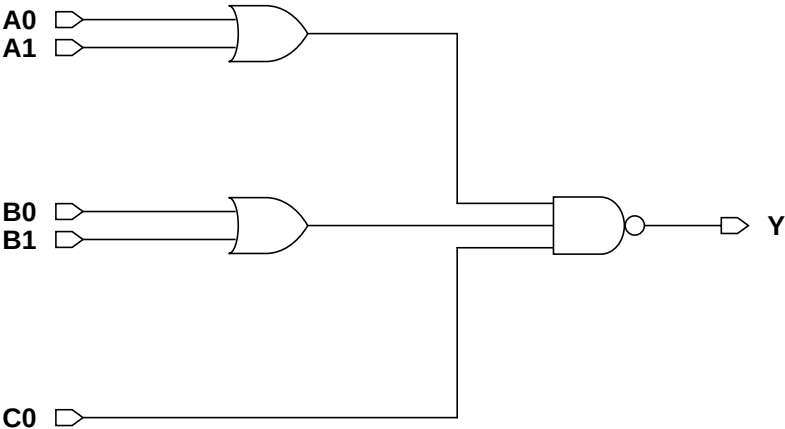
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI221XL	5.04	4.62
OAI221X1	5.04	5.28
OAI221X2	5.04	8.58
OAI221X4	5.04	7.26

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0253	0.0325	0.0622	0.0788
A1	0.0280	0.0366	0.0702	0.0844
B0	0.0328	0.0426	0.0813	0.0866
B1	0.0363	0.0466	0.0895	0.0918
C0	0.0196	0.0266	0.0488	0.0695

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0038	0.0052	0.0100	0.0035
A1	0.0037	0.0050	0.0100	0.0034
B0	0.0037	0.0051	0.0100	0.0035
B1	0.0035	0.0048	0.0099	0.0032
C0	0.0035	0.0045	0.0090	0.0034

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1029	0.0941	0.0888	0.2598	9.5087	6.7530	3.3770	1.1759
A0 $\rightarrow$ Y $\downarrow$	0.0568	0.0541	0.0516	0.1846	4.4017	3.1647	1.5877	0.6411
A1 $\rightarrow$ Y $\uparrow$	0.1157	0.1070	0.1012	0.2723	9.5048	6.7510	3.3758	1.1758
A1 $\rightarrow$ Y $\downarrow$	0.0661	0.0638	0.0610	0.1951	4.4095	3.1689	1.5896	0.6412
B0 $\rightarrow$ Y $\uparrow$	0.1423	0.1334	0.1265	0.2981	9.5101	6.7509	3.3761	1.1758
B0 $\rightarrow$ Y $\downarrow$	0.0659	0.0626	0.0597	0.1936	4.4009	3.1647	1.5876	0.6412
B1 $\rightarrow$ Y $\uparrow$	0.1544	0.1456	0.1390	0.3096	9.5067	6.7492	3.3751	1.1757
B1 $\rightarrow$ Y $\downarrow$	0.0750	0.0720	0.0692	0.2040	4.4091	3.1689	1.5898	0.6411
C0 $\rightarrow$ Y $\uparrow$	0.0458	0.0447	0.0401	0.1741	6.2950	4.5496	2.2751	1.1758
C0 $\rightarrow$ Y $\downarrow$	0.0471	0.0479	0.0444	0.1770	4.4108	3.1695	1.5900	0.6411

Cell Description

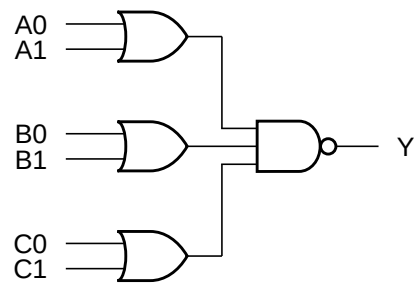
The OAI222 cell provides the logical inverted AND of three OR groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet (B0 + B1) \bullet (C0 + C1)}$$

Functions

A0	A1	B0	B1	C0	C1	Y
0	0	x	x	x	x	1
x	x	0	0	x	x	1
x	x	x	x	0	0	1
x	1	x	1	1	x	0
x	1	x	1	x	1	0
x	1	1	x	1	x	0
x	1	1	x	x	1	0
1	x	x	1	1	x	0
1	x	x	1	x	1	0
1	x	1	x	1	x	0
1	x	1	x	x	1	0

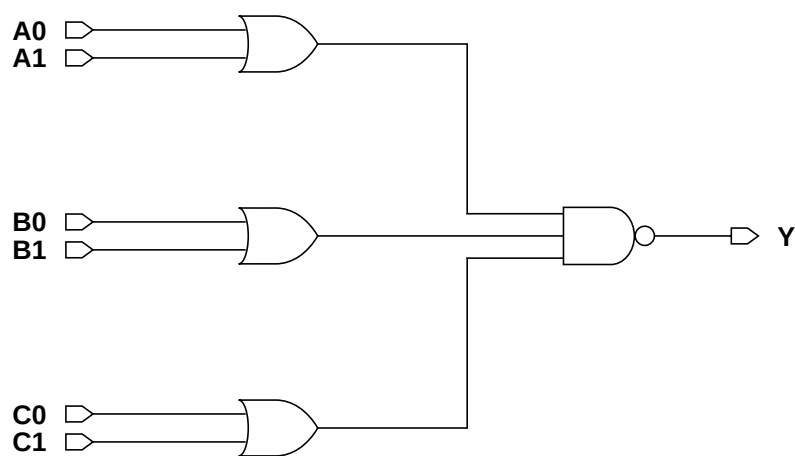
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI222XL	5.04	5.28
OAI222X1	5.04	5.94
OAI222X2	5.04	9.90
OAI222X4	5.04	7.92

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0	0.0335	0.0425	0.0837	0.0859
A1	0.0363	0.0466	0.0917	0.0911
B0	0.0409	0.0525	0.1033	0.0947
B1	0.0443	0.0566	0.1115	0.0990
C0	0.0224	0.0303	0.0589	0.0720
C1	0.0247	0.0340	0.0670	0.0782

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0038	0.0050	0.0101	0.0034
A1	0.0037	0.0050	0.0101	0.0034
B0	0.0037	0.0051	0.0100	0.0034
B1	0.0035	0.0048	0.0099	0.0032
C0	0.0040	0.0053	0.0100	0.0037
C1	0.0036	0.0050	0.0105	0.0033

Delays at 25°C, 1.8V, Typical Process

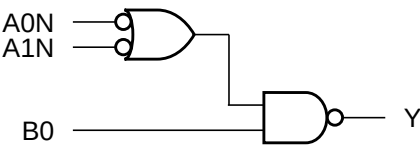
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1323	0.1194	0.1147	0.2905	9.4989	6.7458	3.3731	1.1759
A0 $\rightarrow$ Y $\downarrow$	0.0767	0.0729	0.0696	0.2109	4.4111	3.1700	1.5756	0.6414
A1 $\rightarrow$ Y $\uparrow$	0.1451	0.1328	0.1273	0.3040	9.4944	6.7442	3.3718	1.1759
A1 $\rightarrow$ Y $\downarrow$	0.0862	0.0830	0.0792	0.2204	4.4168	3.1732	1.5771	0.6414
B0 $\rightarrow$ Y $\uparrow$	0.1763	0.1633	0.1584	0.3326	9.4950	6.7441	3.3721	1.1757
B0 $\rightarrow$ Y $\downarrow$	0.0854	0.0813	0.0779	0.2197	4.4110	3.1700	1.5756	0.6414
B1 $\rightarrow$ Y $\uparrow$	0.1884	0.1755	0.1708	0.3450	9.4916	6.7423	3.3712	1.1757
B1 $\rightarrow$ Y $\downarrow$	0.0948	0.0910	0.0875	0.2308	4.4170	3.1731	1.5771	0.6414
C0 $\rightarrow$ Y $\uparrow$	0.0754	0.0717	0.0661	0.2327	9.5550	6.7777	3.3882	1.1759
C0 $\rightarrow$ Y $\downarrow$	0.0533	0.0541	0.0507	0.1876	4.4058	3.1669	1.5741	0.6414
C1 $\rightarrow$ Y $\uparrow$	0.0858	0.0835	0.0799	0.2428	9.5450	6.7737	3.3875	1.1760
C1 $\rightarrow$ Y $\downarrow$	0.0604	0.0626	0.0611	0.1946	4.4222	3.1750	1.5763	0.6414

Cell Description

The OAI2BB1 cell provides the logical inverted AND of one OR group of two inverted inputs (A0N, A1N) and an additional non-inverted input (B0). The output (Y) is represented by the logic equation:

$$Y = (\overline{A0N} + \overline{A1N}) \bullet B0$$

Logic Symbol



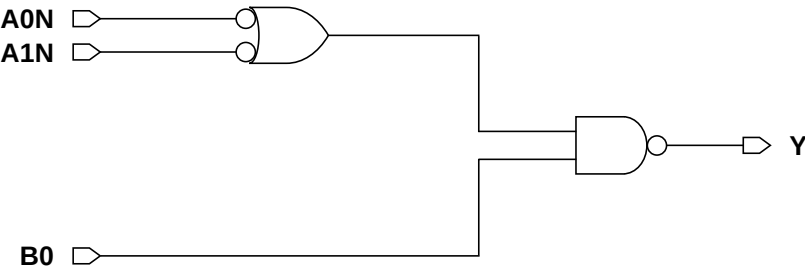
Functions

A0N	A1N	B0	Y
1	1	x	1
x	x	0	1
x	0	1	0
0	x	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI2BB1XL	5.04	3.30
OAI2BB1X1	5.04	3.30
OAI2BB1X2	5.04	4.62
OAI2BB1X4	5.04	6.60

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0N	0.0216	0.0245	0.0421	0.0788
A1N	0.0192	0.0221	0.0374	0.0717
B0	0.0112	0.0149	0.0263	0.0545

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0N	0.0020	0.0020	0.0034	0.0058
A1N	0.0017	0.0016	0.0027	0.0046
B0	0.0027	0.0036	0.0071	0.0130

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0N $\rightarrow$ Y $\uparrow$	0.0891	0.0952	0.0837	0.0917	6.2783	4.5390	2.2108	1.1057
A0N $\rightarrow$ Y $\downarrow$	0.1120	0.1269	0.1110	0.1054	3.8764	2.8155	1.4054	0.7266
A1N $\rightarrow$ Y $\uparrow$	0.0833	0.0908	0.0799	0.0878	6.2715	4.5298	2.2108	1.1057
A1N $\rightarrow$ Y $\downarrow$	0.1007	0.1151	0.1001	0.0960	3.8724	2.8137	1.4045	0.7263
B0 $\rightarrow$ Y $\uparrow$	0.0409	0.0404	0.0359	0.0375	6.2569	4.5281	2.2065	1.1033
B0 $\rightarrow$ Y $\downarrow$	0.0259	0.0270	0.0250	0.0271	3.8352	2.7932	1.3963	0.7230

Cell Description

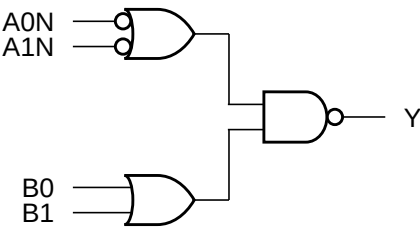
The OAI2BB2 cell provides the logical inverted AND of one OR group of two inverted inputs (A0N, A1N) and one OR group of two non-inverted inputs (B0, B1). The output (Y) is represented by the logic equation:

$$Y = \overline{(A0N + A1N)} \bullet (B0 + B1)$$

Functions

A0N	A1N	B0	B1	Y
1	1	x	x	1
x	x	0	0	1
x	0	x	1	0
x	0	1	x	0
0	x	x	1	0
0	x	1	x	0

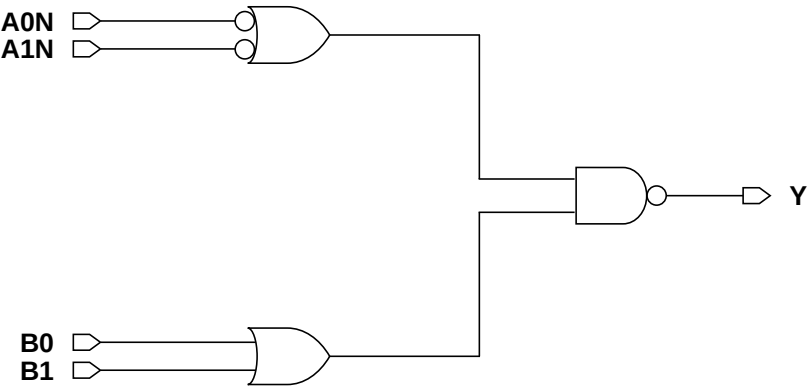
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI2BB2XL	5.04	4.62
OAI2BB2X1	5.04	4.62
OAI2BB2X2	5.04	6.60
OAI2BB2X4	5.04	9.90

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0N	0.0249	0.0289	0.0541	0.1037
A1N	0.0198	0.0227	0.0421	0.0791
B0	0.0153	0.0196	0.0374	0.0695
B1	0.0179	0.0235	0.0462	0.0844

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0N	0.0020	0.0020	0.0036	0.0081
A1N	0.0023	0.0023	0.0040	0.0076
B0	0.0036	0.0048	0.0099	0.0193
B1	0.0035	0.0047	0.0102	0.0182

Delays at 25°C, 1.8V, Typical Process

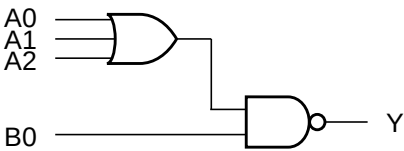
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0N $\rightarrow$ Y $\uparrow$	0.0918	0.0973	0.0929	0.0890	6.2926	4.5472	2.2724	1.1234
A0N $\rightarrow$ Y $\downarrow$	0.1237	0.1392	0.1305	0.1268	3.8721	2.8119	1.4199	0.7307
A1N $\rightarrow$ Y $\uparrow$	0.0882	0.0936	0.0889	0.0840	6.2768	4.5380	2.2678	1.1211
A1N $\rightarrow$ Y $\downarrow$	0.1034	0.1176	0.1089	0.1032	3.8693	2.8120	1.4194	0.7304
B0 $\rightarrow$ Y $\uparrow$	0.0768	0.0714	0.0688	0.0642	9.4749	6.7347	3.3673	1.6402
B0 $\rightarrow$ Y $\downarrow$	0.0349	0.0346	0.0343	0.0343	3.8390	2.7945	1.4124	0.7276
B1 $\rightarrow$ Y $\uparrow$	0.0898	0.0843	0.0826	0.0763	9.4696	6.7320	3.3665	1.6395
B1 $\rightarrow$ Y $\downarrow$	0.0428	0.0430	0.0426	0.0430	3.8494	2.7996	1.4147	0.7287

Cell Description

The OAI31 cell provides the logical inverted AND of one OR group and an additional input. The output (Y) is represented by the logic equation:

$Y = \overline{(A0 + A1 + A2)} \bullet B0$

Logic Symbol



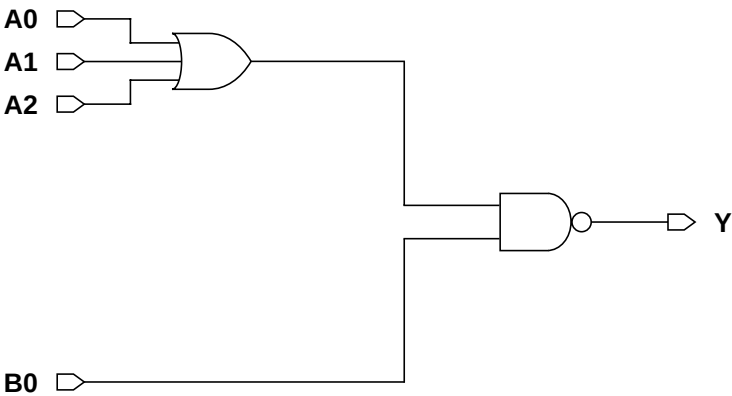
Functions

A0	A1	A2	B0	Y
0	0	0	x	1
x	x	x	0	1
x	x	1	1	0
x	1	x	1	0
1	x	x	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI31XL	5.04	3.96
OAI31X1	5.04	3.96
OAI31X2	5.04	5.94
OAI31X4	5.04	5.94

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0190	0.0253	0.0533	0.0751
A1	0.0231	0.0308	0.0657	0.0785
A2	0.0264	0.0365	0.0768	0.0845
B0	0.0183	0.0254	0.0481	0.0711

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0038	0.0050	0.0100	0.0037
A1	0.0037	0.0049	0.0103	0.0034
A2	0.0035	0.0048	0.0106	0.0034
B0	0.0032	0.0042	0.0081	0.0031

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0932	0.0876	0.0943	0.2787	12.2908	8.5981	4.3008	1.1041
A0 $\rightarrow$ Y $\downarrow$	0.0339	0.0334	0.0349	0.1633	3.8266	2.7863	1.4020	0.6418
A1 $\rightarrow$ Y $\uparrow$	0.1241	0.1187	0.1254	0.3080	12.2830	8.5945	4.2991	1.1041
A1 $\rightarrow$ Y $\downarrow$	0.0436	0.0438	0.0455	0.1748	3.8407	2.7933	1.4053	0.6418
A2 $\rightarrow$ Y $\uparrow$	0.1350	0.1300	0.1386	0.3201	12.2813	8.5940	4.2995	1.1041
A2 $\rightarrow$ Y $\downarrow$	0.0474	0.0486	0.0510	0.1818	3.8875	2.8167	1.4174	0.6419
B0 $\rightarrow$ Y $\uparrow$	0.0355	0.0348	0.0327	0.1663	6.2781	4.5398	2.2695	1.1041
B0 $\rightarrow$ Y $\downarrow$	0.0366	0.0378	0.0362	0.1697	3.8804	2.8133	1.4156	0.6419

Cell Description

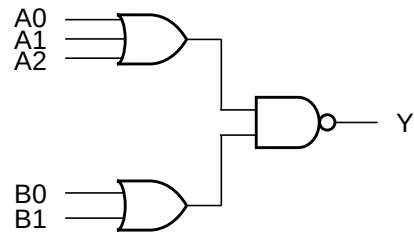
The OAI32 cell provides the logical inverted AND of two OR groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1 + A2) \bullet (B0 + B1)}$$

Functions

A0	A1	A2	B0	B1	Y
0	0	0	x	x	1
x	x	x	0	0	1
x	x	1	x	1	0
x	x	1	1	x	0
x	1	x	1	x	0
x	1	x	x	1	0
1	x	x	1	x	0
1	x	x	x	1	0

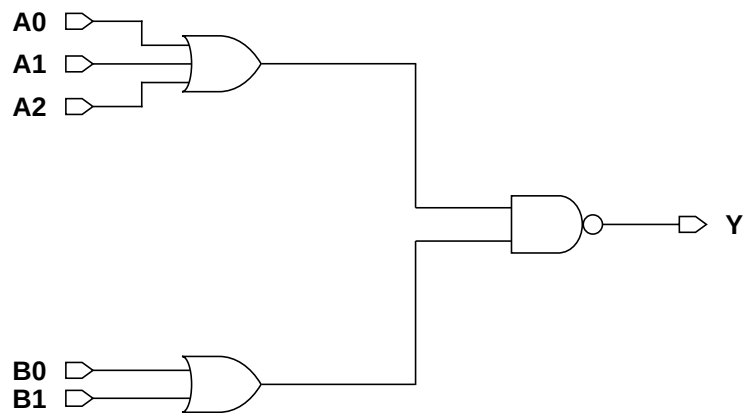
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI32XL	5.04	4.62
OAI32X1	5.04	4.62
OAI32X2	5.04	7.26
OAI32X4	5.04	6.60

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0270	0.0353	0.0690	0.0835
A1	0.0313	0.0415	0.0817	0.0881
A2	0.0348	0.0466	0.0925	0.0928
B0	0.0212	0.0287	0.0586	0.0719
B1	0.0240	0.0327	0.0669	0.0776

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0038	0.0052	0.0097	0.0036
A1	0.0038	0.0051	0.0100	0.0036
A2	0.0037	0.0049	0.0100	0.0032
B0	0.0036	0.0049	0.0089	0.0035
B1	0.0037	0.0048	0.0094	0.0033

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1315	0.1213	0.1178	0.3190	12.2928	8.5989	4.2987	1.1040
A0 $\rightarrow$ Y $\downarrow$	0.0497	0.0486	0.0570	0.1810	3.8412	2.7935	1.6808	0.6419
A1 $\rightarrow$ Y $\uparrow$	0.1626	0.1520	0.1493	0.3498	12.2854	8.5955	4.2974	1.1041
A1 $\rightarrow$ Y $\downarrow$	0.0601	0.0594	0.0707	0.1931	3.8506	2.7982	1.6830	0.6419
A2 $\rightarrow$ Y $\uparrow$	0.1735	0.1630	0.1606	0.3600	12.2838	8.5949	4.2974	1.1040
A2 $\rightarrow$ Y $\downarrow$	0.0649	0.0651	0.0785	0.2002	3.8897	2.8178	1.6937	0.6420
B0 $\rightarrow$ Y $\uparrow$	0.0565	0.0542	0.0564	0.2155	9.5130	6.7580	3.3775	1.1042
B0 $\rightarrow$ Y $\downarrow$	0.0407	0.0420	0.0521	0.1741	3.8819	2.8140	1.6920	0.6420
B1 $\rightarrow$ Y $\uparrow$	0.0714	0.0678	0.0705	0.2288	9.5129	6.7564	3.3770	1.1042
B1 $\rightarrow$ Y $\downarrow$	0.0502	0.0514	0.0635	0.1821	3.8877	2.8161	1.6943	0.6419

Cell Description

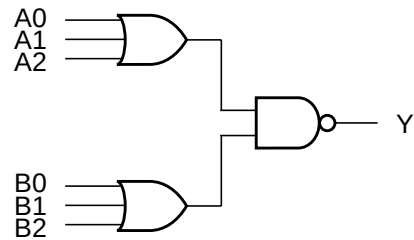
The OAI33 cell provides the logical inverted AND of two OR groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1 + A2) \bullet (B0 + B1 + B2)}$$

Functions

A0	A1	A2	B0	B1	B2	Y
0	0	0	x	x	x	1
x	x	x	0	0	0	1
x	x	1	x	x	1	0
x	x	1	x	1	x	0
x	x	1	1	x	x	0
x	1	x	x	x	1	0
x	1	x	x	1	x	0
x	1	x	1	x	x	0
1	x	x	x	x	1	0
1	x	x	x	1	x	0
1	x	x	1	x	x	0

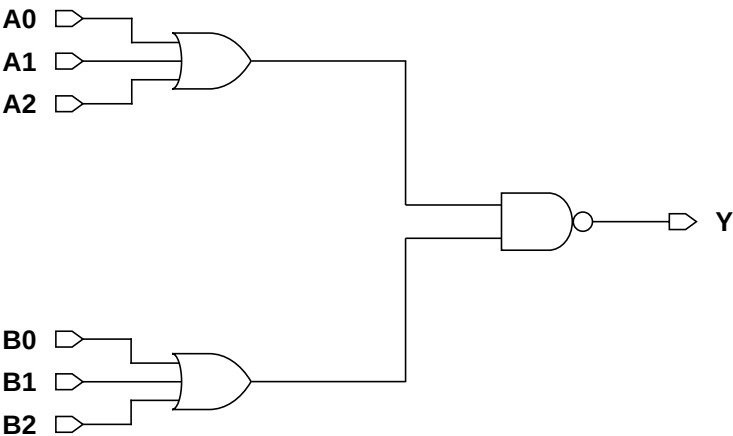
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI33XL	5.04	5.28
OAI33X1	5.04	5.28
OAI33X2	5.04	8.58
OAI33X4	5.04	7.26

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0	0.0252	0.0344	0.0660	0.0773
A1	0.0292	0.0396	0.0779	0.0808
A2	0.0327	0.0454	0.0887	0.0854
B0	0.0350	0.0471	0.0912	0.0911
B1	0.0393	0.0533	0.1042	0.0953
B2	0.0428	0.0585	0.1151	0.0975

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0039	0.0053	0.0099	0.0037
A1	0.0038	0.0051	0.0101	0.0035
A2	0.0038	0.0052	0.0109	0.0033
B0	0.0038	0.0053	0.0096	0.0036
B1	0.0037	0.0051	0.0101	0.0034
B2	0.0036	0.0049	0.0101	0.0034

Delays at 25°C, 1.8V, Typical Process

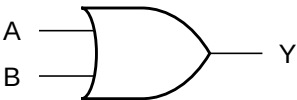
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0811	0.0751	0.0719	0.2672	12.3584	8.6351	4.3162	1.1045
A0 $\rightarrow$ Y $\downarrow$	0.0467	0.0470	0.0554	0.1823	3.8705	2.8114	1.6911	0.6422
A1 $\rightarrow$ Y $\uparrow$	0.1117	0.1076	0.1030	0.2981	12.3503	8.6317	4.3148	1.1045
A1 $\rightarrow$ Y $\downarrow$	0.0570	0.0585	0.0701	0.1928	3.8802	2.8170	1.6917	0.6421
A2 $\rightarrow$ Y $\uparrow$	0.1246	0.1219	0.1166	0.3089	12.3517	8.6331	4.3155	1.1045
A2 $\rightarrow$ Y $\downarrow$	0.0637	0.0671	0.0801	0.2016	3.9122	2.8314	1.7002	0.6423
B0 $\rightarrow$ Y $\uparrow$	0.1595	0.1512	0.1464	0.3517	12.2882	8.5976	4.2987	1.1041
B0 $\rightarrow$ Y $\downarrow$	0.0616	0.0629	0.0734	0.1999	3.8848	2.8183	1.6931	0.6422
B1 $\rightarrow$ Y $\uparrow$	0.1907	0.1814	0.1783	0.3817	12.2819	8.5942	4.2975	1.1041
B1 $\rightarrow$ Y $\downarrow$	0.0729	0.0746	0.0882	0.2123	3.8868	2.8189	1.6934	0.6422
B2 $\rightarrow$ Y $\uparrow$	0.2017	0.1924	0.1895	0.3934	12.2805	8.5936	4.2975	1.1041
B2 $\rightarrow$ Y $\downarrow$	0.0788	0.0811	0.0972	0.2191	3.9179	2.8346	1.7018	0.6423

Cell Description

The OR2 cell provides the logical OR of two inputs (A, B). The output (Y) is represented by the logic equation:

$Y = (A + B)$

Logic Symbol



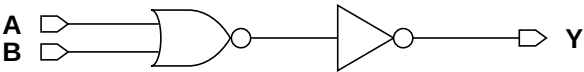
Functions

A	B	Y
0	0	0
x	1	1
1	x	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
OR2XL	5.04	2.64
OR2X1	5.04	2.64
OR2X2	5.04	2.64
OR2X4	5.04	3.96

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0166	0.0194	0.0301	0.0542
B	0.0191	0.0225	0.0350	0.0625

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0025	0.0025	0.0035	0.0063
B	0.0027	0.0027	0.0039	0.0063

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0656	0.0698	0.0659	0.0620
A $\rightarrow$ Y $\downarrow$	0.1268	0.1423	0.1330	0.1353
B $\rightarrow$ Y $\uparrow$	0.0746	0.0788	0.0751	0.0696
B $\rightarrow$ Y $\downarrow$	0.1442	0.1597	0.1494	0.1487

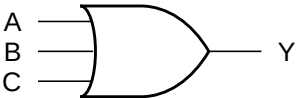
Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	6.2624	4.5303	2.3509	1.1040
A $\rightarrow$ Y $\downarrow$	3.4882	2.4469	1.3006	0.6519
B $\rightarrow$ Y $\uparrow$	6.2652	4.5315	2.3517	1.1042
B $\rightarrow$ Y $\downarrow$	3.4883	2.4470	1.3007	0.6518

Cell Description

The OR3 cell provides the logical OR of three inputs (A, B, C). The output (Y) is represented by the logic equation:

$Y = (A + B + C)$

Logic Symbol



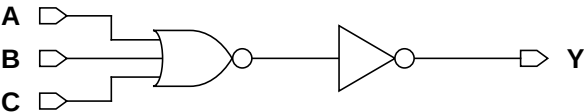
Functions

A	B	C	Y
0	0	0	0
x	x	1	1
x	1	x	1
1	x	x	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
OR3XL	5.04	3.30
OR3X1	5.04	3.96
OR3X2	5.04	3.96
OR3X4	5.04	5.94

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0190	0.0212	0.0340	0.0629
B	0.0214	0.0239	0.0382	0.0709
C	0.0240	0.0264	0.0438	0.0812

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0029	0.0026	0.0040	0.0076
B	0.0026	0.0025	0.0038	0.0079
C	0.0026	0.0025	0.0038	0.0081

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0729	0.0750	0.0720	0.0680
A $\rightarrow$ Y $\downarrow$	0.1739	0.1854	0.1787	0.1663
B $\rightarrow$ Y $\uparrow$	0.0814	0.0843	0.0814	0.0787
B $\rightarrow$ Y $\downarrow$	0.2038	0.2160	0.2097	0.1983
C $\rightarrow$ Y $\uparrow$	0.0850	0.0875	0.0854	0.0830
C $\rightarrow$ Y $\downarrow$	0.2171	0.2285	0.2221	0.2115

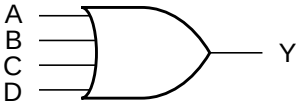
Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	6.2617	4.5300	2.2916	1.1599
A $\rightarrow$ Y $\downarrow$	3.6492	2.5171	1.3335	0.6653
B $\rightarrow$ Y $\uparrow$	6.2647	4.5313	2.2923	1.1604
B $\rightarrow$ Y $\downarrow$	3.6492	2.5170	1.3333	0.6653
C $\rightarrow$ Y $\uparrow$	6.2747	4.5358	2.2948	1.1617
C $\rightarrow$ Y $\downarrow$	3.6489	2.5170	1.3334	0.6653

Cell Description

The OR4 cell provides the logical OR of four inputs (A, B, C, D). The output (Y) is represented by the logic equation:

$Y = (A + B + C + D)$

Logic Symbol



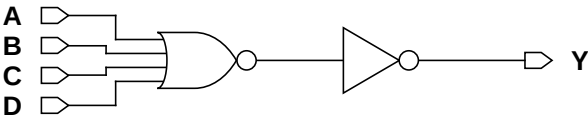
Functions

A	B	C	D	Y
0	0	0	0	0
x	x	x	1	1
x	x	1	x	1
x	1	x	x	1
1	x	x	x	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
OR4XL	5.04	3.96
OR4X1	5.04	3.96
OR4X2	5.04	3.96
OR4X4	5.04	7.26

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0198	0.0241	0.0353	0.0679
B	0.0225	0.0269	0.0407	0.0789
C	0.0259	0.0301	0.0458	0.0885
D	0.0292	0.0338	0.0519	0.1009

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0029	0.0029	0.0044	0.0084
B	0.0026	0.0026	0.0043	0.0087
C	0.0030	0.0030	0.0042	0.0088
D	0.0032	0.0032	0.0042	0.0097

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0778	0.0822	0.0754	0.0745
A $\rightarrow$ Y $\downarrow$	0.1979	0.2239	0.1974	0.1900
B $\rightarrow$ Y $\uparrow$	0.0857	0.0903	0.0871	0.0876
B $\rightarrow$ Y $\downarrow$	0.2466	0.2725	0.2464	0.2413
C $\rightarrow$ Y $\uparrow$	0.0935	0.0984	0.0918	0.0937
C $\rightarrow$ Y $\downarrow$	0.2803	0.3063	0.2754	0.2697
D $\rightarrow$ Y $\uparrow$	0.0994	0.1050	0.0958	0.0980
D $\rightarrow$ Y $\downarrow$	0.2960	0.3223	0.2885	0.2846

Delay Table at 25°C, 1.8V, Typical Process

Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	6.2627	4.5308	2.2916	1.1823
A $\rightarrow$ Y $\downarrow$	3.8220	2.6184	1.3678	0.6702
B $\rightarrow$ Y $\uparrow$	6.2652	4.5323	2.2926	1.1829
B $\rightarrow$ Y $\downarrow$	3.8222	2.6184	1.3678	0.6703
C $\rightarrow$ Y $\uparrow$	6.2758	4.5373	2.2954	1.1844
C $\rightarrow$ Y $\downarrow$	3.8226	2.6184	1.3678	0.6703
D $\rightarrow$ Y $\uparrow$	6.2921	4.5453	2.3001	1.1869
D $\rightarrow$ Y $\downarrow$	3.8217	2.6182	1.3678	0.6703

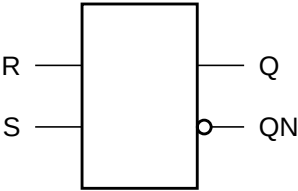
Cell Description

The RSLAT cell is an RS-type latch with active-high set (S) and reset (R).

Function

R	S	Q[n+1]	QN[n+1]
0	0	Q[n]	QN[n]
0	1	1	0
1	0	0	1
1	1	IL	IL

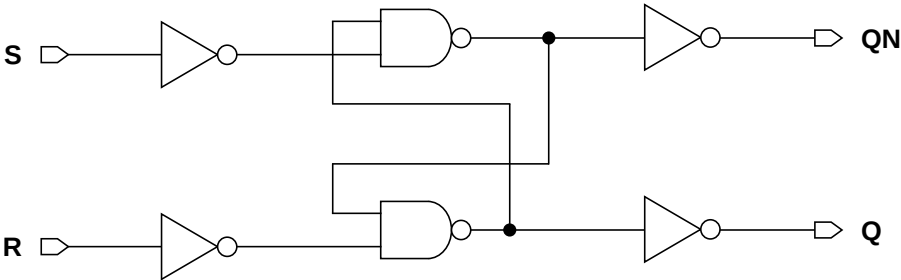
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
RSLATXL	5.04	6.60
RSLATX1	5.04	6.60
RSLATX2	5.04	7.92
RSLATX4	5.04	11.88

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
S	0.0090	0.0101	0.0136	0.0243
R	0.0096	0.0103	0.0149	0.0252
Q	0.0423	0.0536	0.0895	0.1593

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
S	0.0022	0.0023	0.0026	0.0046
R	0.0022	0.0023	0.0026	0.0046

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
S $\rightarrow$ Q $\uparrow$	0.2442	0.2216	0.2110	0.1936	6.2794	4.5373	2.2656	1.1056
R $\rightarrow$ Q $\uparrow$	0.1703	0.1630	0.1603	0.1447	6.2791	4.5373	2.2656	1.1055
R $\rightarrow$ Q $\downarrow$	0.1684	0.1535	0.1455	0.1352	3.4686	2.4128	1.2847	0.6428
S $\rightarrow$ QN $\uparrow$	0.1564	0.1531	0.1600	0.1416	6.2791	4.5359	2.2657	1.1055
S $\rightarrow$ QN $\downarrow$	0.1557	0.1456	0.1469	0.1322	3.4640	2.4127	1.2851	0.6426
R $\rightarrow$ QN $\uparrow$	0.2535	0.2246	0.2123	0.1953	6.2796	4.5361	2.2659	1.1056

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
S	minpwh	0.1286	0.1092	0.1043	0.0995
R	minpwh	0.1286	0.1092	0.1043	0.0995

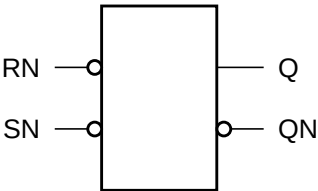
Cell Description

The RSLATN cell is an RS-type latch with active-low set (SN) and reset (RN).

Function

RN	SN	Q[n+1]	QN[n+1]
0	0	IL	IL
0	1	0	1
1	0	1	0
1	1	Q[n]	QN[n]

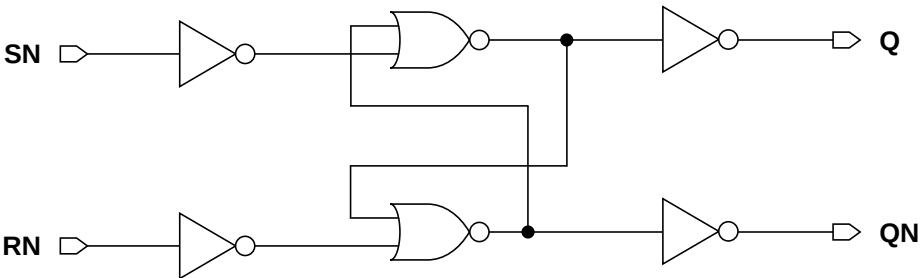
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
RSLATNXL	5.04	6.60
RSLATNX1	5.04	7.26
RSLATNX2	5.04	7.26
RSLATNX4	5.04	11.88

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
SN	0.0103	0.0123	0.0160	0.0317
RN	0.0105	0.0113	0.0177	0.0313
Q	0.0407	0.0496	0.0867	0.1520

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SN	0.0022	0.0022	0.0029	0.0050
RN	0.0023	0.0023	0.0031	0.0051

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
SN $\rightarrow$ Q $\uparrow$	0.1559	0.1593	0.1489	0.1402	6.2756	4.5346	2.2650	1.1326
SN $\rightarrow$ Q $\downarrow$	0.2068	0.1877	0.1725	0.1609	3.3302	2.4575	1.3027	0.6506
RN $\rightarrow$ Q $\downarrow$	0.3135	0.3032	0.2809	0.2589	3.3300	2.4574	1.3027	0.6505
SN $\rightarrow$ QN $\downarrow$	0.3163	0.2989	0.2746	0.2585	3.3360	2.4601	1.3021	0.6503
RN $\rightarrow$ QN $\uparrow$	0.1579	0.1655	0.1521	0.1415	6.2753	4.5344	2.2649	1.1326
RN $\rightarrow$ QN $\downarrow$	0.2091	0.1930	0.1736	0.1617	3.3362	2.4602	1.3021	0.6504

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SN	minpwl	0.1432	0.1432	0.1335	0.1286
RN	minpwl	0.1480	0.1529	0.1383	0.1335

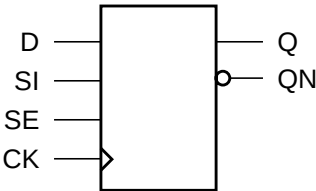
Cell Description

The Sdff cell is a positive-edge triggered, static D-type flip-flop with scan input (SI) and active-high scan enable (SE).

Functions

D	SI	SE	CK	Q[n+1]	QN[n+1]
1	x	0		1	0
0	x	0		0	1
x	x	x		Q[n]	QN[n]
x	1	1		1	0
x	0	1		0	1

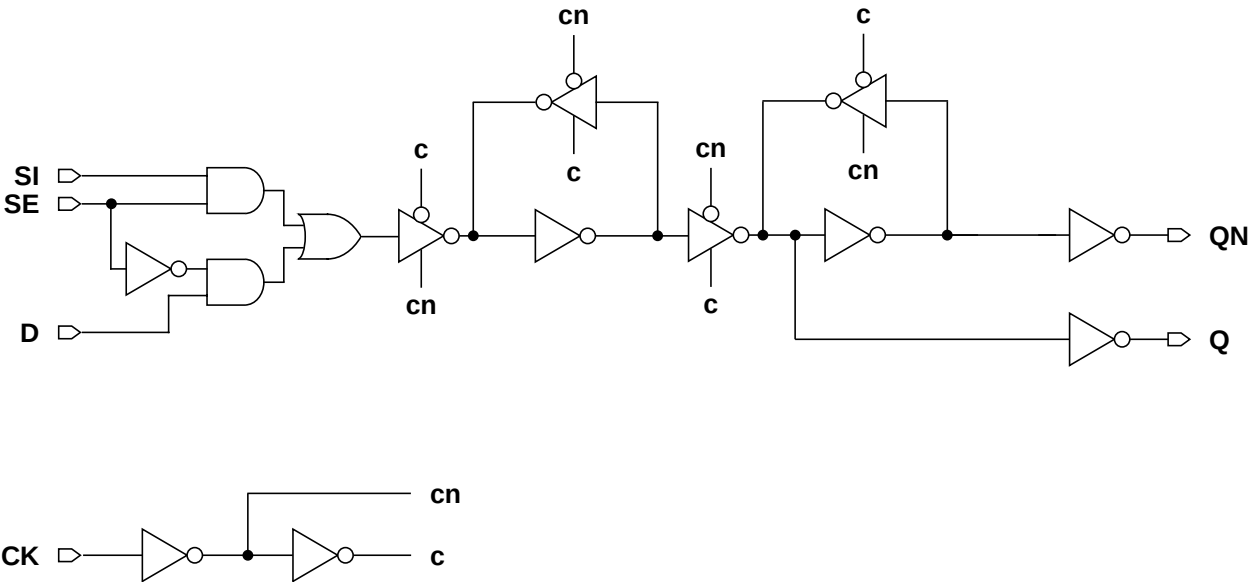
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFXL	5.04	13.20
SDFFX1	5.04	13.86
SDFFX2	5.04	16.50
SDFFX4	5.04	19.14

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
SI	0.0432	0.0404	0.0518	0.0799
SE	0.0496	0.0458	0.0566	0.0823
D	0.0372	0.0355	0.0461	0.0694
CK	0.0414	0.0392	0.0458	0.0702
Q	0.0360	0.0392	0.0675	0.1110

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0024	0.0022	0.0022	0.0022
SE	0.0049	0.0041	0.0043	0.0051
D	0.0030	0.0018	0.0022	0.0032
CK	0.0021	0.0028	0.0039	0.0063

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2572	0.2147	0.1944	0.1722	6.1446	4.4120	2.2691	1.1475
CK $\rightarrow$ Q $\downarrow$	0.2145	0.1788	0.1619	0.1488	3.5483	2.5249	1.2611	0.6490
CK $\rightarrow$ QN $\uparrow$	0.2813	0.2428	0.2156	0.1949	6.2664	4.5329	2.2658	1.1460
CK $\rightarrow$ QN $\downarrow$	0.3485	0.3018	0.2719	0.2431	3.7806	2.4131	1.2035	0.6414

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup $\uparrow$ $\rightarrow$ CK	0.1367	0.1367	0.1641	0.1914
	setup $\downarrow$ $\rightarrow$ CK	0.3438	0.4062	0.4102	0.4922
	hold $\uparrow$ $\rightarrow$ CK	-0.1133	-0.1016	-0.1211	-0.1445
	hold $\downarrow$ $\rightarrow$ CK	-0.2148	-0.2773	-0.2695	-0.3477
SE	setup $\uparrow$ $\rightarrow$ CK	0.3711	0.4336	0.4297	0.5195
	setup $\downarrow$ $\rightarrow$ CK	0.2070	0.4219	0.3516	0.2891
	hold $\uparrow$ $\rightarrow$ CK	-0.0977	-0.0820	-0.1055	-0.1328
	hold $\downarrow$ $\rightarrow$ CK	-0.0820	-0.1484	-0.1680	-0.1406
D	setup $\uparrow$ $\rightarrow$ CK	0.0781	0.1172	0.1445	0.1133
	setup $\downarrow$ $\rightarrow$ CK	0.2109	0.4258	0.3555	0.2930
	hold $\uparrow$ $\rightarrow$ CK	-0.0664	-0.0859	-0.1055	-0.0781
	hold $\downarrow$ $\rightarrow$ CK	-0.0820	-0.3008	-0.2188	-0.1719
CK	minpwh	0.1383	0.1140	0.1043	0.0898
	minpwl	0.2111	0.2111	0.2111	0.1674

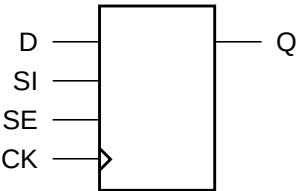
Cell Description

The SDFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI) and active-high scan enable (SE). The cell has a single output (Q) and fast clock-to-out path.

Functions

D	SI	SE	CK	Q[n+1]
1	x	0		1
0	x	0		0
x	x	x		Q[n]
x	1	1		1
x	0	1		0

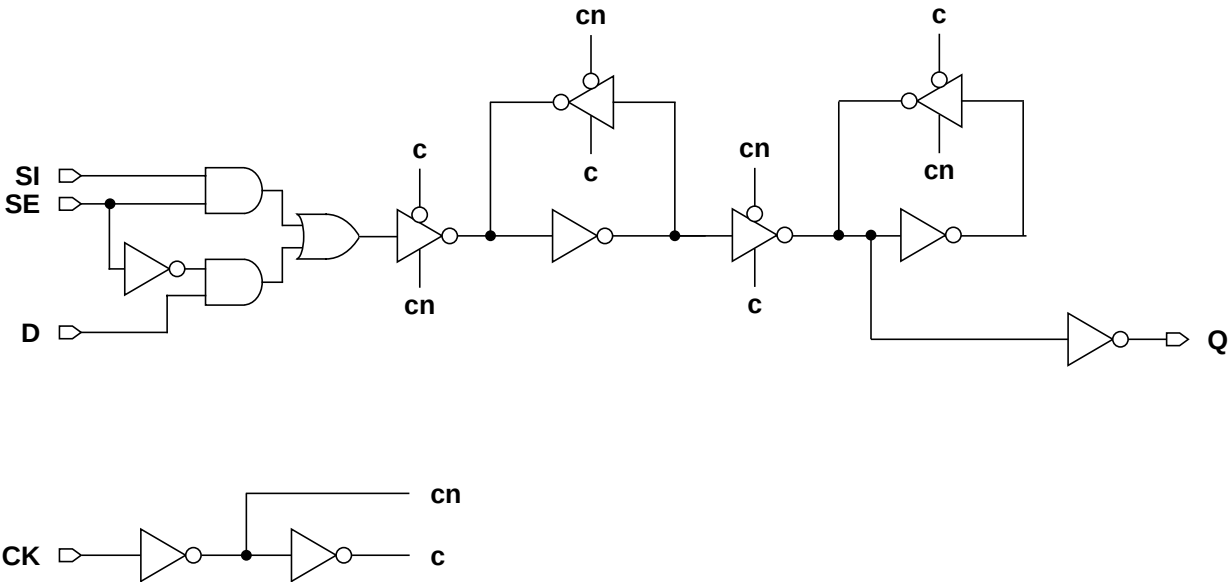
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFHQXL	5.04	13.20
SDFFHQX1	5.04	13.20
SDFFHQX2	5.04	15.84
SDFFHQX4	5.04	17.16

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0471	0.0513	0.0682	0.0939
SE	0.0532	0.0570	0.0734	0.0971
D	0.0405	0.0453	0.0629	0.0826
CK	0.0423	0.0411	0.0460	0.0585
Q	0.0247	0.0259	0.0373	0.0515

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0022	0.0023	0.0023	0.0022
SE	0.0046	0.0042	0.0042	0.0048
D	0.0026	0.0018	0.0021	0.0027
CK	0.0022	0.0029	0.0036	0.0052

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2307	0.1879	0.1822	0.1628	6.2785	4.4085	2.2674	1.1048
CK $\rightarrow$ Q $\downarrow$	0.2329	0.1745	0.1687	0.1508	3.6599	2.5633	1.2815	0.6489

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup $\uparrow$ $\rightarrow$ CK	0.1484	0.1602	0.1680	0.1953
	setup $\downarrow$ $\rightarrow$ CK	0.4141	0.4688	0.4727	0.5000
	hold $\uparrow$ $\rightarrow$ CK	-0.1055	-0.1055	-0.1172	-0.1367
	hold $\downarrow$ $\rightarrow$ CK	-0.2070	-0.2891	-0.2773	-0.3125
SE	setup $\uparrow$ $\rightarrow$ CK	0.4492	0.4883	0.4922	0.5234
	setup $\downarrow$ $\rightarrow$ CK	0.3086	0.4727	0.4336	0.3516
	hold $\uparrow$ $\rightarrow$ CK	-0.0898	-0.0898	-0.0977	-0.1250
	hold $\downarrow$ $\rightarrow$ CK	-0.1133	-0.1562	-0.1641	-0.1523
D	setup $\uparrow$ $\rightarrow$ CK	0.0977	0.1406	0.1484	0.1406
	setup $\downarrow$ $\rightarrow$ CK	0.3047	0.4844	0.4492	0.3516
	hold $\uparrow$ $\rightarrow$ CK	-0.0586	-0.0898	-0.0938	-0.0898
	hold $\downarrow$ $\rightarrow$ CK	-0.1133	-0.3125	-0.2617	-0.1836
CK	minpwh	0.1286	0.1092	0.1092	0.0946
	minpwl	0.2111	0.2111	0.2014	0.1771

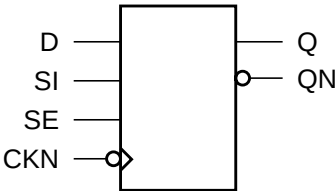
Cell Description

The SDFFN cell is a negative-edge triggered, static D-type flip-flop with scan input (SI) and active-high scan enable (SE).

Function

D	SI	SE	CKN	Q[n+1]	QN[n+1]
1	x	0		1	0
0	x	0		0	1
x	x	x		Q[n]	QN[n]
x	1	1		1	0
x	0	1		0	1

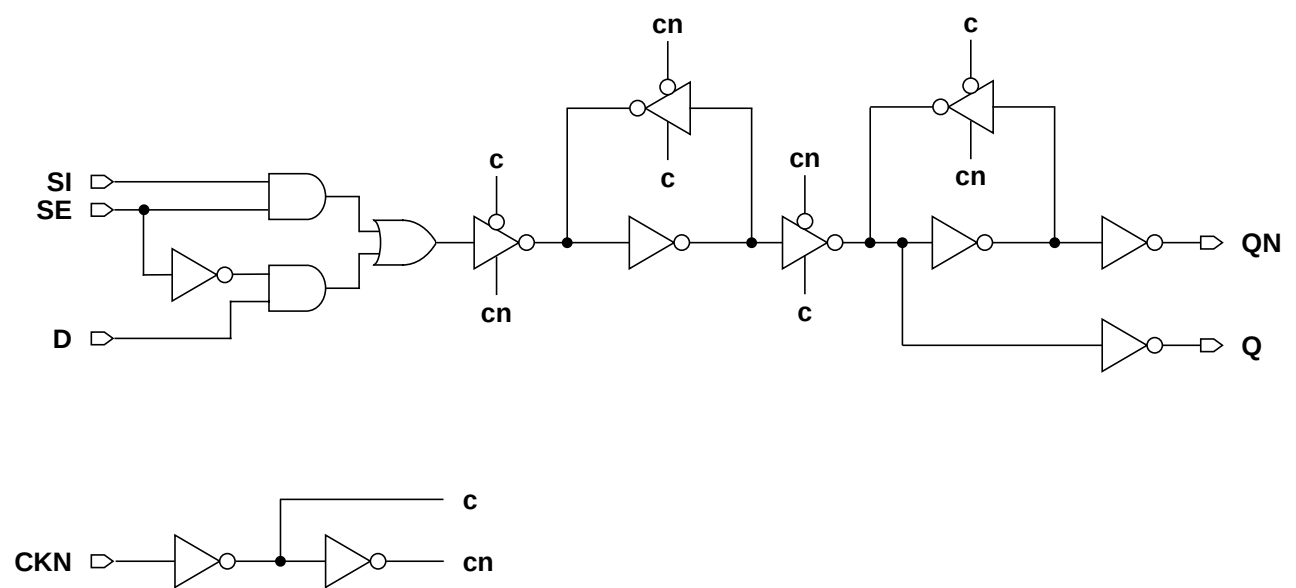
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFNXL	5.04	13.86
SDFFNX1	5.04	13.86
SDFFNX2	5.04	16.50
SDFFNX4	5.04	18.48

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
SI	0.0439	0.0417	0.0529	0.0796
SE	0.0502	0.0477	0.0582	0.0823
D	0.0378	0.0370	0.0472	0.0695
CKN	0.0366	0.0394	0.0542	0.0827
Q	0.0375	0.0445	0.0739	0.1186

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0022	0.0020	0.0021	0.0020
SE	0.0046	0.0040	0.0042	0.0048
D	0.0026	0.0017	0.0020	0.0030
CKN	0.0021	0.0028	0.0039	0.0062

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CKN $\rightarrow$ Q $\uparrow$	0.1941	0.1551	0.1416	0.1211	6.1488	4.4151	2.2704	1.1264
CKN $\rightarrow$ Q $\downarrow$	0.3360	0.2898	0.2595	0.2269	3.5274	2.5245	1.2609	0.6689
CKN $\rightarrow$ QN $\uparrow$	0.4021	0.3536	0.3145	0.2780	6.2672	4.5329	2.2659	1.1248
CKN $\rightarrow$ QN $\downarrow$	0.2839	0.2428	0.2206	0.1944	3.4510	2.4131	1.2038	0.6621

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup $\uparrow$ $\rightarrow$ CKN	0.1289	0.1445	0.1758	0.2031
	setup $\downarrow$ $\rightarrow$ CKN	0.3477	0.3945	0.3945	0.4688
	hold $\uparrow$ $\rightarrow$ CKN	0.0352	0.0039	-0.0195	-0.0508
	hold $\downarrow$ $\rightarrow$ CKN	-0.3047	-0.3398	-0.3320	-0.3906
SE	setup $\uparrow$ $\rightarrow$ CKN	0.3711	0.4180	0.4141	0.4961
	setup $\downarrow$ $\rightarrow$ CKN	0.1992	0.4023	0.3281	0.2617
	hold $\uparrow$ $\rightarrow$ CKN	0.0430	0.0156	-0.0078	-0.0430
	hold $\downarrow$ $\rightarrow$ CKN	-0.0039	-0.0547	-0.0781	-0.0664
D	setup $\uparrow$ $\rightarrow$ CKN	0.0820	0.1328	0.1602	0.1289
	setup $\downarrow$ $\rightarrow$ CKN	0.1992	0.4102	0.3320	0.2617
	hold $\uparrow$ $\rightarrow$ CKN	0.0547	0.0117	-0.0156	-0.0039
	hold $\downarrow$ $\rightarrow$ CKN	-0.1680	-0.3438	-0.2734	-0.2148
CKN	minpwl	0.1626	0.1286	0.1189	0.1043
	minpwh	0.1529	0.1771	0.1820	0.1432

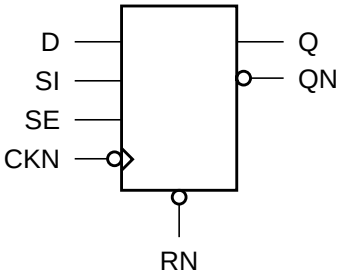
Cell Description

The SDFFNR cell is a negative-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN).

Function

RN	D	SI	SE	CKN	Q[n+1]	QN[n+1]
1	1	x	0		1	0
1	0	x	0		0	1
1	x	x	x		Q[n]	QN[n]
1	x	1	1		1	0
1	x	0	1		0	1
0	x	x	x	x	0	1

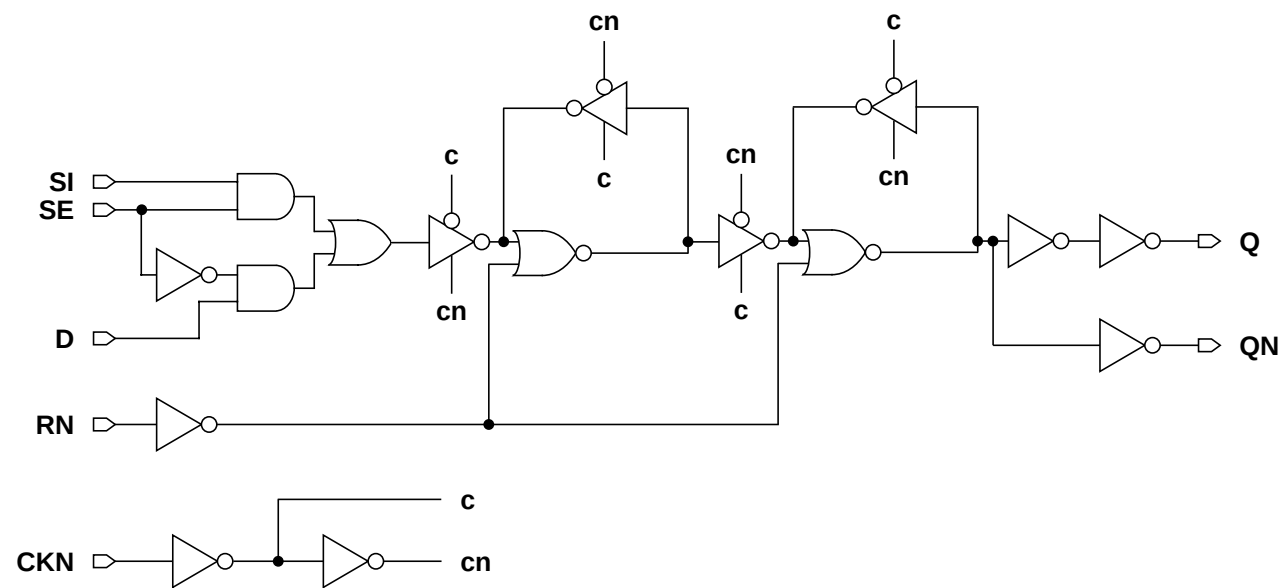
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFNRXL	5.04	18.48
SDFFNRX1	5.04	17.82
SDFFNRX2	5.04	19.80
SDFFNRX4	5.04	23.10

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
SI	0.0454	0.0408	0.0448	0.0563
SE	0.0532	0.0472	0.0530	0.0635
D	0.0395	0.0359	0.0401	0.0500
CKN	0.0393	0.0365	0.0454	0.0568
RN	0.0157	0.0176	0.0215	0.0323
Q	0.0445	0.0530	0.0818	0.1434

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0020	0.0020	0.0021	0.0020
SE	0.0047	0.0040	0.0043	0.0047
D	0.0026	0.0018	0.0018	0.0023
CKN	0.0024	0.0029	0.0031	0.0038
RN	0.0022	0.0027	0.0033	0.0056

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CKN $\rightarrow$ Q $\uparrow$	0.4007	0.3508	0.3340	0.3070	6.2624	4.5328	2.2074	1.1318
CKN $\rightarrow$ Q $\downarrow$	0.4801	0.4550	0.4526	0.4024	3.4021	2.4016	1.4127	0.6467
RN $\rightarrow$ Q $\downarrow$	0.2638	0.2625	0.2467	0.2288	3.4024	2.4017	1.4130	0.6467
CKN $\rightarrow$ QN $\uparrow$	0.4236	0.3866	0.3732	0.3292	6.2586	4.5328	2.2084	1.1324
CKN $\rightarrow$ QN $\downarrow$	0.3660	0.3077	0.2872	0.2560	3.6305	2.4679	1.4360	0.6583
RN $\rightarrow$ QN $\uparrow$	0.2074	0.1947	0.1678	0.1560	6.2686	4.5360	2.2100	1.1333

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CKN	0.1250	0.1445	0.1836	0.1836
	setup↓ → CKN	0.3594	0.4062	0.4453	0.4141
	hold↑ → CKN	0.0469	0.0000	-0.0078	-0.0273
	hold↓ → CKN	-0.3125	-0.3477	-0.3906	-0.3555
SE	setup↑ → CKN	0.3789	0.4336	0.4609	0.4297
	setup↓ → CKN	0.2070	0.3984	0.4492	0.3047
	hold↑ → CKN	0.0547	0.0117	0.0078	-0.0156
	hold↓ → CKN	0.0117	-0.0664	-0.0703	-0.0781
D	setup↑ → CKN	0.0781	0.1328	0.1719	0.1562
	setup↓ → CKN	0.2070	0.4102	0.4492	0.3086
	hold↑ → CKN	0.0703	0.0039	-0.0039	-0.0117
	hold↓ → CKN	-0.1719	-0.3477	-0.3867	-0.2539
CKN	minpwl	0.1917	0.1577	0.1432	0.1286
	minpwh	0.1674	0.1820	0.2111	0.1771
RN	minpwl	0.2062	0.2014	0.2305	0.3518
	recovery	0.0352	0.0742	0.0977	0.0781
	removal	0.1289	0.0664	0.0703	0.0703

Process Technology:
TSMC CL018G

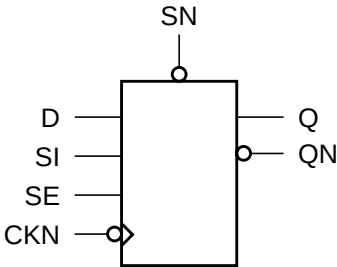
Cell Description

The SDDFFNS cell is a negative-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low set (SN).

Function

SN	D	SI	SE	CKN	Q[n+1]	QN[n+1]
1	1	x	0		1	0
1	0	x	0		0	1
1	x	x	x		Q[n]	QN[n]
1	x	1	1		1	0
1	x	0	1		0	1
0	x	x	x	x	1	0

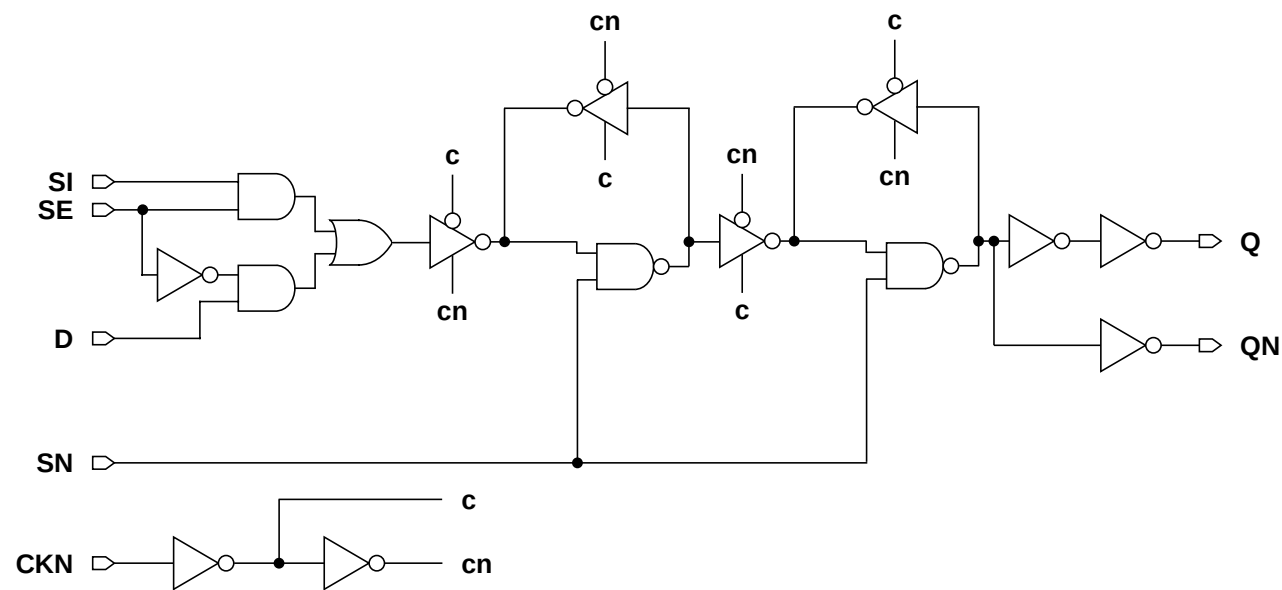
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDDFFNSXL	5.04	15.84
SDDFFNSX1	5.04	15.84
SDDFFNSX2	5.04	16.50
SDDFFNSX4	5.04	21.78

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0443	0.0375	0.0415	0.0516
SE	0.0510	0.0434	0.0476	0.0569
D	0.0383	0.0325	0.0366	0.0457
CKN	0.0357	0.0351	0.0384	0.0508
SN	0.0064	0.0072	0.0101	0.0173
Q	0.0412	0.0493	0.0798	0.1357

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0019	0.0019	0.0019	0.0019
SE	0.0047	0.0041	0.0041	0.0044
D	0.0026	0.0017	0.0017	0.0018
CKN	0.0021	0.0028	0.0028	0.0035
SN	0.0047	0.0052	0.0070	0.0122

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CKN $\rightarrow$ Q $\uparrow$	0.3177	0.2873	0.2892	0.2736	6.2649	4.5317	2.2072	1.1246
CKN $\rightarrow$ Q $\downarrow$	0.4405	0.4194	0.4228	0.3947	3.4005	2.4003	1.2833	0.6417
SN $\rightarrow$ Q $\uparrow$	0.1738	0.1586	0.1615	0.1563	6.2647	4.5317	2.2072	1.1246
CKN $\rightarrow$ QN $\uparrow$	0.3898	0.3577	0.3474	0.3215	6.2711	4.5334	2.2113	1.1267
CKN $\rightarrow$ QN $\downarrow$	0.2843	0.2458	0.2405	0.2228	3.4687	2.4093	1.2878	0.6440
SN $\rightarrow$ QN $\downarrow$	0.1407	0.1173	0.1134	0.1059	3.4570	2.4101	1.2902	0.6464

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CKN	0.1055	0.1289	0.1250	0.1445
	setup↓ → CKN	0.3516	0.3867	0.3984	0.3984
	hold↑ → CKN	0.0352	0.0000	0.0078	-0.0156
	hold↓ → CKN	-0.3203	-0.3438	-0.3516	-0.3477
SE	setup↑ → CKN	0.3750	0.4141	0.4219	0.4141
	setup↓ → CKN	0.1992	0.3906	0.4023	0.3320
	hold↑ → CKN	0.0430	0.0117	0.0156	0.0000
	hold↓ → CKN	0.0000	-0.0586	-0.0547	-0.0742
D	setup↑ → CKN	0.0625	0.1133	0.1133	0.1289
	setup↓ → CKN	0.2031	0.4023	0.4141	0.3359
	hold↑ → CKN	0.0586	0.0039	0.0117	-0.0078
	hold↓ → CKN	-0.1797	-0.3516	-0.3594	-0.2891
CKN	minpwl	0.1577	0.1383	0.1335	0.1238
	minpwh	0.1432	0.1626	0.1723	0.1723
SN	minpwl	0.1383	0.1189	0.1286	0.1577
	recovery	-0.0977	-0.0508	-0.0547	-0.0508
	removal	0.1055	0.0547	0.0586	0.0586

Process Technology:
TSMC CL018G

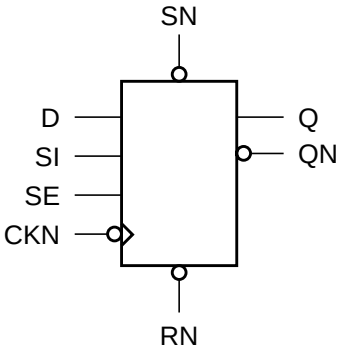
Cell Description

The SDFFNSR cell is a negative-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN) and set (SN). Set (SN) dominates reset (RN).

Functions

RN	SN	D	SI	SE	CKN	Q[n+1]	QN[n+1]
1	1	1	x	0		1	0
1	1	0	x	0		0	1
1	1	x	x	x		Q[n]	QN[n]
1	1	x	1	1		1	0
1	1	x	0	1		0	1
0	1	x	x	x	x	0	1
1	0	x	x	x	x	1	0
0	0	x	x	x	x	1	0

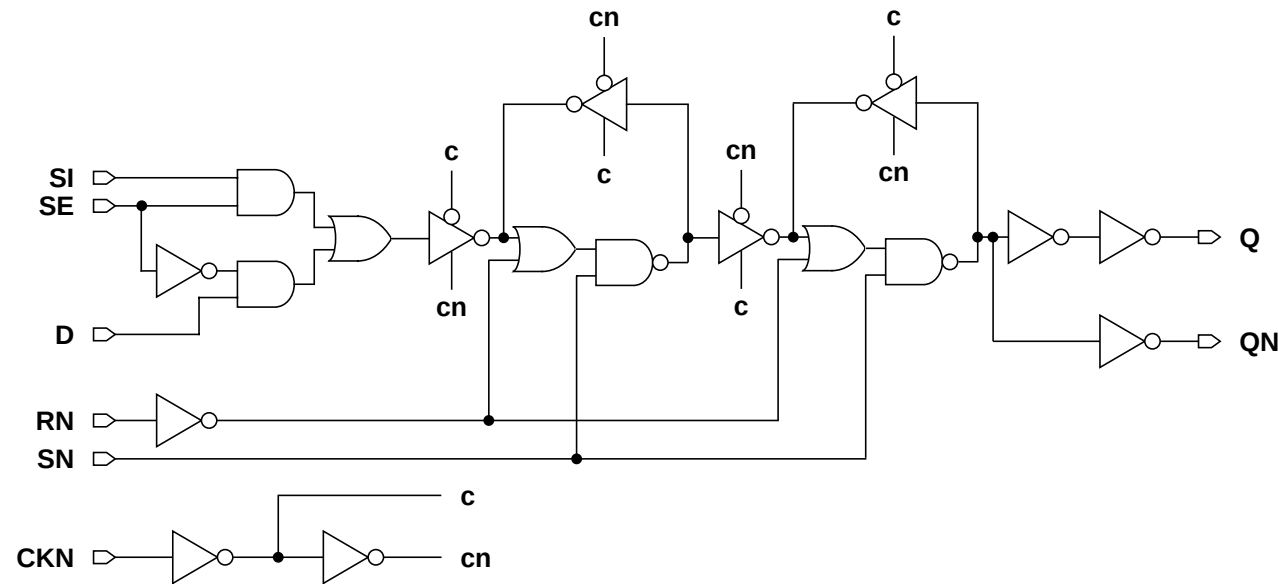
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFNSRXL	5.04	19.14
SDFFNSRX1	5.04	19.14
SDFFNSRX2	5.04	19.14
SDFFNSRX4	5.04	25.08

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
SI	0.0466	0.0405	0.0462	0.0609
SE	0.0539	0.0462	0.0525	0.0671
D	0.0409	0.0356	0.0411	0.0541
CKN	0.0381	0.0377	0.0419	0.0596
SN	0.0084	0.0087	0.0118	0.0181
RN	0.0175	0.0195	0.0242	0.0411
Q	0.0470	0.0549	0.0853	0.1492

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0019	0.0018	0.0018	0.0019
SE	0.0044	0.0040	0.0042	0.0047
D	0.0021	0.0016	0.0017	0.0023
CKN	0.0022	0.0028	0.0028	0.0041
SN	0.0063	0.0067	0.0090	0.0152
RN	0.0025	0.0028	0.0039	0.0061

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CKN $\rightarrow$ Q $\uparrow$	0.3943	0.3524	0.3416	0.3163	6.2616	4.5300	2.2077	1.1318
CKN $\rightarrow$ Q $\downarrow$	0.4774	0.4561	0.4323	0.4020	3.4153	2.4069	1.2839	0.6418
SN $\rightarrow$ Q $\uparrow$	0.2049	0.1900	0.1790	0.1731	6.2615	4.5300	2.2076	1.1317
SN $\rightarrow$ Q $\downarrow$	0.2064	0.1888	0.1771	0.1720	3.4162	2.4071	1.2841	0.6418
RN $\rightarrow$ Q $\downarrow$	0.3079	0.2800	0.2516	0.2457	3.4166	2.4073	1.2840	0.6418
CKN $\rightarrow$ QN $\uparrow$	0.4088	0.3781	0.3554	0.3285	6.2900	4.5367	2.2117	1.1344
CKN $\rightarrow$ QN $\downarrow$	0.3497	0.3022	0.2887	0.2667	3.7136	2.4794	1.3169	0.6580
SN $\rightarrow$ QN $\uparrow$	0.1382	0.1117	0.1009	0.0989	6.3149	4.5474	2.2173	1.1380
SN $\rightarrow$ QN $\downarrow$	0.1585	0.1381	0.1258	0.1233	3.5079	2.4296	1.2954	0.6521
RN $\rightarrow$ QN $\uparrow$	0.2395	0.2026	0.1752	0.1724	6.3121	4.5468	2.2171	1.1378

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CKN	0.1602	0.2070	0.1914	0.2031
	setup↓ → CKN	0.3594	0.4297	0.4258	0.4102
	hold↑ → CKN	0.0078	-0.0312	-0.0195	-0.0352
	hold↓ → CKN	-0.3164	-0.3672	-0.3633	-0.3438
SE	setup↑ → CKN	0.3828	0.4531	0.4453	0.4336
	setup↓ → CKN	0.2773	0.4453	0.4023	0.2891
	hold↑ → CKN	0.0195	-0.0156	-0.0078	-0.0234
	hold↓ → CKN	-0.0273	-0.0898	-0.0820	-0.0625
D	setup↑ → CKN	0.1133	0.1914	0.1758	0.1523
	setup↓ → CKN	0.2734	0.4453	0.3984	0.2852
	hold↑ → CKN	0.0391	-0.0234	-0.0117	0.0000
	hold↓ → CKN	-0.2344	-0.3750	-0.3320	-0.2305
CKN	minpwl	0.1674	0.1480	0.1335	0.1238
	minpwh	0.1626	0.1965	0.2014	0.1626
SN	minpwl	0.1577	0.1432	0.1432	0.1868
	recovery	-0.0742	-0.0352	-0.0391	-0.0430
	removal	0.0820	0.0430	0.0469	0.0508
RN	minpwl	0.2208	0.1965	0.2159	0.3712
	recovery	0.0703	0.1328	0.1016	0.0977
	removal	0.0781	0.0156	0.0508	0.0547

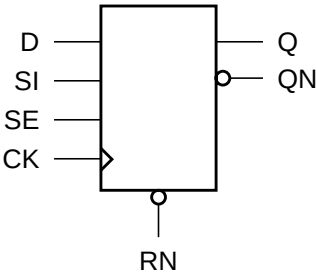
Cell Description

The SDFFR cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN).

Functions

RN	D	SI	SE	CK	Q[n+1]	QN[n+1]
1	1	x	0		1	0
1	0	x	0		0	1
1	x	x	x		Q[n]	QN[n]
1	x	1	1		1	0
1	x	0	1		0	1
0	x	x	x	x	0	1

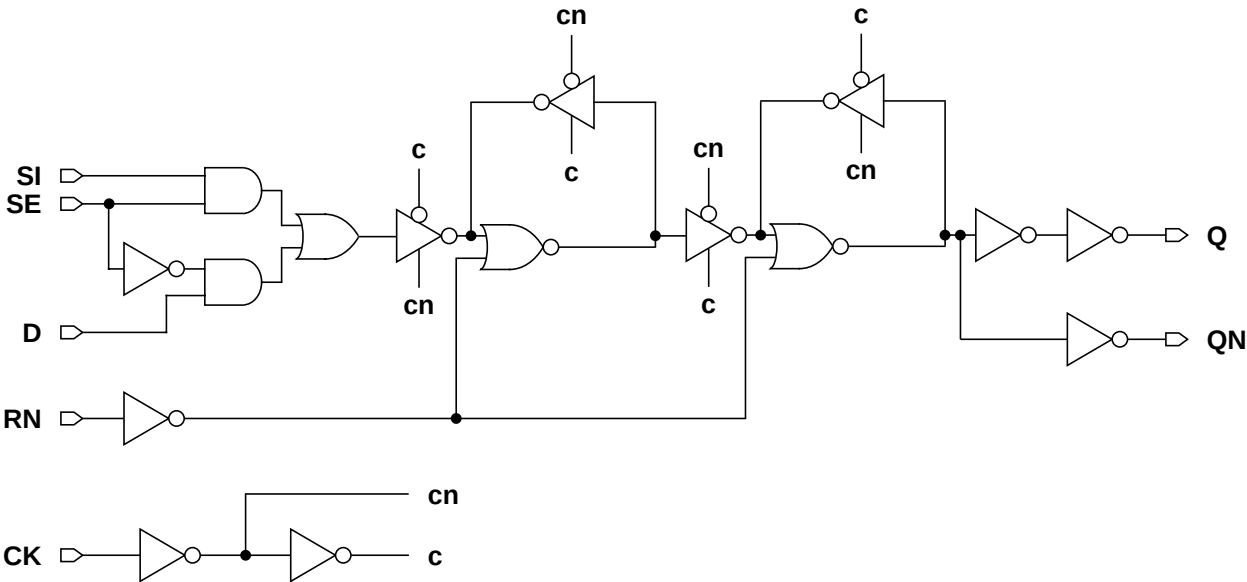
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFRXL	5.04	18.48
SDFFRX1	5.04	18.48
SDFFRX2	5.04	19.80
SDFFRX4	5.04	23.10

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
SI	0.0447	0.0406	0.0435	0.0557
SE	0.0527	0.0469	0.0514	0.0628
D	0.0389	0.0356	0.0387	0.0494
CK	0.0437	0.0406	0.0462	0.0508
RN	0.0153	0.0174	0.0212	0.0318
Q	0.0462	0.0539	0.0828	0.1453

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0022	0.0023	0.0023	0.0023
SE	0.0049	0.0042	0.0044	0.0049
D	0.0029	0.0021	0.0020	0.0027
CK	0.0024	0.0029	0.0031	0.0038
RN	0.0021	0.0027	0.0033	0.0055

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK → Q↑	0.4536	0.4128	0.3982	0.3554	6.2623	4.5310	2.2074	1.1319
CK → Q↓	0.3257	0.3317	0.3133	0.2852	3.4019	2.4023	1.4128	0.6467
RN → Q↓	0.2632	0.2590	0.2462	0.2282	3.4023	2.4022	1.4129	0.6467
CK → QN↑	0.2691	0.2630	0.2334	0.2119	6.2588	4.5300	2.2082	1.1323
CK → QN↓	0.4189	0.3691	0.3527	0.3018	3.6306	2.4669	1.5364	0.6582
RN → QN↑	0.2069	0.1910	0.1668	0.1555	6.2687	4.5334	2.2099	1.1333

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.1562	0.1523	0.1797	0.1797
	setup↓ → CK	0.3672	0.4023	0.4531	0.4336
	hold↑ → CK	-0.1328	-0.1250	-0.1445	-0.1484
	hold↓ → CK	-0.2266	-0.2852	-0.3164	-0.3047
SE	setup↑ → CK	0.3906	0.4336	0.4727	0.4570
	setup↓ → CK	0.2266	0.4023	0.4609	0.3281
	hold↑ → CK	-0.1211	-0.1055	-0.1250	-0.1328
	hold↓ → CK	-0.0898	-0.1758	-0.1953	-0.1797
D	setup↑ → CK	0.0977	0.1328	0.1562	0.1484
	setup↓ → CK	0.2227	0.4180	0.4570	0.3359
	hold↑ → CK	-0.0781	-0.1055	-0.1250	-0.1172
	hold↓ → CK	-0.0898	-0.2969	-0.3203	-0.2109
CK	minpwh	0.1383	0.1335	0.1286	0.0995
	minpwl	0.2548	0.2256	0.2548	0.2208
RN	minpwl	0.2014	0.2014	0.2305	0.3469
	recovery	0.0352	0.0586	0.0781	0.0625
	removal	-0.0156	-0.0391	-0.0430	-0.0234

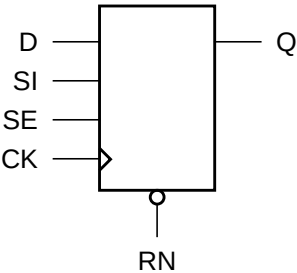
Cell Description

The SDFFRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN). The cell has a single output (Q) and fast clock-to-out path.

Functions

RN	D	SI	SE	CK	Q[n+1]
1	1	x	0		1
1	0	x	0		0
1	x	x	x		Q[n]
1	x	1	1		1
1	x	0	1		0
0	x	x	x	x	0

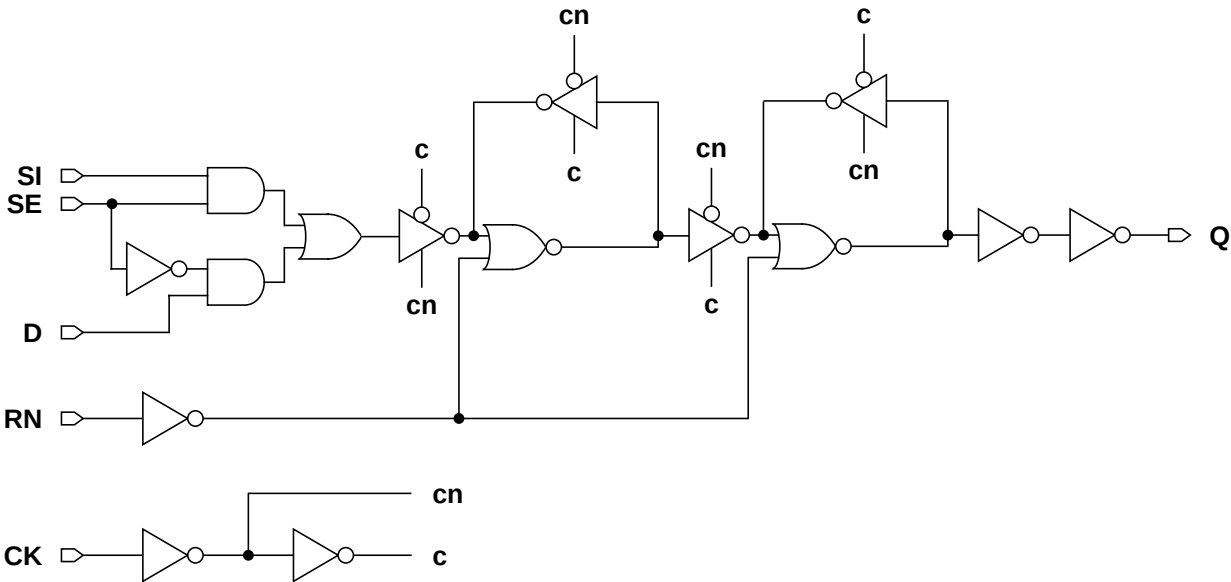
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFRHQXL	5.04	16.50
SDFFRHQX1	5.04	16.50
SDFFRHQX2	5.04	19.80
SDFFRHQX4	5.04	24.42

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0454	0.0551	0.0794	0.1200
SE	0.0554	0.0634	0.0879	0.1258
D	0.0388	0.0490	0.0724	0.1054
CK	0.0455	0.0449	0.0543	0.0717
RN	0.0196	0.0247	0.0341	0.0537
Q	0.0312	0.0338	0.0493	0.0753

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0022	0.0022	0.0022	0.0021
SE	0.0049	0.0045	0.0047	0.0055
D	0.0028	0.0019	0.0023	0.0037
CK	0.0021	0.0029	0.0040	0.0063
RN	0.0025	0.0041	0.0056	0.0097

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2562	0.2118	0.1839	0.1712	9.4900	6.7347	3.3674	1.8472
CK $\rightarrow$ Q $\downarrow$	0.2726	0.1849	0.1627	0.1504	3.7649	2.4424	1.2976	0.7102
RN $\rightarrow$ Q $\downarrow$	0.1876	0.1393	0.1255	0.1053	3.0997	2.0262	1.1790	0.6903

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.1797	0.1992	0.2383	0.2578
	setup↓ → CK	0.4336	0.4961	0.5273	0.5781
	hold↑ → CK	-0.1250	-0.1328	-0.1562	-0.1680
	hold↓ → CK	-0.2266	-0.2969	-0.3164	-0.3633
SE	setup↑ → CK	0.4609	0.5117	0.5391	0.6172
	setup↓ → CK	0.3008	0.4727	0.4180	0.3398
	hold↑ → CK	-0.1094	-0.1172	-0.1406	-0.1602
	hold↓ → CK	-0.0938	-0.1758	-0.1836	-0.1523
D	setup↑ → CK	0.1250	0.1836	0.2070	0.1641
	setup↓ → CK	0.2852	0.4766	0.4180	0.3398
	hold↑ → CK	-0.0742	-0.1133	-0.1250	-0.0859
	hold↓ → CK	-0.0938	-0.2773	-0.2109	-0.1562
CK	minpwh	0.1286	0.1140	0.0995	0.0946
	minpwl	0.2256	0.2256	0.2159	0.1674
RN	minpwl	0.2839	0.2887	0.4003	0.6089
	recovery	0.1016	0.1094	0.1250	0.1133
	removal	-0.0273	-0.0430	-0.0352	-0.0156

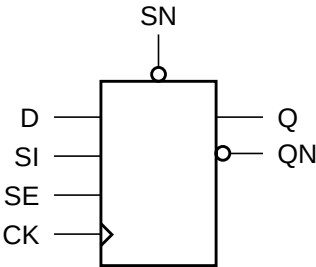
Cell Description

The SDFFS cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low set (SN).

Functions

SN	D	SI	SE	CK	Q[n+1]	QN[n+1]
1	1	x	0		1	0
1	0	x	0		0	1
1	x	x	x		Q[n]	QN[n]
1	x	1	1		1	0
1	x	0	1		0	1
0	x	x	x	x	1	0

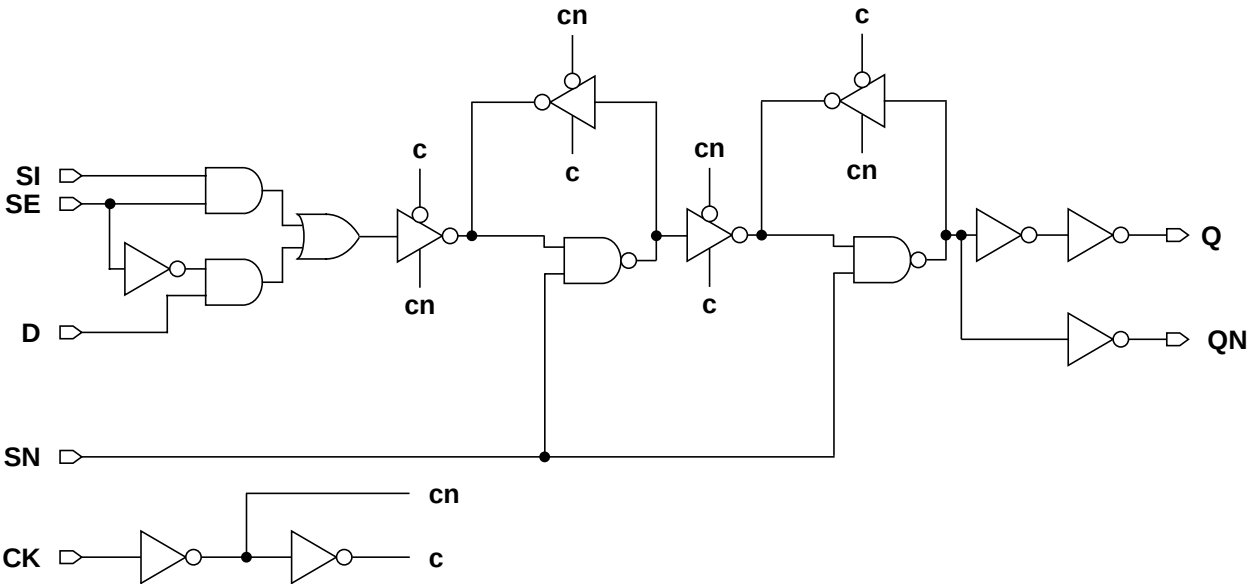
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFSXL	5.04	15.84
SDFFSX1	5.04	15.84
SDFFSX2	5.04	16.50
SDFFSX4	5.04	21.12

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
SI	0.0430	0.0362	0.0405	0.0484
SE	0.0501	0.0417	0.0462	0.0536
D	0.0370	0.0310	0.0357	0.0425
CK	0.0430	0.0402	0.0442	0.0484
SN	0.0069	0.0076	0.0115	0.0179
Q	0.0420	0.0478	0.0787	0.1364

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0021	0.0021	0.0021	0.0022
SE	0.0051	0.0044	0.0044	0.0046
D	0.0027	0.0017	0.0018	0.0020
CK	0.0024	0.0030	0.0030	0.0036
SN	0.0051	0.0056	0.0077	0.0128

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.3692	0.3441	0.3546	0.3231	6.2667	4.5319	2.2625	1.1039
CK $\rightarrow$ Q $\downarrow$	0.3043	0.3100	0.3162	0.2840	3.4016	2.4017	1.2830	0.6413
SN $\rightarrow$ Q $\uparrow$	0.1748	0.1593	0.1646	0.1549	6.2664	4.5319	2.2626	1.1039
CK $\rightarrow$ QN $\uparrow$	0.2532	0.2457	0.2405	0.2117	6.2732	4.7332	2.2667	1.1061
CK $\rightarrow$ QN $\downarrow$	0.3354	0.3022	0.3045	0.2738	3.4702	2.6621	1.2885	0.6440
SN $\rightarrow$ QN $\downarrow$	0.1410	0.1176	0.1149	0.1059	3.4548	2.6628	1.2910	0.6465

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.1406	0.1406	0.1367	0.1445
	setup↓ → CK	0.3750	0.4062	0.4219	0.4102
	hold↑ → CK	-0.1211	-0.1133	-0.1094	-0.1172
	hold↓ → CK	-0.2656	-0.3086	-0.3164	-0.3047
SE	setup↑ → CK	0.4023	0.4336	0.4531	0.4297
	setup↓ → CK	0.2305	0.4102	0.4336	0.3516
	hold↑ → CK	-0.1094	-0.0977	-0.0898	-0.1016
	hold↓ → CK	-0.1211	-0.1602	-0.1641	-0.1641
D	setup↑ → CK	0.0820	0.1211	0.1172	0.1250
	setup↓ → CK	0.2305	0.4219	0.4453	0.3555
	hold↑ → CK	-0.0742	-0.0938	-0.0938	-0.0977
	hold↓ → CK	-0.1250	-0.3281	-0.3398	-0.2500
CK	minpwh	0.1189	0.1140	0.1189	0.1043
	minpwl	0.2305	0.2111	0.2256	0.2111
SN	minpwl	0.1383	0.1189	0.1335	0.1577
	recovery	-0.0273	-0.0078	-0.0039	0.0078
	removal	0.1445	0.0977	0.1055	0.0938

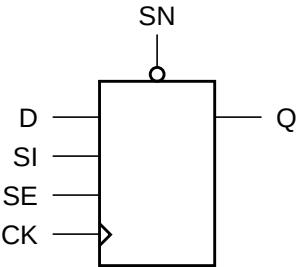
Cell Description

The SDFFSHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low set (SN). The cell has a single output (Q) and fast clock-to-out path.

Functions

SN	D	SI	SE	CK	Q[n+1]
1	1	x	0		1
1	0	x	0		0
1	x	x	x		Q[n]
1	x	1	1		1
1	x	0	1		0
0	x	x	x	x	1

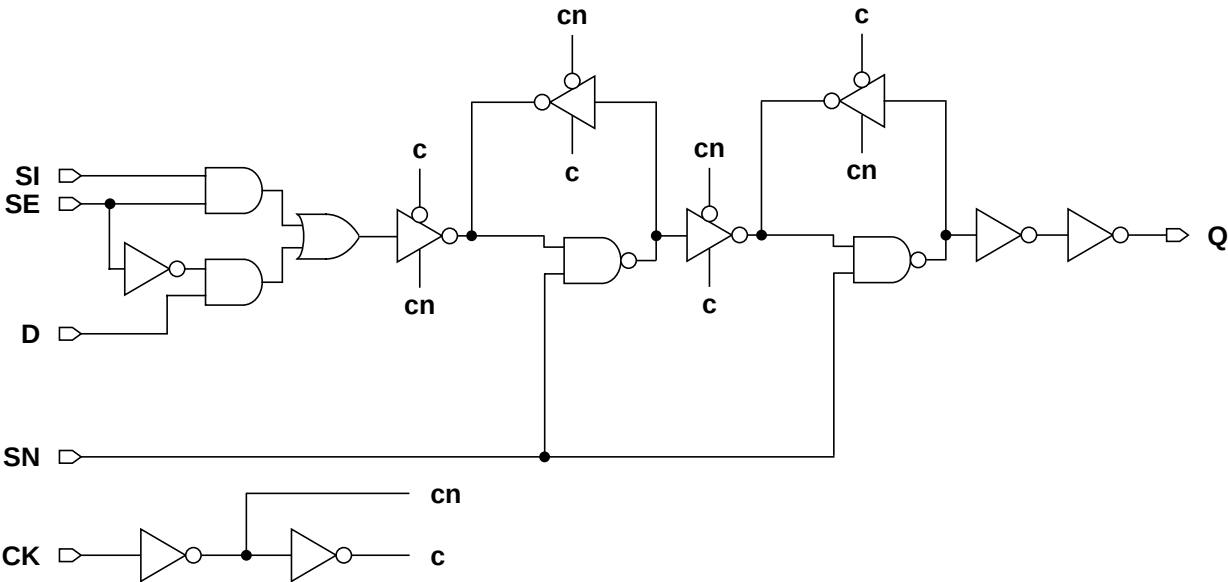
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFSHQXL	5.04	15.84
SDFFSHQX1	5.04	15.84
SDFFSHQX2	5.04	18.48
SDFFSHQX4	5.04	21.12

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0474	0.0535	0.0733	0.1104
SE	0.0551	0.0604	0.0789	0.1139
D	0.0407	0.0482	0.0665	0.0972
CK	0.0437	0.0431	0.0511	0.0685
SN	0.0105	0.0114	0.0194	0.0311
Q	0.0299	0.0316	0.0396	0.0669

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0022	0.0024	0.0022	0.0023
SE	0.0051	0.0045	0.0046	0.0052
D	0.0028	0.0018	0.0021	0.0032
CK	0.0022	0.0029	0.0038	0.0058
SN	0.0084	0.0091	0.0143	0.0224

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2566	0.2071	0.1855	0.1689	6.3031	4.5378	2.2659	1.1328
CK $\rightarrow$ Q $\downarrow$	0.2983	0.1832	0.1607	0.1486	4.2774	2.8460	1.4177	0.7307
SN $\rightarrow$ Q $\uparrow$	0.0781	0.0894	0.0911	0.0951	3.5236	2.4674	1.2844	0.6894

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.1562	0.1680	0.1914	0.2227
	setup↓ → CK	0.3828	0.5156	0.5117	0.5703
	hold↑ → CK	-0.1250	-0.1133	-0.1328	-0.1523
	hold↓ → CK	-0.2500	-0.3203	-0.3164	-0.3711
SE	setup↑ → CK	0.3984	0.5273	0.5273	0.5898
	setup↓ → CK	0.2344	0.5391	0.4297	0.3672
	hold↑ → CK	-0.1094	-0.0977	-0.1172	-0.1445
	hold↓ → CK	-0.1094	-0.1602	-0.1719	-0.1445
D	setup↑ → CK	0.1016	0.1523	0.1641	0.1445
	setup↓ → CK	0.2344	0.5352	0.4258	0.3633
	hold↑ → CK	-0.0742	-0.0977	-0.1094	-0.0859
	hold↓ → CK	-0.1172	-0.3398	-0.2422	-0.1875
CK	minpwh	0.1335	0.1140	0.1043	0.0946
	minpwl	0.2111	0.2111	0.2062	0.1723
SN	minpwl	0.1189	0.1383	0.1723	0.2450
	recovery	-0.0156	0.0664	0.0664	0.0664
	removal	0.1445	0.1016	0.1055	0.1133

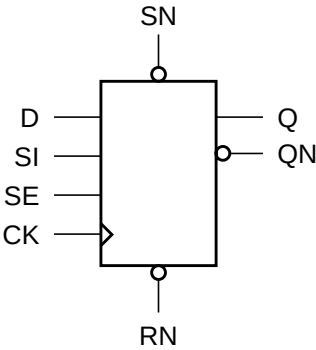
Cell Description

The SDFFSR cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN) and set (SN). Set (SN) dominates reset (RN).

Functions

RN	SN	D	SI	SE	CK	Q[n+1]	QN[n+1]
1	1	1	x	0		1	0
1	1	0	x	0		0	1
1	1	x	x	x		Q[n]	QN[n]
1	1	x	1	1		1	0
1	1	x	0	1		0	1
0	1	x	x	x	x	0	1
1	0	x	x	x	x	1	0
0	0	x	x	x	x	1	0

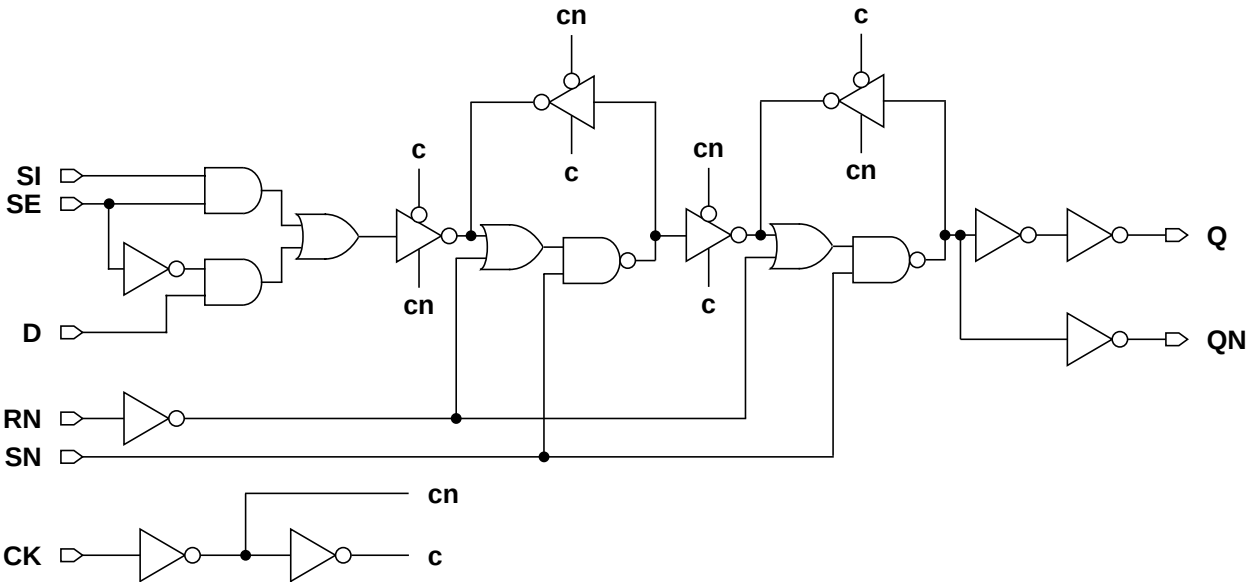
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFSRXL	5.04	19.14
SDFFSRX1	5.04	19.80
SDFFSRX2	5.04	19.14
SDFFSRX4	5.04	25.74

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0464	0.0398	0.0458	0.0604
SE	0.0534	0.0464	0.0524	0.0663
D	0.0404	0.0347	0.0404	0.0535
CK	0.0410	0.0387	0.0401	0.0518
SN	0.0092	0.0095	0.0127	0.0193
RN	0.0169	0.0183	0.0234	0.0409
Q	0.0478	0.0566	0.0844	0.1497

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0021	0.0020	0.0020	0.0021
SE	0.0046	0.0043	0.0044	0.0049
D	0.0024	0.0017	0.0019	0.0026
CK	0.0025	0.0030	0.0031	0.0041
SN	0.0063	0.0067	0.0090	0.0154
RN	0.0024	0.0027	0.0038	0.0062

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.4407	0.4010	0.3979	0.3693	6.2604	4.5291	2.2917	1.1318
CK $\rightarrow$ Q $\downarrow$	0.3396	0.3399	0.3142	0.3006	3.4151	2.4059	1.3141	0.6419
SN $\rightarrow$ Q $\uparrow$	0.2036	0.1902	0.1811	0.1730	6.2597	4.5290	2.2917	1.1317
SN $\rightarrow$ Q $\downarrow$	0.2053	0.1888	0.1789	0.1723	3.4161	2.4061	1.3142	0.6419
RN $\rightarrow$ Q $\downarrow$	0.3067	0.2787	0.2527	0.2459	3.4164	2.4063	1.3140	0.6418
CK $\rightarrow$ QN $\uparrow$	0.2710	0.2628	0.2364	0.2267	6.2890	4.5361	2.2962	1.1345
CK $\rightarrow$ QN $\downarrow$	0.3962	0.3516	0.3440	0.3193	3.7110	2.4795	1.3475	0.6581
SN $\rightarrow$ QN $\uparrow$	0.1371	0.1127	0.1017	0.0987	6.3137	4.5468	2.3017	1.1381
SN $\rightarrow$ QN $\downarrow$	0.1575	0.1390	0.1265	0.1229	3.5074	2.4291	1.3259	0.6521
RN $\rightarrow$ QN $\uparrow$	0.2383	0.2023	0.1754	0.1721	6.3109	4.5462	2.3016	1.1379

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.1758	0.2070	0.1914	0.1953
	setup↓ → CK	0.3711	0.4258	0.4336	0.4297
	hold↑ → CK	-0.1445	-0.1523	-0.1484	-0.1523
	hold↓ → CK	-0.2305	-0.2852	-0.2812	-0.2852
SE	setup↑ → CK	0.3945	0.4492	0.4531	0.4531
	setup↓ → CK	0.2969	0.4414	0.4102	0.3125
	hold↑ → CK	-0.1289	-0.1328	-0.1289	-0.1367
	hold↓ → CK	-0.1562	-0.1992	-0.1953	-0.1641
D	setup↑ → CK	0.1211	0.1875	0.1680	0.1367
	setup↓ → CK	0.2969	0.4414	0.4102	0.3086
	hold↑ → CK	-0.0938	-0.1328	-0.1250	-0.1016
	hold↓ → CK	-0.1523	-0.2969	-0.2578	-0.1719
CK	minpwh	0.1238	0.1140	0.1092	0.0995
	minpwl	0.2208	0.2402	0.2353	0.1965
SN	minpwl	0.1577	0.1432	0.1432	0.1868
	recovery	-0.0078	0.0156	0.0156	0.0156
	removal	0.1328	0.0898	0.1016	0.1055
RN	minpwl	0.2208	0.1965	0.2208	0.3712
	recovery	0.0703	0.1250	0.0898	0.0781
	removal	-0.0430	-0.0898	-0.0586	-0.0391

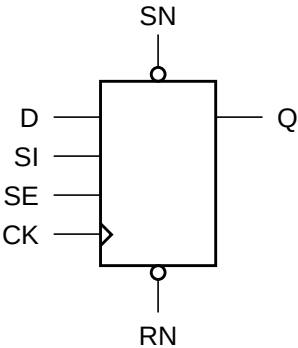
Cell Description

The SDDFSRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN) and set (SN), and set dominating reset. The cell has a single output (Q) and fast clock-to-out path.

Functions

RN	SN	D	SI	SE	CK	Q[n+1]
1	1	1	x	0		1
1	1	0	x	0		0
1	1	x	x	x		Q[n]
1	1	x	1	1		1
1	1	x	0	1		0
0	1	x	x	x	x	0
1	0	x	x	x	x	1
0	0	x	x	x	x	1

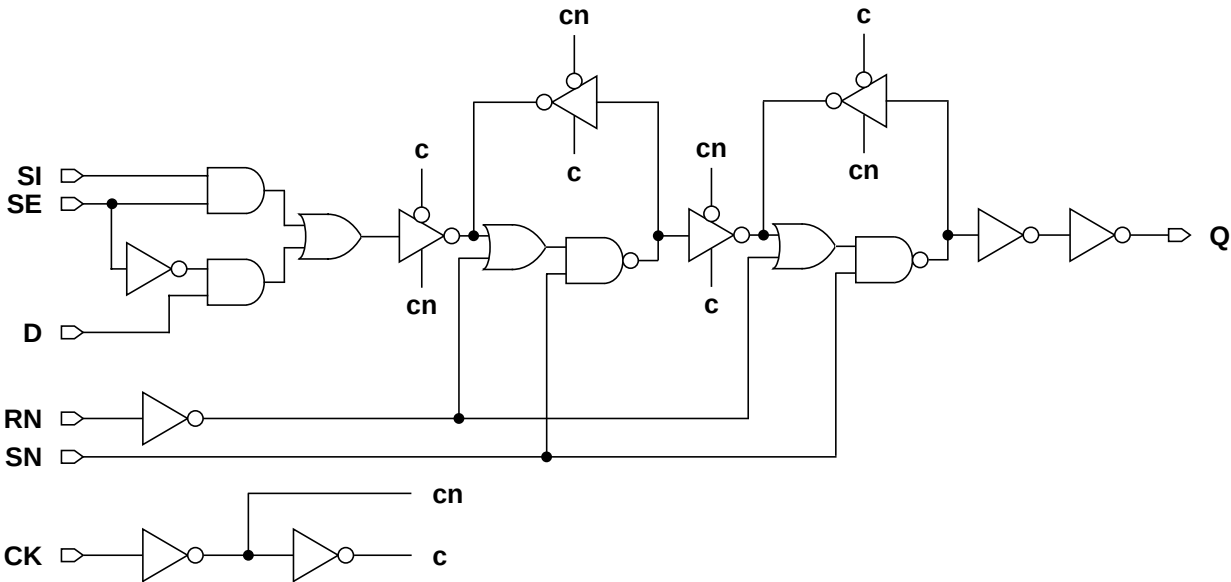
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDDFSRHQXL	5.04	19.80
SDDFSRHQX1	5.04	19.80
SDDFSRHQX2	5.04	25.08
SDDFSRHQX4	5.04	33.66

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
SI	0.0511	0.0574	0.0827	0.1276
SE	0.0588	0.0642	0.0933	0.1400
D	0.0445	0.0502	0.0752	0.1180
CK	0.0446	0.0434	0.0557	0.0791
SN	0.0129	0.0138	0.0197	0.0326
RN	0.0237	0.0281	0.0428	0.0693
Q	0.0344	0.0344	0.0530	0.0852

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0021	0.0021	0.0021	0.0022
SE	0.0057	0.0052	0.0055	0.0060
D	0.0028	0.0018	0.0026	0.0044
CK	0.0023	0.0030	0.0045	0.0067
SN	0.0111	0.0119	0.0172	0.0283
RN	0.0024	0.0038	0.0060	0.0105

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2693	0.2191	0.2010	0.1905	9.4912	6.7390	3.3693	1.6846
CK $\rightarrow$ Q $\downarrow$	0.2649	0.1780	0.1675	0.1568	4.1419	2.8316	1.4606	0.7549
SN $\rightarrow$ Q $\uparrow$	0.0844	0.0933	0.0970	0.0990	4.0667	2.8804	1.4790	0.7908
SN $\rightarrow$ Q $\downarrow$	0.0488	0.0543	0.0505	0.0448	3.7450	2.5963	1.4138	0.7523
RN $\rightarrow$ Q $\downarrow$	0.1913	0.1388	0.1233	0.1116	3.7904	2.5936	1.4123	0.7521

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.2031	0.2148	0.2461	0.2969
	setup↓ → CK	0.4727	0.4922	0.5312	0.6523
	hold↑ → CK	-0.1328	-0.1367	-0.1562	-0.1914
	hold↓ → CK	-0.2461	-0.2852	-0.3203	-0.4375
SE	setup↑ → CK	0.5117	0.5156	0.5547	0.6953
	setup↓ → CK	0.3242	0.4414	0.3867	0.3516
	hold↑ → CK	-0.1133	-0.1172	-0.1406	-0.1836
	hold↓ → CK	-0.0938	-0.1914	-0.1797	-0.1523
D	setup↑ → CK	0.1445	0.1914	0.1992	0.1797
	setup↓ → CK	0.3242	0.4492	0.3867	0.3516
	hold↑ → CK	-0.0781	-0.1172	-0.1133	-0.0859
	hold↓ → CK	-0.1016	-0.2461	-0.1836	-0.1680
CK	minpwh	0.1335	0.1189	0.1043	0.0946
	minpwl	0.2353	0.2305	0.2111	0.1771
SN	minpwl	0.1383	0.1674	0.2014	0.2839
	recovery	0.0547	0.0664	0.0781	0.0742
	removal	0.1445	0.1016	0.1055	0.1094
RN	minpwl	0.2790	0.2159	0.3324	0.5507
	recovery	0.1211	0.1289	0.1367	0.1289
	removal	-0.0352	-0.0508	-0.0391	-0.0078

Logic Symbol

Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFTRXL	5.04	14.52
SDFFTRX1	5.04	14.52
SDFFTRX2	5.04	17.16
SDFFTRX4	5.04	19.80

The diagram illustrates a 3-bit ripple-carry adder. It consists of three full-adder blocks connected in series. The first full-adder has inputs SI, SE, and D. The second full-adder has inputs SE, D, and RN. The third full-adder has inputs D, RN, and the carry-in from the second stage. The outputs are QN (carry-out), Q (sum), and the carry-in to the first stage. The carry signal is labeled 'c' and 'cn'.

AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0432	0.0405	0.0528	0.0788
SE	0.0518	0.0474	0.0582	0.0818
D	0.0370	0.0361	0.0467	0.0669
CK	0.0447	0.0417	0.0520	0.0717
RN	0.0414	0.0394	0.0499	0.0737
Q	0.0315	0.0379	0.0669	0.1117

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0023	0.0022	0.0022	0.0024
SE	0.0052	0.0045	0.0045	0.0056
D	0.0025	0.0017	0.0020	0.0026
CK	0.0021	0.0030	0.0041	0.0067
RN	0.0028	0.0018	0.0021	0.0040

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2352	0.2064	0.2046	0.1744	6.2645	4.5314	2.2656	1.1057
CK $\rightarrow$ Q $\downarrow$	0.1923	0.1675	0.1590	0.1514	3.2471	2.5977	1.3014	0.6501
CK $\rightarrow$ QN $\uparrow$	0.2457	0.2213	0.2126	0.2031	6.2676	4.5321	2.2629	1.1042
CK $\rightarrow$ QN $\downarrow$	0.3084	0.2824	0.2855	0.2478	3.4312	2.4094	1.2847	0.6418

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.1484	0.1445	0.1719	0.1953
	setup↓ → CK	0.4297	0.4648	0.4375	0.5352
	hold↑ → CK	-0.1289	-0.1094	-0.1367	-0.1523
	hold↓ → CK	-0.3047	-0.3359	-0.3047	-0.4023
SE	setup↑ → CK	0.4492	0.4766	0.4453	0.5352
	setup↓ → CK	0.2812	0.4805	0.3828	0.3125
	hold↑ → CK	-0.1172	-0.0938	-0.1172	-0.1406
	hold↓ → CK	-0.1602	-0.1992	-0.2148	-0.1641
D	setup↑ → CK	0.1094	0.1602	0.1875	0.1289
	setup↓ → CK	0.2734	0.4766	0.3828	0.2969
	hold↑ → CK	-0.0938	-0.1250	-0.1484	-0.0977
	hold↓ → CK	-0.1523	-0.3438	-0.2500	-0.1836
CK	minpwh	0.1189	0.1043	0.1092	0.0946
	minpwl	0.2402	0.2305	0.2450	0.1771
RN	setup↑ → CK	0.1250	0.1680	0.1953	0.1406
	setup↓ → CK	0.3203	0.5391	0.4258	0.3477
	hold↑ → CK	-0.1055	-0.1328	-0.1562	-0.1094
	hold↓ → CK	-0.1797	-0.3984	-0.2773	-0.2188

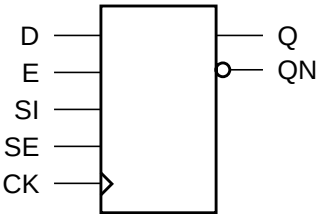
Cell Description

The SEDFF cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and synchronous active-high enable (E).

Functions

D	E	SI	SE	CK	Q[n+1]	QN[n+1]
x	x	1	1		1	0
x	x	0	1		0	1
x	0	x	0		Q[n]	QN[n]
0	1	x	0		0	1
1	1	x	0		1	0
x	x	x	x		Q[n]	QN[n]

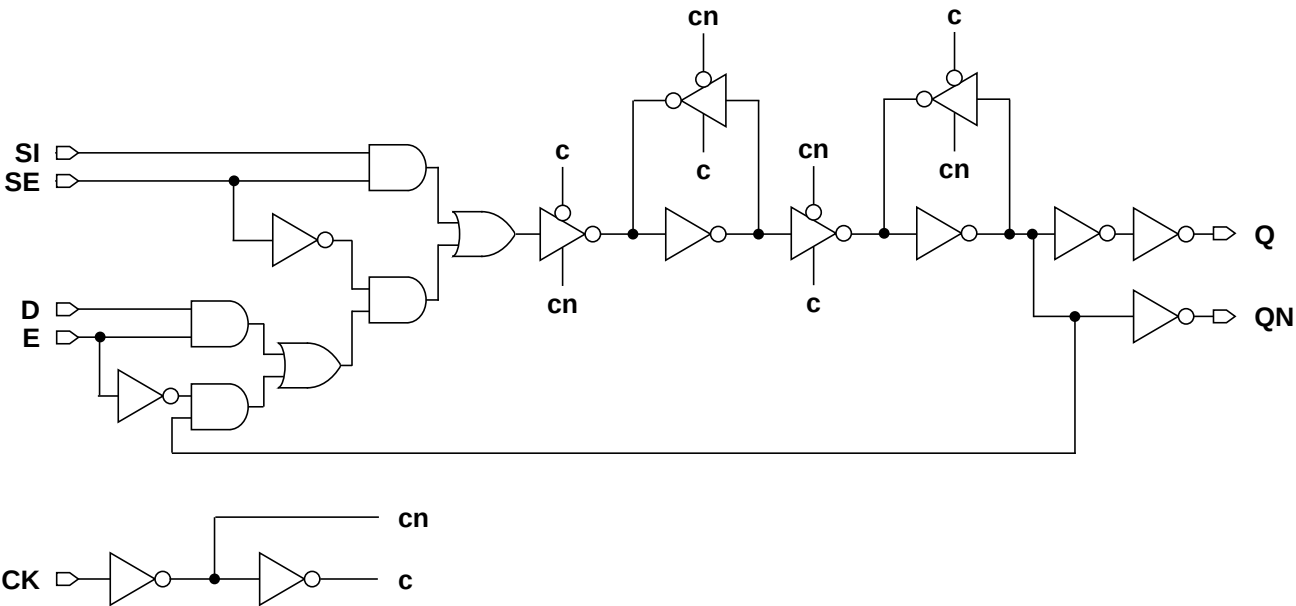
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SEDFFXL	5.04	17.82
SEDFFX1	5.04	17.82
SEDFFX2	5.04	19.80
SEDFFX4	5.04	23.10

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0465	0.0483	0.0644	0.0935
SE	0.0548	0.0578	0.0739	0.1026
D	0.0391	0.0432	0.0564	0.0808
CK	0.0662	0.0716	0.0905	0.1282
E	0.0556	0.0591	0.0726	0.1006
Q	0.0396	0.0452	0.0757	0.1307

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0020	0.0018	0.0022	0.0034
SE	0.0042	0.0040	0.0046	0.0054
D	0.0020	0.0017	0.0022	0.0033
CK	0.0020	0.0029	0.0040	0.0067
E	0.0050	0.0047	0.0051	0.0060

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2469	0.2221	0.1886	0.1698	6.2601	4.9500	2.2655	1.1061
CK $\rightarrow$ Q $\downarrow$	0.2021	0.1695	0.1559	0.1452	3.5317	2.6037	1.3298	0.6323
CK $\rightarrow$ QN $\uparrow$	0.2897	0.2372	0.2157	0.2073	6.2713	4.5326	2.2638	1.1050
CK $\rightarrow$ QN $\downarrow$	0.3603	0.3134	0.2777	0.2522	3.5191	2.4249	1.3189	0.6249

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.1797	0.1719	0.1836	0.1523
	setup↓ → CK	0.3711	0.5625	0.4219	0.3477
	hold↑ → CK	-0.1641	-0.1445	-0.1641	-0.1367
	hold↓ → CK	-0.3438	-0.5664	-0.4023	-0.3281
SE	setup↑ → CK	0.4883	0.7227	0.5469	0.4727
	setup↓ → CK	0.5273	0.7422	0.5586	0.4922
	hold↑ → CK	-0.1289	-0.1211	-0.1367	-0.1094
	hold↓ → CK	-0.2500	-0.2383	-0.2500	-0.2227
D	setup↑ → CK	0.1836	0.1797	0.1953	0.1641
	setup↓ → CK	0.4688	0.6836	0.5117	0.4570
	hold↑ → CK	-0.1484	-0.1445	-0.1523	-0.1250
	hold↓ → CK	-0.3359	-0.5078	-0.3633	-0.3125
CK	minpwh	0.1286	0.1189	0.0995	0.0898
	minpwl	0.2839	0.2402	0.2353	0.1917
E	setup↑ → CK	0.5156	0.7227	0.5586	0.5078
	setup↓ → CK	0.4062	0.6172	0.4414	0.3672
	hold↑ → CK	-0.1719	-0.1641	-0.1719	-0.1367
	hold↓ → CK	-0.2344	-0.2266	-0.2422	-0.2188

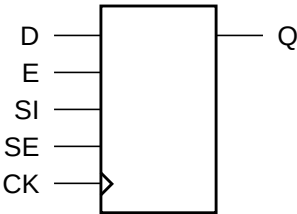
Cell Description

The SEDFFHQ cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and synchronous active-high enable (E). The cell has a single output (Q) and fast clock-to-output path.

Functions

D	E	SI	SE	CK	Q[n+1]
x	x	1	1		1
x	x	0	1		0
x	0	x	0		Q[n]
0	1	x	0		0
1	1	x	0		1
x	x	x	x		Q[n]

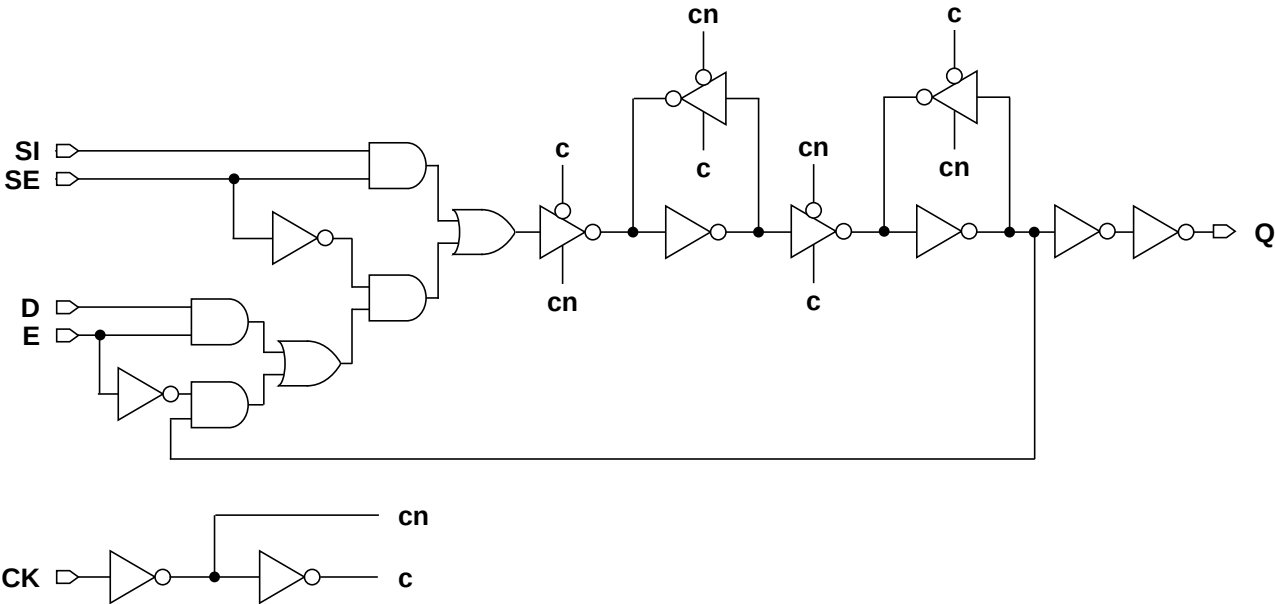
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SEDFFHQXL	5.04	20.46
SEDFFHQX1	5.04	20.46
SEDFFHQX2	5.04	22.44
SEDFFHQX4	5.04	25.08

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
SI	0.0526	0.0648	0.0862	0.1257
SE	0.0750	0.0833	0.1039	0.1429
D	0.0731	0.0844	0.1043	0.1458
CK	0.0615	0.0799	0.1082	0.1582
E	0.0965	0.1006	0.1214	0.1560
Q	0.0332	0.0438	0.0596	0.0863

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0022	0.0022	0.0022	0.0022
SE	0.0021	0.0021	0.0021	0.0021
D	0.0027	0.0031	0.0031	0.0031
CK	0.0022	0.0036	0.0047	0.0066
E	0.0022	0.0023	0.0023	0.0023

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2940	0.1843	0.1767	0.1502	6.2713	4.5295	2.2078	1.1040
CK $\rightarrow$ Q $\downarrow$	0.3398	0.2289	0.2176	0.2004	3.5594	2.4016	1.2823	0.6415

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.2344	0.2617	0.2812	0.3281
	setup↓ → CK	0.2070	0.2344	0.2383	0.2656
	hold↑ → CK	-0.1016	-0.1328	-0.1523	-0.1953
	hold↓ → CK	-0.1875	-0.1992	-0.2070	-0.2266
SE	setup↑ → CK	0.3008	0.3359	0.3555	0.4023
	setup↓ → CK	0.2773	0.2891	0.2930	0.3203
	hold↑ → CK	-0.1719	-0.2109	-0.2305	-0.2578
	hold↓ → CK	-0.1484	-0.1836	-0.2031	-0.2500
D	setup↑ → CK	0.3242	0.3086	0.3320	0.3750
	setup↓ → CK	0.3359	0.2891	0.2930	0.3203
	hold↑ → CK	-0.1836	-0.1836	-0.1992	-0.2500
	hold↓ → CK	-0.3086	-0.2539	-0.2617	-0.2812
CK	minpwh	0.1432	0.0849	0.0801	0.0704
	minpwl	0.2062	0.1917	0.1965	0.2256
E	setup↑ → CK	0.3984	0.3906	0.4141	0.4570
	setup↓ → CK	0.3516	0.3086	0.3086	0.3203
	hold↑ → CK	-0.2578	-0.2695	-0.2891	-0.3281
	hold↓ → CK	-0.1250	-0.1367	-0.1680	-0.2891

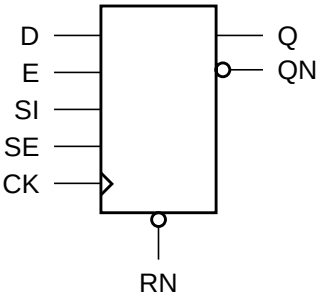
Cell Description

The SEDFFTR cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), synchronous active-high enable (E) and synchronous active low reset (RN). Scan enable (SE) dominates reset (RN) and enable (E).

Functions

RN	D	E	SI	SE	CK	Q[n+1]	QN[n+1]
x	x	x	0	1		0	1
x	x	x	1	1		1	0
1	x	0	x	0		Q[n]	QN[n]
0	x	x	x	0		0	1
1	1	1	x	0		1	0
1	0	1	x	0		0	1
x	x	x	x	x		Q[n]	QN[n]

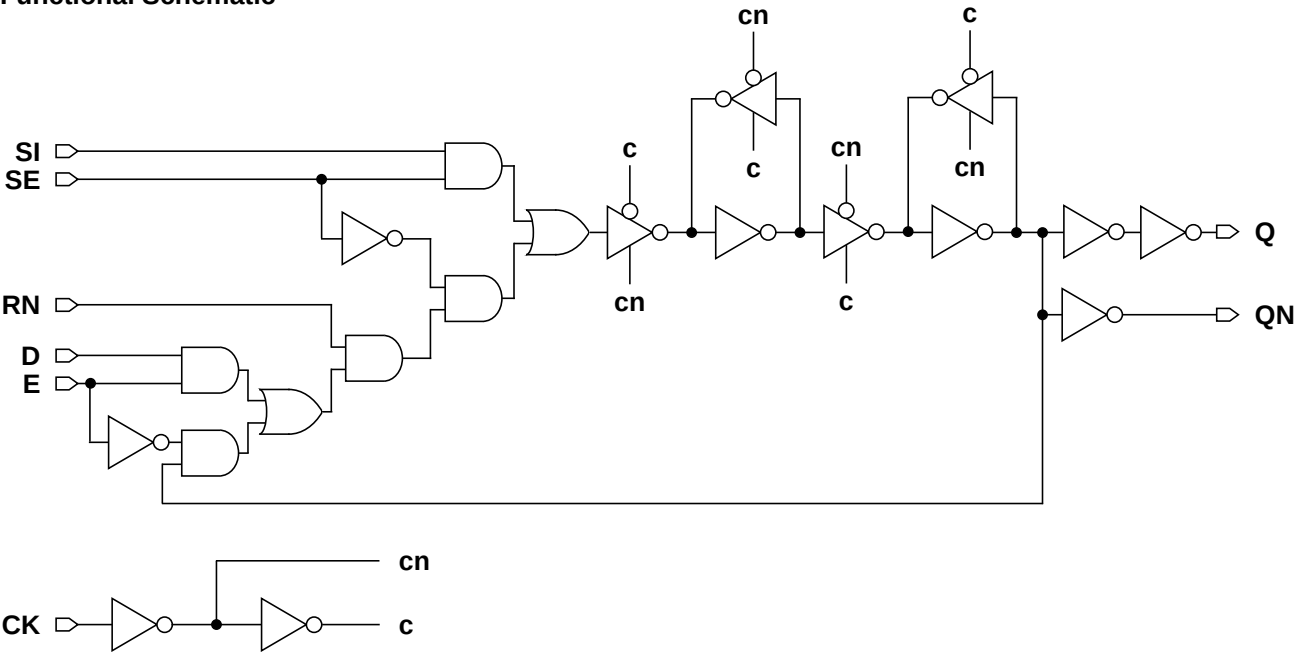
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SEDFFTRXL	5.04	25.08
SEDFFTRX1	5.04	25.74
SEDFFTRX2	5.04	26.40
SEDFFTRX4	5.04	27.72

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
SI	0.0564	0.0716	0.0716	0.0715
SE	0.0815	0.0953	0.0949	0.0948
D	0.0734	0.0939	0.0939	0.0940
CK	0.0655	0.0924	0.0917	0.0917
E	0.0960	0.1157	0.1154	0.1146
RN	0.0509	0.0688	0.0688	0.0685
Q	0.0440	0.0604	0.0831	0.1362

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0023	0.0023	0.0023	0.0023
SE	0.0042	0.0042	0.0042	0.0042
D	0.0021	0.0031	0.0031	0.0031
CK	0.0021	0.0037	0.0037	0.0038
E	0.0022	0.0022	0.0022	0.0022
RN	0.0028	0.0028	0.0028	0.0028

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.3485	0.2089	0.2312	0.2727	6.2745	4.5306	2.2635	1.1045
CK $\rightarrow$ Q $\downarrow$	0.3709	0.2406	0.2513	0.2720	3.5825	2.4031	1.3236	0.6372
CK $\rightarrow$ QN $\uparrow$	0.2208	0.1718	0.1802	0.1984	6.2850	4.5330	2.2674	1.1089
CK $\rightarrow$ QN $\downarrow$	0.2347	0.1574	0.1823	0.2281	3.7449	2.4416	1.3546	0.6668

Timing Constraints at 25°C, 1.8V, Typical Process

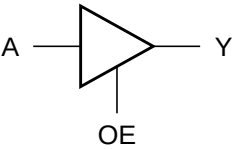
Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.2969	0.3516	0.3477	0.3203
	setup↓ → CK	0.2344	0.2617	0.2617	0.2617
	hold↑ → CK	-0.1562	-0.2070	-0.2070	-0.2031
	hold↓ → CK	-0.2148	-0.2266	-0.2266	-0.2266
SE	setup↑ → CK	0.3672	0.4297	0.4258	0.3984
	setup↓ → CK	0.3945	0.4102	0.4062	0.3789
	hold↑ → CK	-0.2227	-0.2656	-0.2656	-0.2656
	hold↓ → CK	-0.1836	-0.2617	-0.2617	-0.2578
D	setup↑ → CK	0.3945	0.3828	0.3828	0.3516
	setup↓ → CK	0.3516	0.3086	0.3086	0.3086
	hold↑ → CK	-0.2344	-0.2383	-0.2383	-0.2344
	hold↓ → CK	-0.3281	-0.2734	-0.2734	-0.2695
CK	minpwh	0.1917	0.1043	0.1238	0.1626
	minpwl	0.2402	0.2402	0.2450	0.2402
E	setup↑ → CK	0.4727	0.4727	0.4688	0.4414
	setup↓ → CK	0.3672	0.3438	0.3359	0.3281
	hold↑ → CK	-0.3164	-0.3281	-0.3281	-0.3242
	hold↓ → CK	-0.1211	-0.1250	-0.1445	-0.2656
RN	setup↑ → CK	0.2969	0.3281	0.3281	0.2969
	setup↓ → CK	0.2188	0.2500	0.2500	0.2500
	hold↑ → CK	-0.1289	-0.1797	-0.1758	-0.1758
	hold↓ → CK	-0.1992	-0.2148	-0.2148	-0.2148

Cell Description

The TBUF cell provides the logical buffer of a single input (A) with an active-high output enable (OE). When the enable is high, the output (Y) is represented by the logic equation:

$Y = A$

Logic Symbol



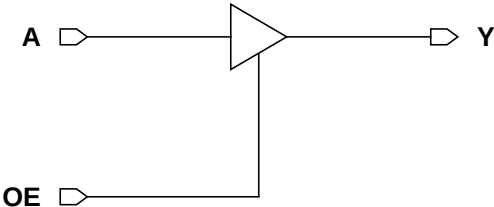
Functions

OE	A	Y
0	x	Z
1	0	0
1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
TBUFXL	5.04	4.62
TBUF1	5.04	4.62
TBUF2	5.04	4.62
TBUF3	5.04	5.28
TBUF4	5.04	5.94
TBUF8	5.04	7.92
TBUF12	5.04	9.90
TBUF16	5.04	11.22
TBUF20	5.04	15.18

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0225	0.0244	0.0338	0.0411	0.0535	0.0977	0.1400	0.1793	0.2273
OE	0.0168	0.0173	0.0242	0.0315	0.0389	0.0752	0.1155	0.1501	0.1819

Pin Capacitance

Pin	Capacitance (pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0024	0.0024	0.0033	0.0046	0.0060	0.0122	0.0173	0.0208	0.0271
OE	0.0042	0.0044	0.0044	0.0044	0.0049	0.0070	0.0101	0.0127	0.0158
Y	0.0020	0.0026	0.0044	0.0046	0.0061	0.0118	0.0173	0.0228	0.0289

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	0.1067	0.1122	0.1104	0.1072	0.0999	0.0967	0.0870	0.0894	0.0895
A $\rightarrow$ Y $\downarrow$	0.1715	0.1792	0.1612	0.1489	0.1463	0.1335	0.1281	0.1299	0.1277
OE $\rightarrow$ Y $\uparrow$	0.0685	0.0740	0.0824	0.0835	0.0773	0.0762	0.0701	0.0705	0.0720
OE $\rightarrow$ Y $\downarrow$	0.1233	0.1238	0.1211	0.1126	0.1152	0.1109	0.1064	0.1048	0.1056

Description	K_{load} (ns/pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	6.2764	4.5419	2.3551	1.5415	1.1494	0.5745	0.3687	0.2766	0.2282
A $\rightarrow$ Y $\downarrow$	4.8316	2.5104	1.3472	0.8345	0.6625	0.3459	0.2358	0.1771	0.1371
OE $\rightarrow$ Y $\uparrow$	6.2703	4.5386	2.3564	1.5412	1.1491	0.5744	0.3687	0.2766	0.2282
OE $\rightarrow$ Y $\downarrow$	4.8233	2.5006	1.3417	0.8308	0.6604	0.3451	0.2353	0.1767	0.1367

Cell Description

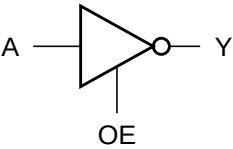
The TBUFI cell provides the logical inversion of a single input (A) with an active-high output enable (OE). When the enable is high, the output (Y) is represented by the logic equation:

$Y = \overline{A}$

Function

OE	A	Y
0	x	Z
1	0	1
1	1	0

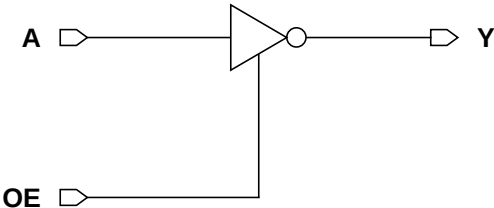
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TBUFIXL	5.04	2.64
TBUFIX1	5.04	2.64
TBUFIX2	5.04	3.96
TBUFIX3	5.04	6.60
TBUFIX4	5.04	6.60
TBUFIX8	5.04	8.58
TBUFIX12	5.04	11.22
TBUFIX16	5.04	12.54
TBUFIX20	5.04	16.50

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0247	0.0245	0.0442	0.0516	0.0640	0.1165	0.1758	0.2276	0.2857
OE	0.0152	0.0152	0.0248	0.0341	0.0398	0.0760	0.1199	0.1551	0.1905

Pin Capacitance

Pin	Capacitance (pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0047	0.0047	0.0093	0.0023	0.0028	0.0050	0.0068	0.0088	0.0109
OE	0.0029	0.0029	0.0040	0.0044	0.0048	0.0071	0.0101	0.0126	0.0158
Y	0.0034	0.0033	0.0044	0.0046	0.0056	0.0115	0.0176	0.0236	0.0294

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	0.0719	0.0716	0.0650	0.1952	0.1839	0.1688	0.1612	0.1583	0.1595
A $\rightarrow$ Y $\downarrow$	0.0322	0.0332	0.0296	0.1954	0.1815	0.1718	0.1698	0.1663	0.1647
OE $\rightarrow$ Y $\uparrow$	0.0710	0.0715	0.0690	0.0850	0.0824	0.0745	0.0714	0.0717	0.0733
OE $\rightarrow$ Y $\downarrow$	0.0215	0.0223	0.0186	0.1153	0.1108	0.1122	0.1074	0.1060	0.1059

Description	K_{load} (ns/pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	6.7377	6.7279	3.3645	1.5174	1.1730	0.5975	0.3691	0.2768	0.2284
A $\rightarrow$ Y $\downarrow$	2.7455	2.7930	1.3973	0.8371	0.6667	0.3152	0.2242	0.1671	0.1293
OE $\rightarrow$ Y $\uparrow$	6.7559	6.7356	3.3689	1.5159	1.1721	0.5970	0.3688	0.2766	0.2283
OE $\rightarrow$ Y $\downarrow$	2.7463	2.7940	1.3982	0.8329	0.6641	0.3142	0.2236	0.1666	0.1289

Cell Description

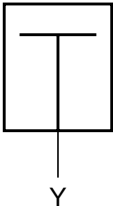
The TIEHI cell drives the output (Y) to a logic high.
The output is driven through diffusion and not tied directly to the power rail to provide some ESD protection. The output (Y) is represented by the logic equation:

$Y = 1$

Function

Y
1

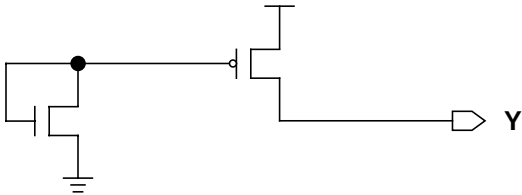
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TIEHI	5.04	1.32

Functional Schematic



Cell Description

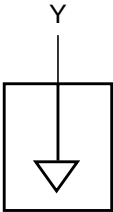
The TIELO cell drives the output (Y) to a logic low. The output is driven through diffusion and not tied directly to the power rail to provide some ESD protection. The output (Y) is represented by the logic equation:

$Y = 0$

Function

Y
0

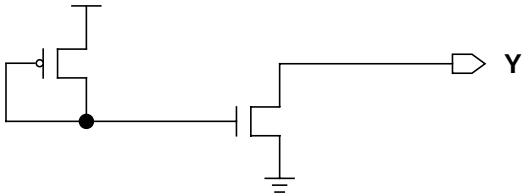
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TIELO	5.04	1.32

Functional Schematic



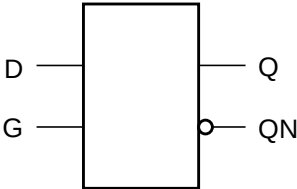
Cell Description

The TLAT cell is an active-high D-type transparent latch. When the enable (G) is high, data is transferred to the outputs (Q, QN).

Functions

G	D	Q[n+1]	QN[n+1]
1	0	0	1
1	1	1	0
0	x	Q[n]	QN[n]

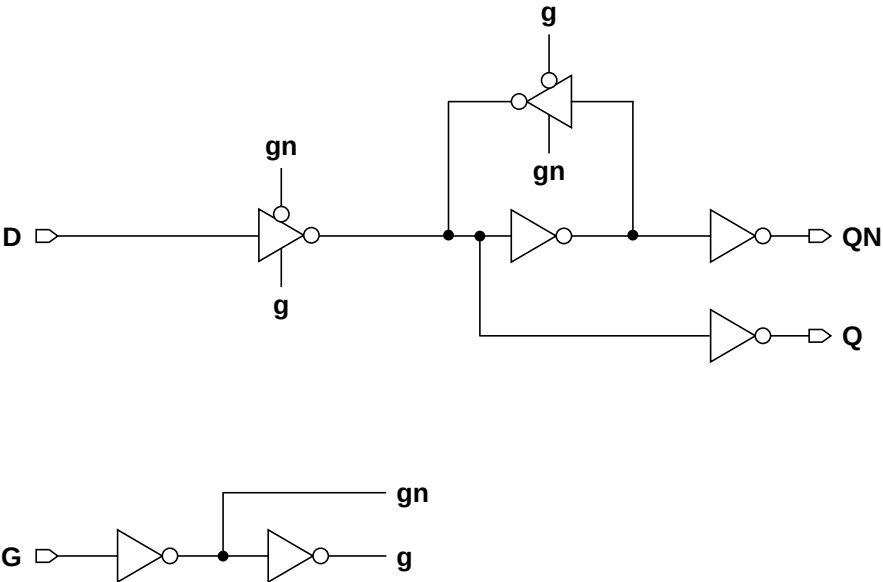
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATXL	5.04	7.26
TLATX1	5.04	7.26
TLATX2	5.04	7.92
TLATX4	5.04	11.22

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0048	0.0058	0.0092	0.0206
G	0.0219	0.0236	0.0282	0.0483
Q	0.0344	0.0420	0.0661	0.1086

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0034	0.0041	0.0062	0.0140
G	0.0020	0.0026	0.0028	0.0046

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.0915	0.0919	0.0891	0.0770	6.2691	4.5343	2.2950	1.1476
D $\rightarrow$ Q $\downarrow$	0.1601	0.1625	0.1528	0.1404	3.5153	2.4471	1.3029	0.6501
G $\rightarrow$ Q $\uparrow$	0.1993	0.1887	0.2023	0.1663	6.2674	4.5338	2.2949	1.1476
G $\rightarrow$ Q $\downarrow$	0.1699	0.1662	0.1655	0.1499	3.5129	2.4459	1.3028	0.6498
D $\rightarrow$ QN $\uparrow$	0.2155	0.2254	0.2215	0.2095	6.2703	4.5332	2.2923	1.1467
D $\rightarrow$ QN $\downarrow$	0.1702	0.1757	0.1880	0.1747	3.4417	2.4142	1.2915	0.6460
G $\rightarrow$ QN $\uparrow$	0.2258	0.2295	0.2345	0.2190	6.2709	4.5335	2.2924	1.1467
G $\rightarrow$ QN $\downarrow$	0.2790	0.2734	0.3021	0.2645	3.4415	2.4144	1.2913	0.6459

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ G	-0.0273	-0.0117	-0.0273	-0.0273
	setup $\downarrow$ $\rightarrow$ G	0.0977	0.1094	0.0938	0.0859
	hold $\uparrow$ $\rightarrow$ G	0.0391	0.0273	0.0391	0.0391
	hold $\downarrow$ $\rightarrow$ G	-0.0938	-0.1055	-0.0898	-0.0781
G	minpwh	0.0995	0.0995	0.1043	0.0849

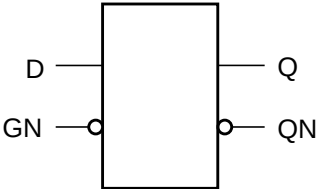
Cell Description

The TLATN cell is an active-low D-type transparent latch. When the enable (GN) is low, data is transferred to the outputs (Q, QN).

Functions

GN	D	Q[n+1]	QN[n+1]
0	0	0	1
0	1	1	0
1	x	Q[n]	QN[n]

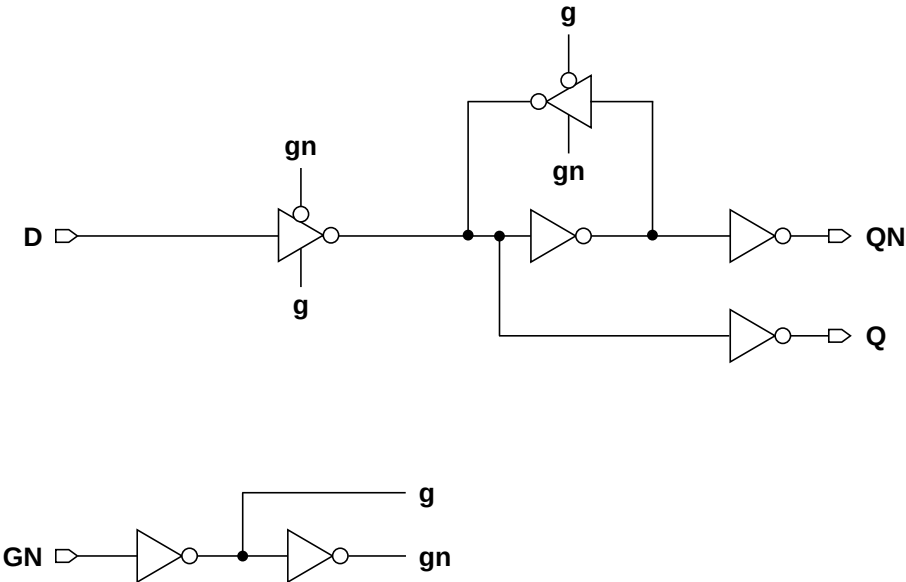
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATNXL	5.04	7.26
TLATNX1	5.04	7.26
TLATNX2	5.04	7.92
TLATNX4	5.04	11.22

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0050	0.0060	0.0090	0.0203
GN	0.0255	0.0281	0.0353	0.0624
Q	0.0358	0.0440	0.0704	0.1159

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0035	0.0041	0.0063	0.0141
GN	0.0020	0.0026	0.0027	0.0046

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.0909	0.0883	0.0881	0.0753	6.2699	4.5333	2.2666	1.1475
D $\rightarrow$ Q $\downarrow$	0.1595	0.1577	0.1554	0.1414	3.5133	2.4423	1.3041	0.6501
GN $\rightarrow$ Q $\uparrow$	0.1660	0.1441	0.1507	0.1238	6.2734	4.5351	2.2677	1.1480
GN $\rightarrow$ Q $\downarrow$	0.2630	0.2461	0.2518	0.2264	3.5119	2.4418	1.3039	0.6501
D $\rightarrow$ QN $\uparrow$	0.2147	0.2213	0.2256	0.2107	6.2702	4.5325	2.2637	1.1467
D $\rightarrow$ QN $\downarrow$	0.1692	0.1729	0.1897	0.1733	3.4404	2.4149	1.2918	0.6461
GN $\rightarrow$ QN $\uparrow$	0.3184	0.3099	0.3220	0.2956	6.2708	4.5327	2.2639	1.1467
GN $\rightarrow$ QN $\downarrow$	0.2458	0.2302	0.2540	0.2231	3.4406	2.4148	1.2918	0.6460

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ GN	0.0508	0.0586	0.0586	0.0508
	setup $\downarrow$ $\rightarrow$ GN	0.0586	0.0703	0.0547	0.0430
	hold $\uparrow$ $\rightarrow$ GN	-0.0430	-0.0469	-0.0469	-0.0391
	hold $\downarrow$ $\rightarrow$ GN	-0.0469	-0.0547	-0.0391	-0.0312
GN	minpwl	0.1383	0.1189	0.1286	0.1043

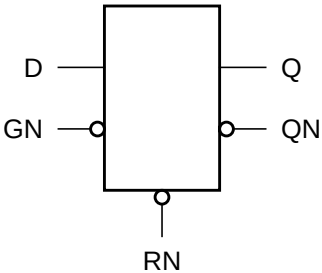
Cell Description

The TLATNR cell is an active-low D-type transparent latch with asynchronous active-low reset (RN). When the enable (GN) is low, data is transferred to the outputs (Q, QN).

Function

RN	GN	D	Q[n+1]	QN[n+1]
1	0	0	0	1
1	0	1	1	0
1	1	x	Q[n]	QN[n]
0	x	x	0	1

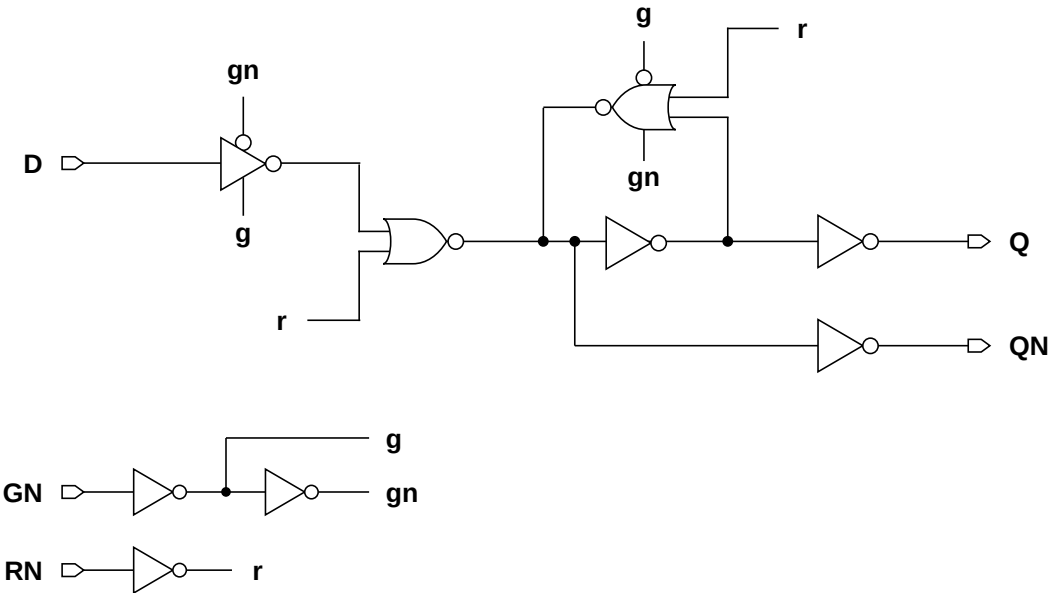
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATNRXL	5.04	8.58
TLATNRX1	5.04	8.58
TLATNRX2	5.04	9.24
TLATNRX4	5.04	11.88

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0036	0.0043	0.0070	0.0139
GN	0.0275	0.0296	0.0388	0.0638
RN	0.0044	0.0045	0.0055	0.0071
Q	0.0394	0.0465	0.0759	0.1297

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0027	0.0032	0.0054	0.0103
GN	0.0021	0.0021	0.0024	0.0037
RN	0.0040	0.0042	0.0053	0.0072

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.1353	0.1282	0.1204	0.1127	6.2979	4.5446	2.2725	1.1091
D $\rightarrow$ Q $\downarrow$	0.2330	0.2287	0.2176	0.1999	3.6712	2.5053	1.3249	0.6598
GN $\rightarrow$ Q $\uparrow$	0.2143	0.2089	0.1884	0.1708	6.2975	4.5450	2.2730	1.1097
GN $\rightarrow$ Q $\downarrow$	0.3414	0.3401	0.3191	0.2976	3.6699	2.5049	1.3248	0.6598
RN $\rightarrow$ Q $\uparrow$	0.1326	0.1243	0.1135	0.1044	6.2976	4.5447	2.2724	1.1091
RN $\rightarrow$ Q $\downarrow$	0.1562	0.1779	0.2705	0.4326	3.5592	2.4991	1.3965	0.7689
D $\rightarrow$ QN $\uparrow$	0.2956	0.2977	0.2883	0.2724	6.2700	4.5336	2.2637	1.1047
D $\rightarrow$ QN $\downarrow$	0.2231	0.2191	0.2248	0.2196	3.4589	2.4224	1.2927	0.6469
GN $\rightarrow$ QN $\uparrow$	0.4048	0.4098	0.3903	0.3704	6.2705	4.5340	2.2638	1.1048
GN $\rightarrow$ QN $\downarrow$	0.3038	0.3014	0.2942	0.2791	3.4610	2.4235	1.2929	0.6470
RN $\rightarrow$ QN $\uparrow$	0.2185	0.2453	0.3415	0.5162	6.2701	4.5339	2.2639	1.1049
RN $\rightarrow$ QN $\downarrow$	0.2210	0.2157	0.2181	0.2115	3.4593	2.4227	1.2927	0.6470

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ GN	0.0977	0.0898	0.0859	0.0781
	setup $\downarrow$ $\rightarrow$ GN	0.1406	0.1328	0.1055	0.0859
	hold $\uparrow$ $\rightarrow$ GN	-0.0898	-0.0781	-0.0742	-0.0703
	hold $\downarrow$ $\rightarrow$ GN	-0.1211	-0.1172	-0.0898	-0.0742
GN	minpwl	0.1868	0.1820	0.1626	0.1432
RN	minpwl	0.1577	0.1771	0.2402	0.3566
	recovery	0.0938	0.0820	0.0742	0.0664
	removal	-0.0898	-0.0820	-0.0703	-0.0625

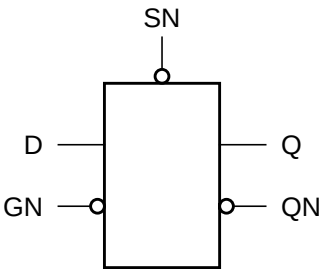
Cell Description

The TLATNS cell is an active-low D-type transparent latch with asynchronous active-low set (SN). When the enable (GN) is low, data is transferred to the outputs (Q, QN).

Function

SN	GN	D	Q[n+1]	QN[n+1]
1	0	0	0	1
1	0	1	1	0
1	1	x	Q[n]	QN[n]
0	x	x	1	0

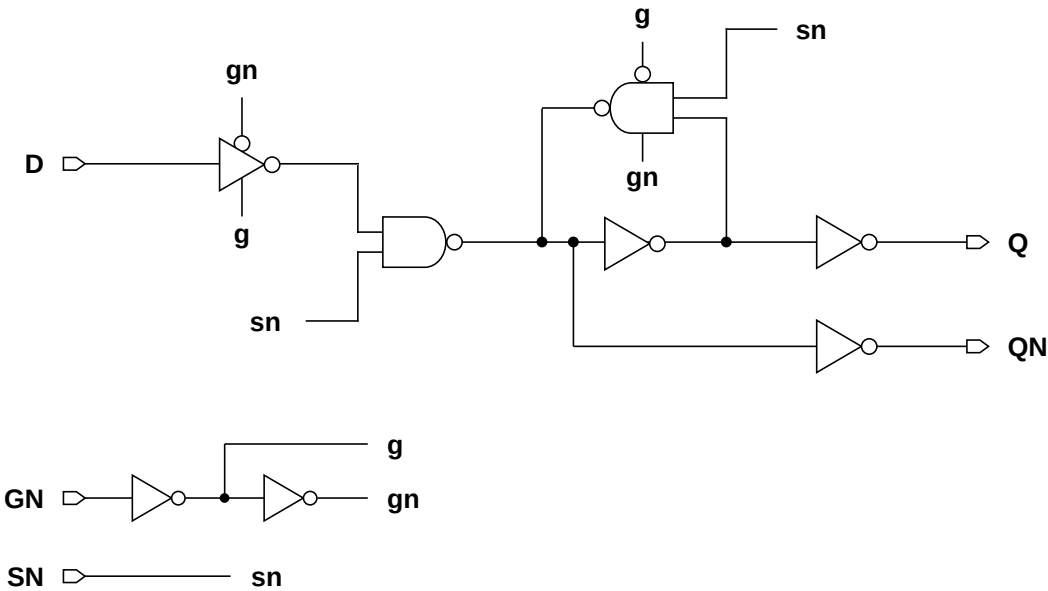
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATNSXL	5.04	10.56
TLATNSX1	5.04	10.56
TLATNSX2	5.04	11.22
TLATNSX4	5.04	13.86

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0039	0.0047	0.0073	0.0135
GN	0.0264	0.0298	0.0352	0.0514
SN	0.0143	0.0146	0.0157	0.0209
Q	0.0502	0.0563	0.0871	0.1361

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0027	0.0032	0.0050	0.0093
GN	0.0021	0.0028	0.0028	0.0040
SN	0.0020	0.0020	0.0026	0.0038

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.1481	0.1335	0.1189	0.1111	6.3057	4.5491	2.2715	1.1093
D $\rightarrow$ Q $\downarrow$	0.3148	0.3062	0.2860	0.2664	3.7161	2.6999	1.3704	0.6812
GN $\rightarrow$ Q $\uparrow$	0.2377	0.1967	0.1838	0.1638	6.3037	4.5508	2.2740	1.1106
GN $\rightarrow$ Q $\downarrow$	0.4084	0.3817	0.3682	0.3262	3.7154	2.6999	1.3704	0.6812
SN $\rightarrow$ Q $\uparrow$	0.2161	0.2312	0.2638	0.3572	6.2696	4.5379	2.2744	1.1187
SN $\rightarrow$ Q $\downarrow$	0.3490	0.3402	0.3106	0.2862	3.7163	2.6999	1.3705	0.6812
D $\rightarrow$ QN $\uparrow$	0.4020	0.3863	0.3589	0.3399	6.2637	4.5324	2.2645	1.1049
D $\rightarrow$ QN $\downarrow$	0.2516	0.2489	0.2298	0.2173	3.2024	2.4352	1.2960	0.6469
GN $\rightarrow$ QN $\uparrow$	0.4963	0.4625	0.4416	0.3999	6.2655	4.5333	2.2648	1.1050
GN $\rightarrow$ QN $\downarrow$	0.3430	0.3143	0.2971	0.2719	3.2029	2.4353	1.2960	0.6469
SN $\rightarrow$ QN $\uparrow$	0.4363	0.4205	0.3836	0.3597	6.2647	4.5329	2.2647	1.1049
SN $\rightarrow$ QN $\downarrow$	0.3188	0.3463	0.3797	0.4786	3.1976	2.4343	1.2964	0.6481

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ GN	0.1211	0.1094	0.0898	0.0859
	setup $\downarrow$ $\rightarrow$ GN	0.2305	0.2109	0.1680	0.1562
	hold $\uparrow$ $\rightarrow$ GN	-0.1133	-0.1055	-0.0820	-0.0781
	hold $\downarrow$ $\rightarrow$ GN	-0.1797	-0.1641	-0.1367	-0.1328
GN	minpwl	0.2159	0.1820	0.1626	0.1432
SN	minpwl	0.1820	0.1917	0.2062	0.2645
	recovery	0.2578	0.2383	0.1914	0.1719
	removal	-0.2578	-0.2383	-0.1875	-0.1719

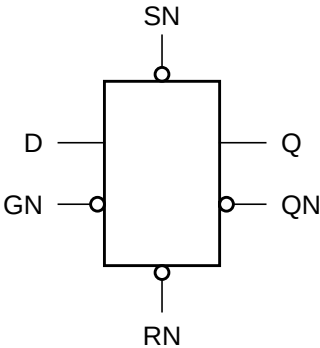
Cell Description

The TLATNSR cell is an active-low D-type transparent latch with asynchronous active-low set (SN) and reset (RN), and set dominating reset. When the enable (GN) is low, data is transferred to the outputs (Q, QN).

Functions

RN	SN	GN	D	Q[n+1]	QN[n+1]
1	1	0	0	0	1
1	1	0	1	1	0
1	1	1	x	Q[n]	QN[n]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0

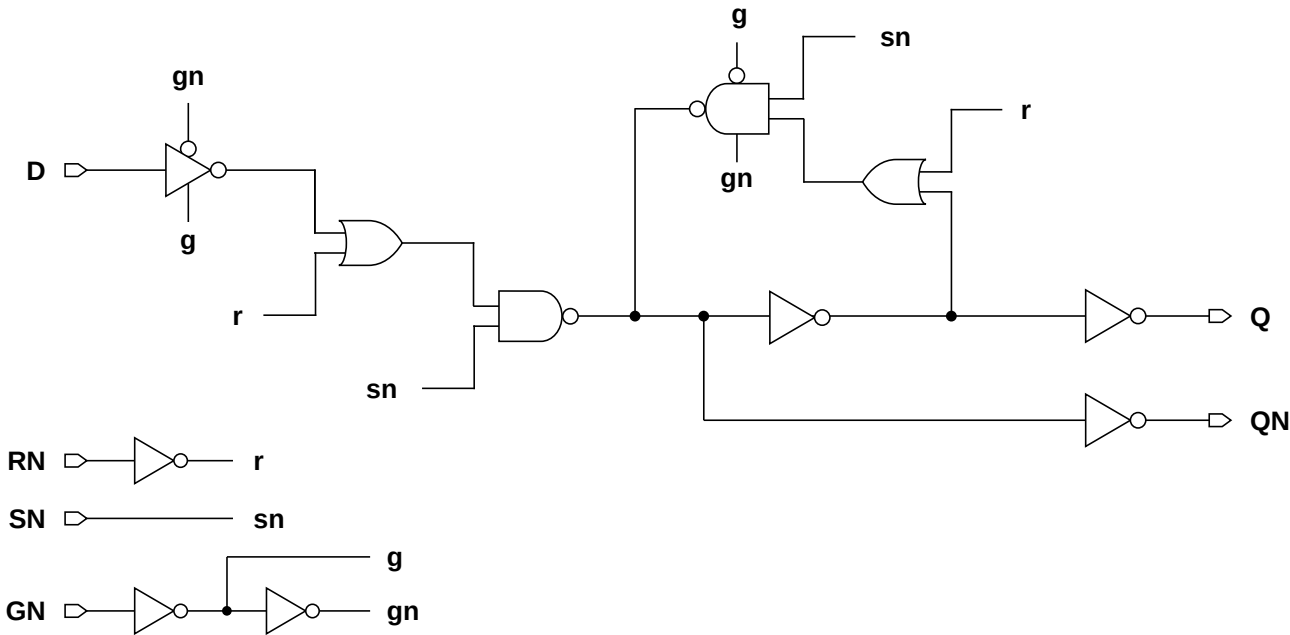
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATNSRXL	5.04	11.22
TLATNSRX1	5.04	11.22
TLATNSRX2	5.04	11.88
TLATNSRX4	5.04	16.50

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0037	0.0044	0.0073	0.0151
GN	0.0275	0.0312	0.0384	0.0683
SN	0.0156	0.0176	0.0232	0.0417
RN	0.0050	0.0057	0.0091	0.0171
Q	0.0549	0.0631	0.0969	0.1619

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0029	0.0034	0.0053	0.0113
GN	0.0020	0.0027	0.0030	0.0042
SN	0.0020	0.0026	0.0040	0.0068
RN	0.0040	0.0045	0.0062	0.0112

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.1853	0.1704	0.1745	0.1588	6.3438	4.5640	2.2812	1.1129
D $\rightarrow$ Q $\downarrow$	0.3638	0.3562	0.3399	0.3158	4.1771	2.8015	1.4261	0.7067
GN $\rightarrow$ Q $\uparrow$	0.2657	0.2223	0.2277	0.2039	6.3418	4.5637	2.2814	1.1133
GN $\rightarrow$ Q $\downarrow$	0.4443	0.4166	0.4078	0.3777	4.1786	2.8013	1.4261	0.7068
SN $\rightarrow$ Q $\uparrow$	0.2484	0.2255	0.1916	0.1847	6.2883	4.5443	2.2697	1.1080
SN $\rightarrow$ Q $\downarrow$	0.3914	0.3774	0.3558	0.3352	4.1573	2.7924	1.4214	0.7043
RN $\rightarrow$ Q $\uparrow$	0.1804	0.1656	0.1699	0.1543	6.3439	4.5639	2.2812	1.1130
RN $\rightarrow$ Q $\downarrow$	0.2651	0.2544	0.2394	0.2189	4.0686	2.7689	1.4097	0.7005
D $\rightarrow$ QN $\uparrow$	0.4516	0.4420	0.4130	0.3882	6.2654	4.5327	2.2644	1.1050
D $\rightarrow$ QN $\downarrow$	0.2954	0.2914	0.2869	0.2679	3.4825	2.4381	1.2952	0.6473
GN $\rightarrow$ QN $\uparrow$	0.5329	0.5032	0.4815	0.4504	6.2672	4.5335	2.2647	1.1050
GN $\rightarrow$ QN $\downarrow$	0.3774	0.3451	0.3414	0.3141	3.4848	2.4391	1.2954	0.6473
SN $\rightarrow$ QN $\uparrow$	0.4786	0.4628	0.4288	0.4073	6.2670	4.5334	2.2647	1.1050
SN $\rightarrow$ QN $\downarrow$	0.3580	0.3450	0.3015	0.2915	3.4750	2.4361	1.2942	0.6469
RN $\rightarrow$ QN $\uparrow$	0.3491	0.3349	0.3083	0.2859	6.2659	4.5342	2.2650	1.1052
RN $\rightarrow$ QN $\downarrow$	0.2910	0.2872	0.2827	0.2636	3.4833	2.4384	1.2952	0.6472

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ → GN	0.1562	0.1484	0.1406	0.1289
	setup $\downarrow$ → GN	0.2891	0.2734	0.2305	0.1953
	hold $\uparrow$ → GN	-0.1445	-0.1367	-0.1328	-0.1211
	hold $\downarrow$ → GN	-0.2383	-0.2266	-0.1953	-0.1719
GN	minpwl	0.2402	0.2014	0.1965	0.1771
SN	minpwl	0.2111	0.1917	0.1529	0.1432
	recovery	0.3047	0.2852	0.2383	0.2070
	removal	-0.3086	-0.2891	-0.2422	-0.2109
RN	minpwl	0.2693	0.2450	0.2014	0.1674
	recovery	0.1523	0.1406	0.1367	0.1211
	removal	-0.1484	-0.1406	-0.1328	-0.1211

Logic Symbol

Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATRXL	5.04	8.58
TLATRX1	5.04	8.58
TLATRX2	5.04	9.24
TLATRX4	5.04	11.88

The diagram shows a 1-bit D flip-flop circuit. It includes three inputs: **D** (data), **G** (enable), and **RN** (reset, active-low). The circuit uses several logic gates: a 2-input NAND gate, a 3-input NAND gate, and two inverters. The **D** input is connected to a 2-input NAND gate along with an enable signal **gn**. The output of this NAND gate is connected to another 2-input NAND gate, which also receives the **r** (reset) signal. The output of this second NAND gate is connected to a 3-input NAND gate, which also receives **gn** and the **r** signal. The output of the 3-input NAND gate is connected to an inverter, which produces the **Q** output. The **Q** output is also connected to another inverter, which produces the **QN** (complement of Q) output. The **G** input is connected to an inverter, and its output is connected to the **gn** input of the 3-input NAND gate. The **RN** input is connected to an inverter, and its output is connected to the **r** input of the 2-input NAND gate and the 3-input NAND gate.

AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0033	0.0041	0.0069	0.0140
G	0.0274	0.0279	0.0325	0.0554
RN	0.0044	0.0044	0.0056	0.0072
Q	0.0360	0.0440	0.0736	0.1257

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0026	0.0031	0.0053	0.0103
G	0.0021	0.0021	0.0025	0.0040
RN	0.0040	0.0041	0.0053	0.0072

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.1380	0.1293	0.1214	0.1146	6.2972	4.5451	2.2726	1.1092
D $\rightarrow$ Q $\downarrow$	0.2357	0.2295	0.2177	0.1994	3.6774	2.5077	1.3254	0.6600
G $\rightarrow$ Q $\uparrow$	0.2513	0.2388	0.2419	0.2318	6.2954	4.5446	2.2722	1.1091
G $\rightarrow$ Q $\downarrow$	0.2383	0.2348	0.2259	0.2018	3.6742	2.5064	1.3261	0.6610
RN $\rightarrow$ Q $\uparrow$	0.1347	0.1246	0.1146	0.1062	6.2971	4.5452	2.2725	1.1092
RN $\rightarrow$ Q $\downarrow$	0.1578	0.1773	0.2713	0.4331	3.5599	2.5013	1.3973	0.7695
D $\rightarrow$ QN $\uparrow$	0.2977	0.2989	0.2892	0.2721	6.2690	4.5336	2.2638	1.1047
D $\rightarrow$ QN $\downarrow$	0.2253	0.2207	0.2265	0.2217	3.4602	2.4224	1.2929	0.6470
G $\rightarrow$ QN $\uparrow$	0.3014	0.3053	0.2985	0.2759	6.2700	4.5341	2.2639	1.1048
G $\rightarrow$ QN $\downarrow$	0.3401	0.3315	0.3482	0.3397	3.4623	2.4233	1.2931	0.6471
RN $\rightarrow$ QN $\uparrow$	0.2195	0.2451	0.3433	0.5169	6.2689	4.5338	2.2639	1.1049
RN $\rightarrow$ QN $\downarrow$	0.2225	0.2164	0.2201	0.2134	3.4610	2.4227	1.2929	0.6470

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow \rightarrow$ G	0.0156	0.0195	-0.0039	-0.0078
	setup $\downarrow \rightarrow$ G	0.1836	0.1797	0.1602	0.1445
	hold $\uparrow \rightarrow$ G	0.0000	0.0078	0.0195	0.0234
	hold $\downarrow \rightarrow$ G	-0.1797	-0.1758	-0.1562	-0.1367
G	minpwh	0.1432	0.1335	0.1335	0.1286
RN	minpwl	0.1577	0.1771	0.2402	0.3615
	recovery	0.0117	0.0000	-0.0156	-0.0234
	removal	-0.0078	0.0000	0.0195	0.0273

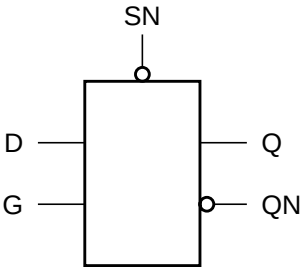
Cell Description

The TLATS cell is an active-high D-type transparent latch with asynchronous active-low set (SN). When the enable (G) is high, data is transferred to the outputs (Q, QN).

Function

SN	G	D	Q[n+1]	QN[n+1]
1	1	0	0	1
1	1	1	1	0
1	0	x	Q[n]	QN[n]
0	x	x	1	0

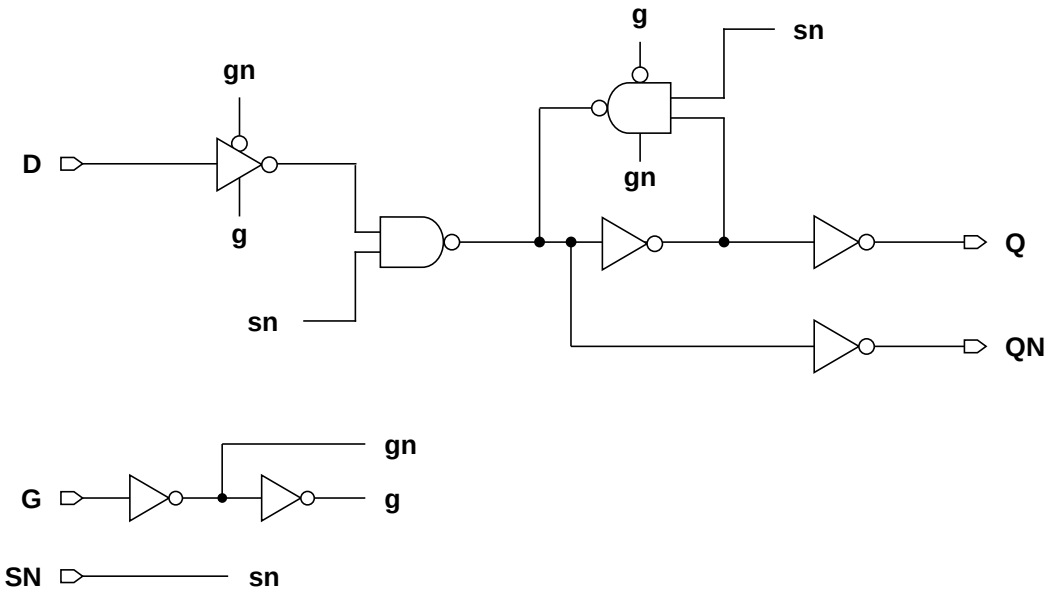
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATSXL	5.04	10.56
TLATSX1	5.04	10.56
TLATSX2	5.04	11.22
TLATSX4	5.04	13.86

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0039	0.0047	0.0075	0.0135
G	0.0289	0.0294	0.0319	0.0427
SN	0.0143	0.0145	0.0157	0.0208
Q	0.0440	0.0497	0.0791	0.1276

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0027	0.0032	0.0052	0.0093
G	0.0022	0.0028	0.0028	0.0041
SN	0.0020	0.0020	0.0026	0.0038

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K _{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.1485	0.1344	0.1209	0.1128	6.3071	4.5499	2.2717	1.1094
D $\rightarrow$ Q $\downarrow$	0.3128	0.3051	0.2849	0.2649	3.7099	2.7784	1.3689	0.6810
G $\rightarrow$ Q $\uparrow$	0.2750	0.2552	0.2416	0.2161	6.3051	4.5491	2.2711	1.1091
G $\rightarrow$ Q $\downarrow$	0.2846	0.2695	0.2584	0.2380	3.7075	2.7772	1.3681	0.6807
SN $\rightarrow$ Q $\uparrow$	0.2159	0.2294	0.2627	0.3568	6.2700	4.5389	2.2744	1.1188
SN $\rightarrow$ Q $\downarrow$	0.3471	0.3386	0.3085	0.2852	3.7099	2.7784	1.3690	0.6811
D $\rightarrow$ QN $\uparrow$	0.4006	0.3840	0.3562	0.3386	6.2640	4.5325	2.2643	1.1049
D $\rightarrow$ QN $\downarrow$	0.2527	0.2505	0.2300	0.2192	3.2027	2.4348	1.2954	0.6469
G $\rightarrow$ QN $\uparrow$	0.3729	0.3489	0.3300	0.3119	6.2662	4.5334	2.2647	1.1050
G $\rightarrow$ QN $\downarrow$	0.3805	0.3728	0.3517	0.3231	3.2029	2.4349	1.2953	0.6469
SN $\rightarrow$ QN $\uparrow$	0.4350	0.4176	0.3798	0.3591	6.2650	4.5330	2.2645	1.1049
SN $\rightarrow$ QN $\downarrow$	0.3196	0.3452	0.3767	0.4784	3.1976	2.4340	1.2958	0.6482

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ G	0.0039	0.0156	-0.0156	-0.0156
	setup $\downarrow$ $\rightarrow$ G	0.2578	0.2500	0.2109	0.1953
	hold $\uparrow$ $\rightarrow$ G	0.0039	-0.0078	0.0273	0.0234
	hold $\downarrow$ $\rightarrow$ G	-0.2500	-0.2422	-0.2031	-0.1836
G	minpwh	0.1432	0.1480	0.1238	0.1092
SN	minpwl	0.1820	0.1917	0.2062	0.2645
	recovery	0.2930	0.2852	0.2344	0.2148
	removal	-0.2891	-0.2812	-0.2305	-0.2109

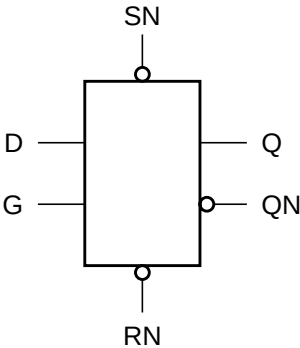
Cell Description

The TLATSR cell is an active-high D-type transparent latch with asynchronous active-low set (SN) and reset (RN), and set dominating reset. When the enable (G) is high, data is transferred to the outputs (Q, QN).

Functions

RN	SN	G	D	Q[n+1]	QN[n+1]
1	1	1	0	0	1
1	1	1	1	1	0
1	1	0	x	Q[n]	QN[n]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0

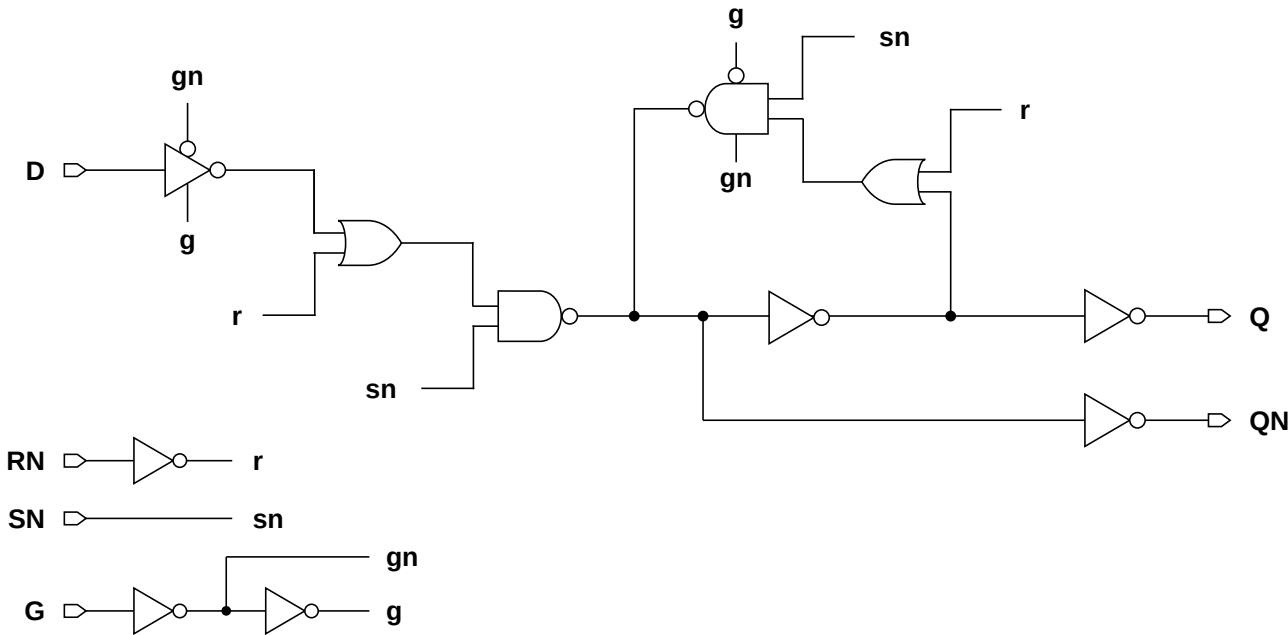
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATSRXL	5.04	11.22
TLATSRX1	5.04	11.22
TLATSRX2	5.04	11.88
TLATSRX4	5.04	16.50

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0037	0.0044	0.0073	0.0150
G	0.0303	0.0309	0.0347	0.0573
SN	0.0157	0.0176	0.0233	0.0419
RN	0.0047	0.0055	0.0089	0.0171
Q	0.0509	0.0588	0.0911	0.1542

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0029	0.0034	0.0053	0.0113
G	0.0020	0.0026	0.0030	0.0042
SN	0.0020	0.0026	0.0040	0.0069
RN	0.0039	0.0045	0.0062	0.0112

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.1857	0.1727	0.1753	0.1601	6.3439	4.5646	2.2813	1.1131
D $\rightarrow$ Q $\downarrow$	0.3657	0.3592	0.3396	0.3145	4.1819	2.8069	1.4260	0.7065
G $\rightarrow$ Q $\uparrow$	0.2981	0.2761	0.2792	0.2523	6.3419	4.5638	2.2809	1.1128
G $\rightarrow$ Q $\downarrow$	0.3575	0.3397	0.3206	0.2903	4.1842	2.8070	1.4258	0.7065
SN $\rightarrow$ Q $\uparrow$	0.2490	0.2266	0.1917	0.1823	6.2889	4.5450	2.2698	1.1081
SN $\rightarrow$ Q $\downarrow$	0.3932	0.3801	0.3556	0.3335	4.1623	2.7979	1.4213	0.7041
RN $\rightarrow$ Q $\uparrow$	0.1806	0.1676	0.1705	0.1555	6.3439	4.5646	2.2813	1.1130
RN $\rightarrow$ Q $\downarrow$	0.2660	0.2568	0.2395	0.2186	4.0738	2.7731	1.4096	0.7004
D $\rightarrow$ QN $\uparrow$	0.4533	0.4447	0.4128	0.3870	6.2654	4.5326	2.2644	1.1049
D $\rightarrow$ QN $\downarrow$	0.2956	0.2937	0.2879	0.2693	3.4821	2.4384	1.2951	0.6473
G $\rightarrow$ QN $\uparrow$	0.4461	0.4262	0.3945	0.3632	6.2674	4.5336	2.2648	1.1051
G $\rightarrow$ QN $\downarrow$	0.4096	0.3986	0.3928	0.3622	3.4845	2.4397	1.2953	0.6474
SN $\rightarrow$ QN $\uparrow$	0.4802	0.4653	0.4287	0.4058	6.2668	4.5333	2.2647	1.1050
SN $\rightarrow$ QN $\downarrow$	0.3582	0.3461	0.3017	0.2892	3.4748	2.4363	1.2941	0.6468
RN $\rightarrow$ QN $\uparrow$	0.3500	0.3370	0.3084	0.2858	6.2659	4.5340	2.2650	1.1052
RN $\rightarrow$ QN $\downarrow$	0.2910	0.2892	0.2834	0.2649	3.4830	2.4389	1.2951	0.6473

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup↑ → G	0.0547	0.0664	0.0469	0.0273
	setup↓ → G	0.3047	0.3047	0.2656	0.2422
	hold↑ → G	-0.0273	-0.0430	-0.0234	-0.0117
	hold↓ → G	-0.2930	-0.2891	-0.2539	-0.2305
G	minpwh	0.1771	0.1771	0.1577	0.1335
SN	minpwl	0.2159	0.1917	0.1529	0.1383
	recovery	0.3281	0.3203	0.2773	0.2539
	removal	-0.3281	-0.3203	-0.2773	-0.2578
RN	minpwl	0.2693	0.2450	0.2014	0.1674
	recovery	0.0430	0.0586	0.0312	0.0156
	removal	-0.0430	-0.0547	-0.0312	-0.0117

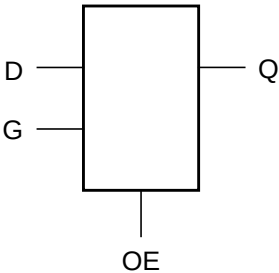
Cell Description

The TTLAT cell is an active-high D-type transparent latch with active-high output enable (OE). When the enable (G) is high and the output enable (OE) is high, data is transferred to the output (Q).

Function

OE	G	D	Q[n+1]
0	x	x	Z
1	1	0	0
1	1	1	1
1	0	x	Q[n]

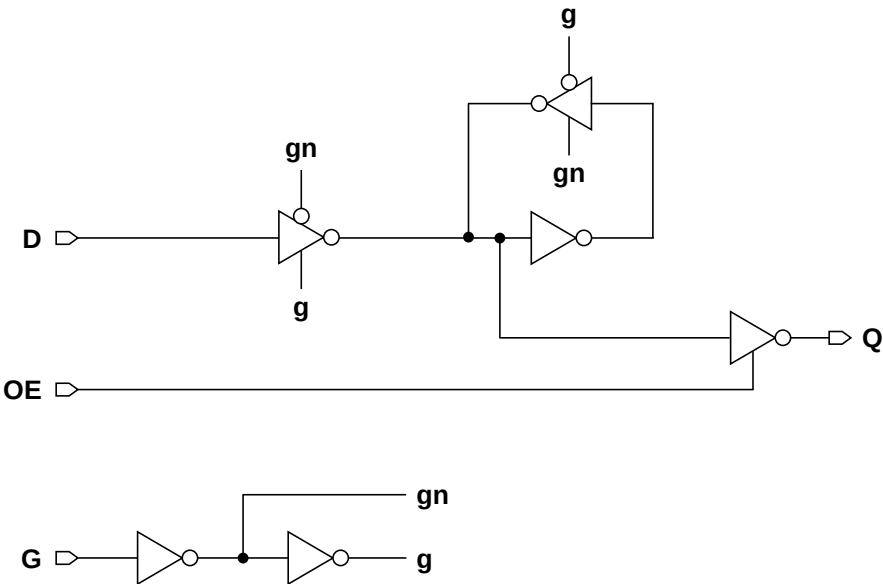
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TTLATXL	5.04	8.58
TTLATX1	5.04	8.58
TTLATX2	5.04	12.54
TTLATX4	5.04	15.18

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0048	0.0091	0.0168	0.0169
G	0.0250	0.0265	0.0428	0.0544
OE	0.0372	0.0661	0.1202	0.1890
Q	0.0358	0.0622	0.1097	0.1710

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0035	0.0062	0.0116	0.0116
G	0.0022	0.0028	0.0042	0.0070
OE	0.0027	0.0035	0.0060	0.0104
Q	0.0020	0.0047	0.0059	0.0110

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.1257	0.1170	0.1189	0.1286	8.6474	2.7495	1.4437	0.7395
D $\rightarrow$ Q $\downarrow$	0.1921	0.1686	0.1668	0.2016	5.5012	1.7302	0.8600	0.4845
G $\rightarrow$ Q $\uparrow$	0.2356	0.2245	0.2218	0.2012	8.6531	2.7504	1.4439	0.7397
G $\rightarrow$ Q $\downarrow$	0.2086	0.1806	0.1771	0.2053	5.4959	1.7293	0.8595	0.4842
OE $\rightarrow$ Q $\uparrow$	0.0334	0.0282	0.0214	0.0262	8.5650	2.6962	1.4179	0.7279
OE $\rightarrow$ Q $\downarrow$	0.0198	0.0184	0.0140	0.0165	5.4139	1.6870	0.8370	0.4693

Timing Constraints at 25°C, 1.8V, Typical Process

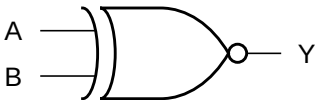
Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ G	-0.0312	-0.0273	-0.0312	0.0039
	setup $\downarrow$ $\rightarrow$ G	0.1094	0.0898	0.0898	0.1328
	hold $\uparrow$ $\rightarrow$ G	0.0430	0.0391	0.0430	0.0078
	hold $\downarrow$ $\rightarrow$ G	-0.0938	-0.0742	-0.0742	-0.1133
G	minpwh	0.1043	0.0995	0.0946	0.0898

Cell Description

The XNOR2 cell provides a logical EXCLUSIVE NOR of two inputs (A, B). The output (Y) is represented by the logic equation:

$$Y = (A \bullet B) + (\bar{A} \bullet \bar{B})$$

Logic Symbol



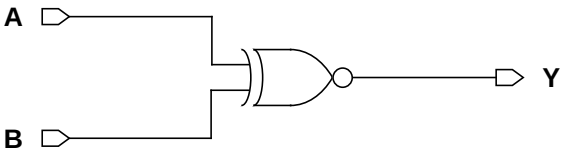
Functions

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
XNOR2XL	5.04	5.28
XNOR2X1	5.04	5.28
XNOR2X2	5.04	7.26
XNOR2X4	5.04	11.22

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A	0.0386	0.0406	0.0687	0.1258
B	0.0364	0.0536	0.1022	0.1921

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0062	0.0058	0.0091	0.0159
B	0.0021	0.0066	0.0144	0.0271

Delays at 25°C, 1.8V, Typical Process

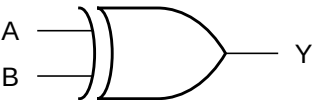
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.1453	0.1394	0.1460	0.1306	6.2740	4.5331	2.2642	1.2088
A $\rightarrow$ Y $\downarrow$	0.1383	0.1208	0.1153	0.1148	3.6328	2.4397	1.2971	0.6078
B $\rightarrow$ Y $\uparrow$	0.2080	0.1334	0.1220	0.1204	6.2786	4.5342	2.2645	1.2089
B $\rightarrow$ Y $\downarrow$	0.2188	0.1525	0.1445	0.1411	3.6182	2.4482	1.3014	0.6104

Cell Description

The XOR2 cell provides a logical EXCLUSIVE OR of two inputs (A, B). The output (Y) is represented by the logic equation:

$$Y = (A \bullet \bar{B}) + (\bar{A} \bullet B)$$

Logic Symbol



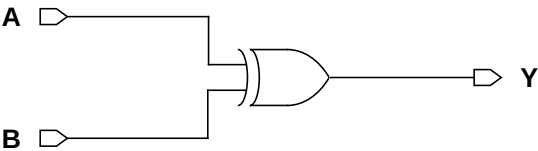
Functions

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
XOR2XL	5.04	5.28
XOR2X1	5.04	5.28
XOR2X2	5.04	6.60
XOR2X4	5.04	11.22

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A	0.0372	0.0378	0.0646	0.1137
B	0.0343	0.0500	0.0980	0.1808

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0062	0.0058	0.0090	0.0167
B	0.0022	0.0065	0.0145	0.0271

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.1447	0.1295	0.1351	0.1163	6.2769	4.5296	2.2626	1.2077
A $\rightarrow$ Y $\downarrow$	0.1419	0.1255	0.1294	0.1199	3.3230	2.4479	1.3006	0.6092
B $\rightarrow$ Y $\uparrow$	0.2013	0.1311	0.1209	0.1149	6.2751	4.5334	2.2644	1.2085
B $\rightarrow$ Y $\downarrow$	0.2081	0.1531	0.1475	0.1432	3.3235	2.4482	1.3007	0.6093

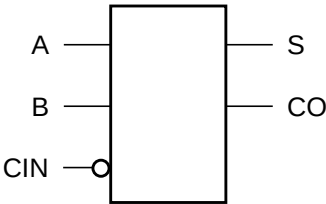
Synthesis Optimized Arithmetic Cells

Cell Description

The AFHCIN cell is a full adder that provides the arithmetic sum (S) and carry-out (CO) of two operands (A, B) with active-low carry-in (CIN). The outputs (S, CO) are represented by the logic equations:

$$S = A \oplus B \oplus \overline{CIN}$$
$$CO = (A \bullet B) + (A \bullet \overline{CIN}) + (B \bullet \overline{CIN})$$

Logic Symbol



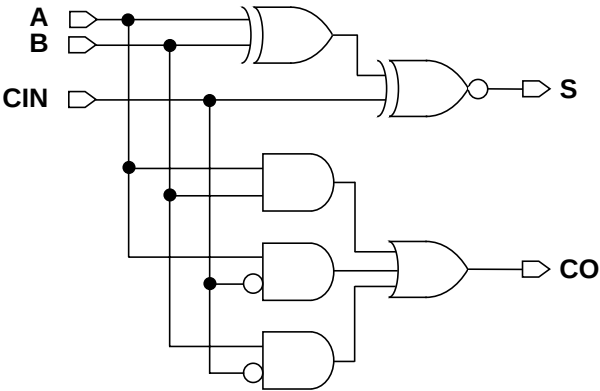
Functions

A	B	CIN	S	CO
0	0	0	1	0
0	0	1	0	0
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
AFHCINX2	5.04	16.50
AFHCINX4	5.04	18.48

Functional Schematic



AC Power

Pin	Power (μ W/MHz)	
	X2	X4
A	0.1714	0.1933
B	0.1796	0.1989
CIN	0.1234	0.1800

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0072	0.0072
B	0.0176	0.0155
CIN	0.0135	0.0265

Delays at 25°C, 1.8V, Typical Process

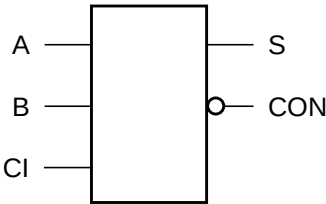
Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ S $\uparrow$	0.2451	0.2610	2.8765	2.2610
A $\rightarrow$ S $\downarrow$	0.2915	0.3139	1.4059	1.2811
B $\rightarrow$ S $\uparrow$	0.2199	0.2363	2.8774	2.2613
B $\rightarrow$ S $\downarrow$	0.2200	0.2404	1.4049	1.2808
CIN $\rightarrow$ S $\uparrow$	0.1401	0.1242	2.8765	2.2607
CIN $\rightarrow$ S $\downarrow$	0.1559	0.1569	1.4102	1.2826
A $\rightarrow$ CO $\uparrow$	0.1742	0.2004	3.0287	3.0215
A $\rightarrow$ CO $\downarrow$	0.2138	0.2498	1.9011	1.8756
B $\rightarrow$ CO $\uparrow$	0.1320	0.1533	3.0092	3.0134
B $\rightarrow$ CO $\downarrow$	0.1809	0.2140	1.8813	1.8676
CIN $\rightarrow$ CO $\uparrow$	0.0517	0.0458	2.7849	1.3993
CIN $\rightarrow$ CO $\downarrow$	0.0422	0.0415	1.6622	0.8366

Cell Description

The AFHCON cell is a full adder that provides the arithmetic sum (S) and active-low carry-out (CON) of two operands (A, B) with carry-in (CI). The outputs (S, CON) are represented by the logic equations:

$$S = A \oplus B \oplus CI$$
$$CON = \overline{(A \bullet B) + (A \bullet CI) + (B \bullet CI)}$$

Logic Symbol



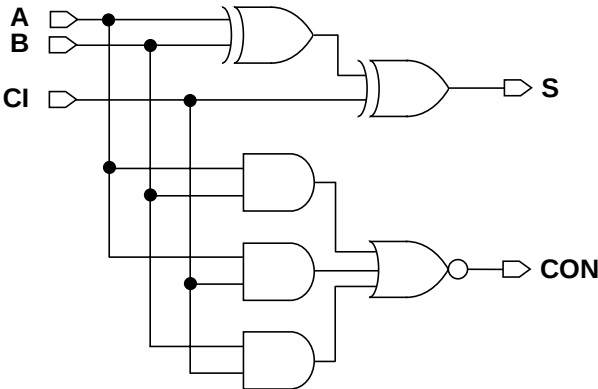
Functions

A	B	CI	S	CON
0	0	0	0	1
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
AFHCONX2	5.04	15.84
AFHCONX4	5.04	17.16

Functional Schematic



AC Power

Pin	Power (μ W/MHz)	
	X2	X4
A	0.1822	0.2036
B	0.1658	0.1854
Cl	0.1194	0.1721

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0071	0.0072
B	0.0213	0.0224
Cl	0.0135	0.0270

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ S $\uparrow$	0.2370	0.2527	2.8757	2.8727
A $\rightarrow$ S $\downarrow$	0.2749	0.2901	1.6105	1.5991
B $\rightarrow$ S $\uparrow$	0.1679	0.1750	2.8761	2.8721
B $\rightarrow$ S $\downarrow$	0.1906	0.2085	1.6115	1.5989
Cl $\rightarrow$ S $\uparrow$	0.1352	0.1219	2.8742	2.8725
Cl $\rightarrow$ S $\downarrow$	0.1504	0.1567	1.6127	1.5995
A $\rightarrow$ CON $\uparrow$	0.2252	0.2493	3.1288	2.9903
A $\rightarrow$ CON $\downarrow$	0.1871	0.2194	1.9358	1.8565
B $\rightarrow$ CON $\uparrow$	0.1505	0.1723	3.1689	3.0104
B $\rightarrow$ CON $\downarrow$	0.1181	0.1466	1.9524	1.8665
Cl $\rightarrow$ CON $\uparrow$	0.0670	0.0519	2.7784	1.3659
Cl $\rightarrow$ CON $\downarrow$	0.0455	0.0389	1.6576	0.8300

Cell Description

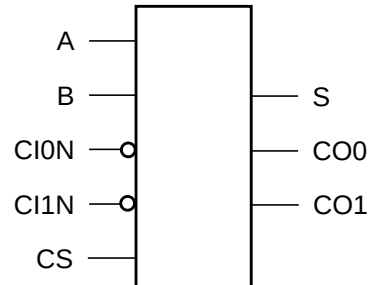
The AFCSHCIN cell provides a carry-select adder function that produces the arithmetic sum (S) and carry-outs (CO0, CO1) of the operands (A,B) with active-low carry-ins (CI0N, CI1N). The three outputs (S, CO0, CO1) are represented by the logic equations:

$$S = CS \bullet (A \oplus B \oplus \overline{CI1N}) + \overline{CS} \bullet (A \oplus B \oplus \overline{CI0N})$$

$$CO0 = (A \bullet B) + (A \bullet \overline{CI0N}) + (B \bullet \overline{CI0N})$$

$$CO1 = (A \bullet B) + (A \bullet \overline{CI1N}) + (B \bullet \overline{CI1N})$$

Logic Symbol



Cell Size

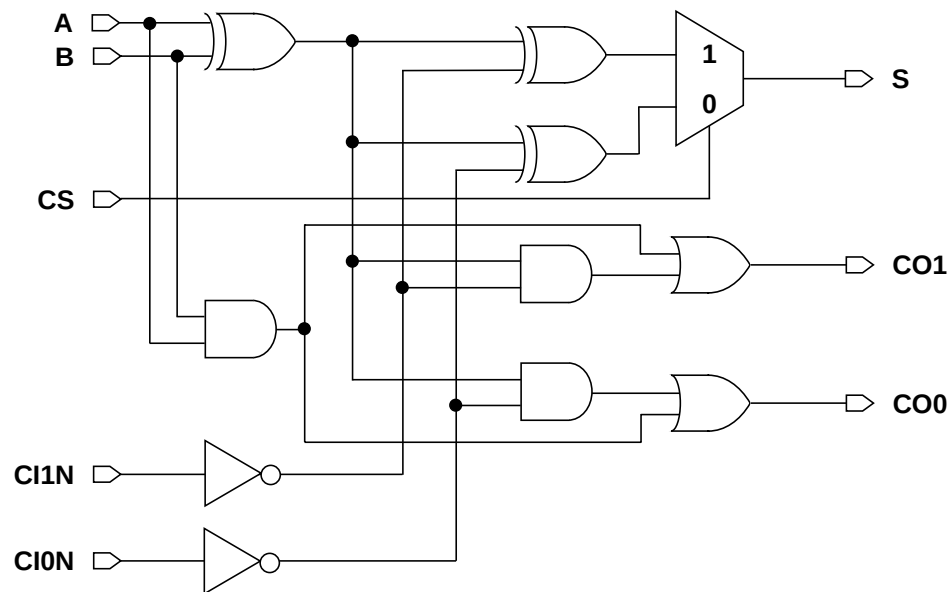
Drive Strength	Height (μm)	Width (μm)
AFCSHCINX2	5.04	33.00
AFCSHCINX4	5.04	38.28

Functions

A	B	CI0N	CI1N	CS	S	CO0	CO1
0	0	0	0	0	1	0	0
0	0	0	0	1	1	0	0
0	0	0	1	0	1	0	0
0	0	0	1	1	0	0	0
0	0	1	0	0	0	0	0
0	0	1	0	1	1	0	0
0	0	1	1	0	0	0	0
0	0	1	1	1	0	0	0
0	1	0	0	0	0	1	1
0	1	0	0	1	0	1	1
0	1	0	1	0	0	1	0
0	1	0	1	1	1	1	0
0	1	1	0	0	1	0	1
0	1	1	0	1	0	0	1
0	1	1	1	0	1	0	0
0	1	1	1	1	1	0	0

A	B	CI0N	CI1N	CS	S	CO0	CO1
1	0	0	0	0	0	1	1
1	0	0	0	1	0	1	1
1	0	0	1	0	0	1	0
1	0	0	1	1	1	1	0
1	0	1	0	0	1	0	1
1	0	1	0	1	0	0	1
1	0	1	1	0	1	0	0
1	0	1	1	1	1	0	0
1	1	0	0	0	1	1	1
1	1	0	0	1	1	1	1
1	1	0	1	0	1	1	1
1	1	0	1	1	0	1	1
1	1	1	0	0	0	1	1
1	1	1	0	1	1	1	1
1	1	1	1	0	0	1	1
1	1	1	1	1	0	1	1

Functional Schematic



AC Power

Pin	Power (μW/MHz)	
	X2	X4
CS	0.0666	0.0666
A	0.3490	0.3860
B	0.3132	0.3444
CI0N	0.1313	0.1646
CI1N	0.1322	0.1636

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
CS	0.0056	0.0056
A	0.0075	0.0075
B	0.0170	0.0170
CI0N	0.0086	0.0146
CI1N	0.0085	0.0144

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)		K _{load} (ns/pF)	
	X2	X4	X2	X4
CS → S↑	0.1470	0.1483	2.2650	2.2610
CS → S↓	0.1315	0.1348	1.3032	1.2816
A → S↑	0.4544	0.4847	2.2659	2.2614
A → S↓	0.4232	0.4468	1.3060	1.2826
B → S↑	0.3717	0.3968	2.2661	2.2614
B → S↓	0.3640	0.3911	1.3064	1.2829
CI0N → S↑	0.2843	0.2908	2.2661	2.2613
CI0N → S↓	0.3011	0.3066	1.3066	1.2827
CI1N → S↑	0.2546	0.2511	2.2652	2.2610
CI1N → S↓	0.2835	0.2789	1.3035	1.2818
A → CO0↑	0.2180	0.2527	3.1298	3.0687
A → CO0↓	0.2698	0.3247	1.6302	1.5970
B → CO0↑	0.1775	0.2075	3.0468	3.0625
B → CO0↓	0.2049	0.2505	1.6077	1.5889
CI0N → CO0↑	0.0908	0.0692	3.1586	1.5900
CI0N → CO0↓	0.0489	0.0415	1.5710	0.8169
A → CO1↑	0.2143	0.2641	3.0293	2.7563
A → CO1↓	0.2853	0.3370	1.6919	1.6649
B → CO1↑	0.1497	0.1920	3.0474	2.7473
B → CO1↓	0.2046	0.2541	1.6801	1.6566
CI1N → CO1↑	0.0861	0.0646	3.0924	1.5432
CI1N → CO1↓	0.0507	0.0405	1.6359	0.8173

Cell Description

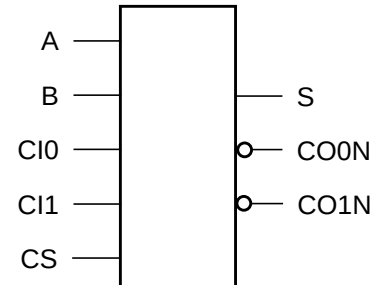
The AFCSHCON cell provides a carry-select adder function that produces the arithmetic sum (S) and active-low carry-outs (CO0N, CO1N) of two operands (A,B) with carry-ins (CI0, CI1). The three outputs (S, CO0N, CO1N) are represented by the logic equations:

$$S = CS \bullet (A \oplus B \oplus CI1) + \overline{CS} \bullet (A \oplus B \oplus CI0)$$

$$CO0N = \overline{(A \bullet B) + (A \bullet CI0) + (B \bullet CI0)}$$

$$CO1N = \overline{(A \bullet B) + (A \bullet CI1) + (B \bullet CI1)}$$

Logic Symbol



Cell Size

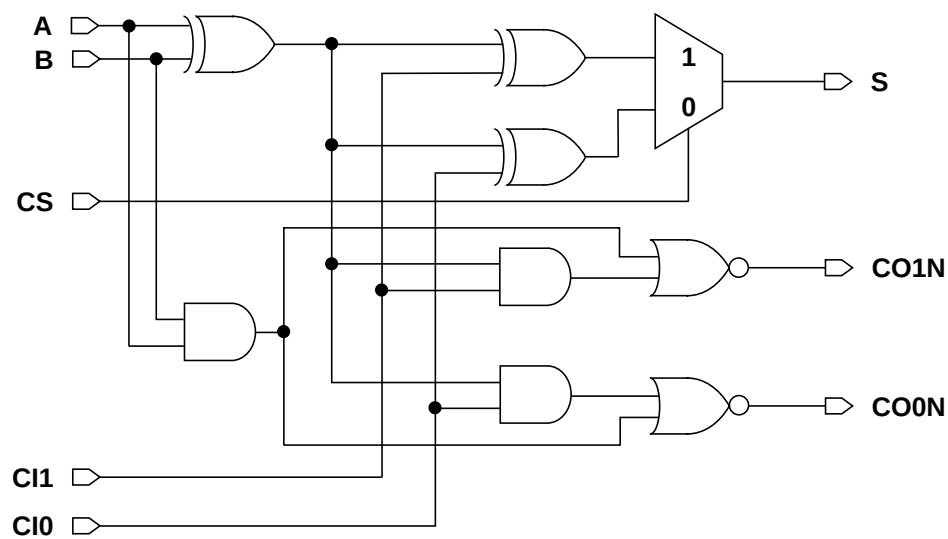
Drive Strength	Height (μm)	Width (μm)
AFCSHCONX2	5.04	33.66
AFCSHCONX4	5.04	38.94

Functions

A	B	CI0	CI1	CS	S	CO0N	CO1N
0	0	0	0	0	0	1	1
0	0	0	0	1	0	1	1
0	0	0	1	0	0	1	1
0	0	0	1	1	1	1	1
0	0	1	0	0	1	1	1
0	0	1	0	1	0	1	1
0	0	1	1	0	1	1	1
0	0	1	1	1	1	1	1
0	1	0	0	0	1	1	1
0	1	0	0	1	1	1	1
0	1	0	1	0	1	1	0
0	1	0	1	1	0	1	0
0	1	1	0	0	0	0	1
0	1	1	0	1	1	0	1
0	1	1	1	0	0	0	0
0	1	1	1	1	0	0	0

A	B	CI0	CI1	CS	S	CO0N	CO1N
1	0	0	0	0	1	1	1
1	0	0	0	1	1	1	1
1	0	0	1	0	1	1	0
1	0	0	1	1	0	1	0
1	0	1	0	0	0	0	1
1	0	1	0	1	1	0	1
1	0	1	1	0	0	0	0
1	0	1	1	1	0	0	0
1	1	0	0	0	0	0	0
1	1	0	0	1	0	0	0
1	1	0	1	0	0	0	0
1	1	0	1	1	1	0	0
1	1	1	0	0	1	0	0
1	1	1	0	1	0	0	0
1	1	1	1	0	1	0	0
1	1	1	1	1	1	0	0

Functional Schematic



AC Power

Pin	Power (μW/MHz)	
	X2	X4
CS	0.0663	0.0664
A	0.3524	0.3942
B	0.3292	0.3722
CI0	0.1636	0.2092
CI1	0.1379	0.1822

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
CS	0.0092	0.0092
A	0.0136	0.0136
B	0.0156	0.0156
CI0	0.0128	0.0256
CI1	0.0125	0.0253

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)		K _{load} (ns/pF)	
	X2	X4	X2	X4
CS → S↑	0.1480	0.1486	2.2651	2.2611
CS → S↓	0.1314	0.1347	1.3028	1.2817
A → S↑	0.4807	0.5261	2.2660	2.2614
A → S↓	0.4308	0.4706	1.3065	1.2832
B → S↑	0.4299	0.4726	2.2660	2.2615
B → S↓	0.3958	0.4206	1.3057	1.2831
CI0 → S↑	0.2776	0.2722	2.2660	2.2615
CI0 → S↓	0.2777	0.2643	1.3058	1.2829
CI1 → S↑	0.2340	0.2240	2.2652	2.2610
CI1 → S↓	0.2470	0.2334	1.3031	1.2818
A → CO0N↑	0.2787	0.3326	2.7172	2.7117
A → CO0N↓	0.2756	0.3375	1.7018	1.6670
B → CO0N↑	0.2354	0.2977	2.9844	2.7105
B → CO0N↓	0.2219	0.2806	1.7003	1.6663
CI0 → CO0N↑	0.0776	0.0662	3.0682	1.6367
CI0 → CO0N↓	0.0525	0.0428	1.7714	0.9147
A → CO1N↑	0.2885	0.3385	3.0418	3.1079
A → CO1N↓	0.2713	0.3376	1.6586	1.6351
B → CO1N↑	0.2553	0.3040	3.0425	3.1078
B → CO1N↓	0.2185	0.2812	1.6497	1.6347
CI1 → CO1N↑	0.0722	0.0580	3.0617	1.5562
CI1 → CO1N↓	0.0490	0.0372	1.7724	0.8295

Cell Description

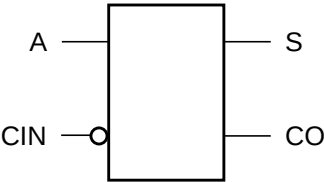
The AHHCIN cell is a half adder that provides the arithmetic sum (S) and carry-out (CO) of the input operand (A) with an active-low carry-in (CIN). The outputs (S, CO) are represented by the logic equations:

$$S = A \oplus \overline{CIN}$$
$$CO = A \bullet \overline{CIN}$$

Functions

A	CIN	S	CO
0	0	1	0
0	1	0	0
1	0	0	1
1	1	1	0

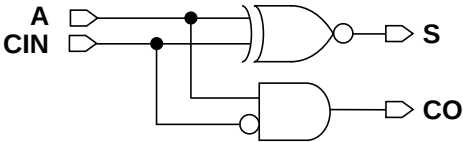
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AHHCINX2	5.04	7.92
AHHCINX4	5.04	9.24

Functional Schematic



AC Power

Pin	Power (μ W/MHz)	
	X2	X4
A	0.0909	0.1114
CIN	0.0762	0.1024

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0081	0.0102
CIN	0.0157	0.0211

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ S $\uparrow$	0.0941	0.0934	3.3230	3.3271
A $\rightarrow$ S $\downarrow$	0.1011	0.0923	1.8679	1.9118
CIN $\rightarrow$ S $\uparrow$	0.0561	0.0552	3.3208	3.3262
CIN $\rightarrow$ S $\downarrow$	0.0734	0.0700	1.9088	1.9293
A $\rightarrow$ CO $\uparrow$	0.0797	0.0732	4.9171	2.4586
A $\rightarrow$ CO $\downarrow$	0.1013	0.0954	1.6649	0.8324
CIN $\rightarrow$ CO $\uparrow$	0.0537	0.0511	4.9147	2.4577
CIN $\rightarrow$ CO $\downarrow$	0.0248	0.0237	1.6551	0.8277

Cell Description

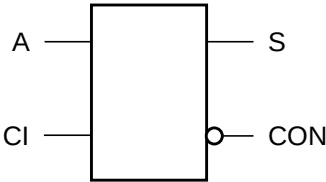
The AHHCON cell is a half adder that provides the arithmetic sum (S) and active-low carry-out (CON) of the input operand (A) with carry-in (CI). The outputs (S, CON) are represented by the logic equations:

$S = A \oplus CI$
 $CON = \overline{A \bullet CI}$

Functions

A	CI	S	CON
0	0	0	1
0	1	1	1
1	0	1	1
1	1	0	0

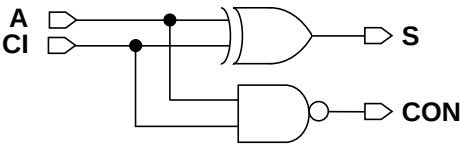
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AHHCONX2	5.04	7.26
AHHCONX4	5.04	8.58

Functional Schematic



AC Power

Pin	Power (μ W/MHz)	
	X2	X4
A	0.0956	0.1188
CI	0.0597	0.0744

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0120	0.0183
CI	0.0170	0.0230

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ S $\uparrow$	0.1008	0.1017	3.3364	3.3332
A $\rightarrow$ S $\downarrow$	0.1085	0.1083	1.9506	1.9478
CI $\rightarrow$ S $\uparrow$	0.0578	0.0567	3.3208	3.3278
CI $\rightarrow$ S $\downarrow$	0.0724	0.0682	1.9067	1.9288
A $\rightarrow$ CON $\uparrow$	0.0387	0.0388	3.0160	1.5084
A $\rightarrow$ CON $\downarrow$	0.0260	0.0244	2.1256	0.9880
CI $\rightarrow$ CON $\uparrow$	0.0310	0.0291	3.0177	1.5090
CI $\rightarrow$ CON $\downarrow$	0.0230	0.0206	2.1258	0.9879

Cell Description

The booth encoder block (BENC) cell performs a 2-bit multiplier recoding per a modified Booth’s algorithm. Each BENC cell examines 3 bits of the multiplier (M0, M1, M2) and generates the appropriate control signals to adjust the multiplicand for subsequent partial product reduction. The outputs (S, A, X2) are represented by the logic equations:

$S = \overline{M2} \bullet (M1 + M0)$

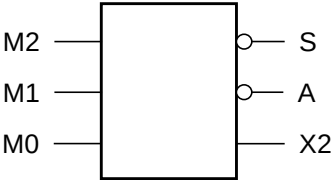
$A = M2 \bullet (\overline{M1} + \overline{M0})$

$X2 = \overline{M1} \oplus \overline{M0}$

Functions

M2	M1	M0	X2	A	S
0	0	0	x	1	1
0	0	1	0	0	1
0	1	0	0	0	1
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	1	0
1	1	1	x	1	1

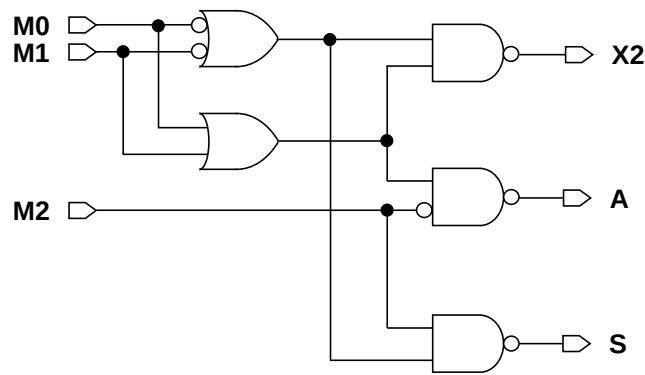
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
BENCX1	5.04	20.46
BENCX2	5.04	27.06
BENCX4	5.04	40.26

Functional Schematic



AC Power

Pin	Power (μW/MHz)		
	X1	X2	X4
M2	0.0760	0.1308	0.2423
M1	0.1581	0.2776	0.5496
M0	0.1762	0.3046	0.5644

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
M2	0.0074	0.0096	0.0155
M1	0.0102	0.0172	0.0303
M0	0.0092	0.0154	0.0243

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			K _{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
M2 → A↑	0.2309	0.2285	0.2161	1.1141	0.5667	0.2803
M2 → A↓	0.2057	0.2190	0.2230	0.6944	0.3442	0.1708
M1 → A↑	0.2474	0.2289	0.2066	1.1139	0.5665	0.2802
M1 → A↓	0.1769	0.1719	0.1864	0.6954	0.3446	0.1710
M0 → A↑	0.2384	0.2168	0.1932	1.1137	0.5665	0.2801
M0 → A↓	0.1694	0.1620	0.1720	0.6954	0.3445	0.1710
M2 → S↑	0.1759	0.1735	0.1683	1.1844	0.5665	0.2784
M2 → S↓	0.1379	0.1301	0.1389	0.6908	0.3431	0.1637
M1 → S↑	0.3103	0.2726	0.2516	1.1841	0.5663	0.2784
M1 → S↓	0.3011	0.2702	0.2815	0.6917	0.3436	0.1639
M0 → S↑	0.2776	0.2423	0.2350	1.1844	0.5665	0.2783
M0 → S↓	0.2412	0.2369	0.2362	0.6913	0.3435	0.1638
M1 → X2↑	0.2139	0.1782	0.1868	1.1128	0.5561	0.2529
M1 → X2↓	0.2582	0.2286	0.2165	0.7430	0.4783	0.1811
M0 → X2↑	0.2580	0.2170	0.2266	1.1132	0.5555	0.2529
M0 → X2↓	0.2682	0.2584	0.2535	0.7431	0.4783	0.1811

Cell Description

The BMX cell performs the shifting and 2's complement inversion of the multiplicand bits (M1 and M0) based on the recode control signals (X2, A, and S) from the BENC cell. The partial product output (PP) is represented by the logic equation:

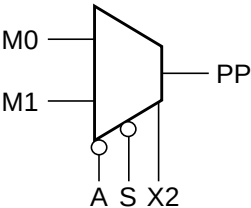
$$PP = X2 \bullet ((M0 \bullet \bar{A}) + (\bar{M0} \bullet \bar{S})) + \bar{X2} \bullet ((M1 \bullet \bar{A}) + (\bar{M1} \bullet \bar{S}))$$

Functions¹

X2	A	S	M0	M1	PP
0	0	0	x	x	x
0	0	1	x	0	0
0	0	1	x	1	1
0	1	0	x	0	1
0	1	0	x	1	0
0	1	1	x	x	0
1	0	0	x	x	x
1	0	1	0	x	0
1	0	1	1	x	1
1	1	0	0	x	1
1	1	0	1	x	0
1	1	1	x	x	0

¹ Shaded areas represent illegal conditions.

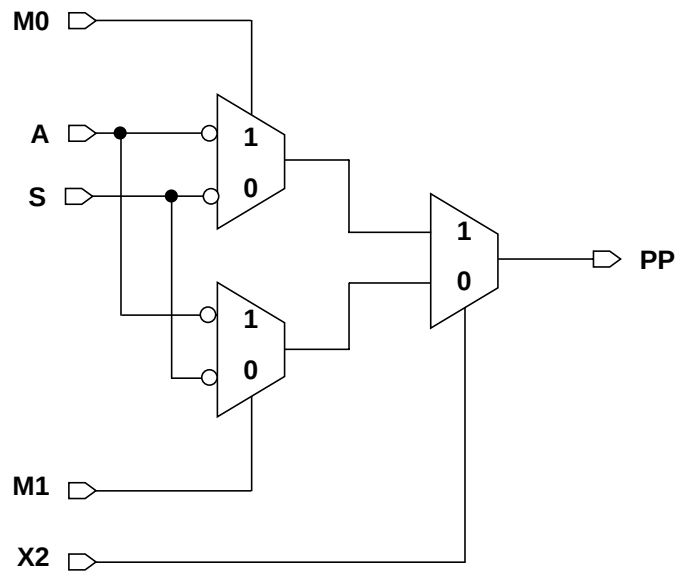
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
BMXX1	5.04	12.54

Functional Schematic



AC Power

Pin	Power (μW/MHz)
	X1
X2	0.0395
M0	0.0573
A	0.0590
S	0.0775
M1	0.0500

Pin Capacitance

Pin	Capacitance (pF)
	X1
X2	0.0033
M0	0.0056
A	0.0045
S	0.0045
M1	0.0052

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)	K _{load} (ns/pF)
	X1	X1
X2 → PP↑	0.1423	4.5294
X2 → PP↓	0.1295	2.4577
M0 → PP↑	0.2002	4.5336
M0 → PP↓	0.2488	2.4611
A → PP↑	0.2208	4.5336
A → PP↓	0.2160	2.4609
S → PP↑	0.2416	4.5340
S → PP↓	0.2307	2.4614
M1 → PP↑	0.1853	4.5338
M1 → PP↓	0.2251	2.4527

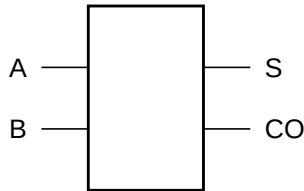
Cell Description

The CMPR22 cell provides the arithmetic sum (S) and carry out (CO) of two operands (A, B). The two outputs (S, CO) are represented by the logic equations:

$S = (\bar{A} \bullet B) + (A \bullet \bar{B})$

$CO = A \bullet B$

Logic Symbol



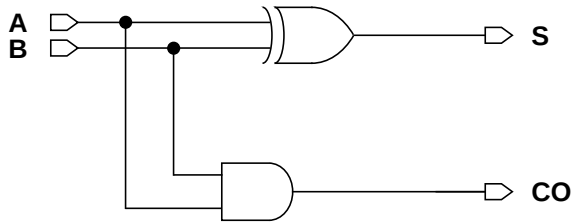
Functions

A	B	S	CO
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
CMPR22X1	5.04	7.92

Functional Schematic



AC Power

Pin	Power (μ W/MHz)
	X1
A	0.1063
B	0.0595

Pin Capacitance

Pin	Capacitance (pF)
	X1
A	0.0108
B	0.0086

Delays (25°C, 1.8V, Typical Process)

Description	Intrinsic Delay (ns)	K _{load} (ns/pF)
	X1	X1
A → S↑	0.0883	2.8422
A → S↓	0.0943	1.7254
B → S↑	0.0589	2.8254
B → S↓	0.0789	1.6436
A → CO↑	0.0895	4.5348
A → CO↓	0.1276	2.5790
B → CO↑	0.0898	4.5349
B → CO↓	0.1219	2.5760

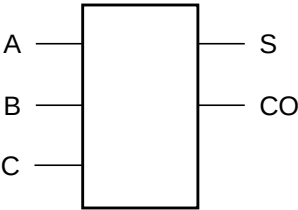
Cell Description

The CMPR32 cell takes in 3 bits of the partial product (A, B, C) and compresses them into 2-bits of partial product (S, CO). The two outputs (S, CO) are represented by the logic equations:

$$S = A \oplus B \oplus C$$

$$CO = (A \bullet B) + (A \bullet C) + (B \bullet C)$$

Logic Symbol



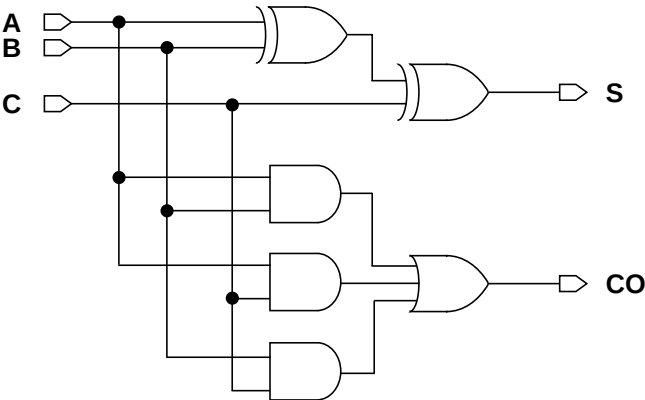
Functions

A	B	C	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
CMPR32X1	5.04	13.86

Functional Schematic



AC Power

Pin	Power (μ W/MHz)
	X1
A	0.1126
B	0.1450
C	0.0608

Pin Capacitance

Pin	Capacitance (pF)
	X1
A	0.0072
B	0.0071
C	0.0065

Delays at 25°C, 1.8V, Typical Process

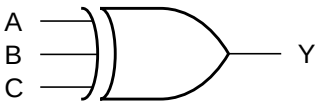
Description	Intrinsic Delay (ns)	K_{load} (ns/pF)
	X1	X1
A $\rightarrow$ S $\uparrow$	0.2193	4.5458
A $\rightarrow$ S $\downarrow$	0.2975	2.5634
B $\rightarrow$ S $\uparrow$	0.2566	4.5491
B $\rightarrow$ S $\downarrow$	0.3410	2.5633
C $\rightarrow$ S $\uparrow$	0.1755	4.5461
C $\rightarrow$ S $\downarrow$	0.1553	2.5665
A $\rightarrow$ CO $\uparrow$	0.2725	4.5263
A $\rightarrow$ CO $\downarrow$	0.2665	2.4760
B $\rightarrow$ CO $\uparrow$	0.3154	4.5259
B $\rightarrow$ CO $\downarrow$	0.2862	2.4285
C $\rightarrow$ CO $\uparrow$	0.1400	4.5394
C $\rightarrow$ CO $\downarrow$	0.1838	2.4977

Cell Description

The XOR3 cell provides a logical EXCLUSIVE OR of three inputs (A,B,C). The output (Y) is represented by the following equation:

$Y = A \oplus B \oplus C$

Logic Symbol



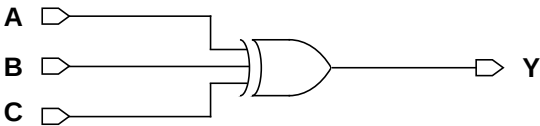
Functions

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
XOR3X2	5.04	11.88
XOR3X4	5.04	19.80

Functional Schematic



AC Power

Pin	Power (μ W/MHz)	
	X2	X4
A	0.1398	0.2672
B	0.1240	0.2288
C	0.0587	0.1084

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0069	0.0138
B	0.0151	0.0286
C	0.0087	0.0164

Delays at 25°C, 1.8V, Typical Process

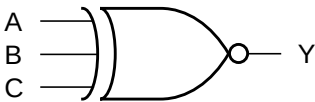
Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ Y $\uparrow$	0.2666	0.2516	2.2171	1.1087
A $\rightarrow$ Y $\downarrow$	0.3016	0.2812	1.3786	0.6854
B $\rightarrow$ Y $\uparrow$	0.1769	0.1644	2.2166	1.1086
B $\rightarrow$ Y $\downarrow$	0.2268	0.2153	1.3688	0.6835
C $\rightarrow$ Y $\uparrow$	0.1464	0.1380	2.2160	1.1082
C $\rightarrow$ Y $\downarrow$	0.1532	0.1460	1.3640	0.6812

Cell Description

The XNOR3 cell provides a logical EXCLUSIVE NOR of three inputs (A,B,C). The output (Y) is represented by the following equation:

$$Y = \overline{A \oplus B \oplus C}$$

Logic Symbol



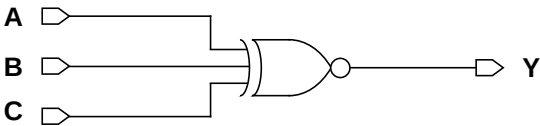
Functions

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
XNOR3X2	5.04	11.88
XNOR3X4	5.04	19.80

Functional Schematic



AC Power

Pin	Power (μ W/MHz)	
	X2	X4
A	0.1382	0.2628
B	0.1197	0.2216
C	0.0579	0.1083

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0069	0.0138
B	0.0151	0.0285
C	0.0051	0.0093

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ Y $\uparrow$	0.2657	0.2518	2.2167	1.1087
A $\rightarrow$ Y $\downarrow$	0.3041	0.2846	1.3764	0.6848
B $\rightarrow$ Y $\uparrow$	0.1779	0.1659	2.2166	1.1086
B $\rightarrow$ Y $\downarrow$	0.2239	0.2116	1.3693	0.6838
C $\rightarrow$ Y $\uparrow$	0.1466	0.1377	2.2155	1.1081
C $\rightarrow$ Y $\downarrow$	0.1512	0.1461	1.3668	0.6813

Cell Description

The CMPR42 cell takes in 4 bits of the partial product (A, B, C, D) and compresses them into 2-bits of partial product (S, CO). The cell requires an intermediate carry-in input (ICI) from the n-1 compressor and an intermediate carry-out output (CO) to the n+1 compressor. The CMPR42 cell also contains an internal sum (IS). The internal sum (IS), carry-in output (ICO), and the two outputs (S, CO) are represented by the logic equations:

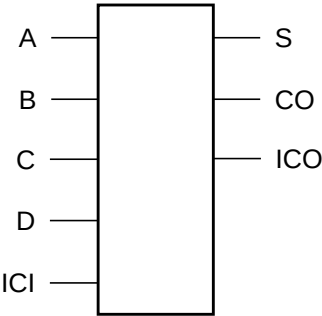
$$IS = A \oplus B \oplus C$$
$$ICO = (A \bullet B) + (A \bullet C) + (B \bullet C)$$
$$S = IS \oplus D \oplus ICI$$
$$CO = (IS \bullet D) + (IS \bullet ICI) + (D \bullet ICI)$$

Functions

A	B	C	IS	ICO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

IS	D	ICI	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

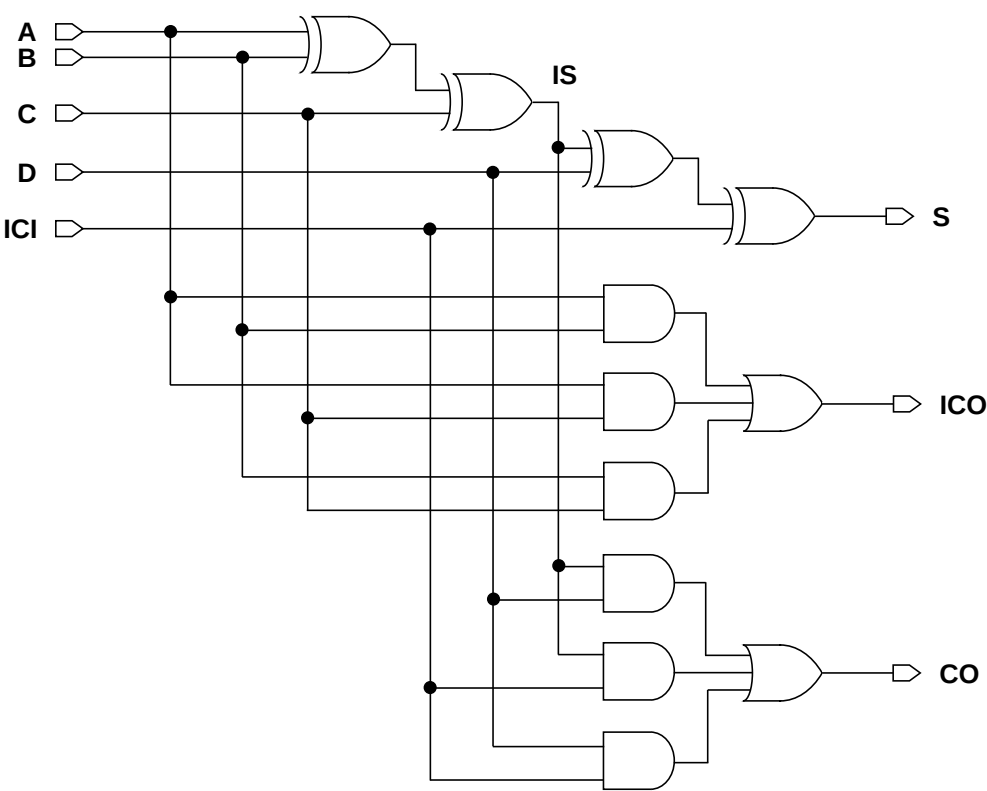
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
CMPR42X1	5.04	22.44
CMPR42X2	5.04	26.40

Functional Schematic



AC Power

Pin	Power (μ W/MHz)	
	X1	X2
A	0.1625	0.2795
B	0.1566	0.2672
C	0.1474	0.2555
D	0.1211	0.2054
ICI	0.0644	0.1128

Pin Capacitance

Pin	Capacitance (pF)	
	X1	X2
A	0.0085	0.0163
B	0.0085	0.0164
C	0.0078	0.0132
D	0.0046	0.0096
ICI	0.0028	0.0061

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)		K _{load} (ns/pF)	
	X1	X2	X1	X2
A → S↑	0.5331	0.5386	4.5444	2.2541
A → S↓	0.6369	0.6355	2.4557	1.3605
B → S↑	0.4656	0.4476	4.5443	2.2539
B → S↓	0.5695	0.5449	2.4556	1.3605
C → S↑	0.4165	0.4052	4.5434	2.2538
C → S↓	0.5221	0.5046	2.4558	1.3606
D → S↑	0.4196	0.3382	4.5399	2.2505
D → S↓	0.4634	0.4096	2.4552	1.3604
ICI → S↑	0.2468	0.1878	4.5424	2.2522
ICI → S↓	0.2601	0.2107	2.4569	1.3608
A → ICO↑	0.1276	0.0903	4.5452	2.2532
A → ICO↓	0.1895	0.1664	2.4583	1.2998
B → ICO↑	0.1268	0.0923	4.5454	2.2534
B → ICO↓	0.1783	0.1604	2.4591	1.3004
C → ICO↑	0.1104	0.0817	4.5408	2.2516
C → ICO↓	0.1550	0.1412	2.4775	1.3047
A → CO↑	0.5178	0.5207	4.5447	2.2523
A → CO↓	0.6046	0.6074	2.4726	1.3000
B → CO↑	0.4587	0.4564	4.5368	2.2523
B → CO↓	0.5375	0.5430	2.4726	1.3000
C → CO↑	0.4074	0.3912	4.5445	2.2522
C → CO↓	0.4520	0.4273	2.4729	1.3001
D → CO↑	0.3775	0.3186	4.5379	2.2503
D → CO↓	0.3785	0.3210	2.4710	1.2938
ICI → CO↑	0.1523	0.1017	4.5469	2.2526
ICI → CO↓	0.1907	0.1602	2.5117	1.3143

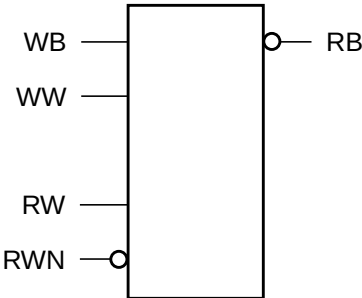
Cell Description

The RF1R1W register file cell is an active-high D-type transparent latch with an active-high tri-state output. The output (RB) is inverted.

Functions for Write Operations

WW	WB	Q[n+1]
0	0	0
0	1	Q[n]
0	0	Q[n]
0	1	Q[n]
1	0	0
1	1	1
1	0	Q[n]
1	1	1

Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
RF1R1WX2	5.04	6.60

Functions for Read Operations¹

RW	RWN	Q	RB
0	0	0	1
0	0	1	Hi-Z
0	1	0	Hi-Z
0	1	1	Hi-Z
1	0	0	1
1	0	1	0
1	1	0	Hi-Z
1	1	1	0

¹ Shaded areas represent operations that are legal only during RW/RWN transitions.

Pin	Power (μ W/MHz)
	X2
WW	0.0352
WB	0.0372
RW	0.0057
RB	0.0239

Pin	Capacitance (pF)
	X2
WW	0.0052
WB	0.0023
RW	0.0032
RWN	0.0019
RB	0.0085

Description	Intrinsic Delay (ns)	K _{load} (ns/pF)
	X2	X2
WW → RB↑	0.2900	4.3921
WW → RB↓	0.1804	1.9397
WB → RB↑	0.2686	4.3920
WB → RB↓	0.1971	1.9397
RW → RB↑	0.0322	4.3877
RW → RB↓	0.0145	1.9331

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)
		X2
WW	minpwh	0.0898
WB	setup $\uparrow$ $\rightarrow$ WW	0.1094
	setup $\downarrow$ $\rightarrow$ WW	0.1445
	hold $\uparrow$ $\rightarrow$ WW	-0.0977
	hold $\downarrow$ $\rightarrow$ WW	-0.1367

Cell Description

The RF2R1W register file cell is an active-high D-type transparent latch with two independently controlled, active-high tri-state outputs. The cell has two read ports and one write port. The outputs (R1B, R2B) are inverted.

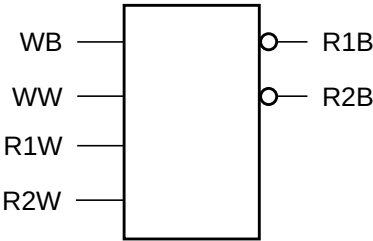
Functions for Write Operations

WW	WB	Q[n+1]
0	0	Q[n]
0	1	Q[n]
1	0	0
1	1	1

Functions for Read Operations

R1W/ R2W	Q	R1B/ R2B
0	0	Hi-Z
0	1	Hi-Z
1	0	1
1	1	0

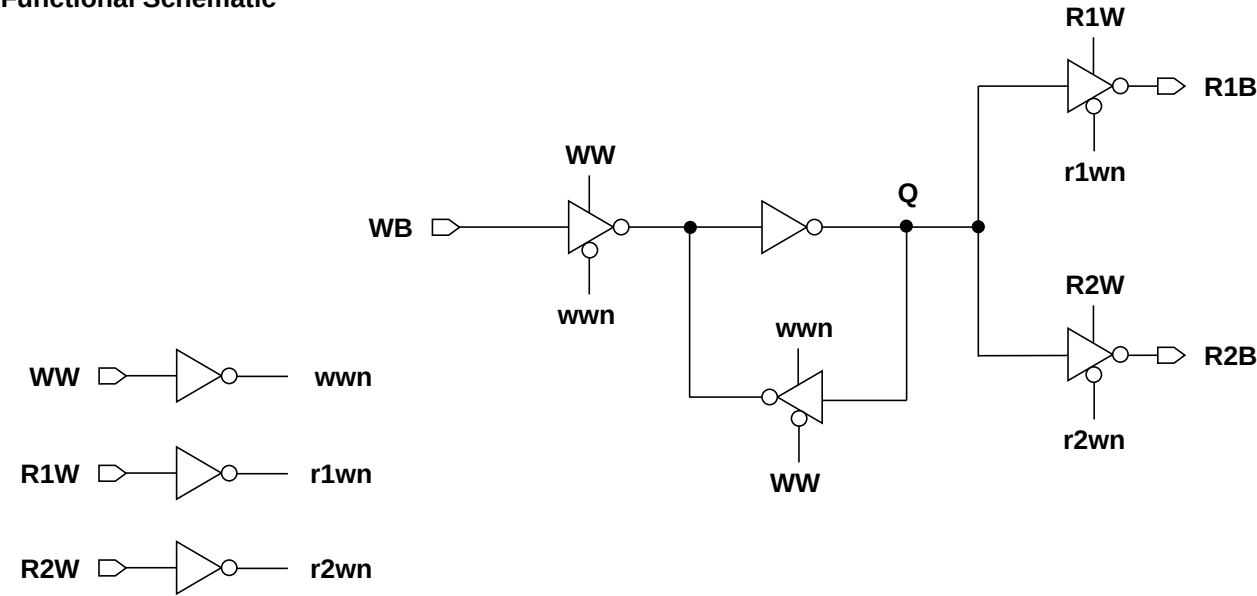
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
RF2R1WX2	5.04	10.56

Functional Schematic



AC Power

Pin	Power (μ W/MHz)
	X2
WB	0.0513
WW	0.0130
R1W	0.0127
R2W	0.0118
R1B	0.0941

Pin Capacitance

Pin	Capacitance (pF)
	X2
WB	0.0024
WW	0.0044
R1W	0.0035
R2W	0.0046
R1B	0.0047
R2B	0.0046

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)	K _{load} (ns/pF)
	X2	X2
WB → R1B↑	0.3065	4.3932
WB → R1B↓	0.2354	1.9463
WW → R1B↑	0.3240	4.3933
WW → R1B↓	0.2199	1.9463
R1W → R1B↑	0.0706	4.3868
R1W → R1B↓	0.0164	1.9326
WB → R2B↑	0.3060	4.3933
WB → R2B↓	0.2351	1.9464
WW → R2B↑	0.3235	4.3934
WW → R2B↓	0.2196	1.9464
R2W → R2B↑	0.0699	4.3869
R2W → R2B↓	0.0163	1.9325

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)
		X2
WB	setup↑ → WW	0.1289
	setup↓ → WW	0.1641
	hold↑ → WW	-0.1094
	hold↓ → WW	-0.1445
WW	minpwh	0.1140

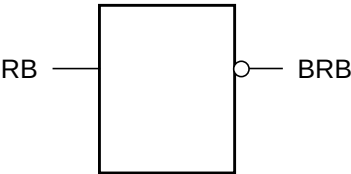
Cell Description

The RFRD output buffer has a “keeper” function that holds the input and output ports at the present level when the input (RB) is in a state of high-impedance.

Functions

RB	BRB
0	1
1	0
Hi-Z	Keep

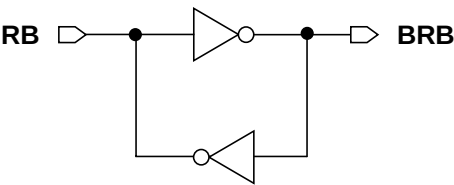
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
RFRDX1	5.04	3.30
RFRDX2	5.04	3.30
RFRDX4	5.04	3.96

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)		
	X1	X2	X4
RB	0.0288	0.0358	0.0529

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
RB	0.0619	0.0744	0.0829

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
RB $\rightarrow$ BRB $\uparrow$	0.0492	0.0339	0.0254	4.5323	2.2477	1.1236
RB $\rightarrow$ BRB $\downarrow$	0.0293	0.0223	0.0161	2.3691	1.2648	0.6332